Beyond the Status Quo, Revisited

A Computational Re-Examination of Lifecycle Asset Allocation, with Twenty-seven Further Studies

A replication and extension study

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Abstract

We reproduce and extend the central result of Anarkulova, Cederburg and O'Doherty (2023, 2024): that a lifecycle investor holding only equities, split evenly between domestic and international markets, delivers higher certainty-equivalent retirement consumption than the age-declining glide path embedded in target-date funds. Working from a 16-country developed-market panel of real asset returns spanning 1890–2020 (2,010 country-years), we build a calendar-joint stationary block bootstrap that samples whole (country, window) blocks and therefore preserves cross-asset covariance, long-horizon persistence and the fat tails of the historical record. Simulating 100,000 lifetimes per strategy under CRRA and Epstein–Zin preferences, we recover the qualitative result: the all-equity portfolio leads the target-date fund by 11.2% and a 60/40 portfolio by 20.3% in certainty-equivalent consumption at $\gamma = 5$, while cutting the probability of exhausting the portfolio from 22% to 12%. The ranking survives 132 parameter settings across 9 dimensions with 0 reversals. Sustainable withdrawal rates are far below the four-percent convention on every strategy (3.2% for all-equity, 2.1% for the target-date fund at a five-percent ruin tolerance). We then push past replication with twenty-seven further studies, taken in the order the paper presents them. Twelve ask whether the result holds up. Re-running the lifetimes the panel actually contains — one country, one birth year, sixty-eight years in the order they happened, with no resampling of any kind — leaves all-international ahead in 13 of 16 markets, so the ordering is not a property of the block bootstrap; that record holds 632 complete lifetimes but only 16 independent ones, and a bootstrap over countries puts the realised lead at [7.9, 31.3]. Deleting each country in turn and rebuilding the panel around its absence leaves the ranking intact in all 16 cases; the same runs form a delete-one jackknife that puts a standard error of 2.92 points on the 5.79-point lead all-international holds over the 50/50 split, a 95% interval of [0.08, 11.51] that is far wider than the Monte Carlo error and is the precision a reader should carry to every other number here — including the headline ones above. Rebuilding the international sleeve under five weighting schemes — equal, real GDP, population, GDP per capita and inverse volatility — narrows that lead to 3.71 points at worst without closing it, so the divergence is not manufactured by the equal weighting. Charging fund fees on the panel, which fall unequally because all-international pays the foreign expense ratio on everything and the 50/50 split on half, needs a differential of 114 basis points to cancel the lead — beyond any index-fund pair, though not beyond the tax code: foreign dividend withholding is a differential of exactly that kind that is neither a fee nor a choice, and at the panel's own dividend share the 30% statutory rate a non-resident pays is worth 115 basis points a year on the international sleeve alone. The 50/50 split overtakes all-international at a withholding rate of 29.2%, just inside that statutory rate, and the certainty-equivalent-maximising home share walks from 10% to 40% across the grid. Currency hedging the international leg loses certainty-equivalent consumption at every ratio tested, even when the hedge is free. Conditioning on what inflation has just done — a state variable an investor observes more reliably than a dividend yield, and one bearing directly on the legs whose payments are fixed in nominal terms — finds the mechanism exactly where theory puts it and then finds it does not survive a lifetime: a high-inflation start costs the bond and bill legs -5.19 points a year over the following year, against

-2.32 for equity, but by 30 years the bond effect has reversed to +3.54 points, and neither the headline ranking nor the optimal equity and home-bias shares move across the inflation terciles. Reading that state variable at the retirement date instead of the birth date turns the null into a result: retiring into the high-inflation third is worth -8.59% of certainty-equivalent retirement consumption against +0.10% for the same lifetimes bucketed by the inflation they began at, 82 times larger and of the opposite sign — the lifetime null was a statement about the horizon rather than about inflation. Correlating labour income with the home market — the textbook reason to hold less of it, and assumed away everywhere else — moves the lead +5.90 points across a correlation range running past anything a labour economist would defend, without changing the order, though correlating it with world equity rather than with the home market specifically keeps only +1.98 points of that, so most of the effect is about home bias and not about human capital being risky; replacing the certain ninety-third birthday with a Gompertz lifespan moves the lead by at most 0.15 points and leaves the ranking identical. The assumption that turns out to matter most is the one the panel makes silently: every result outside one section pays the US social-security schedule in all sixteen countries. Replacing it with Australia's — a means-tested Age Pension alongside a compulsory 12% Superannuation Guarantee that sits on top of voluntary saving rather than instead of it — raises mean retirement consumption +27% and lowers certainty-equivalent consumption -41%, because the fifth percentile falls -27%: compulsory saving buys a larger portfolio and the means test removes the floor it would have sat on. Crossing the two features shows the contribution rate alone is worth +23% and the assets test is what takes it back. It also reverses the allocation ranking — the only place in this paper where that happens: the lead of 5.73% becomes -6.42% and the best strategy becomes the target-date fund, because inside the assets-tested band a dollar of extra wealth costs more pension a year than any asset here reliably earns, so the de-risking schedule is rewarded for staying small. Five ask whether a better portfolio exists inside the same asset menu, and whether solving for one survives data it did not see. Solving the glide path directly by coordinate ascent under common random numbers reproduces the all-equity corner rather than an interior optimum; freeing all four portfolio weights at every age — 204 parameters on the simplex — adds 0.30% over the best fixed benchmark while still producing no glide path; and relaxing the long-only constraint, borrowing to invest is worth +5.10% at a zero borrowing spread over the real bill rate but decays quickly in that spread, breaking even by 3.00%, with the optimiser levering a diversified portfolio rather than a concentrated one. Splitting the record in half and scoring each solved schedule on the half it was never fitted to leaves 2 of 4 ahead of the best constant mix — the ones fitted to the turbulent first half and applied forward (+0.33%), not the ones fitted to the calm post-war half and applied back (-1.25%); the strategy that transfers is a fixed allocation held unchanged for a lifetime. Charging that solved schedule for the trades it makes points the same way: it turns over 8% of the portfolio a year against nothing for a single-asset holding and about three percent for a fixed mix, and its 1.07% edge over the best fixed portfolio is gone by 173 basis points of one-way trading cost. Two widen the menu. Adding housing — de-smoothed to undo the appraisal lag the published index carries — earns 50% of the portfolio when it is free to hold and drops out entirely at an annual holding cost of 4.6%; financing that house with a mortgage — the one asset an ordinary household can borrow against at close to the government's rate — produces a loan-to-value schedule that declines with age, 75% while working against 49% in retirement, reproducing from optimisation the pattern households follow in practice. The last eight leave the portfolio alone and search the rest of the plan, in the order a life meets it. Conditioning a lifetime on how expensive its market was at the moment it began — using the trailing dividend yield an investor could observe, ranked against tercile boundaries computed only from country-years that had already happened — leaves the ranking intact in all 3 valuation buckets while moving the level: lifetimes begun in the dearest third reach retirement with less and run out of money more often than those begun in the cheapest. Conditioning the savings rate on the funded ratio is worth 4.4% once every candidate signal is scored on a common basis, and a deep decomposition of that signal — across functional form, target definition, asymmetry, feasibility bands and eight competing state variables — finds that the strongest available signal is not the portfolio at all but the investor's own pay cheque (6.4%). Making the retirement date a wealth-triggered decision rather than a birthday is worth 3.0% against a date matched on the same mean retirement age, while the decade around that date explains 5.9% of the variation in retirement outcomes — a lottery no allocation rule can diversify away. And comparing 8 families of retirement spending rule, each on its own optimised rate, spreads

certainty-equivalent consumption by 20% between the best (Amortisation) and the worst (Constant real) — a range comparable to the gap between the best and worst allocation strategies. The paper's principal limitation is the breadth of the panel: 16 developed markets, so the international leg spans 15 foreign markets, all advanced economies with long recorded histories that survived. The jackknife interval above is what that breadth costs in precision; what it costs in generality cannot be measured from inside the sample, because deleting a country that is present says nothing about one that was never there.

Keywords: lifecycle asset allocation; target-date funds; block bootstrap; certainty-equivalent consumption; safe withdrawal rate; sequence-of-returns risk; glide path; retirement timing; savings rate; international diversification.

JEL classification: G11, G51, D14, D15, J26, C15.

Contents

1. Introduction

The default investment vehicle of the modern retirement system is the target-date fund. It embodies a single, rarely-examined proposition: that an investor should hold less equity as they age, sliding from a growth portfolio in their twenties toward a bond-heavy portfolio at retirement. That proposition is not a market observation. It is a theoretical inheritance, descended from Samuelson (1969) and Merton (1969) via the human-capital arguments of Bodie, Merton and Samuelson (1992), and it has been institutionalised across trillions of dollars of default-enrolled savings.

Anarkulova, Cederburg and O'Doherty (henceforth ACO) challenged it directly. Their argument is not that equities have higher expected returns — everyone agrees on that — but that the standard evidence against holding them is an artefact of how returns are simulated. Draw returns independently and identically distributed from a normal-ish distribution fitted to post-war US data, and equities look dangerous over short horizons and safe over long ones in exactly the way textbook time diversification predicts. Draw them instead in blocks from the full international historical record — including the markets that closed, the decades that were lost, and the countries whose real returns were negative for a generation — and the shape of the problem changes. ACO's conclusion is that a fixed all-equity portfolio, split between domestic and international markets, beats the glide path on almost every metric an investor would care about.

This paper does three things. First, it reproduces that result from scratch on an independently constructed panel, with the full diagnostic apparatus made visible rather than asserted. Second, it stress-tests the result across a large parameter space, so that the reader can see where the conclusion is robust and where it is thin. Third — and this is where most of the length lies — it takes twenty-seven questions that the original design leaves open and works each of them out, several of which produce results that run against the intuition that motivated them.

1.1 What we find

The replication succeeds. On a 16-country panel covering 1890–2020 (2,010 country-years), the 50/50 domestic/international equity portfolio delivers 11.2% higher certainty-equivalent consumption than a target-date glide path, 20.3% higher than a 60/40 portfolio and 18.8% higher than domestic equity alone, at the baseline risk aversion $\gamma = 5$. It does so while *reducing* the probability of running out of money from 22.2% to 11.7%. Across the 21 distributional criteria we test, the all-equity portfolio wins 20 and loses none.

The mechanism is international diversification, not equity risk per se. Domestic-only equity is beaten by the target-date fund on several tail measures. What makes the all-equity portfolio work is that a bad domestic century and a bad international century are not the same century. This is a stronger claim than "stocks beat bonds" and it is the one the international panel is needed to make.

Four percent is not a safe withdrawal rate on this panel. At a five-percent ruin tolerance the sustainable rate is under three percent for every strategy tested, and under two percent for the conservative ones. The four-percent rule is a US-post-war artefact and does not survive a sample that contains the twentieth century as the rest of the world experienced it.

Solving the glide path directly returns the corner. Rather than testing a handful of candidate schedules, we solve for the equity share at every age by coordinate ascent under common random numbers. The solution is not an interior glide path: it sits at or near the all-equity corner for the entire lifecycle, and the deviation profile shows that most of the apparent age-structure in the solved schedule is worth less than a basis point.

Freeing every weight changes almost nothing. Solving the full four-asset simplex at every age — 204 free parameters, with the bond/bill split no longer imposed — beats the best fixed benchmark by 0.30%. The optimiser holds essentially no fixed income at any age, so the restriction the glide-path solve made for convenience turns out not to have been binding.

When you borrow matters more than whether. A *constant* leverage ratio held for life is worth +5.10% when credit is free, breaks even at a spread of 3.00%, and is already under a tenth of a percent by 3.00% — a null result on margin accounts. Let the ratio *decline with age*, which is what Ayres and Nalebuff actually prescribe, and it is worth more at every price of credit we test, including prices at which the constant ratio is worth nothing. The gain does not depend on the per-age detail: a policy with a single free parameter — one ratio while working, unlevered in retirement — keeps most of it. What the optimiser levers is a diversified portfolio rather than a concentrated one, and the cost of the trade is paid in the left tail.

Timing beats allocation. The single decade around a person's retirement date explains more of the variation in their retirement outcome than the choice between any two of the allocation strategies we test. Making the retirement date itself a decision — retire when wealth reaches a multiple of income, rather than on a birthday — is worth 3.0% of certainty-equivalent consumption against a fixed date matched on the same mean retirement age.

The accumulation side has more room than the allocation side. Conditioning how much you save on whether you are ahead of or behind an age-appropriate wealth target is worth more than any of the allocation refinements we test. Decomposing that signal produces the paper's most counter-intuitive result: the best state variable available to a saver is not their portfolio balance but their own labour income relative to its expected path.

Where you start changes what you get, not what you should hold. Conditioning a lifetime on the dividend yield its market offered the year it began — ranked, as a real investor would have had to, against only the history that had already happened — leaves the allocation ranking intact at every starting valuation while moving the level substantially. An investor starting in an expensive market should expect less from the same portfolio, not a different portfolio.

Housing is a serious asset at a serious price. The historical sources measure a fourth asset class this literature usually skips. Corrected for the appraisal smoothing that makes property indices look artificially calm, it earns a large allocation when it is free to hold and loses it entirely at a plausible annual holding cost. The interesting quantity is not whether housing belongs in a portfolio but the cost at which it stops belonging, and that number is small enough that a real owner's costs are decisive.

1.2 What is new here

Relative to the paper being replicated, this study contributes a methodological discipline and twenty-seven substantive extensions. The discipline is a set of comparison rules applied uniformly: every policy is scored against a *matched* baseline that differs from it in exactly one dimension; every optimiser runs under common random numbers so that differences are policy effects rather than sampling noise; every solved schedule is subjected to a deviation profile before any structure in it is described; and every apparent optimum is checked against the boundary of its own search grid. Several of the results below changed sign or magnitude when these checks were applied, and the paper reports the corrected versions with the diagnostics that forced them.

- **Sensitivity.** A tornado analysis over ten parameter dimensions, reporting the full range of the advantage rather than a point estimate.
- **Retirement spending rules.** Eight families of withdrawal policy — constant real, percentage-of-portfolio, guardrails, endowment smoothing, required-minimum-distribution, amortisation, actuarial and a floor-and-ceiling rule — each optimised over its own rate and compared at its own best setting.
- **The optimal glide path.** Direct numerical solution of the age-by-asset schedule, free-form and parametric, with local-optimum checks from multiple restarts.
- **The whole allocation.** The full four-asset weight simplex solved at every year of the lifecycle — domestic equity, international equity, bonds and bills, with nothing held fixed — by two-stage coordinate ascent on

the simplex.

- **Leverage.** The long-only constraint relaxed: the optimal borrowing ratio and the portfolio that goes with it, swept across the price of credit, with a break-even spread and an age-varying leverage schedule.
- **Currency hedging.** A covered-interest-parity hedged international leg, swept by annual hedging cost, yielding a break-even cost and an optimal hedge ratio as functions of that cost.
- **Endogenous retirement timing.** Retirement as a wealth-triggered decision, plus a formal decomposition of the retirement-date lottery.
- **The savings-rate profile.** Solving for the savings rate at every age with the career average pinned, which separates the shape question from the level question the model cannot answer.
- **Conditioning the savings rate.** Making the contribution respond to portfolio state, scored against a constant rate matched on the realised career average.
- **The accumulation signal, decomposed.** Functional form, target definition, response asymmetry, feasibility bands, age windows, eight competing state variables and their pairwise combination.

1.3 Roadmap

The paper is organised so that each block earns the right to the next. Sections 2–4 set up: the literature, the panel and its construction, and the bootstrap, lifecycle model, preference specification and comparison discipline that everything afterwards runs through.

Sections 5–10 are the result and five ways it could be wrong. Section 5 presents the baseline replication. Section 6 asks whether it survives the preference and lifecycle parameters; Section 8 whether it survives the loss of any one country, and what standard error sixteen countries actually support; Section 9 whether it survives the way the international sleeve is weighted — the construction our one divergence from the replicated study most obviously depends on; Section 13 whether it survives the cost of the funds that implement it; and Section 10 whether that sleeve should be currency-hedged at all.

Sections 19–21 ask whether a better portfolio exists inside the same asset menu, by progressively relaxing what is held fixed: the shape of the glide path (Section 19), then every weight at every age (Section 20), then the long-only constraint itself (Section 21). Sections 24–25 widen the menu instead, adding the asset most households actually hold — housing owned outright (Section 24), then mortgaged (Section 25).

Sections 11–30 leave the portfolio alone and search the rest of the plan, in the order a life meets it: the valuation you start at (Section 11), how much you save (Section 26) and what that saving should respond to (Section 27), when you stop (Section 28), and how you draw down (Section 30). Section 34 draws the results together, Section 35 states the limitations candidly, and Section 36 concludes. Four appendices give the full parameter set, the country panel, supplementary tables and the reproduction instructions.

2. Background and Related Work

2.1 The theoretical case for the glide path

The canonical result is negative: in the Samuelson–Merton framework with constant relative risk aversion, i.i.d. returns and no labour income, the optimal share of wealth in risky assets is independent of the investment horizon. Time does not diversify risk; it multiplies it. Any age-declining allocation therefore has to be justified by something outside that framework.

The standard justification is human capital. Bodie, Merton and Samuelson (1992) observe that a young worker holds a large stock of implicit wealth in the form of future labour income, and that if this asset is bond-like then total wealth is already heavily weighted toward safety; the financial portfolio should compensate by holding more equity. As the worker ages, human capital depletes and the financial portfolio should become more conservative to keep total exposure roughly constant. This produces a glide path, and it is the argument the target-date industry rests on.

The argument is coherent but its quantitative force depends on parameters that are difficult to pin down: how bond-like labour income actually is, how it correlates with equity markets, how much of it remains at each age, and what happens to it in exactly the states of the world where equities fail. It also depends on what the return distribution looks like — which is where the argument this paper reproduces enters.

2.2 The empirical challenge

Almost all of the simulation evidence used to calibrate glide paths draws returns from a parametric distribution fitted to a single market — usually the United States — over a period that begins after the Second World War. That sample has three properties that flatter conservative allocations in a specific direction. It excludes the market closures, expropriations and hyperinflations that other developed markets experienced. It excludes the pre-war period in which the US itself behaved less benignly. And by drawing independently it destroys the multi-decade persistence that makes long-horizon outcomes fat-tailed.

ACO's response is to replace the return-generating process. Their sample is 39 developed countries of monthly returns from 1890 to 2023, drawn from Global Financial Data, and their sampling scheme is a stationary block bootstrap with geometric block lengths averaging 120 months, so runs of good and bad decades survive into the simulated lifetimes. Their household is a couple facing random longevity from Social Security Administration mortality tables, saving 10% of labour income, retiring at 65 and drawing 4% of wealth at retirement thereafter, with utility calibrated to De Nardi, French and Jones (2010) at a risk aversion of 3.84. Their conclusion is that a 50/50 domestic/international equity portfolio, held for life, beats the glide path on wealth, retirement consumption, capital preservation and bequests alike.

This paper reconstructs that process independently and asks whether the claim holds. The sampling scheme here is deliberately the same in form — stationary blocks, geometric lengths, a ten-year mean — so that any divergence is attributable to the panel and the lifecycle assumptions rather than to how returns are drawn. Where this study differs, it differs in ways it can state: annual rather than monthly data, sixteen countries rather than thirty-nine, an individual rather than a couple, and a deterministic horizon rather than random longevity.

2.3 The data source

The underlying return data come from the Jordà–Schularick–Taylor Macrohistory Database and the associated "Rate of Return on Everything" project (Jordà, Knoll, Kuvshinov, Schularick and Taylor, 2019). That database assembles annual total returns on equities, long-term government bonds, short-term bills and consumer price inflation for sixteen advanced economies from the late nineteenth century onward. It is the standard source for long-horizon international asset-return work and it is what makes an exercise of this kind possible at all.

It is also the binding constraint on this study: it carries complete equity, bond and bill total returns for 16 countries, and that set is our panel. Section 3.2 describes it and Section 35.1 sets out what its breadth costs.

2.4 Adjacent literatures this paper touches

- **Bootstrap inference for dependent data.** The sampling scheme here is a stationary block bootstrap in the sense of Politis and Romano (1994), generalised so that a block is a (country, window) pair drawn jointly across all five series.

- **Safe withdrawal rates.** The four-percent convention descends from Bengen (1994) and the Trinity study (Cooley, Hubbard and Walz, 1998), both of which are US-only and historical-sequence based. Section 5.4 re-derives the sustainable rate on this panel.
- **Dynamic withdrawal policy.** Guyton and Klinger (2006) guardrails and the actuarial/amortisation family are represented in the spending-rule comparison of Section 30.
- **Bequest motives.** The utility specification uses the shifted power form of De Nardi (2004), which is what allows a zero terminal balance to be evaluated at all under high risk aversion.
- **Sequence-of-returns risk.** Section 28 formalises the folk observation that the decade around retirement dominates, and prices the option value of choosing when to stop working.
- **Lifecycle leverage.** Ayres and Nalebuff (2010) argue that a young investor should borrow to spread market exposure evenly across a lifetime. Section 21 relaxes the long-only constraint and prices that argument against the cost of credit.

3. Data

3.1 What the panel is

The panel is an annual, country-by-year matrix of five real series — domestic equity, international equity, long-term government bonds, short-term bills and consumer price inflation — for 16 developed markets over 1890–2020. It contains 2,010 usable country-years of domestic equity returns, with a median country history of 125 years and a minimum of 115 years.

All series are *real*: nominal total returns are deflated by each country's own consumer price index in the same year, so that a hyperinflation shows up as a catastrophic real return rather than as a spectacular nominal one. This is not a cosmetic choice. Several of the worst episodes in the sample — Germany in the early 1920s, central Europe in the late 1940s — are invisible in nominal space and dominate the left tail in real space.

The equity series across the panel average 6.8% in arithmetic real terms and 4.3% geometrically, with a mean cross-country standard deviation of 22.2%. The gap between the arithmetic and geometric means — over three percentage points — is itself the story: at this level of volatility, the compounded experience of a lifetime investor is far below the average annual return, and any simulation that draws i.i.d. from the arithmetic mean will overstate what a real investor would have received.

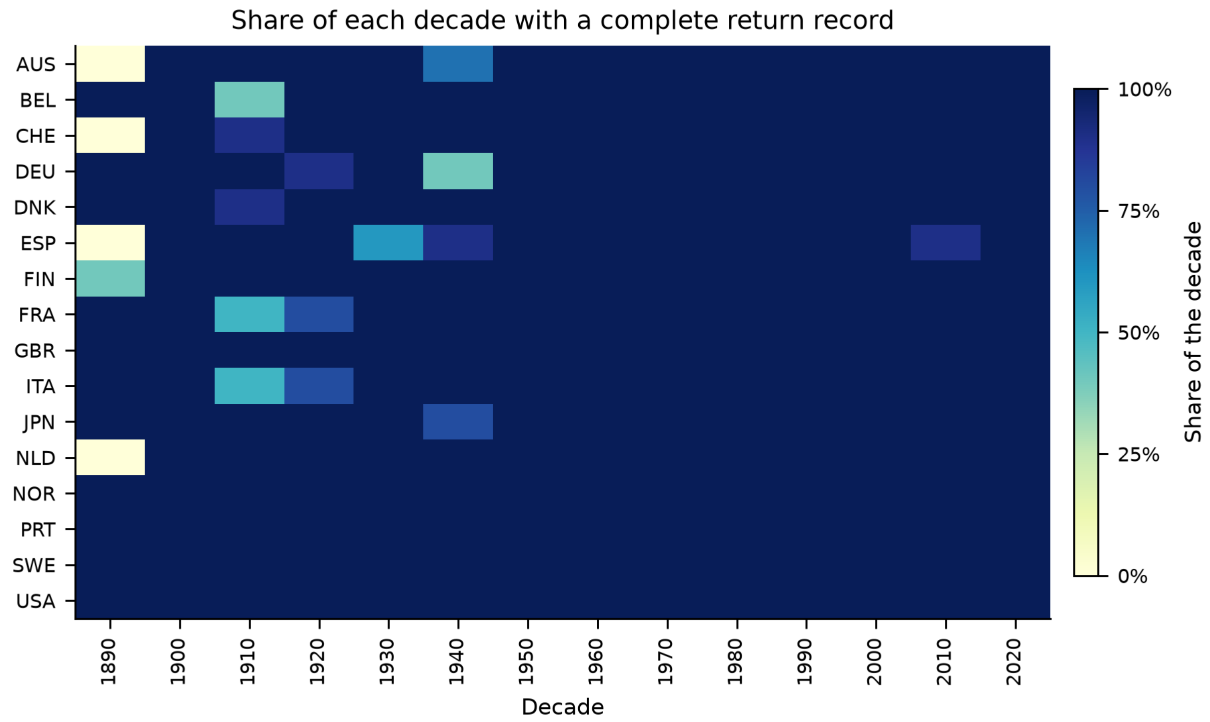

Figure 1.

Share of each decade for which a country has a complete return record. Darker is more complete; the scale is a share so that the final, partial decade compares with the rest. The gaps are not random: they cluster around the two world wars and around the market closures that follow them, which is precisely the period a survivorship-prone sample would drop.

3.2 The country set

The Jordà–Schularick–Taylor database carries complete, independently sourced equity, bond and bill total returns for 16 countries: Australia, Belgium, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States. That set is the panel, and every country-year of every return series in it is a recorded observation.

The wider developed-market universe — IMF advanced economies with an investable domestic equity market, plus Poland — numbers 38. The histories of the other 22 exist in commercial databases such as Global Financial Data and Dimson–Marsh–Staunton, which are proprietary and not redistributable. Working from openly licensed sources, 16 is the extent of the recorded evidence, and Section 3.6.1 reports what the excluded countries do carry.

That breadth is the paper's principal limitation and Section 35.1 develops it. Its most direct consequence is on the international leg, which is a leave-one-out average and therefore spans 15 foreign markets, all advanced economies with long histories. Every statement here about international diversification is a statement about that set.

3.3 Constructing the international leg

An investor in country *i* holding "international equity" does not hold a global index that includes their own market. For each country-year we therefore build the international return as a leave-one-out average across all other countries with data that year, computed in gross terms and in a common currency before being converted back to the home investor's real terms. Leaving the home market out is what makes the domestic and international legs genuinely distinct assets rather than two overlapping slices of the same one.

Three observations in the raw international series are extreme enough to dominate any lifetime that draws them, all of them artefacts of a market reopening after wartime closure with a collapsed price index. We winsorise the international leg at the 0.1th percentile of its own distribution; the affected observations are listed in full in Table 2 so that the reader can judge the intervention rather than take it on trust.

Table 1. Real domestic equity returns by country

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.08
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.10
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	-0.01
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	0.04
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	0.32
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	0.15
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	-0.05
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	0.25
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.13
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.00

Notes. Arithmetic and geometric means are annual real returns deflated by each country's own consumer price index; AR(1) is the first-order autocorrelation of the annual series. Appendix B repeats this with excess kurtosis.

3.4 Cross-asset structure

The correlation structure the bootstrap must preserve is reported below. Domestic and international equity correlate at 0.43, which is high enough that international diversification is not free and low enough that it is worth having. Bonds and bills correlate at 0.72. Inflation correlates negatively with every asset, most strongly with bills (-0.66), which is the mechanism by which cash loses in real terms over long horizons.

Table 2. Pooled cross-asset correlation matrix of annual real returns

	Dom. equity	Intl. equity	Bonds	Bills	Inflation
Domestic equity	1.000	0.429	0.315	0.234	-0.121
International equity	0.429	1.000	0.110	0.117	-0.042
Bonds	0.315	0.110	1.000	0.716	-0.469
Bills	0.234	0.117	0.716	1.000	-0.662
Inflation	-0.121	-0.042	-0.469	-0.662	1.000

Notes. Pooled across all country-years in the panel. These are the moments the block bootstrap is required to reproduce; Table 4 reports how closely it does so.

Table 3. Every winsorised observation in the international leg

Year	Country	Raw return	After winsorisation
1942	Italy	1120%	301%
1945	Japan	-86%	-70%
1946	France	308%	301%

Notes. All three are markets reopening after wartime closure against a collapsed price index. Left unwinsorised, a single one of these draws would dominate the terminal wealth of any lifetime that contained it.

3.5 Market disruptions and the survivorship question

There are 16 contiguous runs of missing data across the countries of the panel, concentrated in the two world wars. The treatment of these gaps is a substantive modelling decision rather than a data-cleaning one. Interpolating across them would smooth away exactly the episodes that motivate the whole exercise; dropping the countries would reintroduce the survivorship bias the panel exists to avoid.

We do neither. Gaps are preserved as gaps, and the bootstrap is constrained to draw only blocks that lie entirely within a contiguous run of available data for the country in question. A country with a six-year wartime hole contributes blocks on either side of it but never across it. The consequence is that the sampler is *less* able to draw from countries with disrupted histories at long block lengths, which we quantify rather than assume: the run-length admissibility statistics are reported in Section 4.1.

3.6 Auditing the data

The database is used here through a redistributed copy rather than obtained from its compilers, which deserves more scepticism than a citation. This section asks whether the file is genuine, reports one test it does not pass, and describes three bodies of recorded data that sit in the sources and stay out of the model.

3.6.1 Why the panel stops at sixteen

The boundary is set by equity. Four countries outside the panel do carry recorded interest-rate histories: long-term yields and short rates in the macro file for Canada and Ireland, Clio-Infra long yields for Austria and New Zealand. Those are enough to build real bond and bill returns — a bond total return follows from a yield and a duration, $r_t = y_{t-1} + D(y_{t-1} - y_t)$; a bill return is a lagged short rate; and deflating both by that country's own price index gives a real return that was measured rather than modelled.

Table 4. Recorded series that exist and still cannot be used

Country	Series	Source	From	To	Observed years
Austria	bond	Clio-Infra bond yields	1966	2010	45
Canada	bill	Jordà–Schularick–Taylor yields and short rates	1935	2020	86
Canada	bond	Jordà–Schularick–Taylor yields and short rates	1891	2020	130
Ireland	bill	Jordà–Schularick–Taylor yields and short rates	1923	2020	98
Ireland	bond	Jordà–Schularick–Taylor yields and short rates	1923	2020	98
New Zealand	bond	Clio-Infra bond yields	1934	2010	74

Notes. Rebuilt from published rates and deflated by each country's own price index. None of these countries has an equity return series in any source available to us, which is why none is in the panel.

None of it gets them into the panel. A lifecycle investor needs a domestic *equity* return, and the macro file carries none for these four — not even the interpolated variants it supplies elsewhere. Rates alone cannot support a portfolio choice. That is why the panel is sixteen countries and not twenty.

3.6.2 The fourth asset class

The macro file carries a fourth asset the source project measured: **housing total returns**, empirical for all 16 observed countries over 1,805 country-years (1891–2020). It is held out of the headline results, which use the same four-asset set as the paper being replicated, and audited here so the reason is visible rather than assumed. Section 24 then puts it into the opportunity set and prices it.

Table 5. Housing total returns: observed, and why they stay out

Country	Years	Mean real	s.d. published	s.d. de-smoothed	Equity s.d.	Autocorr.
Australia	120	6.2%	11.7%	11.7%	16.2%	-0.02
Belgium	119	7.9%	15.2%	21.8%	22.9%	0.34
Switzerland	119	5.7%	6.5%	10.8%	18.9%	0.47
Germany	96	6.6%	8.5%	8.5%	33.5%	-0.13
Denmark	130	8.0%	7.8%	12.1%	17.2%	0.42
Spain	120	5.3%	11.8%	12.2%	20.8%	0.03
Finland	99	9.4%	15.2%	24.0%	30.0%	0.47
France	130	6.3%	10.3%	21.5%	23.1%	0.63
United Kingdom	118	5.4%	9.0%	12.5%	18.7%	0.33
Italy	86	4.7%	9.3%	9.3%	26.6%	-0.05
Japan	75	6.6%	8.1%	12.0%	25.8%	0.46
Netherlands	130	7.4%	9.0%	14.9%	21.2%	0.46
Norway	130	7.8%	8.6%	10.3%	20.1%	0.17
Portugal	73	6.6%	8.5%	11.5%	26.0%	0.33
Sweden	130	8.1%	8.9%	12.8%	20.3%	0.35
United States	130	6.2%	7.9%	7.9%	18.7%	-0.17

Notes. Real annual total returns on the national housing stock, from the same source as the equity series. De-smoothing is first-order Geltner using each country's own lag-one autocorrelation.

The comparison is the one the source project is known for. Median real housing returns are **6.6%** against **6.9%** for the same countries' equity — indistinguishable — at a published standard deviation of 8.9% versus 21.0% — a ratio of 0.43. Equity-like returns at that volatility would look dominant in any mean-variance comparison in this paper, which is precisely why the series deserves scrutiny before adoption.

A house price index is built from appraisals and sparse transactions, and that smooths it. The median lag-one autocorrelation of housing returns is 0.34 against 0.06 for equity, and housing is the more autocorrelated series in 12 of 16 countries. Undoing that to first order raises the median standard deviation from 8.9% to 12.0%: most of the apparent free lunch is a measurement artefact.

Even de-smoothed the series is not investable as written. It is an unlevered, untaxed, frictionless total return on an entire national housing stock, with no transaction costs, no vacancy, no maintenance and none of the single-property concentration a real household bears. Adopting it as a fourth sleeve at face value would therefore overstate what a household can buy. Section 24 adds it anyway, but only after de-smoothing it and charging an explicit annual holding cost — and it is the size of that cost, not the raw return, that decides the answer.

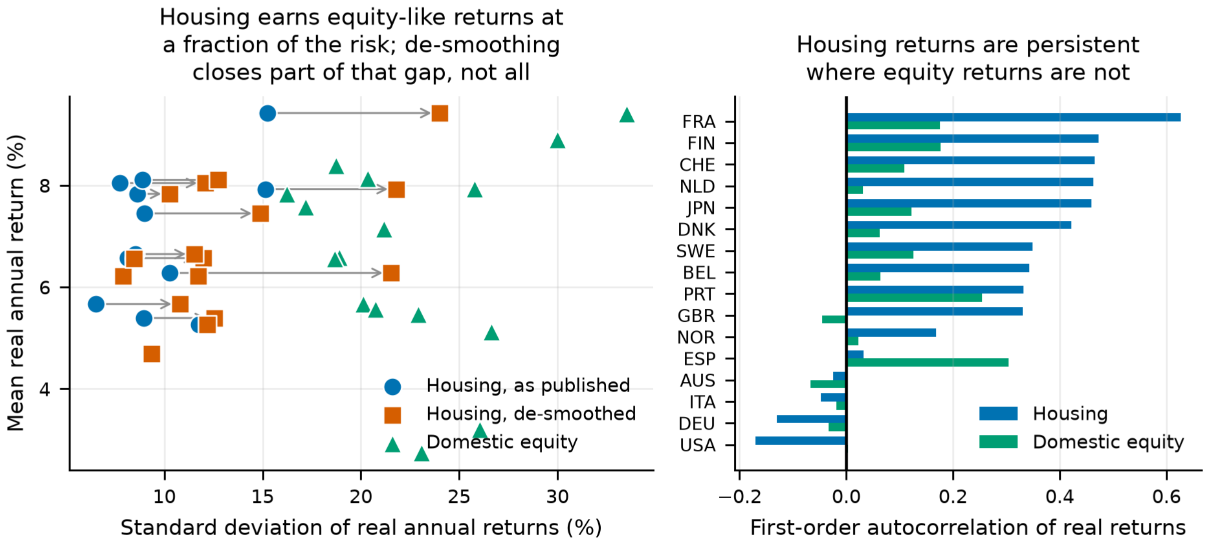

Figure 2.

Left: risk and return by country for housing as published, for housing with its own first-order smoothing undone (arrows), and for that country's domestic equity. Right: the lag-one autocorrelation that does the work. De-smoothing closes part of the volatility gap and not all of it, which is why the holding cost of Section 24 has to carry the rest of the argument.

3.6.3 The series that bears on our income model

The same file carries a nominal wage index for all 18 of its countries, including the two with no return series. Deflated by each country's own consumer prices it measures economy-wide **real wage growth** over 2,269 country-years (1891–2020). The median country compounded real wages at **1.41% a year**, from -0.02% (Belgium) to 3.16% (Finland) — which over the 37-year career we simulate compounds to 1.68×.

Table 6. Measured real wage growth, and what our income profile assumes

Country	Years	Real wage growth p.a.	Excl. war years	s.d.	Compounded over a career
Australia	130	1.43%	1.61%	4.0%	1.72
Belgium	124	-0.02%	1.20%	17.3%	0.99
Canada	130	1.36%	1.27%	2.5%	1.67
Switzerland	130	1.62%	1.75%	3.8%	1.84
Germany	128	0.88%	1.03%	7.9%	1.39
Denmark	130	2.01%	2.05%	6.1%	2.13
Spain	122	1.98%	2.68%	10.3%	2.11
Finland	130	3.16%	3.81%	15.5%	3.26
France	130	1.11%	1.85%	5.2%	1.52
United Kingdom	130	1.29%	1.41%	4.6%	1.63
Ireland	77	2.83%	2.75%	3.2%	2.89
Italy	128	0.98%	2.26%	10.7%	1.45
Japan	130	2.37%	2.66%	13.8%	2.43
Netherlands	130	1.12%	1.18%	4.7%	1.53
Norway	130	1.83%	1.79%	3.8%	1.99
Portugal	130	0.74%	1.03%	9.9%	1.32
Sweden	130	2.43%	2.56%	4.3%	2.49
United States	130	1.40%	1.31%	2.5%	1.69

Notes. Geometric means, because they are what compound across a career. Our deterministic profile implies 1.18% a year over the same span.

The war years carry more of that spread than the economics does, which is why the table reports the series both ways. The two largest observations in the panel are Belgium 1919 (+151%) and Finland 1918 (-66%): a wage index spanning occupation, rationing, suppressed prices and post-war repricing is measuring those at least as much as it is measuring wages. Dropping 1914–1919 and 1939–1946 lifts the median to 1.77% (+0.36 percentage points) and moves the lowest country, Belgium, from -0.02% to 1.20% — from an implausible claim about a century of that country's wages to an unremarkable one. We keep them in the headline number, because a worker alive then lived through them, and note that excluding them only widens the gap we are about to describe.

Our income profile has no term for it. Section 4.3 sets real labour income as a deterministic hump peaking at age 50 at 1.76× starting income and ending the career at 1.54× — an average of 1.18% a year. That is an *age* effect: the progression a worker earns by getting older. Economy-wide wage growth is a different quantity, lifting the whole distribution regardless of age, and in the Cocco–Gomes–Maenhout estimation our profile is taken from the two are separated by construction and are therefore additive. A worker facing both would see roughly 2.61% a year.

That caveat decides the size of the gap rather than its existence: whether the components add depends on how the source profile was estimated, and one fitted to a panel that still carried time effects would already absorb part of the growth. What is not in doubt is that the quantity is measurable, that eighteen countries measure it, and that nothing in our pipeline reads it.

The direction is the one that matters for reading this paper. Understating income growth understates human capital throughout the career, and human capital is the bond-like asset in the standard lifecycle argument — so having less of it *weakens* the case for holding equity when young. The bias runs against our conclusion rather than toward it, as most of the known biases here do. The effect on the savings analysis of Sections 26 and 27 is genuinely ambiguous, because faster income growth raises both what a given savings rate accumulates and the consumption it must replace, and this audit does not resolve it. Re-estimating the income process is a modelling change rather than a data one, so we record it as a quantified limitation rather than apply it silently.

3.6.4 Is the source we kept genuine?

The workbook was obtained from a redistributed copy rather than downloaded from the compilers, so it is audited rather than assumed. Two tests, in opposite directions.

Table 7. The workbook against independently known annual returns

Observation	This workbook	Independently known	Difference	Passes
US equities in the worst year of the slump	-0.4030	-0.4300	0.0270	yes
US equities in the global financial crisis	-0.3875	-0.3700	-0.0175	yes
US equities in the 1974 bear market	-0.2544	-0.2600	0.0056	yes
US equities rebounding in 1933	0.5264	0.5400	-0.0136	yes
US equities in 2013	0.2956	0.3200	-0.0244	yes

Notes. Nominal equity total returns for years whose magnitude is not in dispute. A reconstructed or synthetic file would not reproduce both directions of the 1931-33 swing and the 2008 drawdown.

All 5 of 5 land within tolerance. The second test runs the other way. Equity total return should satisfy the accounting identity $eq\_tr = (1 + \textit{capital gain})(1 + \textit{dividend yield}) - 1$; across 2,163 observations it fails by more than one part in a million 38% of the time. **That failure rate is evidence of authenticity.** The components in the real database are sometimes spliced from different underlying indices, so they do not always reconcile; a file that satisfied the identity everywhere to machine precision would be one whose components had been back-solved from its totals.

3.6.5 One finding that does not pass

The last five years of every equity series are smoother than that country's own history. **All 16 of 16 countries** show a lower standard deviation over 2016–2020 than over 1950–2015, with a median ratio of 0.54. A five-year standard deviation is far too noisy to read one country at a time, which is why this is a sign test: under a fair-coin null the probability of all 16 pointing the same way is **3.1e-05**. Spot-checking the United States sharpens it — the workbook records +14.2% for 2018, a year in which every broad US index fell, and +8.2% for 2019 against roughly +31% for the S&P 500.

The likely explanation is that the redistributed copy was extended past the compilers' own end date by someone else. **We therefore treat 2016–2020 as unverified.** Those five years are under four percent of the panel's country-years and cannot move a 68-year lifecycle result materially, but the finding is recorded rather than left for a reader to discover.

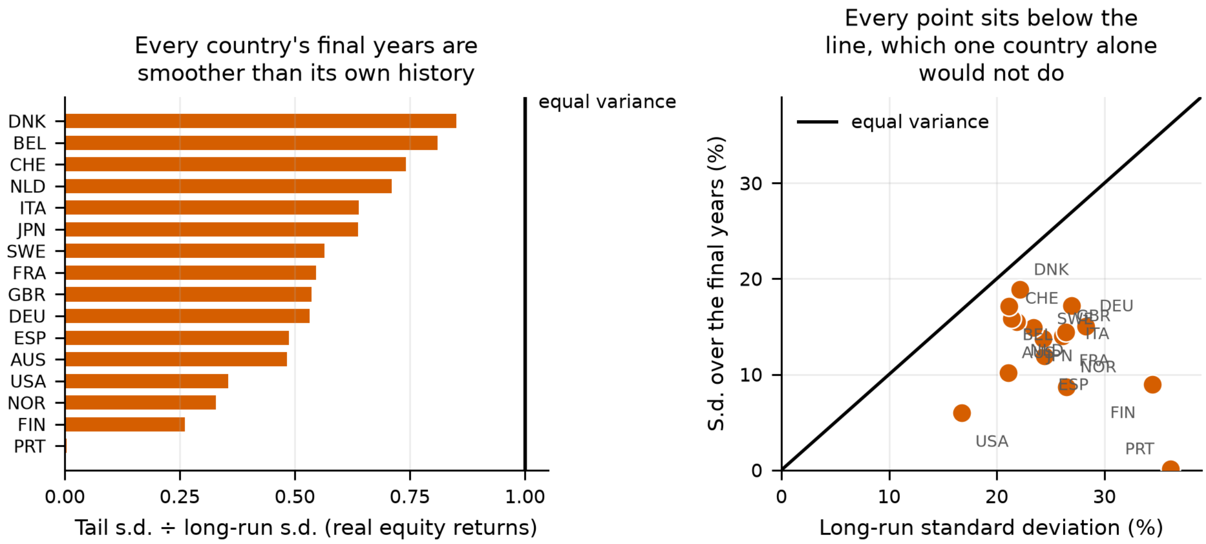

Figure 3.

The variance test, read two ways. Left: each country's standard deviation over the final five years as a ratio of its own 1950–2015 standard deviation. Right: the two standard deviations plotted against each other, with the 45-degree line. One country below the line would be unremarkable; every country below it is a property of the file rather than of the markets.

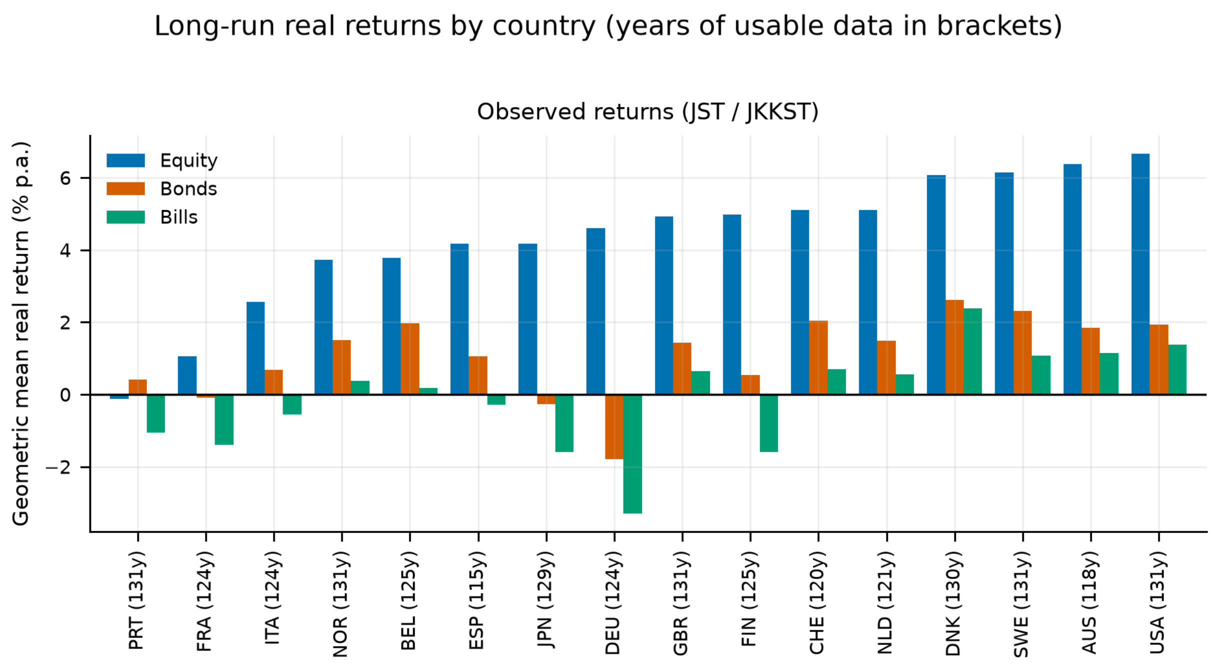

Figure 4.

Distribution of annual real returns by country and asset class. The left tails are the point of the panel: several countries record single-year real equity losses beyond −60%, and these are not outliers to be trimmed but the events a lifecycle investor is exposed to.

4. Methodology

4.1 The cross-country stationary block bootstrap

A simulated lifetime is 68 years long and is assembled from contiguous blocks of history. The sampler is a stationary block bootstrap in the sense of Politis and Romano (1994): block lengths are drawn from a geometric distribution with mean 10 years, truncated to [1, 40], and blocks are laid end to end until the horizon is filled.

Two features distinguish it from a textbook block bootstrap and both matter for the result.

- **Blocks are calendar-joint.** A block is a (country, start-year, length) triple, and the same triple is applied to all five series simultaneously. Domestic equity, international equity, bonds, bills and inflation for a given simulated year therefore come from the same real country in the same real year. This is what preserves the cross-asset correlation matrix of Table 2 without any of it being imposed parametrically.
- **Blocks respect data gaps.** Admissibility is enforced through pre-computed run-lengths: for each (country, year) the sampler knows how many consecutive years of complete data follow, and only draws a block that fits inside a run. A wartime hole is never bridged.

Across the production run the sampler drew 30,848 blocks with a mean realised length of 8.82 years against a target of 10, a median of 6, a ninetieth percentile of 20 and a maximum of 40. The realised mean falls short of the target for a mechanical reason worth stating: truncation at the maximum and the gap-admissibility constraint both bite from above, so the effective block length is always slightly below the nominal one.

The country for a lifetime is drawn once (*per_lifetime*) with probability proportional to the length of that country's usable history (*history*). Both choices are varied in Appendix C: redrawing the country at every block, and weighting countries uniformly, both leave the ranking unchanged. The weighting choice has little room to bite here, because every country carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. The choice would matter on a panel with short histories, where a country covering only one era could otherwise speak as loudly as one covering a century and a half.

Table 8. Does the bootstrap reproduce the panel it samples from?

Series	Bootstrap mean	Panel mean	Gap (bp)	Bootstrap s.d.	Panel s.d.	s.d. ratio	Bootstrap kurtosis
Domestic equity	6.86%	6.77%	8.3	22.87%	22.50%	1.016	2.9
International equity	8.52%	8.33%	19.0	22.27%	21.69%	1.027	42.5
Bonds	2.02%	1.98%	3.9	11.51%	11.81%	0.974	6.0
Bills	0.58%	0.44%	14.0	7.37%	7.99%	0.922	24.0
Inflation	5.23%	5.97%	-74.1	18.76%	37.66%	0.498	1282.5

Notes. Means agree to within twenty basis points on every series and standard deviations to within eight percent. The inflation row has an excess kurtosis in the thousands: that is the hyperinflation episodes surviving into the simulated paths, which is the intended behaviour rather than a defect.

The correlation matrix is reproduced to a maximum absolute deviation of 0.031 across the four asset-to-asset pairs that drive portfolio behaviour. The largest deviation anywhere in the matrix is 0.068 and it sits on the inflation row, where the excess kurtosis reported above makes the sample correlation itself unstable. We report both figures rather than the flattering one.

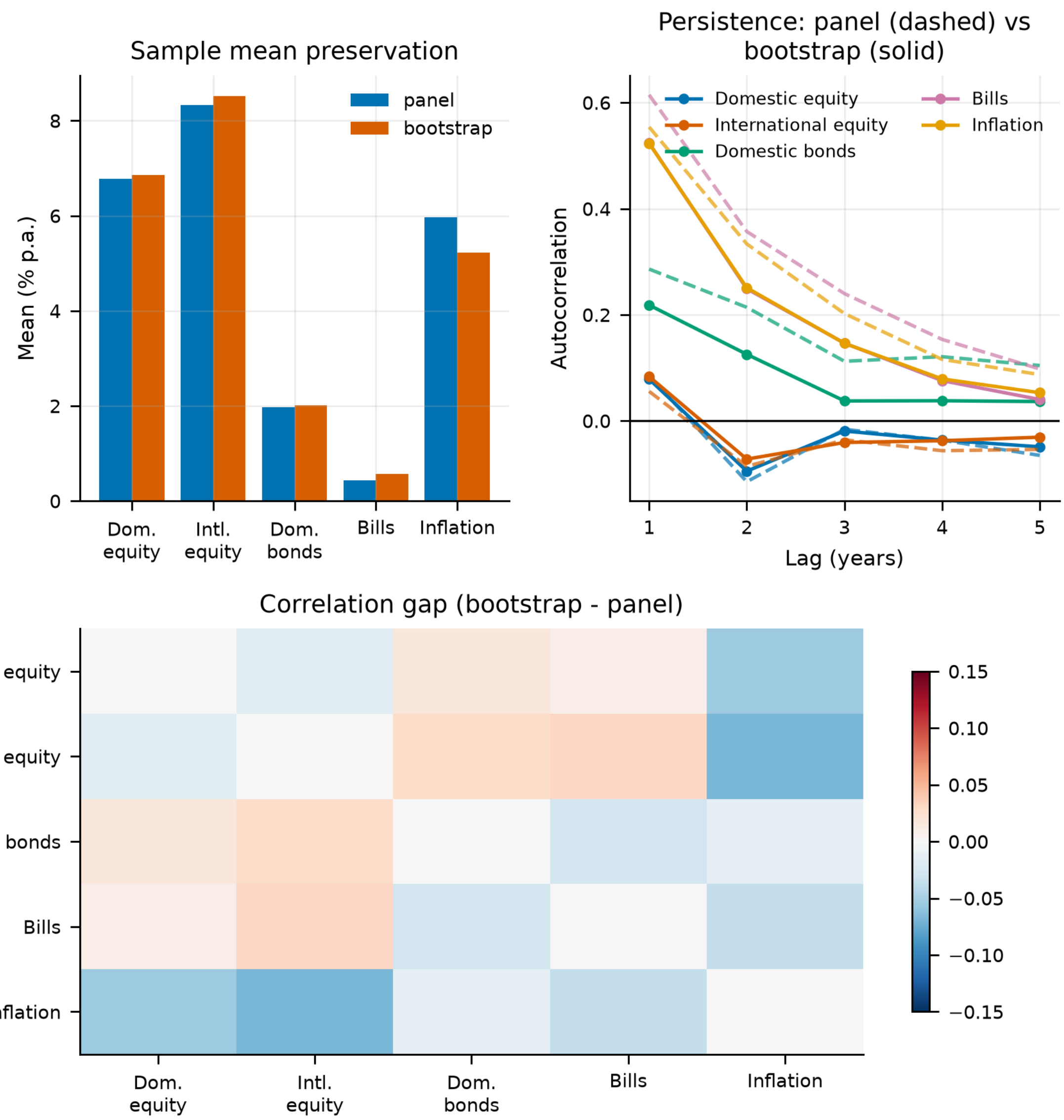

Figure 5.

Bootstrap diagnostics. Simulated marginal distributions against the panel they are drawn from, the preservation of the cross-asset correlation structure, and the within-path autocorrelation that block sampling is designed to retain.

Block length and long-horizon dispersion

Block length is the single most consequential tuning parameter in the design, because it controls how much multi-decade persistence survives into a simulated lifetime. The naive expectation is that longer blocks produce more dispersion in long-horizon outcomes. The data say the opposite, and the reason is instructive.

Table 9. Sensitivity of 68-year outcomes to the mean block length

Mean block (years)	Mean annualised	S.d. annualised	5th pct	95th pct	Within-path AR(1)
1	4.41%	3.31%	-1.52%	9.49%	-0.014
5	4.49%	3.27%	-1.41%	9.25%	0.064
10	4.43%	3.20%	-1.53%	8.89%	0.079
20	4.37%	3.14%	-1.69%	8.68%	0.088

Notes. Longer blocks raise the within-path autocorrelation, as intended, but slightly *reduce* the dispersion of annualised 68-year outcomes. Over a horizon this long the historical series mean-reverts, so preserving more of its internal structure preserves that mean reversion too.

Within-path autocorrelation rises monotonically from -0.014 at one-year blocks to 0.088 at twenty-year blocks, confirming that the sampler is doing what block sampling is for. But the standard deviation of annualised 68-year returns *falls* slightly over the same range. This is a long-horizon mean-reversion finding, not a bug: a lifetime that inherits real historical sequences inherits their tendency to revert, whereas one assembled from independent annual draws does not. It is also a caution against reading the block length as a simple dial for "more risk".

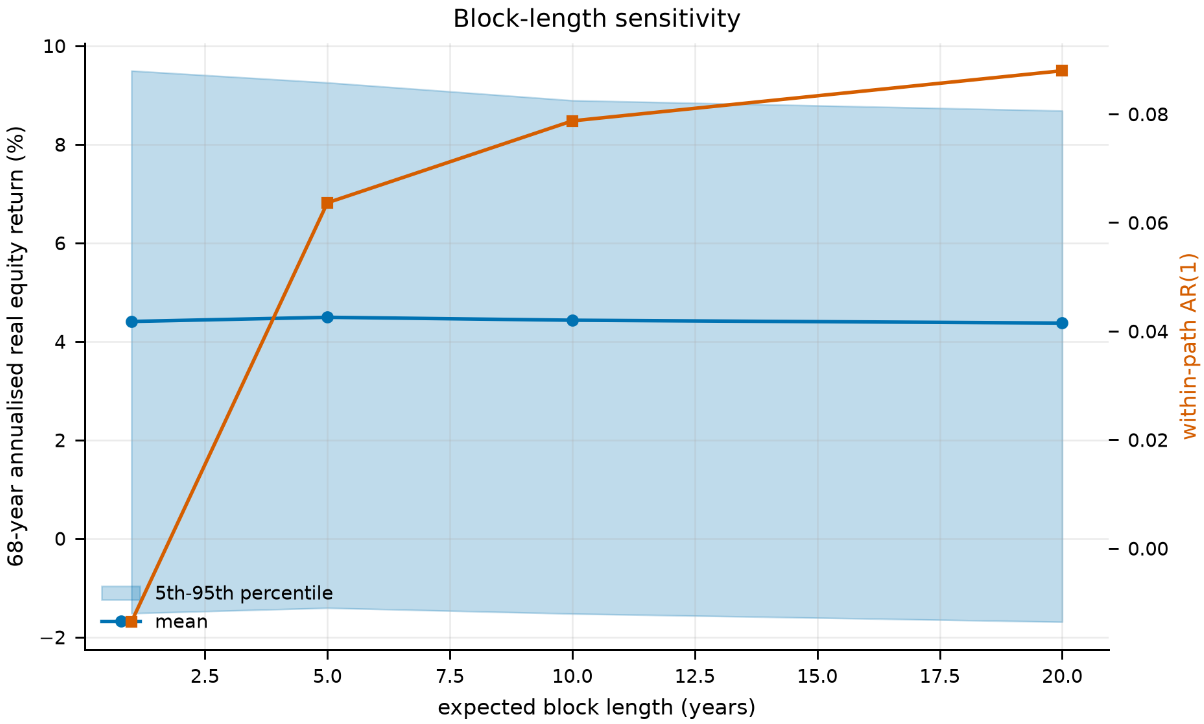

Figure 6.

Block-length sensitivity. Long-horizon dispersion is close to flat in the mean block length even as within-path persistence rises steadily, which is the signature of mean reversion in the underlying series.

4.2 The lifecycle model

A simulated investor begins work at age 25, retires at 63 and dies at 93, giving a 68-year horizon of which 38 are working years. They save a fixed fraction s = 10% of labour income, rebalance annually to the strategy's target weights, and on retirement switch to a withdrawal rule.

Wealth evolves as

$$W_{h+1} = (W_h + c_h - x_h) \cdot (1 + r^p_h), \quad r^p_h = \Sigma_a w_{a,h} r_{a,h} \quad (1)$$

where c_h is the contribution in working years and zero afterwards, x_h is the withdrawal in retirement and zero before, $w_{a,h}$ is the strategy's weight on asset a at age h, and all returns are real. Consumption is labour income net of saving while working, and the sum of the social-security benefit and the portfolio withdrawal in retirement:

$$C_h = (1 - s)\, Y_h \ (working) \quad C_h = B + x_h \ (retired) \quad (2)$$

Labour income follows a deterministic hump-shaped profile (b_1 = 0.045, b_2 = -0.0009 in age and age squared) multiplied by permanent and transitory shocks with standard deviations 0.10 and 0.25. The permanent shock is a cumulative sum, so income dispersion widens over a career; the transitory shock is i.i.d. The profile is an age effect and carries no economy-wide wage growth; Section 3.6.3 measures what that leaves out and which way it biases the result.

The social-security benefit uses the US primary-insurance-amount bend-point schedule rather than a flat replacement rate, applying 90%, 32% and 15% to successive tranches of career-average earnings. This is not decoration. A flat replacement rate scales with the outcome and therefore insures nothing; a progressive schedule supplies a genuine real consumption floor, which is exactly what determines how much the left tail of a risky strategy actually costs.

Table 10. The six benchmark strategies, as constant weights

Strategy	Domestic eq.	Intl. eq.	Bonds	Bills
50/50 domestic/international equity	50%	50%	—	—
100% domestic equity	100%	—	—	—
100% international equity	—	100%	—	—
60/40 domestic equity/bonds	60%	—	40%	—
100% bills (cash)	—	—	—	100%
Target-date fund (glide path)	see below	see below	see below	see below

Notes. Held constant at every age and rebalanced annually, except the target-date fund.

The target-date fund is the one benchmark whose weights move. Its equity share is piecewise-linear in age through the knots below, interpolated between them; the equity sleeve is split 60/40 domestic/international at every age and the fixed-income sleeve 70/30 bonds/bills. The shape is the industry standard — flat and growth-heavy while young, declining through the decade before retirement, and still declining gently after it.

Table 11. The target-date fund's glide path, at its knots

Age	25	40	55	63 (retire)	75	93
Equity share	90%	90%	65%	50%	35%	30%

Notes. Linear between knots. At age 63 the fund therefore holds 30% domestic equity, 20% international, 35% bonds and 15% bills.

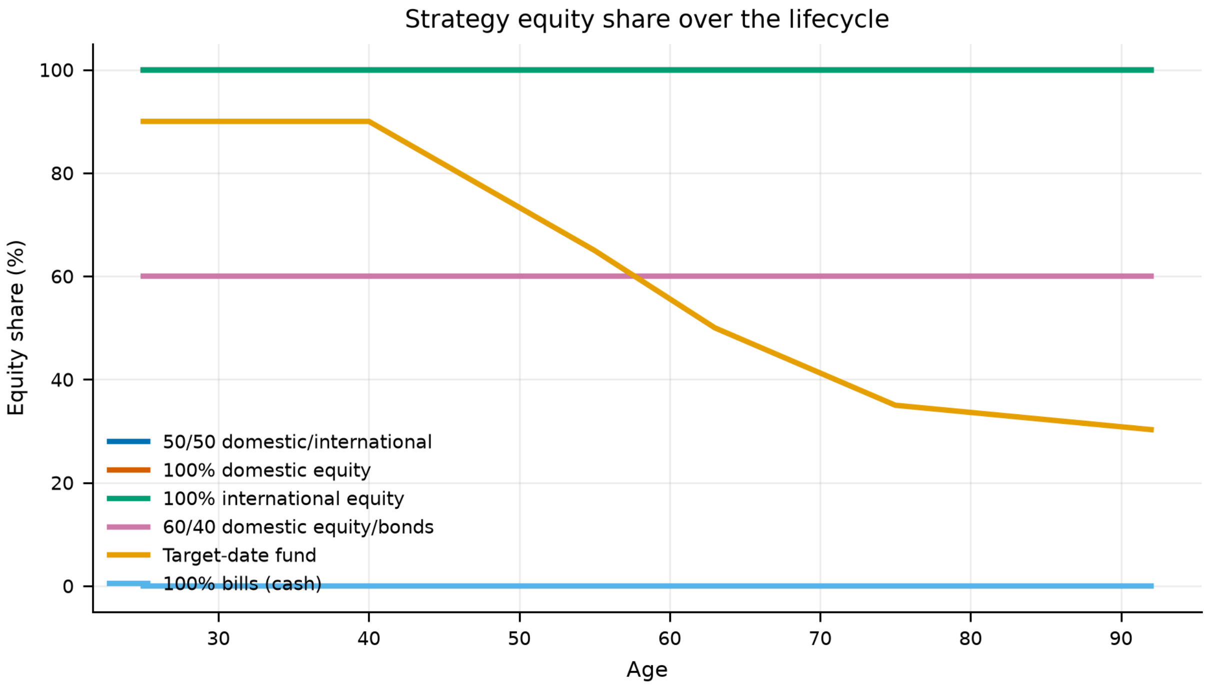

Figure 7.

The six benchmark allocation strategies as equity share by age. The target-date fund is the industry-standard declining glide path; the comparison strategies are held at fixed weights.

4.3 Preferences and the certainty equivalent

Strategies are ranked by certainty-equivalent consumption: the constant, riskless consumption stream that would leave the investor indifferent to the risky one their strategy produces. Under constant relative risk aversion with discount factor β and curvature γ, lifetime utility over the evaluation window is

$$V = \Sigma_h \beta^h u(C_h) + \beta^H \theta \cdot u(\kappa + W_H), \quad u(c) = c^{1-\gamma} / (1-\gamma) \quad (3)$$

and the certainty equivalent inverts that utility back into consumption units. We report risk aversions of {2, 5, 10} with $\gamma = 5$ as the baseline, $\beta = 0.96$, and a bequest weight $\theta = 2$.

The bequest term uses the shifted-power specification of De Nardi (2004) with $\kappa = 1$. The shift is not cosmetic. Without it, any path that dies with exactly zero wealth contributes $u(0) = -\infty$ at $\gamma \geq 1$, one such path in a hundred thousand collapses the entire certainty equivalent, and the ranking becomes a statement about which strategy avoids exact zeros rather than about which delivers better consumption. This is a genuine trap in lifecycle work and we flag it because we walked into it.

We also report Epstein–Zin preferences in their ex-ante, early-resolution form, which separates risk aversion from the elasticity of intertemporal substitution, with ψ taking the values {0.5, 1.5}. Under CRRA these two are locked together at $\psi = 1/\gamma$, and a reader is entitled to ask whether the result is being driven by the intertemporal channel rather than the risk channel. Section 6.2 shows it is not.

One further specification choice deserves emphasis. Utility is evaluated over the **retirement** window rather than the whole lifetime for the allocation comparisons. The reason is mechanical: with a fixed savings rate, working-life consumption is *identical* across allocation strategies by construction, so including it adds a large constant to every strategy's utility and compresses the differences that the exercise is about. Where a policy *does* change working-life consumption — the retirement timing and savings-rate studies of Sections 11 to 28 — we switch to the whole-lifetime window and say so explicitly.

4.4 The comparison discipline

Most of the extensions in this paper compare policies that differ in more than one way at once, and the arithmetic of a certainty equivalent will happily credit the wrong difference. Four rules are applied throughout.

- **Matched baselines.** A rule that retires people later is scored against a fixed date matched on the same *mean* retirement age, not against the original age 63. A savings rule that drifts to saving more is scored against a constant rate interpolated at its own realised career average. Without this, a policy is rewarded for working longer or saving more rather than for being smarter.
- **Common random numbers.** Every sweep and every optimiser reuses the same bootstrap draws and the same income shocks, so a difference between two settings is a policy effect and not a sampling artefact. This also makes the objective a deterministic function of the policy, which is what licenses coordinate ascent.
- **Deviation profiles.** Before any structure in a solved schedule is described in words, each element is reset to a neutral reference and the cost measured in basis points. A solved glide path can look highly structured while most of its structure is worth nothing; the profile separates the two.
- **Grid-edge checks.** An optimum sitting on the boundary of its own search grid is a truncation, not an optimum. Every such case is flagged in the text and, where it mattered, the grid was widened and the analysis rerun.

These are not stylistic preferences. Section 28 reports a result that fell by roughly sixty percent when a matched baseline replaced an unmatched one, Section 19 reports apparent glide-path structure that the deviation profile dissolved, and Section 27 reports a functional-form ranking that reversed its interpretation once the grid-edge check was applied.

4.5 Implementation and verification

The model is implemented in Python with NumPy and pandas. Portfolio returns for a whole cohort are formed as a batched matrix product and the wealth recursion runs vectorised across paths, which is what makes a hundred thousand lifetimes per strategy — and the tens of thousands of policy evaluations the optimisers require — tractable on a single core.

Correctness is enforced by 1244 automated tests. The load-bearing ones are equivalence tests: the path-dependent engine used for flexible retirement and conditional saving must reproduce the fixed-date engine *bit for bit* when given a fixed date and a constant rate, and the batched glide-path evaluator must reproduce the reference simulator to stated floating-point tolerance. Every extension in this paper is built on a simulator that is asserted equal to the one that produced the baseline.

Appendix D sets out how the computation is organised, what the test suite establishes, and the discipline by which the prose in this paper is held to the tables it describes.

5. Baseline Results

All results in this section use 100,000 simulated lifetimes per strategy, drawn from the same bootstrap sample so that strategies face identical market histories. Certainty equivalents are over the retirement window, for the reason given in Section 4.3.

5.1 The headline comparison

Table 12. Headline lifecycle outcomes by strategy

Strategy	CEC γ=2	CEC γ=5	CEC γ=10	P(ruin)	Median wealth at 63	Median cons.	5th pct cons.	Median bequest
50/50 domestic/international equity	1.406	1.048	0.756	11.7%	21.2	1.483	0.707	44.7
100% domestic equity	1.183	0.882	0.668	25.4%	16.1	1.241	0.579	16.7
100% international equity	1.521	1.108	0.782	9.0%	24.4	1.613	0.755	64.2
60/40 domestic equity/bonds	1.105	0.871	0.671	24.2%	13.3	1.149	0.585	10.2
Target-date fund (glide path)	1.214	0.943	0.704	22.2%	16.4	1.273	0.630	9.4
100% bills (cash)	0.847	0.739	0.618	56.4%	6.4	0.856	0.521	0.0

Notes. Consumption and wealth are expressed as multiples of the investor's real income at age 25. P(ruin) is the probability of exhausting the portfolio with retirement years still to fund. 100,000 paths per strategy on common bootstrap draws.

The ordering is unambiguous at every risk aversion tested. At γ = 5 the 50/50 all-equity portfolio delivers a certainty equivalent of 1.048 against 0.943 for the target-date fund, an advantage of 11.2%. Against the 60/40 portfolio the advantage is 20.3%, and against bills 41.8%.

What makes the result more than a restatement of the equity premium is the tail behaviour. The all-equity portfolio has a *lower* ruin probability (11.7%) than the target-date fund (22.2%), a higher fifth-percentile retirement consumption (0.707 against 0.630), and a lower probability of falling below a seventy-percent replacement target (13.6% against 22.6%). The conservative portfolio is not buying downside protection on this panel. It is paying for the appearance of it.

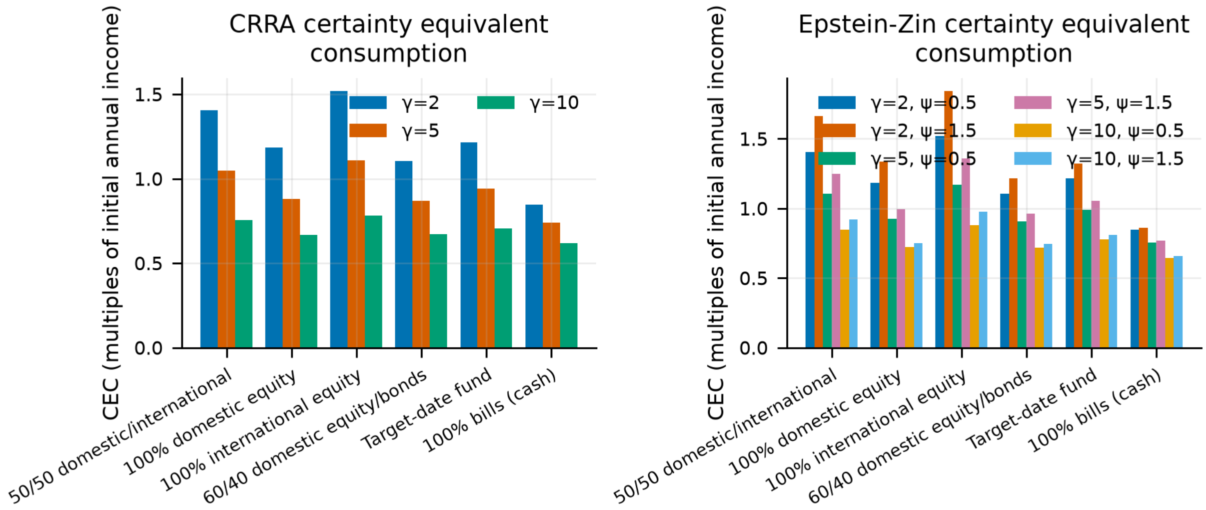

Figure 8.

Certainty-equivalent consumption by strategy and risk aversion. The ranking is stable in γ; the gap narrows as risk aversion rises but does not close within the range tested.

5.2 The mechanism is international, not equity

Domestic equity alone is *not* the winning strategy. Its certainty equivalent at $\gamma = 5$ is 0.882, below the 50/50 portfolio's 1.048, and its ruin probability is 25.4% — nearly double the diversified portfolio's. Pure international equity does better still (1.108), which is a consequence of the leave-one-out construction rather than a recommendation: an investor holding "international" equity in this model holds a 15-country average, and no single real investor has that opportunity set without also holding their own market.

This is the one place the replication does not reproduce ACO's headline. Their recommended portfolio is an even 50/50 split, held for life; on this panel that split is beaten by the international leg alone. We read the difference as a property of a 16-country panel rather than a correction to them. The domestic leg here is a single draw from a set that includes several century-long underperformers, while the international leg averages away exactly that risk; with thirty-nine countries and a tradeable international index the two legs are far closer in character, and the case for holding both is correspondingly stronger. Nothing measurable here contradicts the 50/50 recommendation — the narrower claim this panel supports is that a diversified sleeve dominates a concentrated one.

The practical reading is that the case for equities here is a case for *diversified* equities. A single national market can and did deliver multi-decade real losses; the cross-section of national markets did not do so simultaneously. This is a claim about the covariance structure of the panel, and it is the reason the exercise needs an international sample rather than a longer US one.

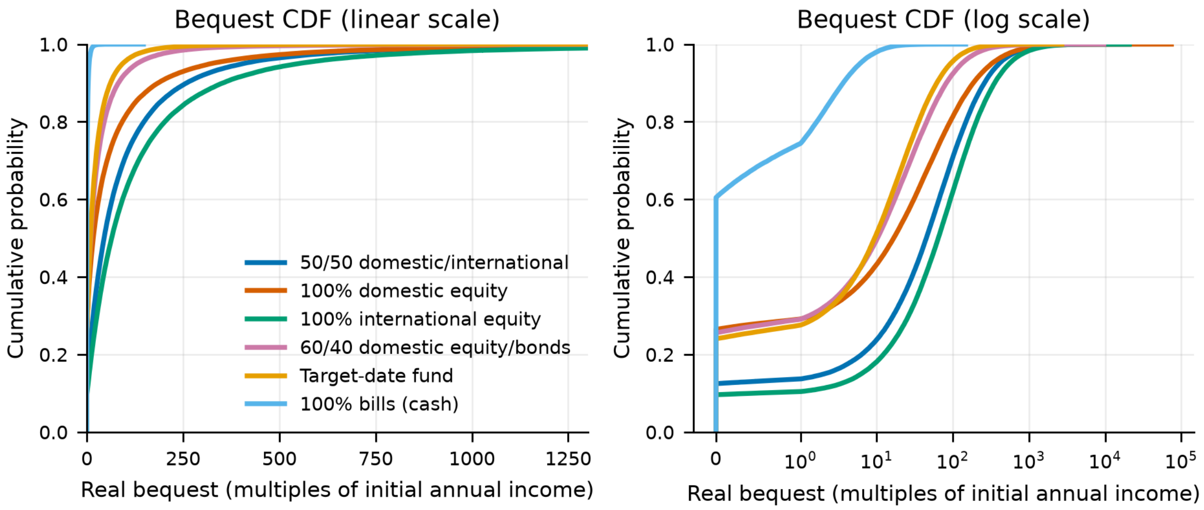

Figure 9.

Cumulative distribution of wealth at retirement. The all-equity distributions stochastically dominate the conservative ones over almost the entire range, including the left tail where the conservative case is usually made.

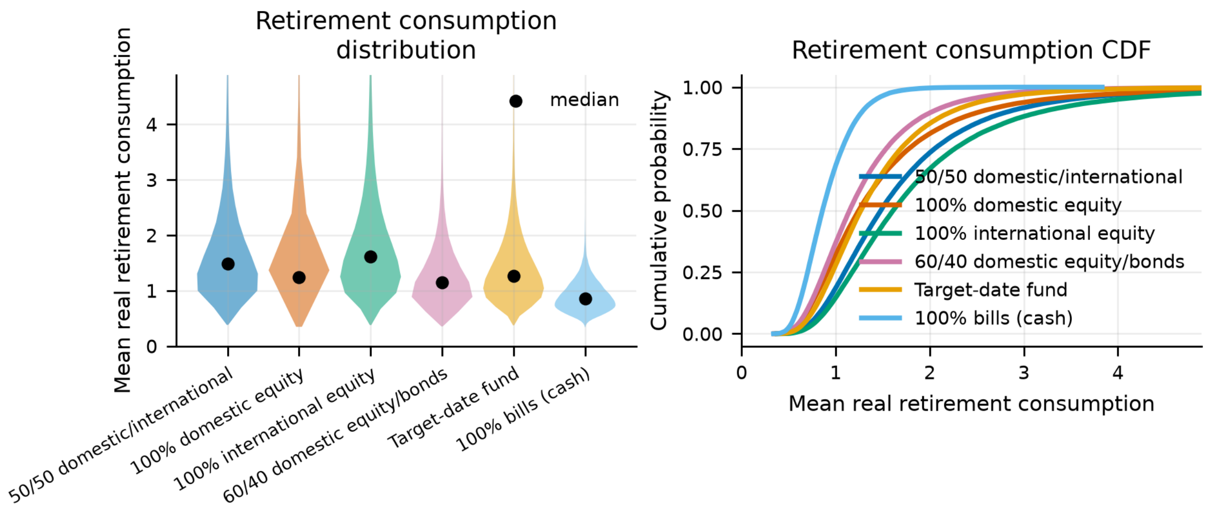

Figure 10.

Distribution of average real retirement consumption by strategy. The vertical line marks the seventy-percent replacement target used in the shortfall statistics.

5.3 Distributional dominance

A certainty equivalent is a single scalar and can hide a strategy that wins on average while losing where it matters. We therefore test the all-equity portfolio against each rival across 21 separate distributional criteria — percentiles of wealth at retirement, of retirement consumption and of bequest, plus ruin and shortfall probabilities — counting a tie as neither a win nor a loss.

Table 13. Distributional dominance of the 50/50 all-equity portfolio

Beaten strategy	Criteria	Won	Tied	Lost	Strict dominance
Target-date fund (glide path)	21	20	1	0	yes
60/40 domestic equity/bonds	21	20	1	0	yes
100% domestic equity	21	20	1	0	yes
100% bills (cash)	21	20	1	0	yes

Notes. A tie is counted separately rather than as a loss, since counting it either way would misstate the comparison. The single tie in each row is the zero-bequest floor, which several strategies reach.

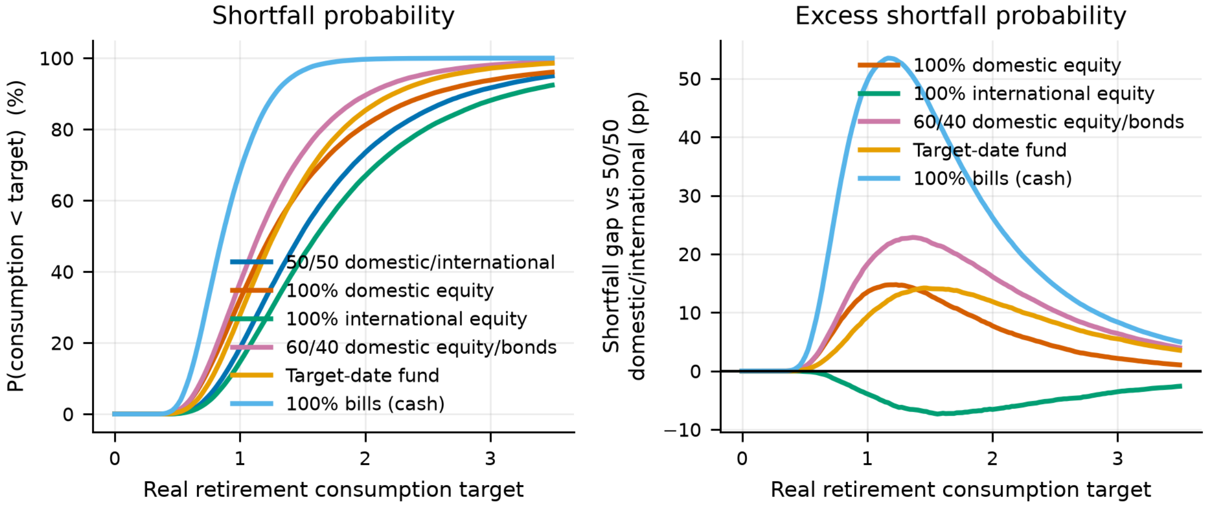

Figure 11.

Shortfall probabilities against a fixed consumption target. Measuring each strategy against a common target rather than against its own median is essential: a strategy-specific target rewards strategies with low medians.

5.4 Sustainable withdrawal rates

The four-percent rule is the most widely used number in retirement planning. It comes from US historical sequences and it does not survive this panel.

Table 14. Sustainable withdrawal rates on the international panel

Strategy	Sustainable rate at 5% ruin	Ruin probability at 4%
100% international equity	3.48%	9.1%
50/50 domestic/international equity	3.17%	11.7%
Target-date fund (glide path)	2.10%	22.0%
60/40 domestic equity/bonds	1.69%	24.0%
100% domestic equity	1.57%	25.2%
100% bills (cash)	1.07%	56.0%

Notes. The sustainable rate is the highest initial real withdrawal rate, inflation-adjusted thereafter, that leaves at most a five-percent probability of exhausting the portfolio over a 30-year retirement.

At a five-percent ruin tolerance the sustainable rate is 3.17% for the all-equity portfolio and 2.10% for the target-date fund. Withdrawing four percent produces a ruin probability of 11.7% and 22.0% respectively. Two things follow. First, the ordering of strategies is the same here as everywhere else in the paper — the equity portfolio sustains a higher rate, not a lower one. Second, the level is a serious warning: on a sample that includes the twentieth century as most countries lived it, four percent is roughly a one-in-seven proposition even on the best strategy tested.

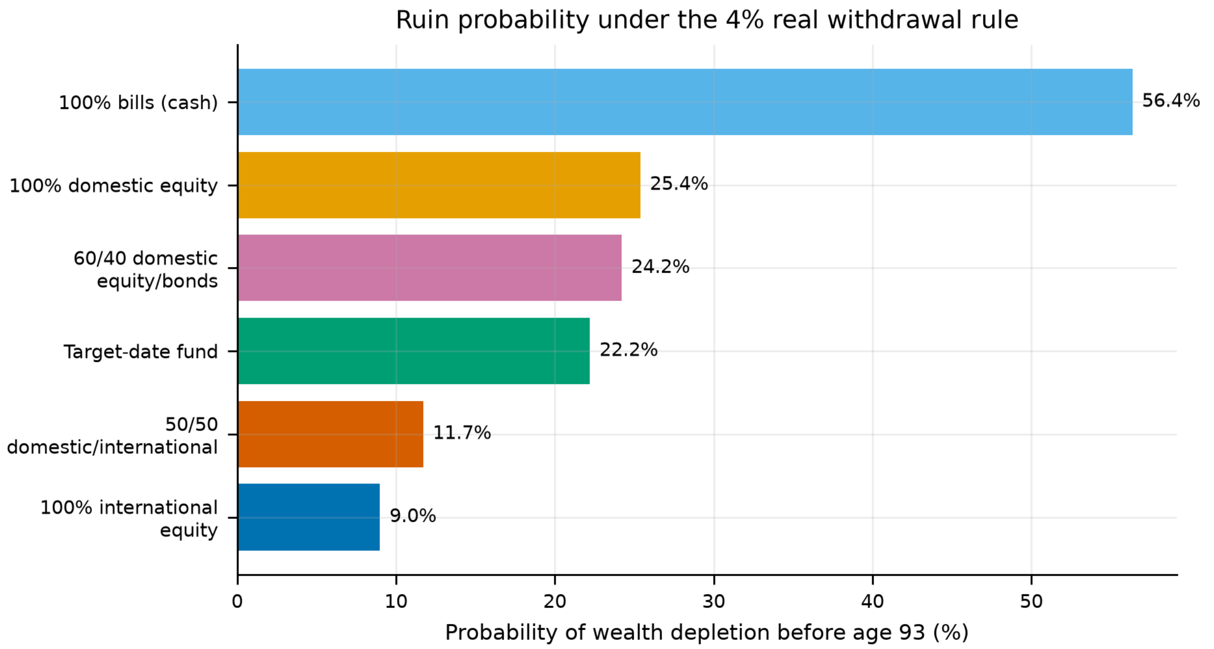

Figure 12.

Probability of portfolio exhaustion as a function of the initial withdrawal rate. The horizontal line is the five-percent tolerance used to define the sustainable rate.

5.5 How the countries are drawn

Two choices in the sampler decide how the cross-section enters a simulated life, and neither is obviously right. A lifetime's domestic country can be drawn once and held, or redrawn at every block; and countries can be weighted by the length of their recorded history or treated as equally likely. Appendix C reports both variants in full. Neither reverses a ranking.

The weighting choice has little room to bite, because every country in the panel carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. It would matter on a panel containing short histories, where a country covering a single era could otherwise speak as loudly as one covering a century and a half.

Redrawing the country at every block is the more consequential variant, and it is the design the original study uses. It implicitly assumes an investor can be resident in a different country every decade, which is not a description of anybody, but it pools the cross-section more aggressively and so gives the international mechanism its most favourable reading. Holding the country fixed for a lifetime — the specification behind every headline number here — is the more conservative of the two.

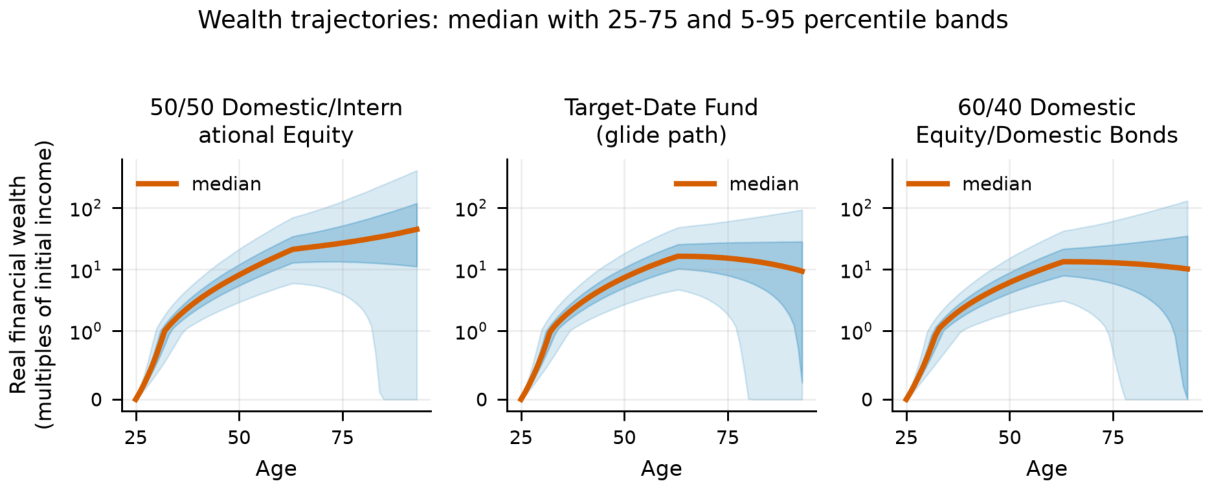

Figure 13.

Wealth trajectories by percentile for the all-equity portfolio and the target-date fund. The fan makes visible what the scalar comparison cannot: the conservative path is not narrower at the bottom, it is lower throughout.

6. Sensitivity Analysis

The headline is one point in a large parameter space. This section sweeps that space along equity share, domestic share, risk aversion, elasticity of intertemporal substitution, bequest weight, longevity, retirement age, savings rate, withdrawal rate, social-security design, block length and return panel. Every sweep uses common random numbers, so differences between settings are parameter effects rather than sampling noise.

The tornado below summarises 132 settings across the 9 of those dimensions along which a like-for-like advantage over a fixed incumbent strategy is defined; the remainder vary the portfolio itself and are reported in their own subsections.

The ranking reverses in 0 of them. That is the single most useful sentence in this section, and the tornado below reports the range of the advantage rather than its point estimate so the reader can see how much room there is.

6.1 Tornado analysis

A tornado analysis varies one parameter at a time across its whole plausible range, holding everything else at baseline, and records how far the result moves. Sorting the dimensions by that range — widest at the top — shows at a glance which assumptions the conclusion depends on and which are incidental. The name comes from the funnel shape the sorted bars make. The column that matters most here is the last one: a *reversal* is a setting at which the all-equity portfolio stops winning, and a dimension with none of them cannot overturn the result however it is set.

Table 15. Tornado: range of the all-equity advantage by parameter dimension

Dimension	Compared against	Settings	Min adv.	Median adv.	Max adv.	Range (pp)	Reversals
Risk aversion	60/40	12	5.4	19.3	33.8	28.4	0
Risk aversion	Target-date fund	12	3.5	10.5	22.2	18.7	0
Elasticity of substitution	60/40	9	20.2	25.3	34.2	14.0	0
Retirement age	60/40	6	15.7	21.4	28.7	13.0	0
Savings rate	60/40	5	14.0	20.2	26.5	12.5	0
Elasticity of substitution	Target-date fund	9	11.0	14.4	22.8	11.8	0
Withdrawal rate	60/40	13	9.0	16.3	20.2	11.3	0
Retirement age	Target-date fund	6	7.9	11.8	17.1	9.2	0
Social security design	60/40	4	20.2	24.6	29.3	9.0	0
Social security design	Target-date fund	4	11.0	13.5	18.2	7.2	0
Savings rate	Target-date fund	5	7.8	11.0	14.8	7.0	0
Bootstrap block length	60/40	6	16.2	18.9	22.9	6.7	0

Notes. Advantage is the percentage certainty-equivalent lead of the 50/50 all-equity portfolio over the named incumbent, at $\gamma = 5$ unless the dimension is risk aversion itself. Ordered by the width of the range, so the dimensions the result is most sensitive to appear first.

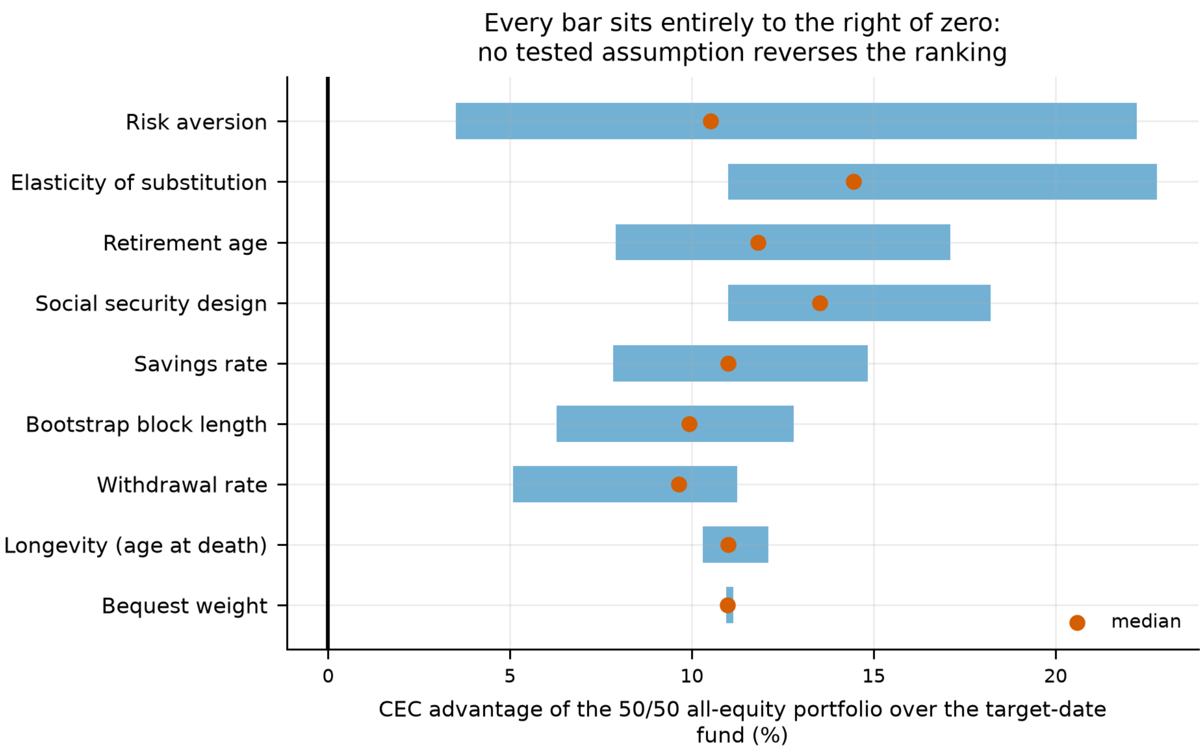

Figure 14.

Tornado plot of the all-equity advantage. The bar spans the minimum and maximum advantage observed across the settings in that dimension; no bar crosses zero.

Risk aversion is the widest dimension, as it should be: the advantage over a 60/40 portfolio ranges from 5.4% to 33.8% across γ over [1, 20]. It narrows as γ rises but does not close. Fitting the crossover point directly, the certainty-equivalent lines do not intersect within the tested range for either the target-date fund or the 60/40 portfolio; at the highest γ tested the all-equity lead is still 3.5% and 5.4% respectively.

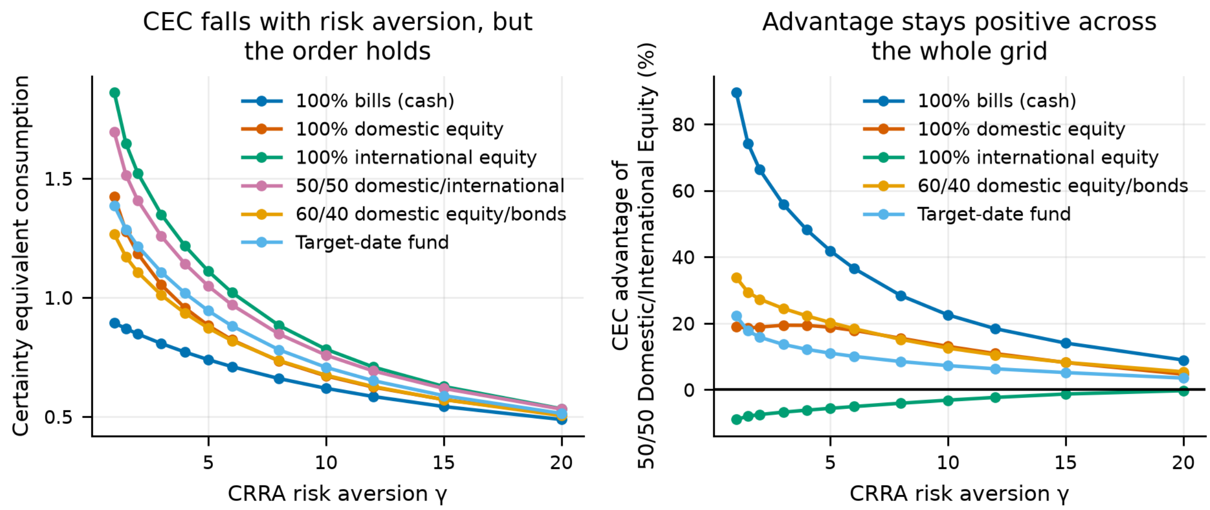

Figure 15.

Certainty equivalent by strategy across risk aversion. The conservative strategies converge toward the equity portfolios as γ rises but do not overtake them within the tested range.

6.2 Allocation sweeps and the corner solution

Sweeping the equity share continuously rather than testing discrete strategies asks a sharper question: is the optimum interior?

Table 16. Optimal equity share by risk aversion

Risk aversion	Optimal equity share	CEC at optimum	P(ruin) at optimum
γ = 2	100%	1.408	11.7%
γ = 5	100%	1.048	11.7%
γ = 10	100%	0.758	11.7%

Notes. The optimum is at the corner for every risk aversion tested. Section 19 asks the same question of the full age-by-asset schedule and reaches the same answer.

Table 17. Optimal domestic share within the equity allocation

Risk aversion	Optimal domestic share	CEC at optimum	P(ruin) at optimum
γ = 2	0%	1.522	9.1%
γ = 5	10%	1.119	8.3%
γ = 10	20%	0.793	8.1%

Notes. The home-bias question, asked of the model rather than assumed. The optimum is well below the 50% used in the headline strategy and rises with risk aversion — an artefact of the leave-one-out international construction, which gives the international leg a diversification advantage no individual investor can replicate.

The equity optimum sits at 100% at every risk aversion tested. The domestic-share optimum is more interesting: it is 0% at γ = 2 and rises to 20% at γ = 10. We do not read this as a recommendation to hold almost no domestic equity. The international leg in this model is a 15-country leave-one-out average, which is better diversified than any tradeable international index; the honest reading is that the model prefers *more* diversification than the 50/50 headline strategy provides, not that a specific number is optimal.

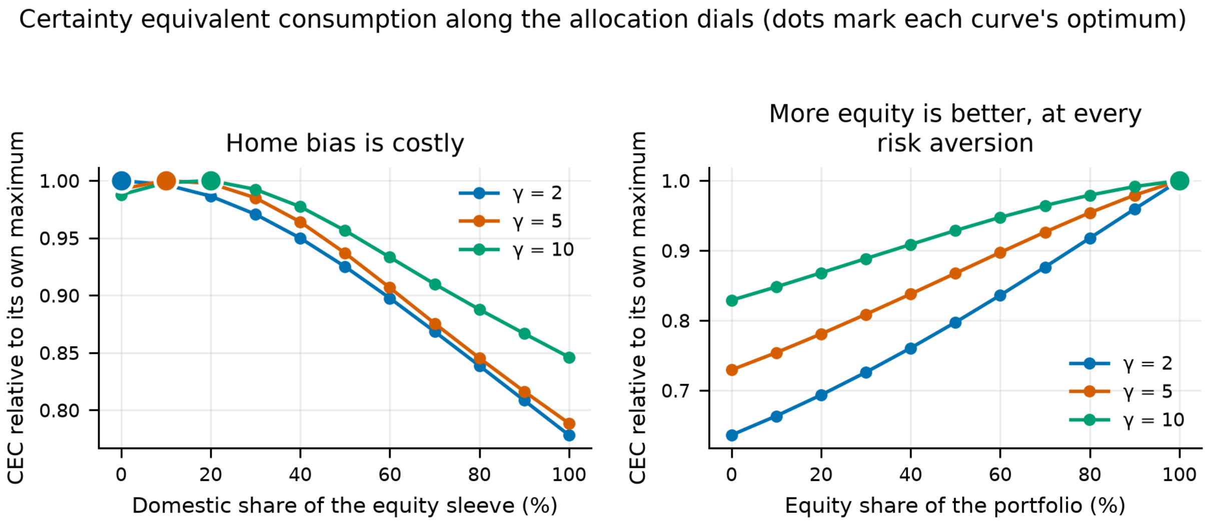

Figure 16.

Certainty equivalent across the equity share and the domestic share. The equity dimension is monotone to the corner; the domestic dimension has an interior optimum well below the 50% headline weight.

6.3 Separating risk aversion from intertemporal substitution

Under CRRA, γ and the elasticity of intertemporal substitution are the same parameter inverted, so a sceptic can reasonably ask whether the result is about attitudes to risk or about willingness to substitute consumption over time. Epstein–Zin preferences separate them. Holding γ at 5 and sweeping ψ from 0.2 to 2.5, the advantage over the 60/40 portfolio ranges from 20.2% to 34.2% and never reverses. The result is a risk result, not a substitution result.

6.4 Planning parameters

Retirement age, longevity, savings rate, withdrawal rate and social-security design are the levers an individual actually controls, and none of them changes the ranking. They do change the *size* of the advantage substantially — the withdrawal-rate dimension alone spans several percentage points — which is worth keeping in view: the all-equity lead is not a constant of nature but a quantity that depends on how the rest of the plan is set up.

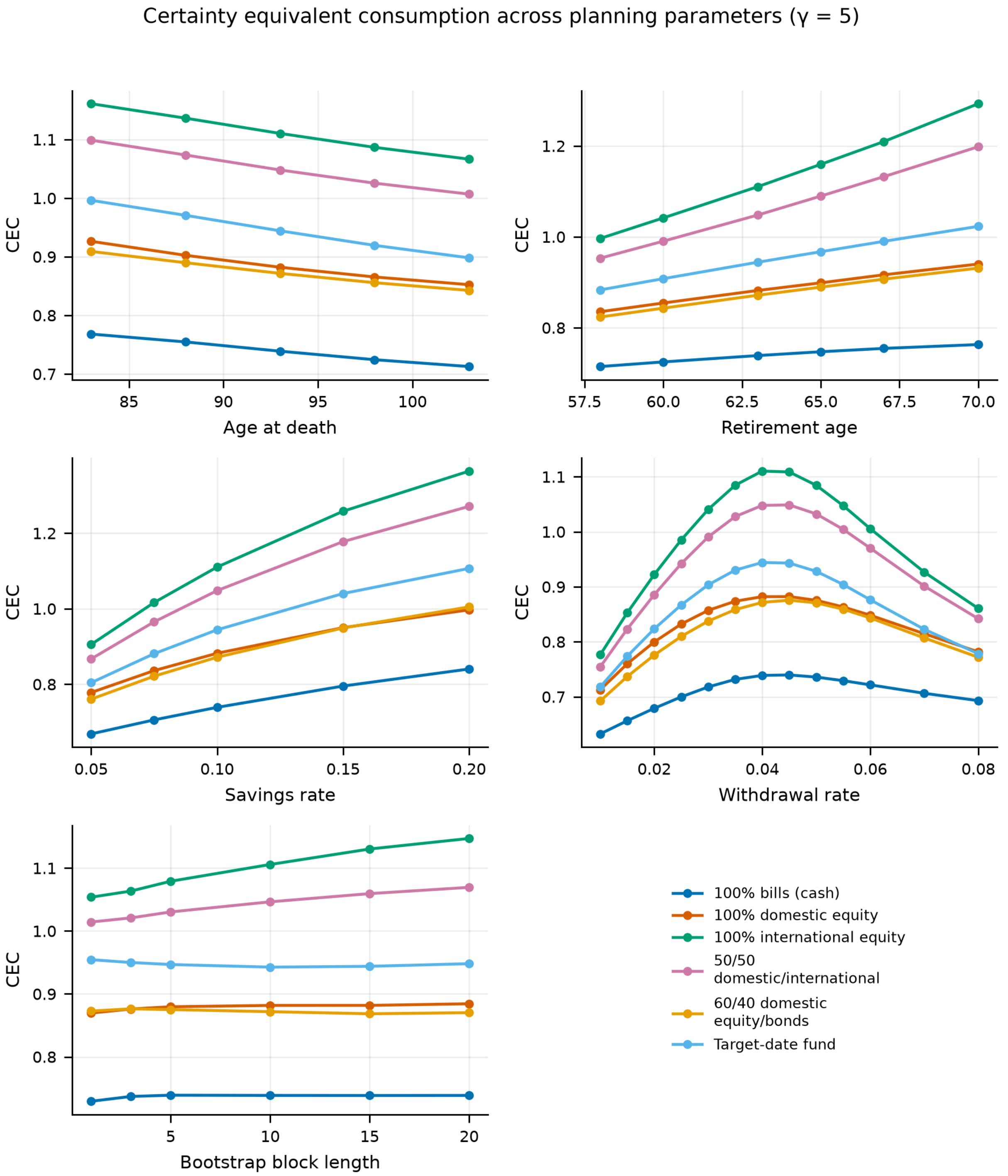

Figure 17.

Planning-parameter sweeps: retirement age, longevity, savings rate and social-security design. Ranking is preserved throughout; the magnitude of the advantage is not.

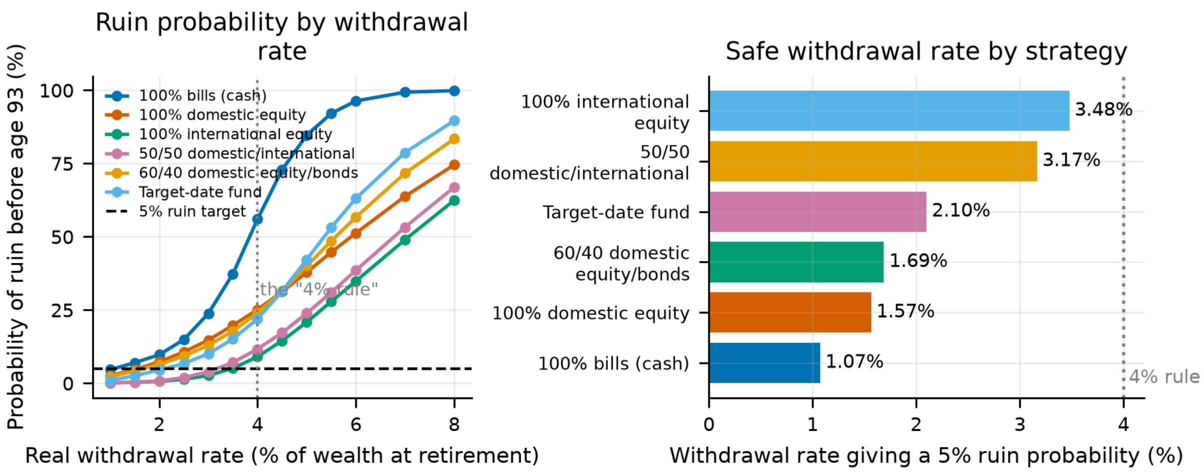

Figure 18.

The advantage as a function of the withdrawal rate. Higher withdrawal rates widen the gap, because a portfolio being drawn down faster is more exposed to the left tail the conservative strategies fail to protect against.

7. The Realised Record, Without the Bootstrap

Every number to this point comes from resampled histories. That is the right way to get a distribution out of 131 years of annual data, and it leaves one question open that no amount of resampling can close: is the result a property of the world, or of the sampler? This section answers it by not resampling at all.

A **cohort** is one country and one birth year. The investor turns 25 in the first year of the window, holds the strategy through that country's realised returns and the realised leave-one-out sleeve of every other market, in calendar order, and dies 68 years later. Nothing is drawn. There is exactly one such lifetime for every (country, birth year) pair the panel can support, and labour income is deterministic here so that the whole cross-sectional spread is about returns.

7.1 How thin the realised record is

The panel supports **632 complete lifetimes**. It does not contain 632 pieces of evidence. Adjacent cohorts in the same country share 67 of their 68 years, and cut into non-overlapping lifetimes the same record yields **16** — one per market. Every cohort also lives through the same two world wars, so even those are not independent of one another. Inference below is therefore a bootstrap over *countries*, never a standard error over cohorts.

Table 18. What each market contributes to the realised record.

Market	Usable years	Longest unbroken run	Lifetimes	Earliest birth year
NOR	131	131	64	1890
PRT	131	131	64	1890
SWE	131	131	64	1890
USA	131	131	64	1890
GBR	131	131	64	1890
FIN	125	125	58	1896
NLD	121	121	54	1900
CHE	120	105	38	1916
DNK	130	105	38	1916
BEL	125	101	34	1920

Market	Usable years	Longest unbroken run	Lifetimes	Earliest birth year
ITA	124	99	32	1922
FRA	124	99	32	1922
ESP	115	77	10	1941
AUS	118	73	6	1948
JPN	129	73	6	1948
DEU	124	71	4	1950

Notes. A lifetime cannot step over a market closure, so a gap costs sixty-eight cohorts rather than the handful of years it spans.

A market closure removes a lifetime, not a year. The bootstrap refuses blocks that span a gap and keeps drawing; a cohort that spans one cannot be run at all. NOR contributes 64 lifetimes and DEU contributes 4, the earliest of them beginning in 1950. Every runnable German, Japanese and Spanish lifetime therefore *begins after* that country's catastrophe and rides the recovery. This design is structurally kinder to domestic equity than the bootstrap is, and that should be held in mind for everything that follows.

It is worth being precise about what is selected, because it is more than the usual complaint. The panel is already restricted to markets that survived; this section restricts it again, to *windows* that ran unbroken for 68 years. Those are not the same filter. The second one removes every lifetime that straddles a catastrophe while keeping the ones that follow it, so the markets with the worst equity histories in the panel appear here only through their recoveries. The consequence is that the lead below is conservative in *direction* — the missing lifetimes are ones a home market would have lost badly — while the spread across markets is not the spread a randomly chosen saver faced, and should not be read as one.

7.2 What the realised lifetimes paid

Table 19. Certainty equivalent over 632 realised lifetimes, γ = 5. No resampling of any kind.

Strategy	CEC	Median retirement consumption	5th percentile	P(ruin)
100% International Equity	1.5449	1.680	1.196	0.9%
50/50 Domestic/International Equity	1.3008	1.464	1.006	6.2%
Target-Date Fund (glide path)	1.1185	1.217	0.889	19.3%
100% Domestic Equity	1.0362	1.189	0.782	19.8%
60/40 Domestic Equity/Domestic Bonds	1.0020	1.084	0.784	22.3%
100% Bills (cash)	0.8669	0.877	0.765	55.1%

The ordering survives with no resampling at all. All-international delivers a certainty equivalent 18.8% above the 50/50 split over the lifetimes the panel actually contains, and it is ahead in 73% of them. Whatever else the headline is, it is not an artefact of the block bootstrap.

Resampling *countries* rather than cohorts gives a mean lead of 18.5% with a 95% interval of [7.9, 31.3], which excludes zero. Counting by market rather than by cohort — which gives the United States sixty-four votes instead of one — the lead holds in 13 of 16 markets.

That pooled mean is weighted by history, not by market. A country enters it once per runnable lifetime, so the five markets with a full unbroken run carry sixteen times the weight of Germany's 4. Giving each market one vote instead moves the lead to 16.2% (-2.3 points), with an interval of [4.3, 28.7] that still excludes zero. A percentile interval on 16 clusters under-covers, so the Student-t interval on the between-market spread is the conservative reading: [2.3, 30.2], which excludes zero.

The exceptions are the interesting rows: **JPN** (-37%, 6 lifetimes from 1948), **USA** (-5%, 64 lifetimes from 1890), **DEU** (-2%, 4 lifetimes from 1950). Two of them are markets whose only runnable cohorts start at the bottom of a post-war recovery, which is the composition bias of the previous subsection showing up as a result.

7.3 Why the realised lead is larger than the resampled one

The bootstrap headline and the number above are not the same quantity and should not be read as an estimate and a re-estimate of one thing. A bootstrapped lifetime is a mosaic of blocks drawn from different decades, so a market that underperformed for forty years running contributes a few of those blocks and the rest of the lifetime is spliced in from better eras. A cohort cannot splice.

The realised record makes the consequence visible. Over the 68-year windows the panel supports, a home market's annualised real return ranges from -4.0% to 9.1%. The international sleeve over the identical years ranges from 3.6% to 9.4%, with a standard deviation of 0.97 points against 2.75 for the single market.

That is the argument of this paper stated without a single resampled path: there is no realised 68-year window in which the diversified sleeve did badly, and there are several in which a single home market was destroyed. The sleeve was ahead in 87% of realised lifetimes, by 2.71 points a year.

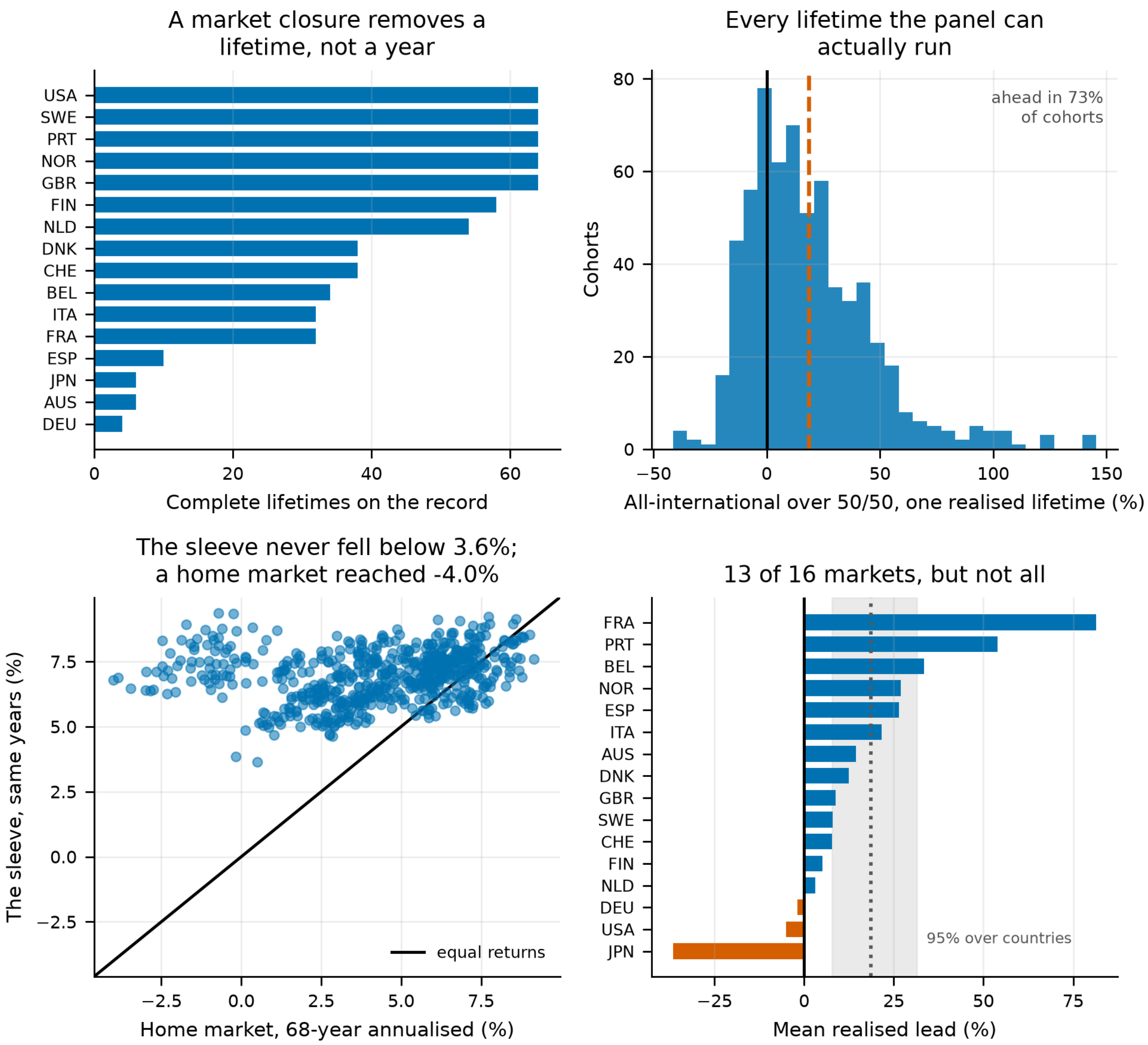

Figure 19.

Top left: how many complete lifetimes each market contributes, which a market closure reduces by a lifetime rather than by a year. Top right: the realised lead, one observation per lifetime the panel can run. Bottom left: the two legs' annualised returns over the same sixty-eight years, cohort by cohort. Bottom right: the mean lead by market, against the interval a bootstrap over countries supports.

7.4 What this section adds, and what it does not

- The ordering is not an artefact of the block bootstrap. It is in the realised record, without resampling, in 13 of 16 markets.
- The honest interval on the realised lead is [7.9, 31.3] pooled, [4.3, 28.7] with each market given one vote, and [2.3, 30.2] on the parametric reading — all from 16 independent lifetimes. Take the widest. This section adds confidence in the *direction* and almost none in the magnitude.
- The design cannot see the worst domestic histories in the panel, because a lifetime cannot step over a market closure. The bootstrap can, which is one of the reasons it stays the headline.

8. How Much Rests on One Country, or One Era

Section 6 varies the preferences and the lifecycle, and Section 9 will vary how the international sleeve is built. Neither touches what a sceptical reader asks about first: the panel is 16 developed markets that survived, and a result assembled from 16 histories could be one history wearing a disguise. If dropping the United States overturns the ranking, this paper is about the United States. This section asks three questions on the same machinery, each scored by the summariser that produces Section 5's own table.

- **Influence.** Rebuild the panel once per country with that country removed, and re-run the headline.
- **Uncertainty.** Those runs form a delete-one jackknife, which gives a standard error reflecting the *panel's* size rather than the Monte Carlo's.
- **Stability.** Re-run on expanding windows of history, and on the two halves of the sample.

The deletion is genuine. A dropped market disappears from every other country's international sleeve as well as from the set its own domestic leg is drawn from, so each run answers what this paper would say had that market's history never been recorded — not the much weaker question of what it would say had we chosen not to invest there.

8.1 The noise floor

Every delete-one run draws its own bootstrap paths, so it differs from the full panel even when the country it dropped carried no information. Re-running the **unmodified** panel under 3 seeds moves the lead across a range of 0.201 points, and that is the floor any shift below has to clear. 11 of 16 countries clear it.

8.2 Removing one country at a time

Table 20. The headline with each country removed, at γ = 5

Country removed	Lead	Change in the lead	CEC, all-international	CEC, 50/50
DEU	3.46%	-2.33 pp	1.0509	1.0158
DNK	5.22%	-0.58 pp	1.0882	1.0342
PRT	5.39%	-0.41 pp	1.1369	1.0788
AUS	5.49%	-0.30 pp	1.0897	1.0329
SWE	5.58%	-0.21 pp	1.1025	1.0442
FRA	5.61%	-0.18 pp	1.1417	1.0811
FIN	5.72%	-0.08 pp	1.1062	1.0463
ESP	5.73%	-0.06 pp	1.1148	1.0543
GBR	5.80%	+0.00 pp	1.1028	1.0424
NOR	5.87%	+0.08 pp	1.1167	1.0548
CHE	6.02%	+0.23 pp	1.1074	1.0445
NLD	6.05%	+0.25 pp	1.1098	1.0465
USA	6.18%	+0.39 pp	1.1002	1.0362
JPN	6.27%	+0.48 pp	1.0910	1.0266
BEL	6.37%	+0.57 pp	1.1252	1.0578
ITA	7.26%	+1.47 pp	1.1450	1.0675

Notes. 100,000 lifetimes per run. The full-panel lead is 5.79%. A negative change means the removed country was holding the result up.

No single country carries the result. All-international leads the 50/50 split in all 16 delete-one panels. Removing DEU costs the most, taking the lead from 5.79% to 3.46%, and it still leads. Notably the United

States is not the load-bearing market: removing it moves the lead by +0.39 points.

8.3 Why that market, and not the obvious one

The natural guess is that the load-bearing market is the one with the best domestic history, and that removing it makes the domestic half of the 50/50 split look worse. The data says otherwise, and the reason is a general point about what an averaged sleeve does to a constituent's volatility.

Every country sits in the panel twice over: once as somebody's home market, and once inside the fifteen-market average that everybody *else* holds as their international leg. The two strategies weight those roles differently — all-international is entirely sleeve, the 50/50 split is half sleeve and half domestic — so writing S for a deletion's effect on the first and D for the implied effect on the domestic half, the change in all-international is S, the change in the 50/50 split is $(S + D)/2$, and the change in the gap is $(S - D)/2$. A market whose value lies mostly in other people's sleeves narrows the gap when removed; one whose value lies mostly in being its own home market widens it.

Table 21. Each market's own returns, and what removing it does to the sleeve

Market	Own arithmetic mean	Own geometric mean	Volatility drag	Effect on the sleeve
DEU	8.72%	4.60%	4.13 pp	-0.63 pp
AUS	7.76%	6.37%	1.40 pp	-0.13 pp
USA	8.39%	6.66%	1.72 pp	-0.09 pp
FIN	8.91%	4.98%	3.93 pp	-0.08 pp
JPN	7.94%	4.16%	3.77 pp	-0.08 pp
DNK	7.46%	6.08%	1.38 pp	-0.06 pp
SWE	8.13%	6.14%	1.99 pp	-0.06 pp
NLD	7.15%	5.10%	2.05 pp	+0.01 pp
GBR	6.55%	4.93%	1.62 pp	+0.02 pp
CHE	6.77%	5.10%	1.67 pp	+0.03 pp
ESP	6.18%	4.16%	2.01 pp	+0.04 pp
NOR	5.67%	3.73%	1.93 pp	+0.09 pp
BEL	5.91%	3.77%	2.13 pp	+0.12 pp
ITA	6.08%	2.56%	3.53 pp	+0.13 pp
FRA	3.58%	1.05%	2.53 pp	+0.25 pp
PRT	3.21%	-0.12%	3.33 pp	+0.25 pp

Notes. The volatility drag is the wedge between a market's arithmetic and geometric mean — what its own residents lose to its volatility. The last column is measured, not argued: the panel is rebuilt without the market and the pooled sleeve's compound return recomputed.

A deletion's effect on the headline tracks what it does to the **sleeve's compound return** (correlation +0.68), not the removed market's own average return (correlation -0.20). The load-bearing market is not the one with the best domestic history; it is the one the fifteen-way average could least afford to lose.

DEU is that market, and the reason is the gap between its two means: an arithmetic mean of 8.72% against a geometric mean of 4.60% — the largest volatility drag in the panel. An equal-weighted average of fifteen markets adds up *arithmetic* returns each year and diversifies the volatility away, so the sleeve collects DEU's high mean without paying its drag while its own residents pay it in full. It is an excellent constituent of somebody else's sleeve and a mediocre thing to hold alone, so removing it hits the all-sleeve strategy hardest.

The United States is the mirror image, which answers the question most readers arrive with. It is the third-best market in the panel by arithmetic mean (8.39%) and the best by geometric mean (6.66%) — one of the smallest drags here. Almost all of its return survives being held on its own, so its value is concentrated in the role the 50/50 split holds and the all-international strategy does not. Removing it costs the sleeve only -0.09 points of compound return while taking a good home market out of the pool, so it hurts the 50/50 split more and widens the lead by 0.39 points. The United States is not what the result rests on; it is what the *comparison* rests on.

The reading should not be pushed past the column it rests on. Across the panel the correlation between a deletion's effect and the removed market's volatility drag is only -0.17: the drag explains why DEU in particular is worth more to others than to itself, not the pattern as a whole. FIN makes the point — a higher arithmetic mean still (8.91%), and removing it costs the sleeve only -0.08 points. How a market's good years line up against the fourteen others it shares any given sleeve with matters as much as its average, and that is what the measured column captures.

8.4 What sixteen countries actually support

The delete-one runs give a jackknife standard error of **2.92 points** on a lead of 5.79%, for a 95% interval of [0.08, 11.51].

The interval excludes zero, but only just: the implied t-statistic is 1.99 against a threshold of 1.96, and the lower bound clears zero by 0.08 points. The ordering is supported by the panel rather than only by the simulation, and it is supported *thinly*. The direction should be read as established and the magnitude as barely resolved.

This is the number to quote, and it is far wider than the Monte Carlo error. Every certainty equivalent in this paper is reported to four decimal places because that is what the simulation resolves; the panel resolves much less, and a hundred thousand bootstrap paths add precision to the former without adding a single country of evidence to the latter. Nothing else in this paper carries an interval of this kind, and the reader should assume every gap reported elsewhere is estimated no more precisely than this one.

8.5 Is the ranking stable through time?

Table 22. The headline on expanding windows of history, and on the two halves

Window	Country-years	Lead
1890-1950	891	4.41%
1890-1970	1,211	5.26%
1890-1990	1,531	5.82%
1890-2020	2,010	5.79%
1890-1955 (first half)	971	4.68%
1956-2020 (second half)	1,039	6.15%

Notes. Each window masks the country-years outside it; the sleeve in any year is the sleeve that year had, so a 1930 lifetime can still see 1929.

The ordering holds in every window. The weakest is 1890-1950 at 4.41%. An investor standing in 1950, with only the record to that date, would have reached the same ranking as one standing today — which is the strongest of the three checks here, because it is the one that could have been acted on contemporaneously.

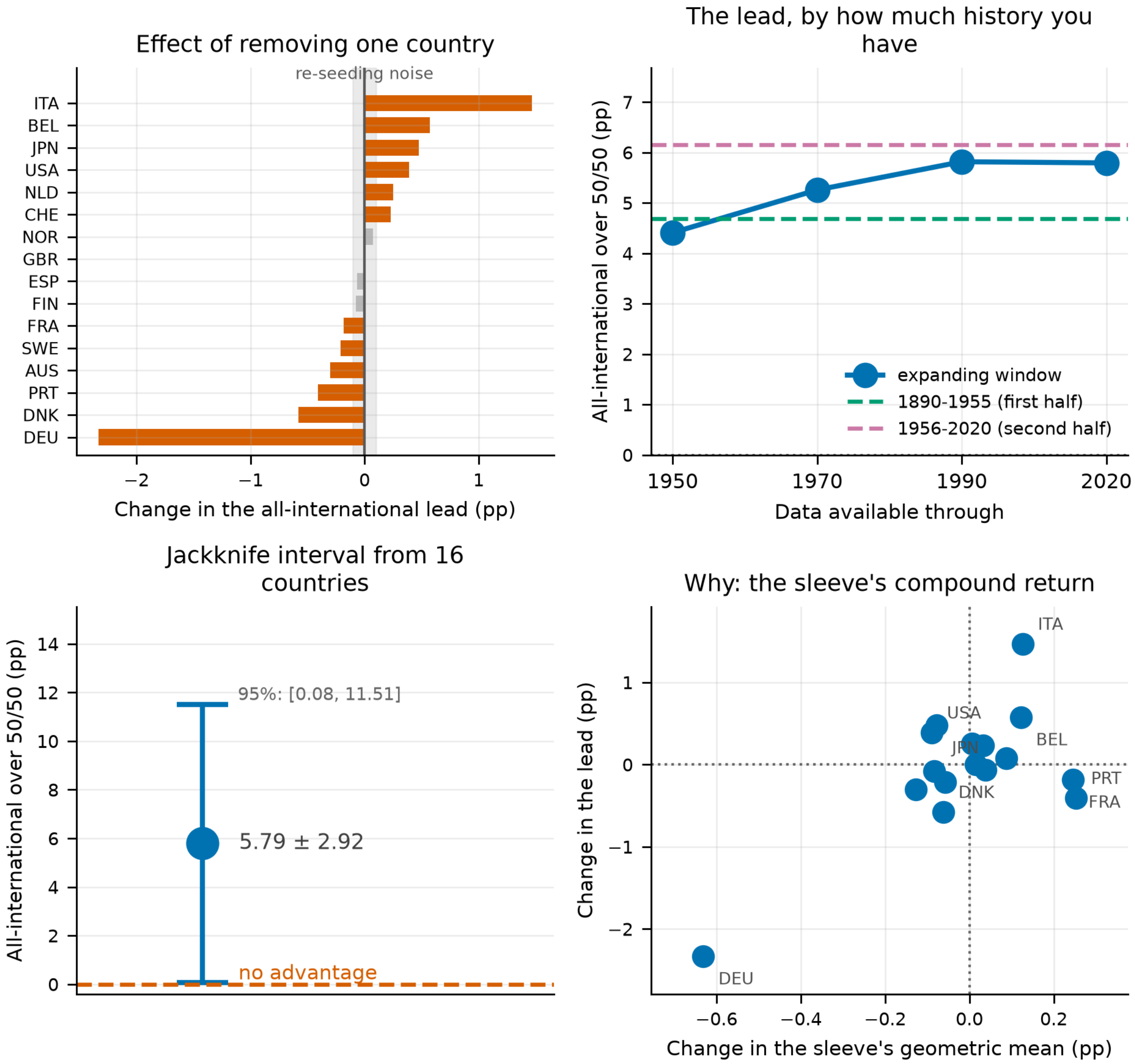

Figure 20.

Top left: what removing each country does to the headline lead, against the band a re-seeded run produces on an unchanged panel. Top right: the lead by how much history is available. Bottom left: the jackknife interval the delete-one runs support. Bottom right: why the deletions land where they do — each country's effect on the lead against its effect on the sleeve's own compound return.

8.6 What this changes

- No single market carries the headline, so it is not a restatement of one country's history — and in particular it is not a restatement of the United States'.
- The honest precision on the headline gap is ±2.92 points, not the fourth decimal place of a certainty equivalent. This is the single largest caveat in the paper and Section 35.1 carries it.
- The ranking is stable across every window of history tested, including windows an investor could have stood in.

9. How the International Sleeve Is Weighted

Every result to this point rests on an international equity leg built as a leave-one-out **equal-weighted** average of the other 15 markets. That construction is defensible and unusually favourable. An equal-weighted portfolio of 16 national markets is a more diversified object than any index a person could have bought: it holds as much Portugal as it holds the United States, it rebalances into whatever has fallen, and it never lets one market grow to dominate. A real investor holds something closer to a capitalisation-weighted index, which is concentrated by construction and grows *more* concentrated exactly when one market has run.

That matters here specifically. The one place this paper diverges from the study it re-implements — all-international beating the 50/50 domestic/international split — is a claim about how much diversification the foreign sleeve delivers. If the equal weighting is doing that work, the divergence is an artefact of panel construction rather than a finding about the world. This section rebuilds the panel under 5 weighting schemes and re-runs the headline comparison through the same summariser, so the answers differ in the weighting and in nothing else.

9.1 The schemes, and why these ones

The database supports very few honest cross-country weights. A series has to be in comparable units across countries and complete enough not to change the sleeve's membership from one year to the next. That rules out the nominal GDP, exports, loans and wage columns, which carry country-specific currency units — pesetas for Spain, lire for Italy — and the ratio columns, which are gappy and are not sizes in any case. What survives is population, Maddison real GDP per head, their product, and one scheme estimated from the returns themselves.

Table 23. The weighting schemes, ordered by how much they concentrate the sleeve

Weighting	Tilts towards	Effective markets	Sleeve SD	Corr. with home	All-intl over 50/50
Real GDP	large economies	4.6	24.3%	0.357	+3.71%
Population	populous countries	6.3	26.0%	0.333	+4.36%
Inverse volatility	historically stable markets	13.8	20.8%	0.435	+4.41%
GDP per capita	rich countries	14.1	21.3%	0.436	+6.75%
Equal-weighted	nothing	15.7	21.7%	0.429	+5.79%

Notes. Effective markets is the panel average of one over the Herfindahl index. The final column is the advantage all-international holds over the 50/50 split under that scheme — the quantity this section exists to stress.

- **Every weight is lagged.** Sizes are read from the prior year and the inverse-volatility estimate uses a strictly prior window, so no weight is formed from a number the investor could not yet have seen — the same discipline Section 11 applies to valuation terciles.
- **Every panel is paired.** All schemes share their years, countries and availability mask, so the bootstrap draws identical calendar history for each.
- **None of them is capitalisation weighting.** These are proxies. They reproduce the concentration that makes a real index a real index; they do not reproduce the wedge between an economy's size and its listed market's, and the PPP-based ones understate a market whose currency is temporarily strong — Japan in the late 1980s most of all.

The set is deliberately two-dimensional. Real GDP and population concentrate the sleeve heavily, to an effective 4.6 markets at the tightest. GDP per capita and inverse volatility barely concentrate it at all while tilting it somewhere quite different — towards rich countries and towards historically stable ones respectively. That is what lets Section 9.3 separate the effect of concentration from the effect of the tilt.

9.2 The headline under every weighting

Table 24. Certainty-equivalent consumption for every strategy under every sleeve construction, at γ = 5

Strategy	Equal-weighted	Real GDP	GDP per capita	Inverse volatility	Population
100% International Equity	1.1085	1.1098	1.1230	1.0745	1.1264
50/50 Domestic/International Equity	1.0478	1.0701	1.0520	1.0291	1.0793
Target-Date Fund (glide path)	0.9425	0.9541	0.9426	0.9329	0.9594
100% Domestic Equity	0.8818	0.8818	0.8818	0.8818	0.8818
60/40 Domestic Equity/Domestic Bonds	0.8707	0.8707	0.8707	0.8707	0.8707
100% Bills (cash)	0.7390	0.7390	0.7390	0.7390	0.7390

Notes. 100,000 lifetimes per weighting, drawn from identical calendar history. The strategies holding no international equity are unchanged across every column by construction, which is the cheapest available check that the panels differ only in the sleeve.

The ordering survives every weighting tested. All-international leads the 50/50 split by +5.79% on the equal-weighted sleeve. The harshest alternative is Real GDP, which leaves +3.71% — 64% of the reference gap. The divergence from the replicated study narrows under a realistic sleeve but does not close.

Worth noting against our own interest in the other direction too: the equal weighting is **not** the construction most favourable to the finding. GDP per capita produces a larger gap still, +6.75% against +5.79%, so the headline is not resting on the kindest sleeve available to it.

9.3 Concentration, or the tilt?

The obvious hypothesis is that concentration alone drives the result: the more the sleeve collapses onto a few markets, the less diversification it delivers and the smaller all-international's advantage. The schemes are chosen to test that, because two of them barely concentrate the sleeve at all while tilting it somewhere quite different. Across the 5 schemes the gap correlates +0.75 with the effective number of markets.

But the 3 schemes that leave concentration essentially intact still span 2.34 points of gap between them, against 3.04 points across the whole set. So the direction of the tilt matters roughly as much as the degree of concentration: a sleeve tilted towards rich countries and one tilted towards historically stable countries have almost the same Herfindahl index and materially different consequences. Concentration is not a sufficient statistic for what a weighting scheme does to a lifetime.

Table 25. Pooled moments of the international sleeve under each weighting

Sleeve	Mean	SD	Return per unit risk	Correlation with home market
Equal-weighted	8.33%	21.69%	0.384	0.429
Real GDP	9.05%	24.32%	0.372	0.357
Population	9.36%	26.04%	0.360	0.333
GDP per capita	8.30%	21.31%	0.390	0.436
Inverse volatility	7.79%	20.80%	0.375	0.435

Notes. Pooled over every available country-year. The correlation is with the investor's own domestic market.

The correlation column carries a result that reads as a surprise and is not one. Concentrating the sleeve by economy size moves its correlation with the home market down by 0.072. Because the sleeve is leave-one-out, loading it onto the largest economies makes the typical investor's foreign holding *less* like their own market, not more: a Danish investor's equal-weighted sleeve is mostly other small European markets that move with Denmark, while their GDP-weighted sleeve is mostly the United States and Japan, which do not. That effect works against the loss of diversification, which is why the certainty equivalents move less than the Herfindahl indices do.

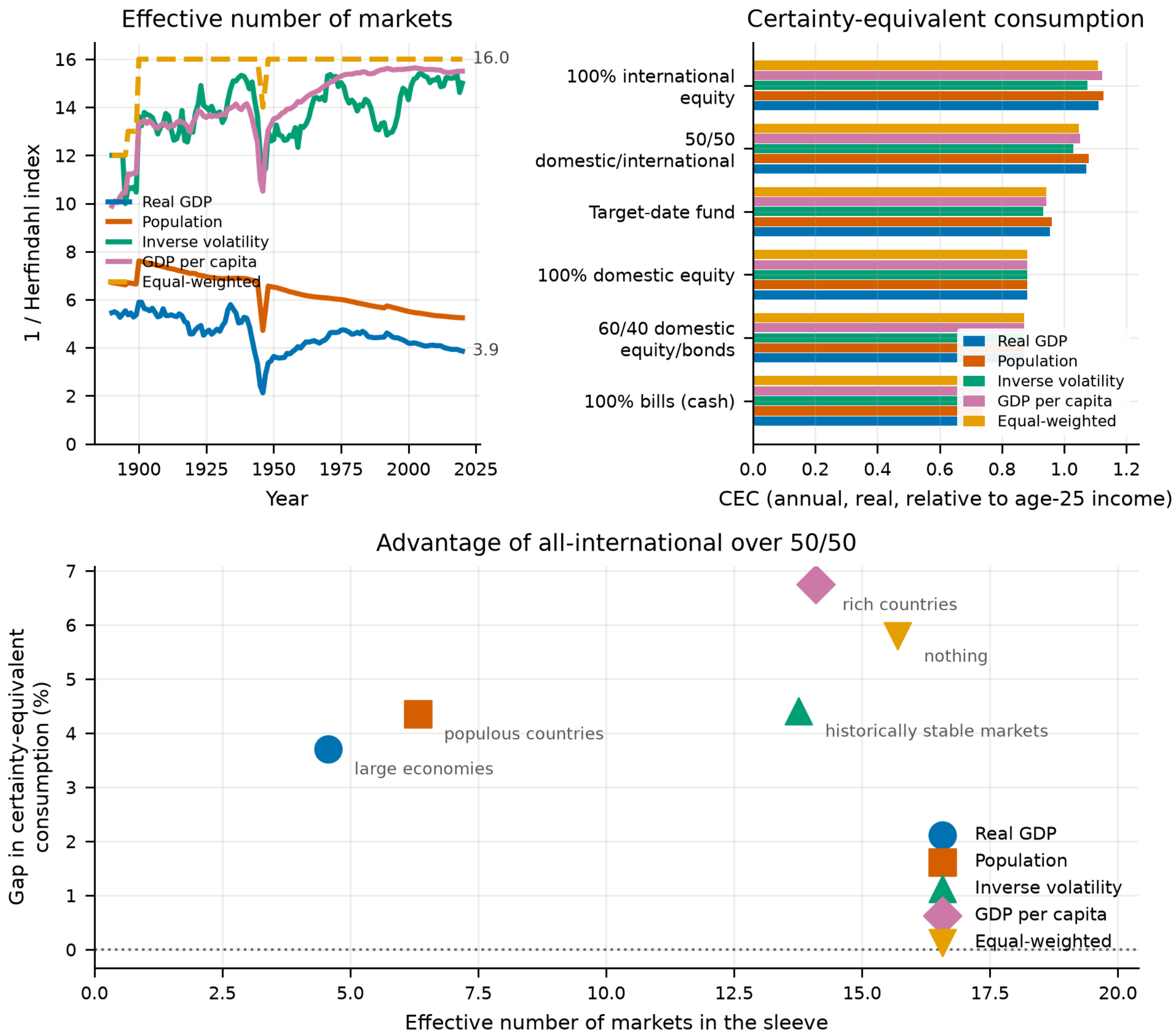

Figure 21.

Top left: the effective number of markets in the sleeve under each weighting. Top right: certainty-equivalent consumption for every strategy under all of them. Bottom: whether all-international's advantage over the 50/50 split tracks concentration or the tilt.

9.4 What this changes

- The equal weighting is not the construction most favourable to the finding — GDP per capita produces a larger gap — so the headline is not resting on the kindest sleeve available. The levels do move with the construction, and should be read as construction-dependent.
- The ordering this paper diverges from the replicated study on **does** survive every reweighting tested, so the divergence is not manufactured by the equal weighting — though the margin narrows.
- Concentration is not a sufficient statistic for a weighting scheme. Two sleeves with nearly the same Herfindahl index and different tilts give materially different answers, which is a caution against reading any single alternative construction as *the* robustness check.

- None of these is capitalisation weighting. This section brackets the answer rather than settling it: the truth sits between an equal-weighted sleeve nobody could buy and PPP-based proxies that misprice strong-currency markets. Section 35.1 carries this as a limitation.

10. Currency Hedging the International Leg

Section 13 priced one cost of holding the foreign sleeve. This section prices the other, which is a choice rather than a fee: the international leg of the headline strategy is unhedged, so a domestic investor holding it bears both foreign equity risk and foreign currency risk. Whether that currency exposure is a cost or a diversifier is an old question, and it has a clean answer in this framework because the panel contains real exchange rates.

We construct a hedged international leg using covered interest parity: the hedged return replaces the realised currency movement with the interest-rate differential implied by the two countries' bill rates, less an explicit annual hedging cost. The cost is the parameter of interest. Hedging is not free — it consumes bid-offer, collateral and roll — and the question a practitioner faces is not "is hedging good?" but "is hedging good at the price I can get it for?" Setting the cost to zero therefore gives the decision its most favourable possible reading, which is where we start.

Table 26. Break-even annual hedging cost by hedge ratio

Hedge ratio	Unhedged CEC	CEC at zero cost	Gain at zero cost (%)	Break-even annual cost
25%	1.049	1.047	-0.135	never
50%	1.049	1.038	-1.021	never
75%	1.049	1.023	-2.432	never
100%	1.049	1.004	-4.240	never

Notes. The break-even cost is the annual charge at which hedging exactly offsets its benefit. "Never" means the hedged position loses even at zero cost, so no price makes it worthwhile.

The answer here is that no price is low enough. Every hedge ratio tested loses certainty-equivalent consumption before a single basis point of cost is charged. The mildest is a 25% hedge, which gives up 0.13%; the fully hedged position gives up 4.24%. The loss grows monotonically in the ratio, so there is no break-even cost to report: the decision is settled at zero, and adding a realistic charge only widens the gap.

Tracing the optimal ratio as a function of cost adds nothing, because it is 0% at every cost including free. For a lifecycle investor holding this international leg, the model's answer is not “hedge if you can get it cheaply” but “do not hedge.”

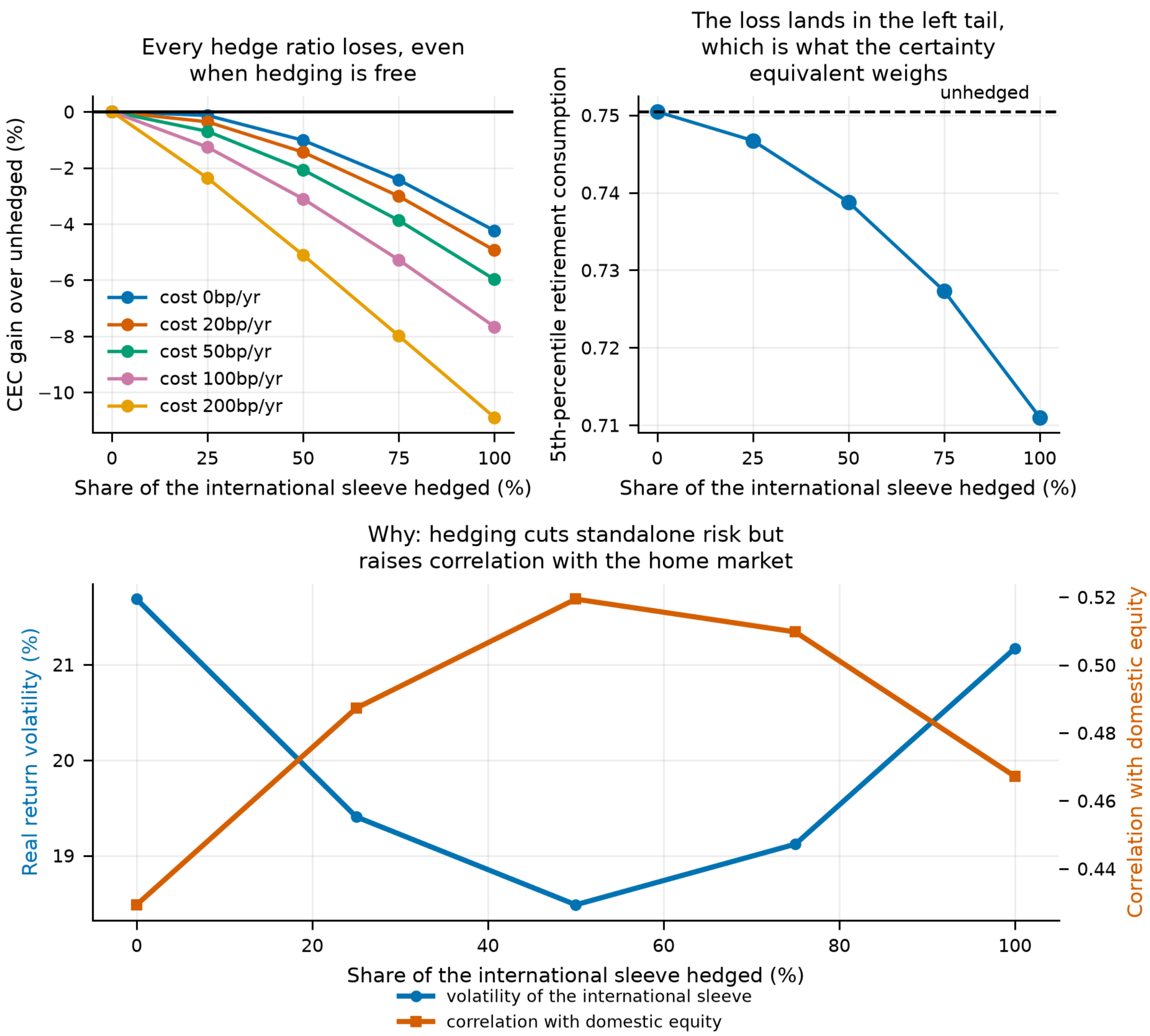

Figure 22.

Top left: the certainty-equivalent cost of hedging by ratio, at five annual hedging costs; every line is below zero, so hedging loses even when free. Top right: where the loss lands — fifth-percentile retirement consumption falls monotonically in the hedge ratio, and the certainty equivalent weighs that tail heavily. Bottom: why — hedging lowers the standalone volatility of the foreign sleeve up to a half hedge, but raises its correlation with the home market over the same range.

11. What the Market Costs When You Start

Every result so far treats one starting point as interchangeable with another. The bootstrap draws calendar windows without regard to how expensive equities were when the window opened, so a lifetime that begins at a market peak is statistically identical to one that begins at a trough. That is a strong assumption, and it is the one a reader is least able to accept: they know what today's market costs, and the model does not.

This section supplies the missing state variable. It is constrained throughout by a single rule — nothing may condition on information the investor could not have had at the moment they started.

11.1 The observable, and why it is not the obvious one

The natural candidate is the dividend-price ratio the source workbook records. It cannot be used directly. That series is a dividend *return* — the dividend paid during year t over the price at the start of it — and its numerator is unknown until the year is over. Conditioning on it would build look-ahead into every number that followed.

What an investor standing at the start of year t can actually see is the trailing yield on the current price, D(t-1) / P(t-1), which we recover as the previous year's dividend return divided by one plus the previous year's capital gain. Both terms are last year's, so the value is fully formed before the year being predicted begins.

We verify this structurally rather than statistically, because a correlation cannot tell the two cases apart: a yield built from the current year's dividend would predict the current year's return, and a correctly lagged one still correlates with the *previous* year's return because a bad year lowers the price in the denominator. Both are expected; neither distinguishes a leak from a signal. Instead we overwrite everything the workbook records for a probe year, rebuild the series, and confirm that the row for that year does not move. It does not, at six probe years spread across the panel, and the pipeline aborts rather than reporting anything if that check ever fails.

The international leg needs its own answer, since it holds many markets at once. We use the equal-weighted leave-one-out mean of the other countries' yields rather than the median, because the leg holds equal money in each market and the dividend yield of an equally weighted portfolio is the plain mean of its constituents'. The median is retained as a robustness check on the pull of any one distressed market.

11.2 The yield forecasts returns

Table 27. Trailing dividend yield and subsequent real equity returns

Horizon (years)	Observations	Correlation	Started expensive (%)	Started cheap (%)	Gap (pp)
1	1,925	0.12	4.24	9.93	5.69
10	1,772	0.24	3.17	6.32	3.15
20	1,607	0.23	3.26	5.53	2.27
30	1,447	0.28	3.02	5.20	2.18

Notes. Annualised real return over the years following the observation, split at the terciles of the yield distribution. The conditioning in this section is only worth doing if these gaps are positive.

The relationship is present at every horizon and strengthens with it. Over 30 years the correlation is 0.28 against 0.12 at 1 year, and the third of history that began cheapest returned 2.18 percentage points a year more than the third that began dearest. Compounded over 30 years that is a factor of 1.91 on terminal wealth — not a rounding difference.

11.3 Which tercile, on what was knowable at the time

A look-ahead-free yield is only half of an implementable signal. The *boundaries* are the other half, and the obvious way to draw them fails: terciles of the pooled sample put the cut-points at fixed values estimated from 1890 through 2020, so a lifetime beginning in 1910 is called cheap or dear against a threshold that already knows what happened a century later. Nobody in 1910 could have known which tercile they were standing in.

We therefore compute the boundaries recursively. A lifetime beginning in year t is ranked against every country-year strictly before t — the distribution its own investor could have seen — with a minimum-history requirement below which lifetimes are left unclassified rather than classified badly.

Table 28. Tercile boundaries an investor could have computed at the time

Lifetime begins	Country-years of history	Expensive / middling	Middling / cheap
1920	410	4.30%	4.94%
1940	697	4.33%	5.10%
1960	984	4.08%	4.89%
1980	1,303	3.85%	4.75%
2000	1,612	3.51%	4.61%
2020	1,926	3.22%	4.46%

Notes. Quantiles of every country-year strictly before the lifetime's first year. The pooled boundaries used by a hindsight split are fixed at a single pair of values for the whole sample.

The boundaries move a long way — the expensive/middling cut falls from 4.35% in 1903 to 3.22% by 2020. That movement is the whole point: early cohorts faced higher yields, and ranking them against a sample dragged down by the low-yield modern era makes them look cheaper than they could possibly have known themselves to be.

Two consequences follow, and both are results rather than defects. The correction is **large**: the pooled and recursive labellings disagree on close to a third of the lifetimes both can classify. And the buckets no longer come out balanced — a majority of lifetimes land in the expensive third. Terciles of a fixed sample are balanced by construction; terciles of an expanding one are not. Yields fell across this panel, so a lifetime drawn in year t typically begins below the average of everything before t and is expensive relative to its own history. An investor running this rule in real time would have called the market dear for most of a century while yields went on falling — which is precisely the gap between a signal fitted to a sample and one a person could have executed.

11.4 What it does to the allocation decision

Each simulated lifetime takes the blended portfolio yield at the country-year its first block opened, and is bucketed against the boundaries in force that year. Only the first block carries a starting condition: the rest of the chain is the future, which no investor chooses.

Table 29. The headline comparison, conditioned on starting valuation

Started	Lifetimes	All-equity CEC	Glide-path CEC	Advantage (%)	All-equity ruin (%)	Glide-path ruin (%)
Expensive	49,504	1.039	0.934	11.25	12.32	23.13
Middling	18,938	1.058	0.949	11.54	11.29	21.83
Cheap	17,939	1.089	0.975	11.67	9.98	20.25

The ranking survives at every starting valuation. The all-equity portfolio leads the glide path in all 3 buckets, and the advantage varies by only 0.42 percentage points across them. An investor starting at a market peak faces the same allocation answer as one starting at a trough.

What does change is the level. The valuation an investor starts at does not tell them what to hold; it tells them what to expect from holding it. That distinction matters for planning — a withdrawal rate calibrated on unconditional averages is calibrated on a mixture of starting points, most of which are cheaper than the one a reader in an expensive market actually faces.

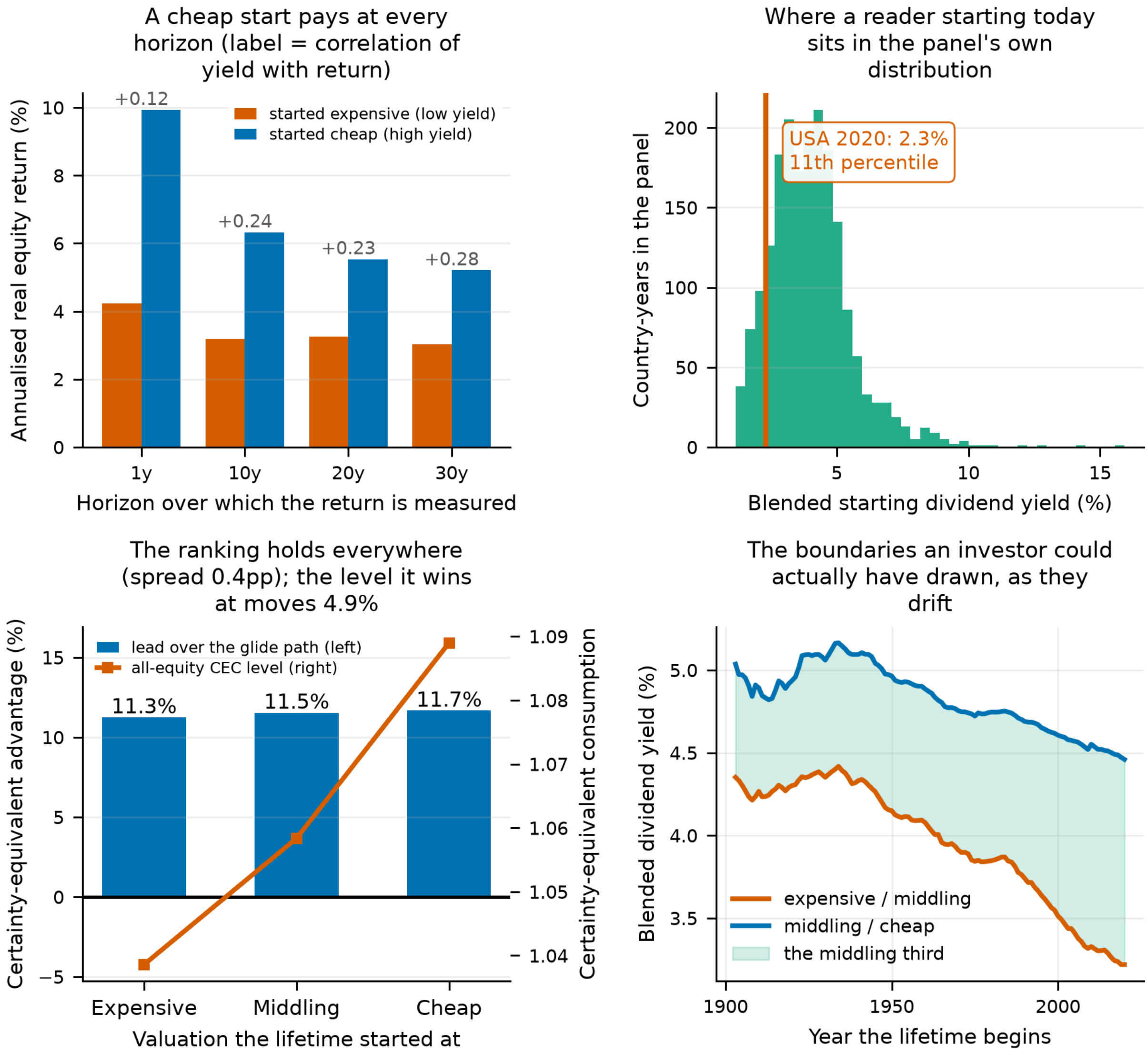

Figure 23.

Top left: annualised real equity returns following cheap and expensive starting yields, by horizon, with the correlation printed above each pair. Top right: the distribution of blended starting dividend yields across the panel's country-years, with the most recent United States observation marked so a reader can place themselves in it. Bottom left: the all-equity advantage over the glide path within each valuation bucket (bars, left axis) against the level of certainty-equivalent consumption it wins at (line, right axis) — the advantage is flat and the level is not, which is the section's whole result. Bottom right: the tercile boundaries as an investor could have computed them at each date, drifting down across the century; a pooled split replaces both lines with a single pair of values and is what the recursive construction exists to avoid.

11.5 The same yield, read at the retirement date

Everything above asks what a twenty-five-year-old should make of the market they are about to start saving into. That is not the only person who can read a dividend yield, and Section 12 gives the reason to doubt they are the one for whom it matters most: a sixty-eight-year lifetime has time to average a starting condition away, and a thirty-year decumulation does not. So the same yield is read again at the retirement date, against the tercile boundaries in force then. Both reads are look-ahead free for the person standing there. What differs is who can act on it.

Table 30. Outcomes for *50/50 domestic/international* by the valuation its investor retired into, γ = 5.

Yield at retirement	Lifetimes	CEC	P(ruin)
Expensive	53,097	1.0591	14.5%
Middling	19,382	1.0571	9.6%
Cheap	18,231	1.0011	7.6%

The same yield points opposite ways at the two dates. A lifetime that *began* in a cheap market ends +4.85% better off than one that began in a dear one — high forward returns, and sixty-eight years to compound them. A lifetime that *retired* into a cheap market ends -5.48%, the wrong side of zero, because a cheap market at sixty-three means the portfolio being handed over is small and thirty years is not long enough to make that back. The magnitudes are close (1.1 times), so it is the sign that carries the result. A dividend yield is a buy signal to a saver and a warning to a retiree, and nothing in the birth-date reading above says so.

Consumption and failure point at different buckets, which is the substance of the result rather than a footnote. Retiring into the *Expensive* bucket gives the higher certainty-equivalent consumption — the portfolio is simply worth more the day the wage stops — while the lower probability of running out belongs to *Cheap*: -5.48% in consumption against -47.67% in ruin across the same buckets. The valuation a retiree faces is a transfer between the early years of their retirement and the late ones. High prices pay out now and are repaid in the forward returns they imply.

What a retiree can choose is not a lifetime allocation — the accumulation has happened — but what to hold from the day they stop working. Sweeping those weights alone, with *50/50 domestic/international* held through the working years, gives an instruction they could follow.

Table 31. What to hold from the retirement date, by the valuation read there.

Yield at retirement	Optimal equity share	Optimal domestic share	Smallest margin (%)
Expensive	100%	20%	0.09
Middling	100%	20%	0.09
Cheap	100%	30%	0.03

Notes. The margin column is the smaller of the two, and is what stops a flat maximum being read as an identification.

The retiree's equity share does not move across the buckets, holding at 100%, and the domestic share moves 20% to 30%. Either way the instruction is a schedule read against a number the retiree can look up, not a rule of thumb about age.

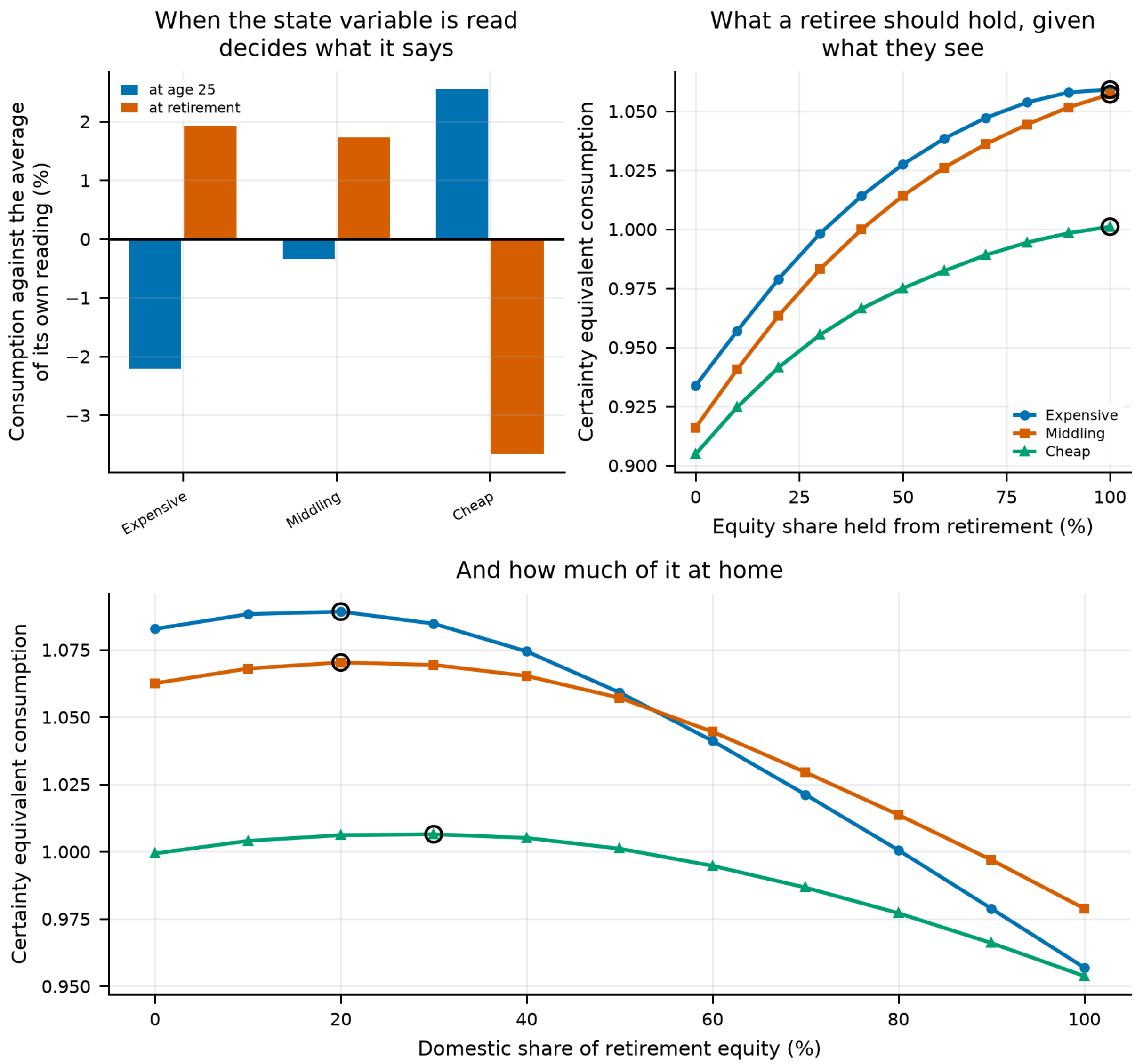

Figure 24.

Left: certainty-equivalent consumption by valuation bucket, each bucket measured against the average of its own reading, with the same lifetimes classified by the yield at age 25 and by the yield at retirement; the two readings lean opposite ways, which is the subsection's result. Middle and right: what a retiree should hold from the retirement date — equity share and then the domestic share of it — one curve per bucket, with the argmax circled.

12. What Inflation Has Just Done

Section 11 conditions a lifetime on how expensive the market was when it opened. This section conditions it on something an investor knows better than the dividend yield and reads about every month: what the price level has just done.

The two questions are not the same, and the second has a sharper mechanism behind it. A dividend yield is a claim about expected returns that has to be argued for. Trailing inflation is a fact about the price level, and every return in this paper is already deflated by it — so the channel is direct. A nominal bond promises a fixed number of currency units; if inflation is high and persistent that promise is worth less in real terms, and the asset whose real return inflation eats is the one this paper’s rivals hold most of. Equity is a claim on cash flows that reprice, which is a partial hedge over a long horizon and a famously poor one over a short.

That gives this section something Section 11 lacked: a reason to expect the *allocation* to move rather than only the level. A valuation is a statement about every asset at once. Inflation is not — it falls hardest on the legs whose payments are fixed in the currency that is losing value, and it is a domestic phenomenon, which gives the foreign leg a claim to be a hedge against it. Whether either effect is large enough to move an optimum is the question.

12.1 The observable, and the constraint

Inflation in year t is unknown until year t is over, so the quantity an investor observes on its first day is the annualised rate over the k years already finished — built from rows $t-k$ through $t-1$ and nothing later. The headline uses $k = 3$. The property is checked structurally rather than assumed: corrupting one year's inflation must leave every earlier row untouched. A correlation test could not establish it, because a leak and an honest signal look identical in a correlation.

Every correlation in this section is a **rank** correlation, and that is not a stylistic preference. The panel contains real hyperinflations — Germany 1923 at 1.06×10^9, Japan 1945 at 976%, Italy 1944 at 344%. They are observations, not errors, and deleting them would remove exactly the episodes an inflation study exists to look at. But a Pearson correlation on a series containing a billion-per-cent observation describes that observation and nothing else: it reports −0.0004 for the persistence of inflation, a series whose rank correlation with its own recent past is 0.58. The Pearson figure is carried in its own column rather than suppressed.

12.2 Does it predict anything?

The mechanism first, because without it nothing downstream has a reason to work. Trailing inflation can only predict returns if it predicts inflation.

Table 32. Trailing 3-year inflation against the inflation that followed it.

Years ahead	Rank correlation	After low inflation	After high inflation
1	0.582	0.88%	155913983.22%
5	0.497	2.05%	125.66%
10	0.354	5.21%	23.51%
30	0.116	8.29%	7.89%

Inflation is persistent, and the persistence decays. The rank correlation with next year is 0.58 and with the 30-year average 0.12. That shape — strong at short range, weak at long — is what sets the boundary on everything below, and it is the reason a sixty-eight-year lifetime turns out to be a poor place to look for an inflation effect.

Table 33. Annualised real returns over the 1 year(s) after a lifetime's start, by the trailing 3-year inflation it began at.

Asset	Rank correlation	After low inflation	After high inflation	Difference
Dom. bonds	-0.216	3.82%	-1.75%	-5.56 pp
Bills	-0.266	2.47%	-2.34%	-4.81 pp
Dom. equity	-0.100	7.62%	4.92%	-2.69 pp
Intl. equity	-0.118	9.43%	7.49%	-1.94 pp

Notes. Sorted worst-affected first. A negative difference means the high-inflation third was followed by poorer real returns.

Inflation hurts the nominal legs and largely spares equity. Over the following year the bond and bill legs give up -5.19 points a year after a high-inflation start, against -2.32 points for equity. That is the ordering the mechanism predicts: a fixed number of currency units is worth less when the currency is losing value, and a claim on repricing cash flows is not.

But the effect is a short-horizon one, and it does not survive a lifetime. The same comparison run over 30 years gives +3.54 points on bonds and +1.63 on domestic equity — the damage has decayed and in places reversed, because a long window beginning in a high-inflation year captures the disinflation that followed it. Inflation is a risk to the portfolio a retiree is holding now, not to the one a twenty-five-year-old is starting. Everything in the rest of this section should be read against that.

12.3 Which lookback window

Table 34. One, three and five years of trailing inflation, against the same forward domestic equity returns.

Lookback (years)	Observations	Rank correlation	Pearson	High minus low third
1	1,537	0.126	-0.0677	+1.40 pp
3	1,513	0.165	-0.0362	+1.63 pp
5	1,489	0.172	-0.0418	+1.56 pp

Notes. The Pearson column is the hyperinflation problem made visible: it is near zero at every window on a relationship the rank statistic finds without difficulty.

The headline uses 3 years, named in the configuration rather than chosen by search. With three candidates and one panel, picking the best-performing window and then reporting its performance would be a selection effect wearing a result; the table above is here so a reader can see what the other two would have given.

12.4 Conditioning a lifetime

Lifetimes are bucketed into terciles of the trailing rate they began at, against boundaries computed from country-years strictly before each lifetime started — the same discipline as Section 11, and for the same reason. The rate itself is look-ahead-free, but a pooled tercile boundary would not be: a lifetime beginning in 1910 would be called high-inflation against a threshold that already knew about the 1970s.

Table 35. The headline comparison inside each inflation tercile, γ = 5.

Starting inflation	Lifetimes	All-equity over target-date (%)	P(ruin), all-equity	P(ruin), target-date
Low inflation	16,223	+10.85	11.2%	20.3%
Moderate	33,621	+10.72	11.2%	21.2%
High inflation	41,573	+11.82	12.2%	23.6%

The ranking survives every inflation regime. The lead runs from 10.85% after calm years to 11.82% after inflationary ones, a spread of 1.10 points, and never changes sign. Starting inflation moves what an investor should expect; it does not move which of these two they should hold.

12.5 The optimal portfolio

Comparing two fixed portfolios answers a narrower question than the section set out to ask. Two grids are therefore scored on the same lifetimes and the same buckets — how much equity to hold, and how much of that equity to hold at home — and the certainty-equivalent maximum is read off within each regime. The composition inside each sleeve is held fixed while the parameter moves, so the argmax means what it appears to mean.

Table 36. The equity share that maximises certainty-equivalent consumption in each inflation regime.

Starting inflation	Optimal equity share	CEC there	Best over worst (%)	Margin over runner-up (%)
Low inflation	100%	1.0491	+37.7	2.218
Moderate	100%	1.0548	+36.9	2.045
High inflation	100%	1.0502	+37.9	2.367

Notes. The margin column is the honesty check: a winner that beats the runner-up by a rounding error has not identified an optimum, and a shift between buckets is only a finding if the grid can resolve it.

Table 37. And how much of that equity belongs at home.

Starting inflation	Optimal domestic share of equity	CEC there	Over all-international (%)	Over the next grid point (%)
Low inflation	10%	1.1111	+0.99	0.031
Moderate	10%	1.1200	+0.66	0.257
High inflation	10%	1.1262	+0.78	0.226

Notes. The fourth column is the comparison that matters — the optimum against the all-international corner this paper otherwise treats as the winner. The fifth is the narrower question of whether the grid can separate one step from the next.

The equity optimum sits on the boundary of its grid in every regime — the search wanted 100% equity and was not allowed more. That is a limit of the grid rather than an interior solution, and it is consistent with Section 21, which finds borrowing to buy more equity worth +5.10% when credit is free. What matters here is that the boundary is the same boundary in all three regimes: inflation does not move it.

The home/abroad optimum is interior, and it is not zero. The grid runs from nothing at home to everything, in steps of ten points, and in every inflation regime the maximum sits at 10% domestic — worth +0.66% over the all-international corner that the six fixed strategies of Section 5 treat as the winner. That corner was only ever the best of six; offered a finer grid, the panel wants a little of the home market back.

Three things say this is real rather than a lucky grid point, and none of them involve inflation. The same sweep run on all 16 markets' lifetimes at once, with no conditioning of any kind, peaks at 10% domestic. Section 20, which solves the whole four-asset simplex at every age with no grid at all and by an entirely different procedure, lands on 9.6%. And the panel holds 16 markets, so an equal-weighted world portfolio puts 6.2% in the home market — which is where all of these sit.

The reading is not that home bias pays. It is that the leave-one-out sleeve this paper uses as its international leg is, for any one investor, the world *minus their own market* — a slight underweight of home rather than a neutral position — and that adding back roughly the missing sixteenth restores it. The strategy this paper calls "100% international" is a corner of a six-item menu that offers nothing between nought and a half; it wins that menu on merit, and it is not the optimum. What this section adds is that the correction does not depend on the inflation regime: the same 10% in all three.

Neither optimum moves. The equity share that maximises the certainty equivalent is 100% in every regime, and the domestic share is 10% in every regime. Recent inflation changes what a lifetime is worth and does not change what it should hold — the same answer Section 11 reached about starting valuation, arrived at through a variable with a much more direct mechanism. The optimum's *level* is a finding; its *invariance to inflation* is this section's finding.

12.6 The same question asked of the retiree

Everything above conditions a lifetime on the inflation its investor saw at twenty-five. That is the implementable version of the question — it is what a saver can act on — and it produces a null, for a reason this section has already given: the damage is short-horizon, and a sixty-eight-year window averages it away.

But the utility window in this paper is *retirement consumption alone*, and a retiree drawing down over 30 years cannot wait a shock out. So the state variable is read a second time, at the retirement date rather than the birth date. A twenty-five-year-old cannot know what it will say — this is not an instruction to them — but a 63-year-old standing there observes it exactly as reliably, and is bucketed against tercile boundaries built from history before *their* retirement rather than before their birth.

Table 38. The headline comparison inside each retirement-date inflation tercile, γ = 5.

Inflation at retirement	Lifetimes	All-equity over target-date (%)	P(ruin), all-equity	P(ruin), target-date
Low inflation	17,333	+7.49	8.9%	15.4%
Moderate	36,693	+11.24	10.8%	20.2%
High inflation	41,953	+12.07	14.2%	27.2%

When the state variable is read decides whether it matters at all. Retiring into the high-inflation third rather than the low one is worth -8.59% of certainty-equivalent retirement consumption. Conditioning the same lifetimes on the inflation they *began* at is worth +0.10% — 82.3 times smaller, and of the opposite sign. The null above was a statement about the horizon, not about inflation. This is the same panel, the same state variable and the same terciles; only the date it is read at has changed.

12.7 What a retiree should hold

Sweeping a lifetime allocation answers a question no retiree can act on: they cannot go back and hold something else from twenty-five. What they can choose is what to hold from the day they stop working, with whatever accumulation they arrived with. The portfolios swept here are therefore identical to the baseline through the working years and differ only afterwards, so the maximum in each bucket is an instruction a retiree could actually follow.

Table 39. How much equity to hold from retirement, by the inflation observed at that date.

Inflation at retirement	Optimal equity share	CEC there	Over holding no equity (%)	Over the next grid point (%)
Low inflation	90%	1.0845	+11.15	0.171
Moderate	100%	1.1144	+13.93	0.035
High inflation	100%	0.9890	+13.40	0.458

Table 40. And how much of that equity at home.

Inflation at retirement	Optimal domestic share	CEC there	Over all-international (%)	Over the next grid point (%)
Low inflation	30%	1.0899	+0.88	0.023
Moderate	20%	1.1374	+0.41	0.046
High inflation	20%	1.0104	+0.65	0.142

The retiree's optimum moves with the regime. The equity share runs from 90% in the calm third to 100% in the hot one, and the domestic share from 30% to 20%. At least one of them sits inside the grid's resolution and should be read against the margin columns rather than as a clean identification.

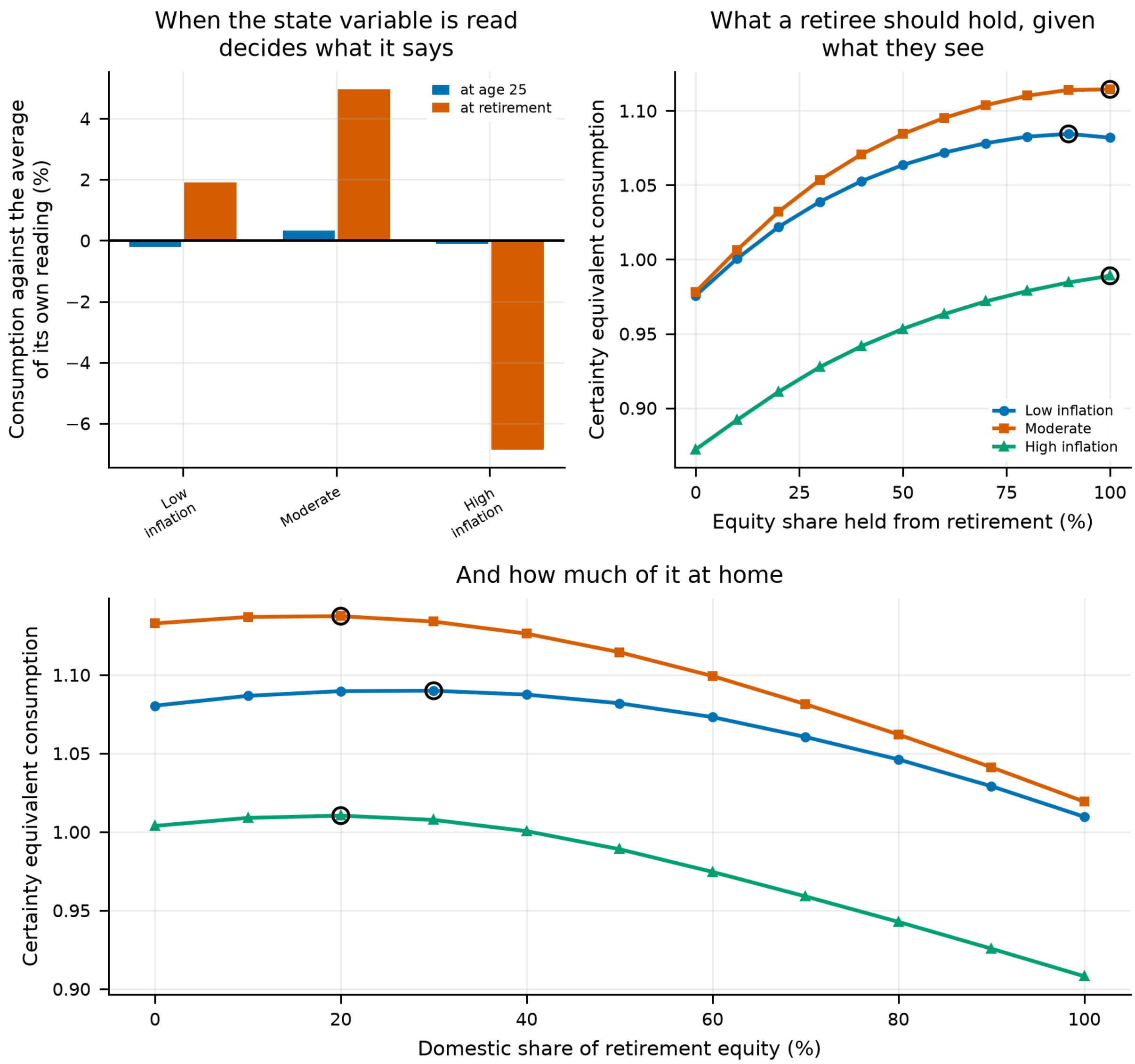

Figure 25.

Left: the same portfolio's certainty-equivalent consumption by inflation tercile, each bucket measured against the average of its own reading, with the tercile assigned at age 25 and again at retirement — a difference rather than a level, because the levels differ by too little to see and truncating the axis would exaggerate them. Middle and right: what a retiree should hold from the day they stop working, given the inflation they observe then, with the maximum circled.

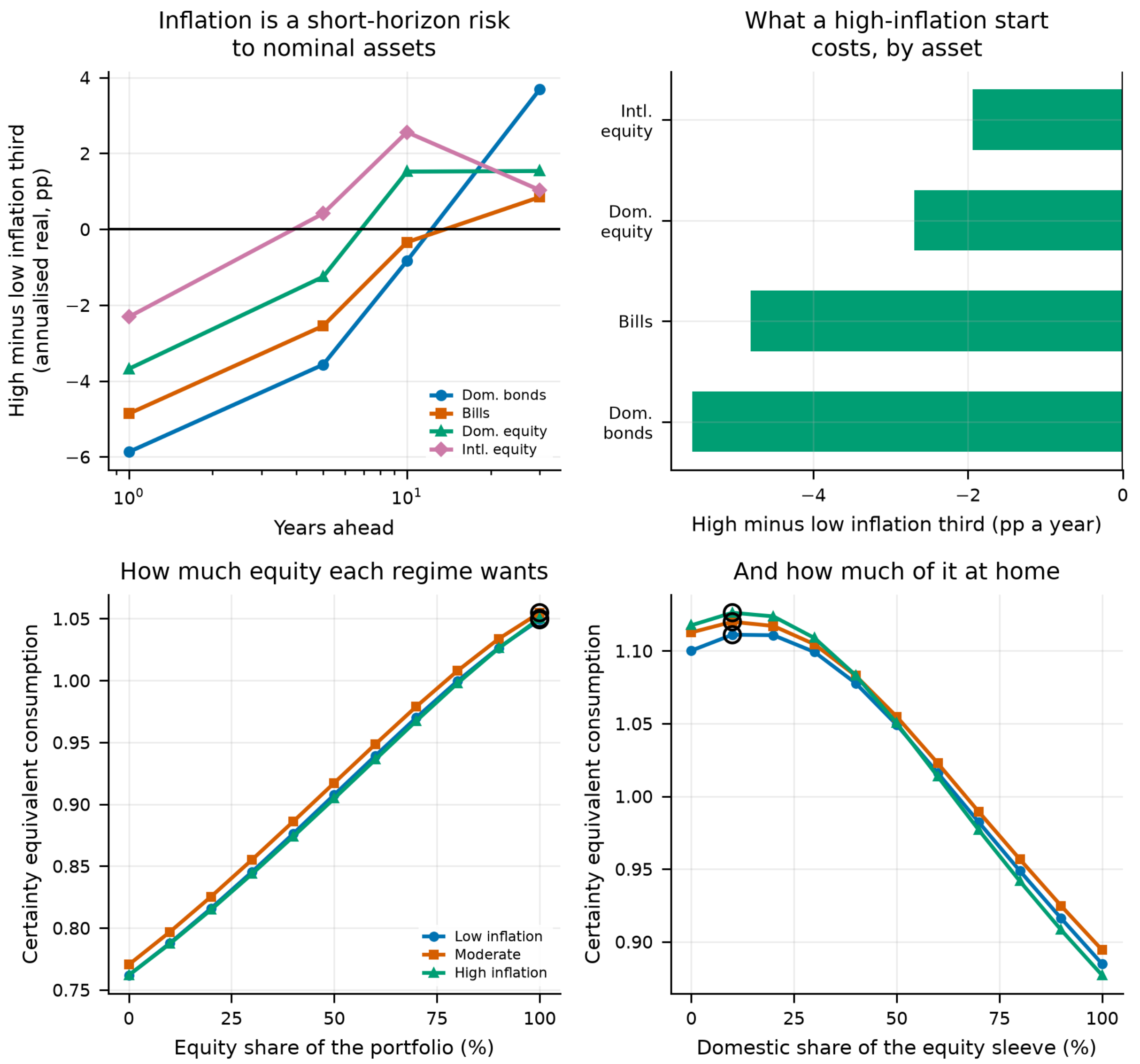

Figure 26.

Top left: how far a high-inflation start sets each asset back, by horizon — the effect is large at one year and gone by thirty. Top right: the same comparison at the headline horizon, by asset. Bottom left: certainty-equivalent consumption against the equity share, one curve per inflation regime, with the maximum circled. Bottom right: the same against the domestic share of the equity sleeve.

12.6 What this changes

- **Inflation is a short-horizon risk to nominal assets and a long-horizon non-event.** Over the following year it costs the bond leg -5.56 points a year; over the following 30 years it is worth +3.54. A lifecycle study is the wrong instrument for measuring it, and that is a finding about the instrument as much as about inflation.
- The headline ranking is unaffected: the lead spans 1.10 points across the terciles without changing sign.
- The optimal portfolio is unaffected on both axes tested. Where Section 11 could be accused of testing a weak signal, this section tests a strong one with a direct mechanism and reaches the same conclusion, which makes the pair of them better evidence than either alone.
- **What is not modelled**: inflation-linked bonds, which are the instrument this section's mechanism most obviously calls for and which did not exist over most of the panel; and any policy response to inflation, since

the withdrawal rules here are nominal-blind by construction. Both would raise the value of conditioning on inflation rather than lower it, so the null above is a floor.

13. Fees, and the Differential That Would Undo the Headline

Every return in this paper is gross. Nobody earns a gross return. The omission is defensible for a comparison between strategies drawn from the same panel — a cost common to all of them largely cancels — but it is not defensible for the one divergence from the study we re-implement, because the strategies at stake hold different amounts of the sleeve that costs more. All-international pays the international fund's expense ratio on the whole portfolio; the 50/50 split pays it on half. A **differential** falls on them unequally and compounds over a sixty-eight-year lifetime.

An expense ratio is levied on assets rather than returns, so a gross real return r net of a fee f is $(1 + r)(1 - f) - 1$ rather than $r - f$; the difference is second-order in one year and not in sixty-eight. Fees are charged on the panel before the bootstrap sees it and the availability mask is left alone, so every fee level draws the same blocks.

13.1 The control: a fee on everything

Table 41. A fee charged on all four sleeves alike

Fee on every asset	Lead
0 bp	5.79%
10 bp	5.72%
25 bp	5.61%
50 bp	5.41%
75 bp	5.20%
100 bp	4.99%

Notes. The control. A cost common to every strategy should largely cancel out of a comparison between them.

The gap never closes on this grid, which is the control working as intended. It does not cancel entirely — 100 bp on everything still costs 0.80 points of lead, because the two strategies accumulate different amounts of wealth for the fee to be charged on — but that is a fraction of what the same fee does when it falls on the foreign leg alone.

13.2 The question: a fee on the foreign leg alone

Table 42. All-international over the 50/50 split, by fee differential, at γ = 5

Extra fee on the foreign leg	Lead
0 bp	5.79%
5 bp	5.51%
10 bp	5.24%
15 bp	4.97%
20 bp	4.70%
30 bp	4.17%
50 bp	3.12%
75 bp	1.85%
100 bp	0.64%
150 bp	-1.62%
200 bp	-3.63%

Notes. 100,000 lifetimes per level. The grid runs past anything realistic on purpose: the distance between the break-even and what an investor can actually pay is the margin of safety.

Table 43. The lead at expense ratios an investor has actually faced

Reference	Differential	Lead at that differential
modern index funds	5 bp	5.51%
index funds circa 2000	19 bp	4.75%
an active international fund	75 bp	1.85%

Notes. These are configured anchors, not findings. Nothing in this paper's data can verify them; they are here to put the swept grid on a scale a reader recognises.

The result survives every fund cost a real investor has faced. Cancelling a lead of 5.79% takes a differential of 114 basis points, wider than any reference expense ratio above — including an active international fund at 75 bp, which still leaves 1.85%.

The erosion is real even so, and worth quoting. Each basis point of differential costs roughly 0.047 points of lead, so an investor paying 75 bp over a domestic fund keeps 32% of the advantage this paper reports. Fees do not decide the question, but they are not free either, and nothing in the gross-return tables elsewhere shows that.

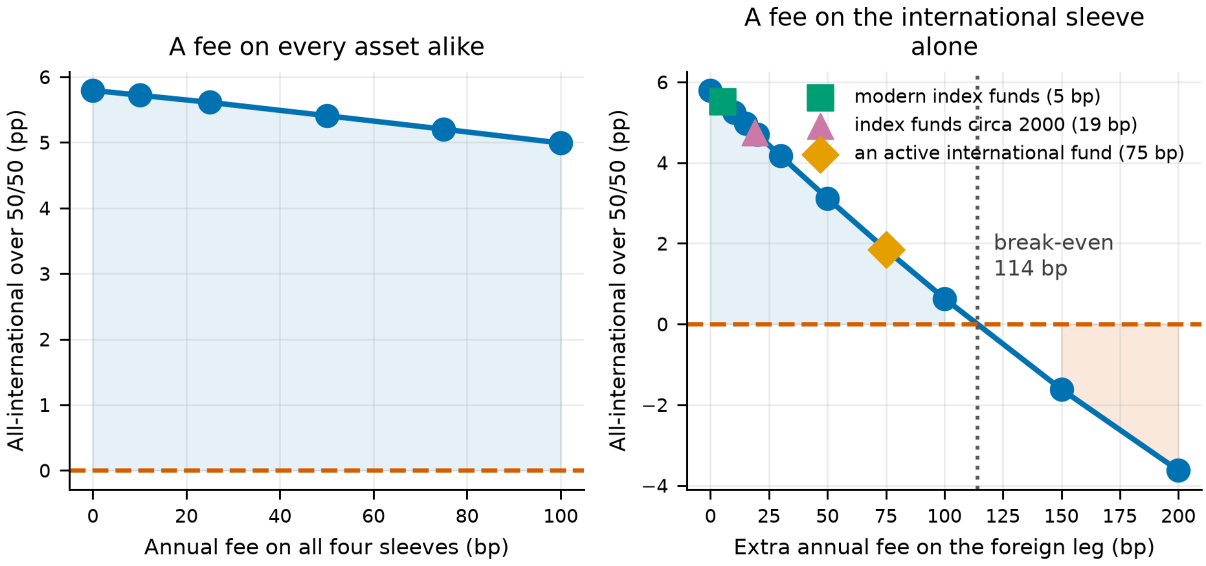

Figure 27.

Left: a fee charged on every asset alike, the control. Right: a fee on the international sleeve alone, with the break-even and the expense ratios a real investor has faced marked on it.

13.3 What this changes

- Implementation cost does not decide this comparison. The break-even differential of 114 bp is far outside what index funds charge, so the ranking can be taken at face value on cost grounds.
- The rest of the paper works in gross returns, which is defensible for comparisons between strategies holding the same sleeves in different proportions and would not be for a comparison between asset classes of different cost. No result elsewhere turns on the latter.
- What is **not** modelled: trading costs, spreads, taxes, platform fees, and the fact that index funds did not exist for most of this sample. Before roughly 1975 a diversified foreign portfolio was expensive and often impossible to hold, so this is a sensitivity rather than a history of what was available.

14. The Cost Differential That Is Not a Fee

Section 13 asks how large a cost differential between a domestic and an international fund would have to be to cancel the headline, and answers 114 basis points — a number it describes as beyond any index-fund pair. That framing has a hole in it. There is a cost differential between holding your own market and holding everyone else's that is not a fund's fee, is not negotiable, and is not a choice: **foreign dividend withholding tax**.

A government taxes dividends leaving its borders. A resident holding that same market pays nothing on them, or receives a full credit; a foreigner pays and, in an ordinary retail structure, cannot reclaim. The statutory United States rate on dividends to non-resident individuals is 30%, most treaties reduce it to 15% for portfolio investors, and a broad developed-markets index fund bears a weighted average of roughly 7.5% across its constituents. None of those is a price an investor can shop around for.

Two things make this a different experiment from a fee, and both cut against the sleeve this paper recommends. The **base is dividends, not assets**: a fee is levied on the whole portfolio, withholding only on the part of the return that arrives as a dividend — so the drag tracks the dividend yield, which in this panel has fallen by roughly a third since the war. And it falls on **exactly one leg**: all-international pays it on every dividend it receives, the 50/50 split on half, a domestic-only portfolio on none. Section 13 builds that asymmetry by hypothesis. Here it is the law.

14.1 Putting a tax rate in fee units

The panel's source data compounds rather than adds — one plus the total return equals one plus the capital gain times one plus the dividend, verified to eight decimal places, with the dividend measured against the ending price. Withholding at rate τ leaves the investor $(1 - \tau)$ of that dividend, so the after-tax return is the gross return multiplied by $(1 - \tau q)$, where $q = dp/(1 + dp)$ is the share of the year's ending value that arrived as a taxable dividend. That is exactly the multiplicative form Section 13 uses for an expense ratio, with a time-varying rate in place of a constant — which is what makes a statutory tax rate comparable with a fund's fee at all.

Table 44. What a 15% withholding rate actually costs, by era.

Era	Mean dividend share of the return	Drag on the sleeve (bp a year)
whole panel	0.0383	57.5
1890-1949	0.0438	65.6
1950-1989	0.0382	57.3
1990-2020	0.0279	41.9

Notes. A statutory rate is a constant; the drag it produces is not, because it is levied on a dividend yield that has fallen. An investor facing the same law in 1930 and in 2010 paid materially different amounts for it.

That translation is the section's first result, and it is worth pausing on. At the panel's own mean dividend share of 0.0383, the 30% statutory rate is worth **115 basis points a year** on the international sleeve. The break-even differential Section 13 reports — the one it calls beyond any index-fund pair — is 114 basis points. They are the same order of magnitude, and one of them is not optional.

14.2 Who wins, and at what rate

Table 45. All-international's lead over each rival, γ = 5.

Rate	vs 50/50 domestic/international	vs Target-date fund	vs 100% domestic equity	Best strategy
0.0%	+5.79%	+17.61%	+25.71%	100% international equity
5.0%	+4.72%	+15.83%	+23.31%	100% international equity
7.5%	+4.20%	+14.97%	+22.14%	100% international equity
10.0%	+3.69%	+14.11%	+20.99%	100% international equity
15.0%	+2.67%	+12.44%	+18.74%	100% international equity
20.0%	+1.69%	+10.83%	+16.57%	100% international equity
25.0%	+0.76%	+9.28%	+14.48%	100% international equity
30.0%	-0.15%	+7.80%	+12.48%	50/50 domestic/international
35.0%	-1.02%	+6.37%	+10.55%	50/50 domestic/international
40.0%	-1.84%	+5.01%	+8.71%	50/50 domestic/international
50.0%	-3.37%	+2.46%	+5.24%	50/50 domestic/international

Table 46. Where each rival overtakes all-international.

Rival	Lead at a zero rate	Overtakes at	which is
50/50 domestic/international	+5.79%	29.2%	112 bp a year
Target-date fund	+17.61%	never on this grid	—
100% domestic equity	+25.71%	never on this grid	—

Notes. The last column converts the crossing rate into the annual drag it represents at the panel's mean dividend share, so it can be read against the fee differential of Section 13.

The headline does not survive a statutory withholding rate. *50/50 domestic/international* overtakes all-international at 29.2%, which is inside the 30% a non-resident pays without a treaty and outside the 15% a documented one pays. 1 of 3 rivals overtake somewhere on the grid.

Table 47. The rates that actually exist, placed on the swept curve.

Rate	%	Drag (bp a year)	vs 50/50 domestic/international	vs Target-date fund	vs 100% domestic equity
a recovering pension fund	0.0	0	+5.79%	+17.61%	+25.71%
a broad developed-markets index fund	7.5	29	+4.20%	+14.97%	+22.14%
the typical treaty rate on portfolio dividends	15.0	57	+2.67%	+12.44%	+18.74%
the US statutory rate for non-residents	30.0	115	-0.15%	+7.80%	+12.48%

Notes. These are configured anchors, not findings: nothing in this project's data can verify a statutory rate.

14.3 What to hold at each rate

Asking which of two fixed portfolios wins presumes the investor is choosing between two fixed portfolios. They are not — they are choosing a weight. The same domestic-share grid Section 12 uses is scored at every rate, on the same paths, and the certainty-equivalent maximum read off. This is the more useful of the two answers, because it is a schedule an investor can look up against the rate they actually face.

Table 48. The certainty-equivalent-maximising domestic share at each rate.

Withholding rate	Optimal domestic share of equity	CEC there	Over all-international (%)	Over the next grid point (%)
0.0%	10%	1.1168	+0.75	0.209
5.0%	10%	1.0970	+0.89	0.017
7.5%	20%	1.0882	+1.04	0.073
10.0%	20%	1.0797	+1.21	0.164
15.0%	20%	1.0629	+1.52	0.331
20.0%	20%	1.0467	+1.83	0.377
25.0%	20%	1.0309	+2.12	0.176
30.0%	30%	1.0158	+2.42	0.021
35.0%	30%	1.0029	+2.88	0.202
40.0%	30%	0.9903	+3.31	0.373
50.0%	40%	0.9663	+4.12	0.017

Notes. The last column is the honesty check: a winner that beats its neighbour by a rounding error has not identified an optimum.

The optimal portfolio walks home as the tax rises. The domestic share that maximises the certainty equivalent moves from 10% at a zero rate to 40% at 50%, +30 points. Nothing about the return panel has changed between those rows — only who is allowed to keep the dividends.

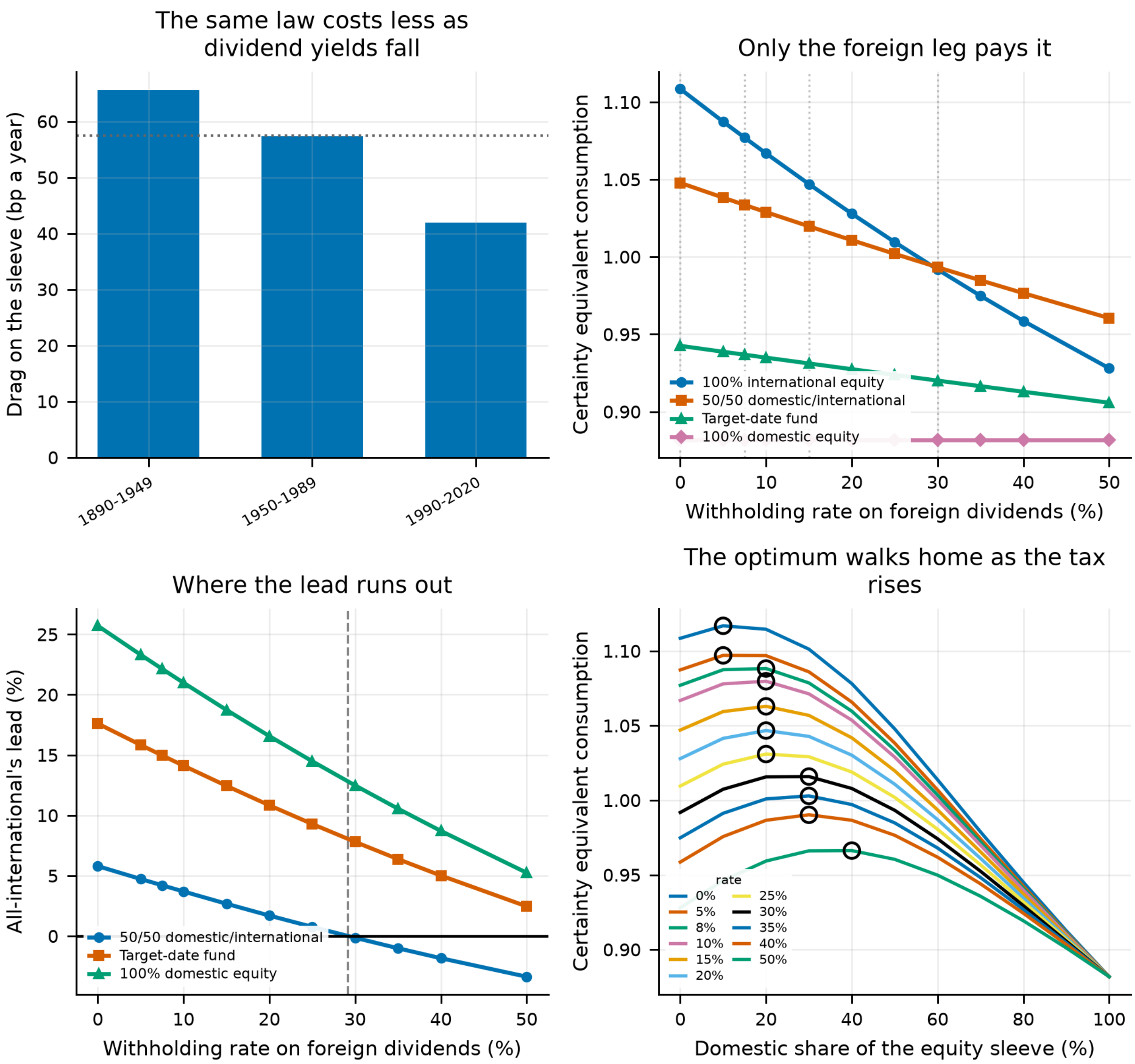

Figure 28.

Top left: what one statutory rate costs by era, as dividend yields fall. Top right: each fixed strategy's certainty equivalent against the rate, with the real rates marked. Bottom left: all-international's lead over each rival, and where it runs out. Bottom right: the certainty-equivalent surface over the domestic share, one curve per rate, with the maximum circled.

14.4 What this changes

- **Section 13 understates the cost differential an international investor faces, because it looks only at fees.** At the panel's own dividend share the 30% statutory rate is worth 115 basis points a year against that section's 114-point break-even. The break-even is not beyond reach; it is roughly the law.
- The headline ranking changes inside the statutory range, so it is conditional on the investor’s tax position and residence. A reader who can hold foreign equity through a vehicle that reclaims the tax — many pension wrappers can — faces the top row of these tables. A reader in an ordinary taxable account does not.
- The optimal domestic share is the number to carry rather than the winner of a two-horse race, and it moves with the rate. That is the practical form of this section's answer.
- **What is not modelled here**: dividend imputation, which refunds corporate tax to *domestic* shareholders and widens the home market's advantage further — Section 15 takes it up, and it changes the answer this section reaches; the second layer of withholding an investor suffers by holding a US-domiciled fund of

foreign stocks rather than the stocks themselves; reclaim procedures, which recover part of the tax at a paperwork cost most retail investors do not pay; and the investor's own income tax on the dividend afterwards. Every one of those runs the same way, against the international sleeve, so the rates here are a floor on the real burden rather than an estimate of it.

15. Dividend Imputation, and the Other Blade

Section 14 prices the tax that falls on the international leg and finds it large enough to matter — 57 basis points a year at the treaty rate. It closes by listing what it does not model, and the first item on that list is the one this section takes up, because it runs the other way and lands on the other leg.

Under a classical tax system a company pays corporate tax and its shareholder is then taxed again on what is distributed. Under an *imputation* system the corporate tax counts as tax the shareholder already paid: the dividend arrives with a credit attached, and a shareholder taxed below the corporate rate can be refunded the difference in cash. Australia has run such a system since 1987 and has made the credit fully refundable since 2000, so a superannuation fund in pension phase — which pays no tax at all — collects the whole credit as a cheque.

The credit is available to residents only, which makes it the exact mirror of withholding: one falls only abroad, one lands only at home, and both are levied on dividends rather than on assets. That symmetry is what allows the two to be quoted in the same units and then added. Adding a credit on top of a pre-tax return is not double-counting the same relief: an equity total return is already net of corporate tax, so the baseline here is an investor who keeps the cash dividend and pays no *personal* tax on it, and imputation refunds the *corporate* tax on top of that.

15.1 What a credit is worth

Write q for the dividend's share of the gross return — the quantity Section 14 already builds — and let $t\text{c}$ be the corporate rate imputed, $t\text{f}$ the rate the holder's own fund pays, and φ the franked fraction. A franked dollar of dividend grosses up to $1/(1 - t\text{c})$, is taxed at $t\text{f}$ and returns $(1 - t\text{f})/(1 - t\text{c})$; an unfranked one simply returns $(1 - t\text{f})$. So a dollar of cash dividend is worth $1 + c$ with $c = (1 - t\text{f})(1 + \varphi\, t\text{c}/(1 - t\text{c})) - 1$, and the after-credit return is $(1 + r)(1 + c\, q)$ — Section 14's expression with $-\tau$ replaced by $+c$.

Table 49. The credit implied by each holder's position, y = 5.

Who holds it	Company rate	Fund's own rate	Franked	Credit (% of the dividend)
no imputation, no fund tax	30%	0%	0%	+0.00
a taxed fund, nothing franked	30%	15%	0%	-15.00
an Australian fund still accumulating	30%	15%	100%	+21.43
an Australian fund paying a pension	30%	0%	100%	+42.86

Notes. Nothing in the last column is written down; each is the three numbers beside it put through the formula above. The row worth pausing on is the taxed fund holding unfranked stock, which comes back *negative*: a fund paying 15% on a dividend carrying no credit is worse off than the untaxed baseline this paper otherwise assumes. The formula is not built to produce a credit, and the sign is the first check that it is a tax code being modelled.

Partial franking scales it, and no market's franking level is observable in this project's data, so it is swept rather than assumed. Inside a fund taxed at 15% the credit has that tax to make back before it is worth anything at all, and solving for where it does gives 41.2% of dividends franked. Below that a fund holding its own market is behind this paper's untaxed baseline rather than ahead of it; the +21.4% in the paragraph above is what full franking delivers, and it is an upper bound on what any real market does.

15.2 What it is worth in basis points

A credit rate means nothing beside a fee until it has been put in the same units. At the headline credit of 42.9% — fully franked, pension phase — the home leg gains 164 basis points a year across the panel, against the 57 the foreign leg is losing to withholding at 15%. The wedge between the two legs is therefore 221 basis points, which is larger than either half and larger than the fee differential Section 13 identifies as the break-even.

Table 50. The credit at 42.9%, by era.

Era	Mean dividend share	Worth of the credit (bp a year)
whole panel	0.0382	+164
1890-1949	0.0439	+188
1950-1989	0.0382	+164
1990-2020	0.0281	+120

Notes. The same tax code delivers less as time passes, for the same reason the withholding drag falls in Section 14: both are levied on dividends, and this panel's dividend yields have fallen by roughly a third.

15.3 Both blades, closed

Neither section on its own describes anybody. An investor collecting franking credits is also paying withholding, simultaneously, and the two push the same way. The table below scores the handful of positions someone can actually stand in, each on identical paths.

Table 51. Certainty-equivalent consumption at each position, γ = 5.

Where the investor stands	Credit	WHT	All intl.	50/50	Glide path	All dom.	Best
neither tax	+0.0%	0%	1.1085	1.0478	0.9425	0.8818	All intl.
withholding only	+0.0%	15%	1.0470	1.0197	0.9311	0.8818	All intl.
a taxed fund, nothing franked	-15.0%	15%	1.0470	0.9958	0.9159	0.8553	All intl.
an Australian fund still accumulating	+21.4%	15%	1.0470	1.0556	0.9536	0.9215	50/50
an Australian fund paying a pension	+42.9%	15%	1.0470	1.0934	0.9769	0.9635	50/50

Notes. WHT is the withholding rate the foreign leg pays. The first row is this paper's own baseline and the second isolates withholding, so the table reads as a decomposition rather than as a list of scenarios.

The wedge overturns the headline, and neither blade does it alone. With withholding abroad and no credit at home *100% international equity* still wins, which is Section 14's finding restated. Add the credit and *50/50 domestic/international* wins instead. Nothing about the returns changed between those rows; what changed is which leg the tax code is standing on.

The swept curve locates the reversal exactly. *50/50 domestic/international* overtakes at a credit of 16.4% of the dividend, and a fully franked dividend is worth 21.4% inside a fund still accumulating and 42.9% inside one paying a pension. Both clear it, which is why the reversal is not a statement about an extreme parameter.

15.4 What to hold, rather than who wins

Table 52. The certainty-equivalent-maximising domestic share at each credit, swept on top of 15% withholding abroad.

Credit	Optimal domestic share	CEC at the optimum	Margin over the runner-up (%)
-15.0%	10%	1.0537	0.21
+0.0%	20%	1.0629	0.33
+5.0%	20%	1.0668	0.41
+10.0%	20%	1.0706	0.24
+15.0%	20%	1.0746	0.06
+21.4%	30%	1.0812	0.15
+25.0%	30%	1.0853	0.27
+30.0%	30%	1.0912	0.44
+35.0%	30%	1.0971	0.37
+42.9%	30%	1.1065	0.13
+50.0%	40%	1.1163	0.10
+60.0%	40%	1.1320	0.41

Notes. The margin column is what stops a flat maximum being read as an identification.

The optimum walks home as the credit rises, from 20% with no credit at all — this paper's own baseline — to 40% at the top of the grid. That is the practical form of the answer: not which of two portfolios wins, but how much of the home market a given tax position justifies.

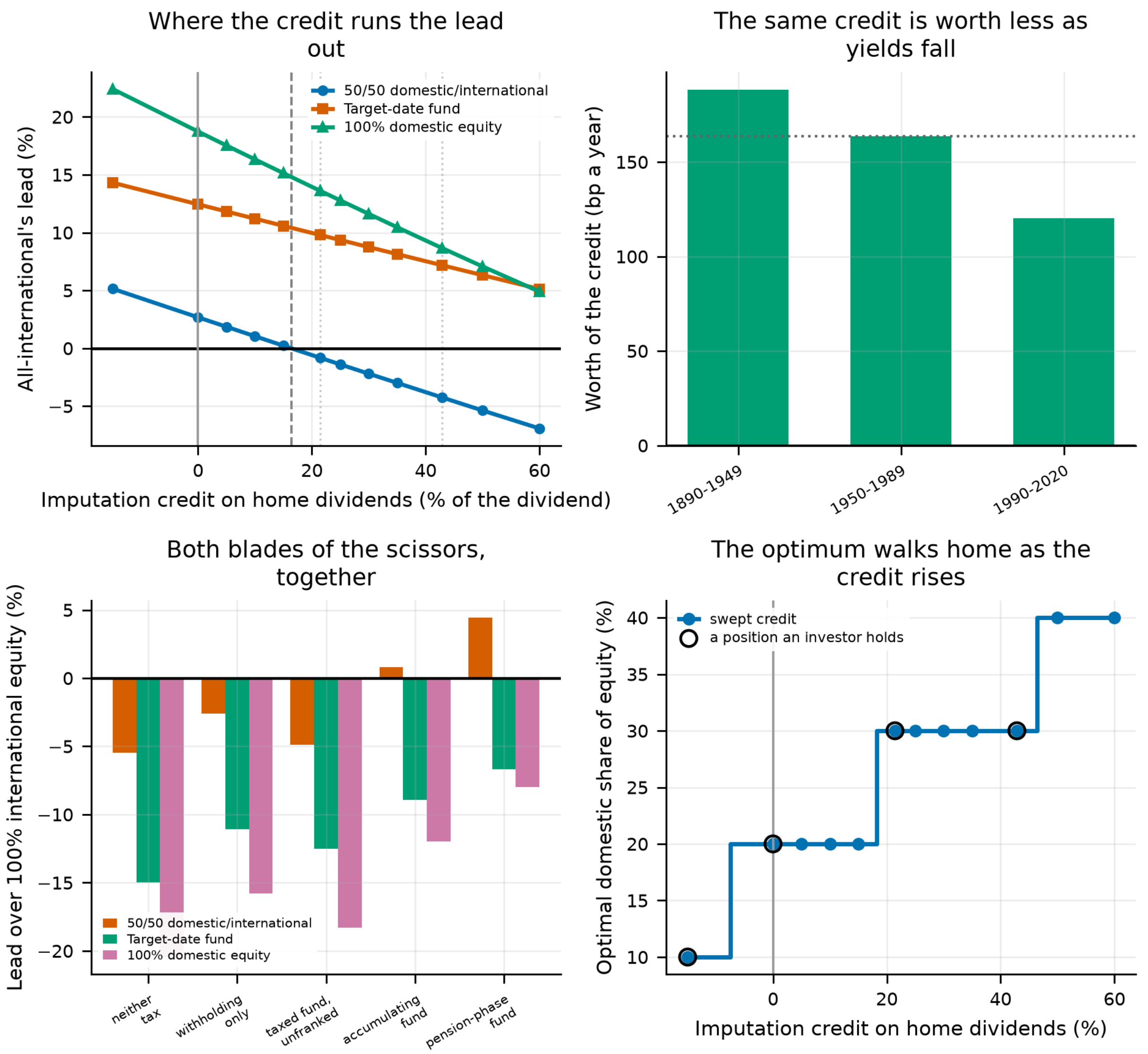

Figure 29.

Top left: all-international's lead against the credit, with the crossing and the real anchors marked. Top right: what the credit is worth by era. Bottom left: each rival's lead over all-international at every position an investor can occupy — a bar above the line is a reversal. Bottom right: the optimal domestic share as the credit rises.

15.5 What this changes

- **The tax code moves the answer, and only when both halves of it are modelled.** Withholding alone leaves *100% international equity* ahead. The credit at home turns a 57 basis-point drag on one leg into a 221 basis-point wedge between the two, and that is enough.
- **It is not a case for going home.** *50/50 domestic/international* wins at the largest credit tested, not *100% domestic equity*. The correction to this paper's headline is a smaller foreign allocation, not none.
- **Whose law this is.** The credit is applied to whichever market an investor holds as their own, which models a world where every country operates imputation rather than the one that exists. That is the same convention Section 18 uses when it pays Australia's Age Pension to an investor drawing sixteen countries' returns: the question is what the mechanism is worth, not what a population-weighted average of sixteen tax codes comes to. Several of this panel's countries ran an imputation system during the twentieth century and abolished it; Australia did not. That is a fact from the tax literature rather than from this project's data, and

no number above rests on it.

- **What is not modelled**: the personal tax a classical system would then levy on the dividend, which runs against the home leg and would narrow the wedge; any franking level other than the swept grid; and the years before 1987, when Australia's own investors had no credit at all. The last is the largest of the three and runs the same way as the first.

16. When the Pay Cheque Is a Claim on the Home Market

Labour income in this model is a hump-shaped real profile multiplied by a permanent random walk and a transitory shock, and every result so far has drawn those shocks independently of the return panel. That independence is a gift to domestic equity. A worker whose employer, industry and tax base are the index is long that index twice: once in the portfolio and once in the pay cheque. Holding less of the home market than its weight in a global index is the standard prescription that follows, and the model has been assuming the premise away.

The parameter swept here is a **correlation, not a loading**. The permanent innovation becomes $\rho \cdot u + \sqrt{(1 - \rho^2)} \cdot z$, with u the standardised domestic equity return of the same year. That rotation preserves unit variance, so raising ρ changes which part of a career's risk is systematic without changing how much risk a career carries. Any movement below is therefore the correlation and not extra income volatility smuggled in alongside it; at $\rho = 0$ every other result in this paper is unchanged to the last bit.

16.1 The sweep

Table 53. γ = 5, 50,000 lifetimes per level. Correlated with the home market; the foreign correlation is whatever the two markets' own co-movement implies.

Correlation with the home market	All-international over 50/50 (%)	Best strategy
0.0	5.73	100% international equity
0.1	6.63	100% international equity
0.2	7.57	100% international equity
0.3	8.54	100% international equity
0.4	9.54	100% international equity
0.6	11.63	100% international equity

Correlated human capital widens the lead, and nothing changes places. Going from independence to a correlation of 0.6 moves the lead from 5.73% to 11.63% — +5.90 points, or about +0.98 points per 0.1 of correlation. The direction is the one theory predicts, so the independence assumption used everywhere else in this paper is conservative.

Table 54. Certainty equivalent for every strategy at every correlation.

Strategy	ρ = 0.0	ρ = 0.1	ρ = 0.2	ρ = 0.3	ρ = 0.4	ρ = 0.6
50/50 Domestic/International Equity	1.0493	1.0365	1.0236	1.0108	0.9979	0.9724
100% Bills (cash)	0.7387	0.7367	0.7345	0.7322	0.7298	0.7247
100% Domestic Equity	0.8830	0.8719	0.8609	0.8499	0.8389	0.8174
100% International Equity	1.1094	1.1052	1.1011	1.0971	1.0931	1.0855
60/40 Domestic Equity/Domestic Bonds	0.8721	0.8624	0.8528	0.8432	0.8336	0.8145
Target-Date Fund (glide path)	0.9446	0.9340	0.9234	0.9127	0.9020	0.8806

Notes. Every strategy loses ground as the correlation rises, because a career correlated with the market is a riskier career; what matters here is whether the gaps between them move.

16.2 A claim on which market?

The sweep above answers a narrower question than it appears to. It correlates the pay cheque with the *home* market and leaves the foreign correlation to fall wherever the two markets' own co-movement puts it. But the argument for tilting away from home rests on the difference between the two correlations, not on the level of either. A worker whose income is a claim on world equity rather than on their own market has no home market to tilt away from, and whatever the correlation was buying should largely cancel. That is the objection, and it is answerable.

Three readings are run over the same grid. **Home only** is the sweep above. **Strict** pins the foreign correlation to zero, which — because the markets move together — requires loading negatively on the foreign market, and is therefore the most favourable version of the argument the data allow. **Diagonal** correlates the pay cheque equally with both markets, which is the objection stated as a specification. All three preserve unit variance, so they differ in what the career's risk is a claim on and not in how much of it there is.

Table 55. The same correlation grid under three assumptions about the foreign market.

Reading	Lead at ρ = 0 (%)	Lead at the top of the grid (%)	Change (points)	Share of the home-only effect
home market only (foreign correlation left free)	5.73	11.63	+5.90	1.00×
home market only (foreign correlation pinned to zero)	5.73	15.11	+9.39	1.59×
both markets equally	5.73	7.71	+1.98	0.34×

Notes. All three start at the same place, because at ρ = 0 they are the same specification.

The objection is substantially right, and the direction survives it. Correlating the pay cheque with both markets equally keeps only 0.34 of the widening that correlating with the home market alone produced — +1.98 points against +5.90. Most of what this section measures is a statement about home bias specifically, not about human capital being risky. What does not change is the sign or the ranking: under every reading the lead still widens, and the winner still never changes.

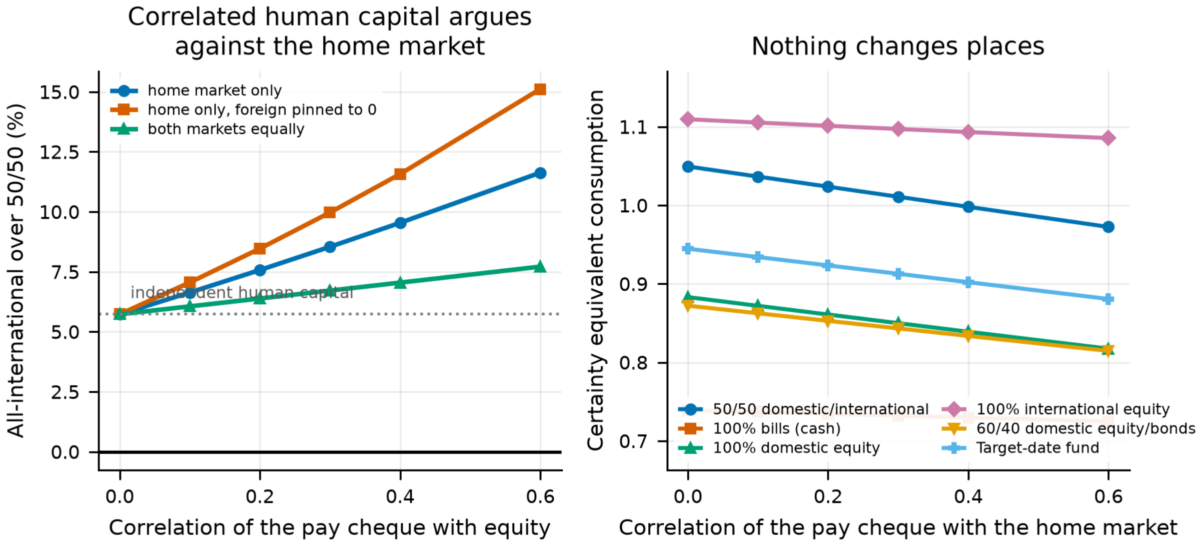

Figure 30.

Left: the lead of all-international over the 50/50 split as human capital becomes a claim on equity, under each of the three readings of which market it is a claim on. Right: every strategy's certainty equivalent over the same range, correlated with the home market.

16.3 What this changes

- The independence assumption used everywhere else in this paper is conservative: relaxing it makes the case for the home market worse, not better.
- The size is the useful part. Over a correlation range that runs past anything a labour economist would defend, the lead moves +5.90 points — a second-order lever against the fee differential of section 13 and the country-deletion range of section 8.
- What is not modelled: unemployment spells, industry, and any correlation between labour income and the international sleeve, which is not zero either and would push the other way. This is a bound on one channel, not a calibrated model of human capital.

17. Death at a Random Age

Every result before this section kills the investor on schedule at 93. That is a modelling convenience, and it distorts in two directions at once: it understates longevity risk, because nobody knows they have exactly 30 retired years to fund; and it overstates the far tail, because a strategy is rewarded for consumption at ninety-two that most investors never live to spend.

The treatment here is a **re-weighting, not a re-simulation**. Under the headline withdrawal rule the policy does not depend on the death age, so a random lifespan changes only which years of an already-simulated path are experienced and with what probability. Consumption in each year is weighted by the probability of being alive to enjoy it, and the estate is whatever wealth is left in the year the investor dies. A certain stream c paired with the matching bequest still returns exactly c, so these certainty equivalents are in the same units as every other one in the paper — and at a degenerate law that kills everyone at 93, the arithmetic reproduces the paper's own certainty equivalent to machine precision.

Exact for any policy that does not condition on the death age — the fixed real rule, the constant-percentage rule, every constant-weight strategy. Approximate for the horizon-based spending rules of section 30, which amortise over a planning horizon and would themselves change if they knew the mortality table.

That design decides how much this section can find, and the reader should discount it accordingly. Re-weighting a policy that does not itself respond to longevity is close to guaranteed to return a small number: the investor is not allowed to buy an annuity, to spend faster because the table says they are unlikely to reach ninety, or to hold back because it says they might. What is being tested here is therefore whether the *allocation ranking* is robust to the horizon assumption — a narrow question, and one worth answering, because the ranking is what the paper claims. It is not a finding that longevity risk is unimportant. The value of longevity risk lives almost entirely in the response to it, and this model makes no response.

17.1 The sweep

Table 56. Gompertz survival, γ = 5, 100,000 lifetimes.

Mortality	E[age at death]	All-international over 50/50 (%)	P(outlives the money): international equity	P(outlives the money): balanced all equity
fixed horizon	93	5.79	9.0%	11.7%
shorter lives (m=83)	78	5.65	3.0%	4.0%
baseline (m=88)	82	5.70	4.5%	5.9%
longer lives (m=93)	85	5.73	5.8%	7.6%
less dispersed (m=88, b=7)	84	5.69	4.5%	6.0%

Notes. Expected ages are within the model: mass beyond the simulated horizon is carried at the final age, so these are lower bounds on each law's own life expectancy.

Nothing changes places. Across a fixed horizon and 4 mortality laws spanning expected death ages from 78 to 93, the ordering is identical and the lead moves by at most 0.15 points from its fixed-horizon value of 5.79%.

Outliving the portfolio becomes *less* likely under every law, which is not the portfolio getting safer: it is the investor having fewer years in which to outlive it. A certain ninety-third birthday is a pessimistic longevity assumption, and the ruin probabilities quoted elsewhere in this paper inherit that pessimism.

Table 57. Rank under each mortality assumption.

Strategy	baseline (m=88)	fixed horizon	less dispersed (m=88, b=7)	longer lives (m=93)	shorter lives (m=83)
50/50 Domestic/International Equity	2	2	2	2	2
100% Bills (cash)	6	6	6	6	6
100% Domestic Equity	4	4	4	4	4
100% International Equity	1	1	1	1	1
60/40 Domestic Equity/Domestic Bonds	5	5	5	5	5
Target-Date Fund (glide path)	3	3	3	3	3

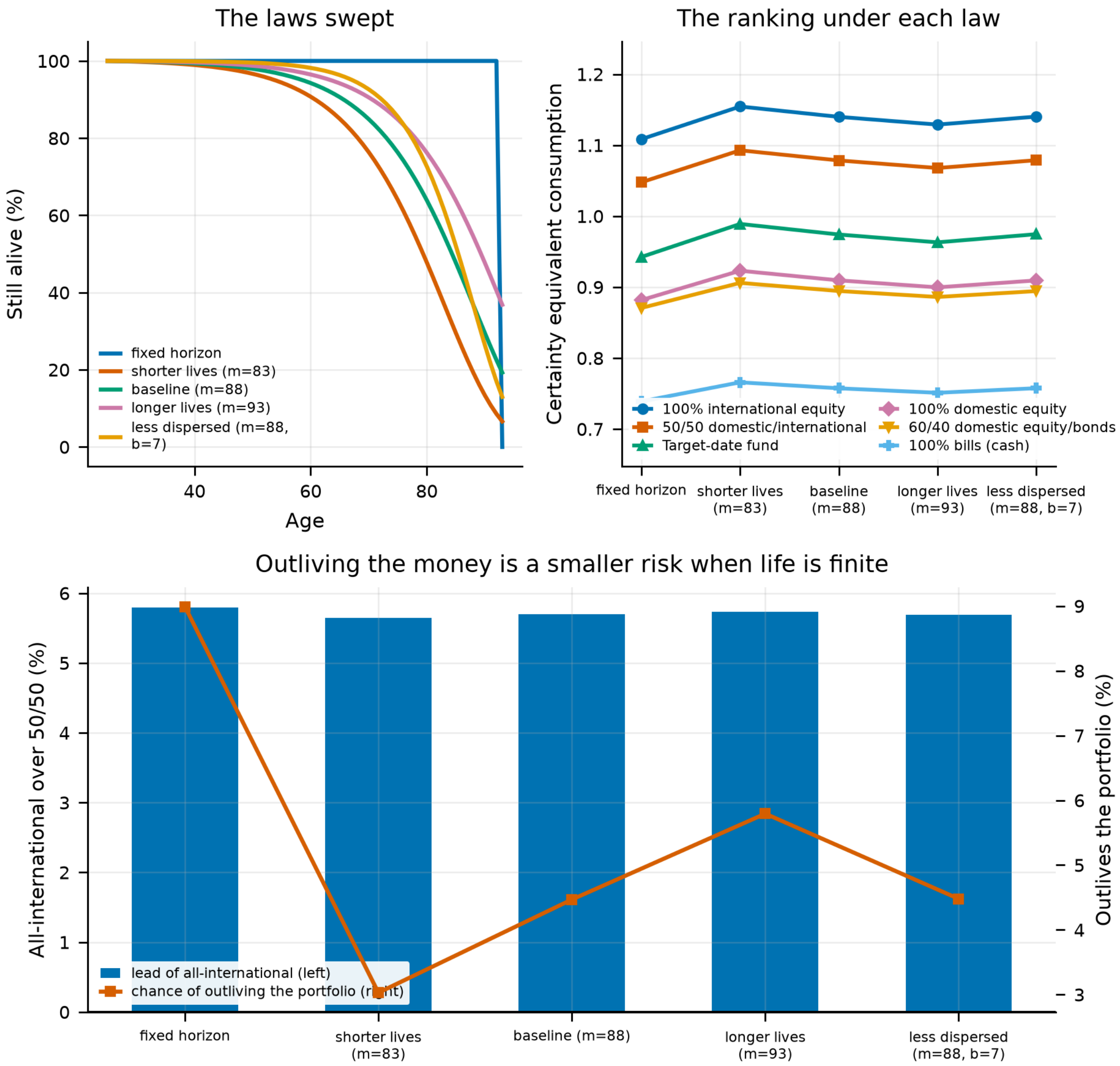

Figure 31.

Top left: the survival curves swept, against the certain death the rest of the paper assumes. Top right: every strategy's certainty equivalent under each. Bottom: the lead, and what a finite life does to the chance of outliving the portfolio.

17.2 What this changes

- The results elsewhere in this paper are not resting on the fixed horizon.
- The lead spans 5.65% to 5.79% across every assumption tested, against 5.79% at the fixed horizon.
- Ruin probabilities quoted elsewhere are computed against a certain 93rd birthday. Under any of these laws they are too high.
- What is not modelled: a couple rather than an individual, mortality correlated with wealth, and a policy that adapts to the mortality table. The last would raise every certainty equivalent here, so these numbers are a floor.

18. One Country's Pension Behind Sixteen Countries' Returns

Every result to this point pays the same public pension in all 16 markets: the US primary-insurance-amount schedule, progressive in career earnings and paid in full whatever else the retiree owns. It is the schedule of exactly one country in the panel, and the panel's whole point is that countries differ.

It is also the shape most flattering to the argument. An earnings-related pension paid regardless of wealth is a **risk-free real annuity**: it arrives whatever the portfolio does, which puts a floor under retirement consumption, crowds fixed income out of the financial portfolio and pushes the optimal equity share up. Nothing in this paper has tested how much of the result that floor is carrying.

Australia is the developed world's counter-example, and it differs in two ways at once. The **pension** is flat rather than earnings-related and is **means-tested**: assessable assets above a free area withdraw it on a taper until it cuts out. And the **contribution** is compulsory: the Superannuation Guarantee pays 12% of ordinary time earnings into a separate fund, on top of whatever the worker saves voluntarily. The two push in opposite directions, so the sweep crosses them rather than running one Australia-versus-America row that would confound them.

18.1 Calibration

Table 58. The Age Pension for a single retiree, March 2026 indexation, with assets-test thresholds from July 2026, against Australian average weekly ordinary time earnings of $2,051.10 (ABS, November 2025).

Quantity	Statutory	In average earnings
Maximum single rate	$1,200.90 a fortnight	0.293
Assets-test free area (homeowner)	$321,500	3.014
Assets-test free area (non-homeowner)	$600,000	5.625
Taper	$3 a fortnight per $1,000	0.078 a year
Cut-out (homeowner)	$722,000	6.767
Superannuation Guarantee	12% of earnings, taxed 15% on entry	0.102 reaching the fund

Notes. Rates are held as multiples of economy-wide average earnings so the schedule travels across the panel's currencies unchanged.

The guarantee is a second contribution stream, not a larger first one. The worker still saves this paper's own 10% out of take-home pay; the employer pays 12% on top, 15% of it is taken as contributions tax, and the remaining 10.2% is invested in the same strategy and assessed by the same means test. Total contributions are 20.2% of income against the American saver's 10%. Because both pots hold the same strategy and face no differential tax on earnings in this model, they are financially one pot and are simulated as one; the guarantee supplies 50% of everything contributed.

Working-life consumption is unchanged by the guarantee here, because its statutory incidence is on the employer. If the true incidence is on workers through lower wages — which is what most of the empirical literature finds — an Australian is paying for it in forgone pay and the comparison below is generous to them by that amount. It does not touch the certainty equivalents, because the utility window in this paper is retirement only. It does mean this is a comparison of *systems as legislated* rather than of two workers with the same lifetime resources, which is why the matched-contribution rows are there.

The taper is the other mechanism, and it is steep. A dollar of assessable assets inside the tapered band costs 7.8% of pension every year it is held. No asset in this panel earns that reliably in real terms, so inside the band the marginal return on wealth is negative — and run backwards, a retiree whose portfolio falls is met by a pension that rises. A means test is a floor and a ceiling at once, which is why it is assessed on assets as they stand, year by year, rather than settled once at retirement.

18.2 Crossing the two features

Table 59. Retirement consumption relative to the US baseline, γ = 5, 50,000 lifetimes per regime.

System	Certainty equivalent (%)	Mean consumption (%)	5th percentile (%)	All-intl over 50/50 (%)	Best strategy
US Social Security (the paper's baseline)	+0.0	+0.0	+0.0	+5.73	100% international equity
US Social Security, saving 20.2%	+23.1	+63.1	+32.8	+6.98	100% international equity
Age Pension rate, paid to everyone (no means test)	-12.9	-12.1	-10.1	+6.33	100% international equity
Australian Age Pension, saving 10%	-49.9	-34.5	-51.6	-8.06	100% bills (cash)
Australia as legislated: 10% voluntary + 12% SG, means-tested pension	-41.0	+26.8	-26.5	-6.42	Target-date fund
Australia as legislated, non-homeowner thresholds	-34.3	+27.7	-20.3	-5.28	Target-date fund

Notes. The first three columns are levels — how much retirement consumption a system delivers. The fifth is the ranking question — which portfolio it leads its investor to hold. They are different questions and here they have different answers.

The Australian system raises the average and lowers the certainty equivalent. As legislated it delivers +26.8% mean retirement consumption against the American baseline and -41.0% certainty-equivalent consumption. Both are right, and the distance between them is the whole result: the fifth percentile is -26.5%. Compulsory saving buys a bigger portfolio and the means test takes away the floor that portfolio would otherwise sit on, so the good outcomes get better and the bad ones get worse. A criterion as averse to the lower tail as a certainty equivalent at γ = 5 prefers the annuity.

The crux is one number. Under the American schedule this investor collects a public pension worth 44% of economy-wide average earnings, every year of retirement, unconditionally and for life. The Australian maximum rate is 29% to begin with — and this investor collects a mean of 2.1%, because the guarantee has made them wealthy enough that the assets test withdraws almost all of it. They have swapped a large risk-free real annuity for a larger portfolio. That is a good trade on the average and a bad one in the tail, which is exactly what the three level columns above report.

The crossing says where each piece of that comes from. Holding the American pension and paying the Australian contribution rate is worth +23.1%: compulsory saving on its own is a large gain. Holding the contribution rate and cutting the pension to the Age Pension's flat rate, still paid to everybody, costs -12.9% and changes nothing about the ordering — it is a smaller annuity, not a different kind of one. Adding the assets test to that is what does the damage. Together they land at -41.0%: the guarantee does not buy back what the means test removes.

18.3 What it does to the ranking

The ranking does not survive the means test. All-international leads the 50/50 split by 5.73% under the American schedule and by -6.42% under the Australian one, where the best strategy becomes *Target-date fund*. This is the only place in this paper where the headline ordering fails, and it fails for a reason that has nothing to do with the return panel.

The mechanism is the taper. Inside the assets-tested band every extra dollar of assets costs 7.8% of pension a year, which is more than any asset in this panel earns reliably in real terms. A portfolio that stays small keeps an entitlement that a portfolio that grows loses, so the means test pays a retiree to hold less risk — and it pays them at a rate the risk premium cannot match. Nothing about the returns has changed between these rows. The objective has.

How far it goes depends on the balance, not the country. Run the same means test on a saver with no guarantee behind them — 10% voluntary saving and nothing else — and the lead falls further, to -8.06%, with *100% bills (cash)* taking first place rather than *Target-date fund*. The poorer the saver, the deeper into the taper they sit and the further the distortion goes: at 20.2% of income the answer is a de-risking schedule, at 10% it is cash. The compulsory contribution does not remove the distortion, it softens it.

Two things follow, and they should be kept apart. The first is about this model: its investor contributes 20.2% of a rising income for 38 years and compounds it at historical real equity returns with no tax on fund earnings, which makes them far wealthier than a median Australian and puts them at the top of the taper rather than in the middle of it. The second is about the paper as a whole: every other section models a retiree whose public income is an unconditional annuity, and a retiree whose public income is withdrawn at the margin is solving a different problem. The answer here is not that Australians should hold a glide path. It is that a means test changes what the portfolio is for.

The assets test has a higher free area for retirees who do not own a home, and the model's investor owns nothing outside the portfolio. Running the non-homeowner thresholds — a free area of 5.63 times average earnings rather than 3.01 — moves the certainty equivalent to -34.3%, which is enough to matter and is reported alongside.

Table 60. Where retirees sit on the taper under the system as legislated, holding the 50/50 portfolio.

Age	Full rate	Part rate	No pension	Mean pension (× average earnings)
70	0.8%	4.2%	95.0%	0.008
80	5.2%	5.3%	89.5%	0.023
90	13.2%	3.8%	83.0%	0.045

Notes. A means test that never binds is a flat pension in disguise; one that always binds to zero is no pension at all. This table says which of the two this saver faces.

Table 61. How much of retirement the state is paying for.

System	Mean pension	Share of retirement consumption
Age Pension rate, paid to everyone (no means test)	0.449	29.7%
Australian Age Pension, saving 10%	0.063	5.6%
Australia as legislated: 10% voluntary + 12% SG, means-tested pension	0.033	1.5%
Australia as legislated, non-homeowner thresholds	0.048	2.2%

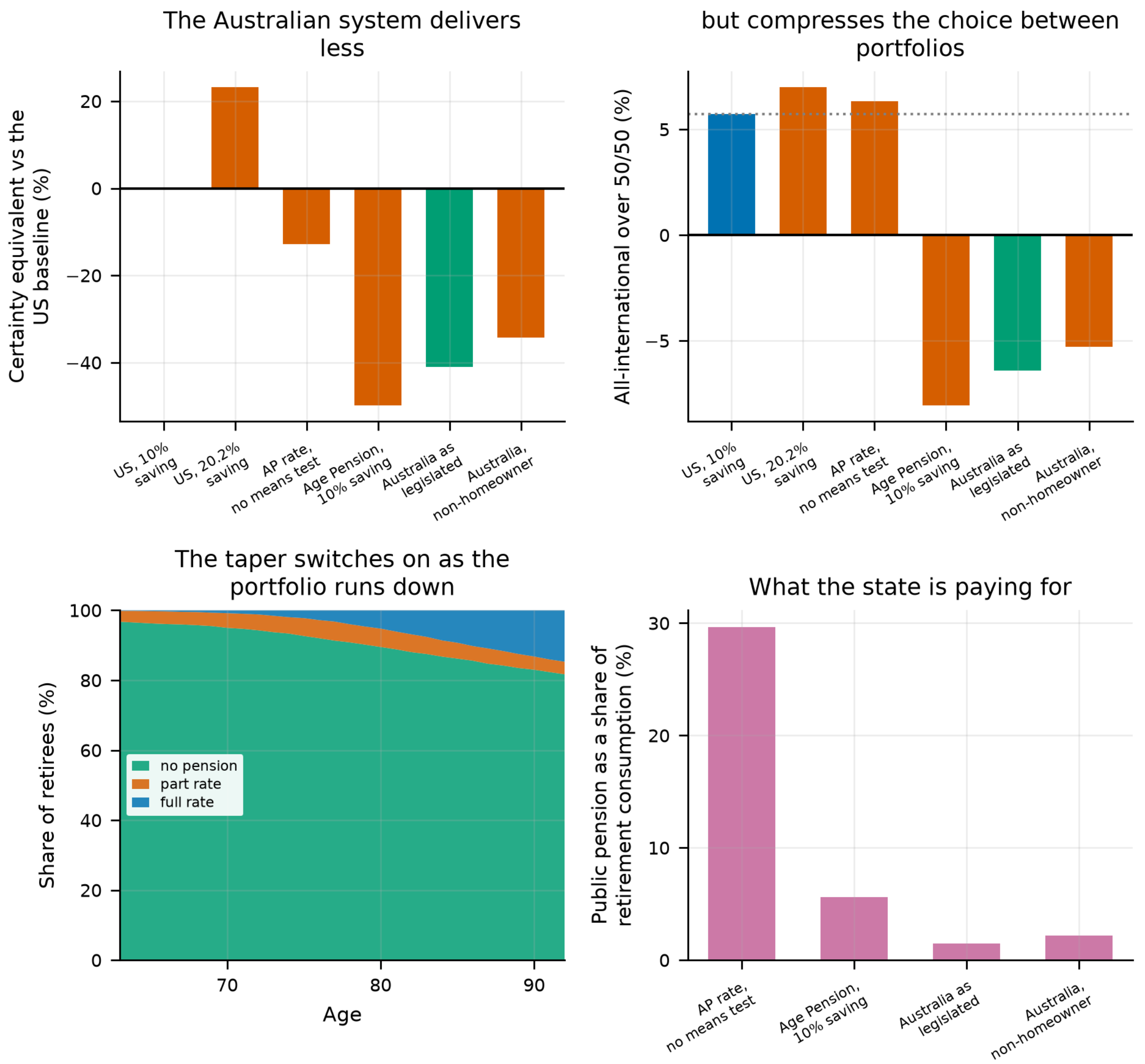

Figure 32.

Top left: retirement consumption under each regime against the American baseline. Top right: what each regime does to the choice between portfolios. Bottom left: where retirees sit on the assets-test taper as the portfolio draws down. Bottom right: how much of retirement consumption the public pension supplies.

18.4 What this changes

- **A pension schedule is not an innocent modelling choice.** Swapping one developed country's for another's moves certainty-equivalent retirement consumption by -41% while the return panel, the preferences and the portfolios stay exactly where they were. Every level quoted elsewhere in this paper is conditional on the American schedule.
- **The average and the certainty equivalent disagree, and both are true.** Compulsory saving raises mean retirement consumption +27% and lowers the fifth percentile -27%. A reader who cares about expected consumption should read the first; one who cares about the bad case should read the second. The paper's own criterion reads the second.
- **The allocation ranking reverses.** This is the only place in this paper where it does. Under the means test the best strategy becomes *Target-date fund*, and for a saver poorer than this one it goes further still. Every other section models a retiree whose public income arrives regardless of what they own; that is the

assumption the ranking rests on.

- **What is not modelled**: the income test, the family home's exemption from the assets test — the largest single feature of the real system — the tax on fund earnings in accumulation, and every other tax in this paper. Nor is an annuity, which is what the American schedule is and what an Australian retiree would have to buy to match it.

19. Solving for the Optimal Glide Path

Sections 5 and 6 compare a handful of candidate allocation schedules. That is a test, not an answer. This section solves the problem directly: what equity share should the investor hold at each of the 68 ages, if the schedule is free to be anything at all?

The objective is the certainty equivalent under common random numbers, which makes it a deterministic function of the schedule. That licenses coordinate ascent — sweeping one age at a time over a grid of weights, holding the rest fixed — because each per-age search is then exact and each full sweep is monotone. To guard against local optima the search is restarted from several initial schedules, and a relative improvement threshold is imposed so that the reported solution does not encode year-to-year jitter worth a fraction of a basis point.

Table 62. Solved schedules against the benchmark strategies at y = 5

Schedule	CEC	Gap to best (%)
free form optimal	1.117	0.00
parametric optimal	1.115	-0.20
100% international equity	1.106	-0.98
50/50 domestic/international equity	1.045	-6.46
Target-date fund (glide path)	0.941	-15.76
100% domestic equity	0.881	-21.16
60/40 domestic equity/bonds	0.870	-22.10
100% bills (cash)	0.738	-33.93

Notes. "Free-form optimal" solves an unconstrained weight at every age; "parametric optimal" fits a three-parameter logistic glide. The benchmarks are the same six strategies used throughout.

The free-form solution beats every benchmark, but by a margin that matters less than its *shape*. Across the three restarts the solved schedule holds a mean equity share of 99.4% and sits at a full equity allocation in 96% of ages. The restarts agree to within 0.012% of each other, so this is not a local optimum artefact.

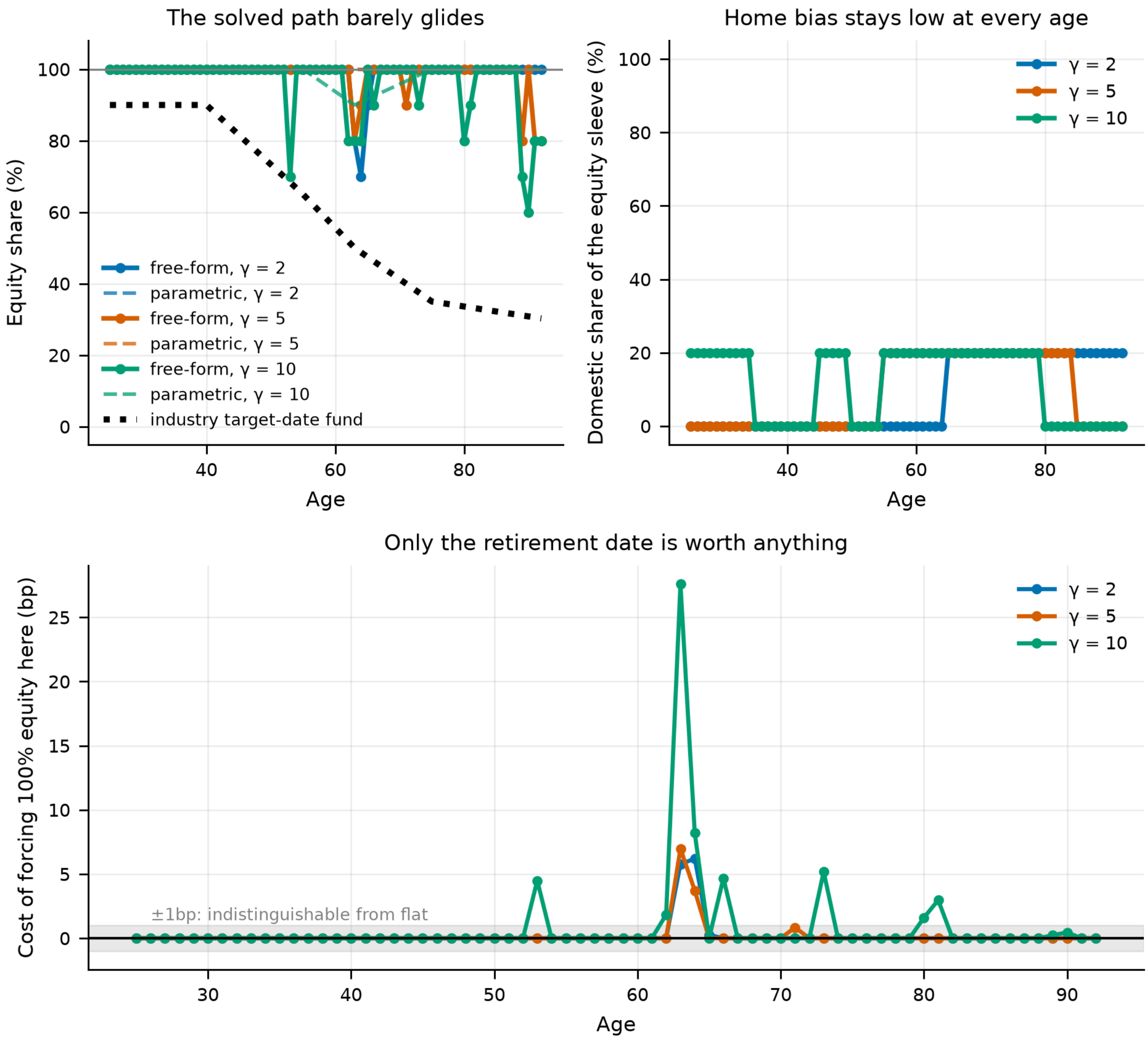

Figure 33.

The solved equity share by age, from three restarts, against the target-date fund's prescribed glide. The solution is flat at or near the corner; it is not a glide path.

19.1 How much of the solved structure is real?

A solved schedule always looks structured. The deviation profile tests whether it is: each age's solved weight is reset to a neutral reference and the certainty-equivalent cost measured in basis points. Of the 68 ages at γ = 5, only 2 move the objective by more than a single basis point. The remaining 66 are on a flat part of the surface, and any narrative built on their apparent pattern would be a narrative about search noise.

This is the reason the paper does not describe the solved glide path as having interesting age structure. It does not. The honest summary is that the optimiser goes to the equity corner and stays there, and that the small departures it makes near the ends of the horizon are worth almost nothing.

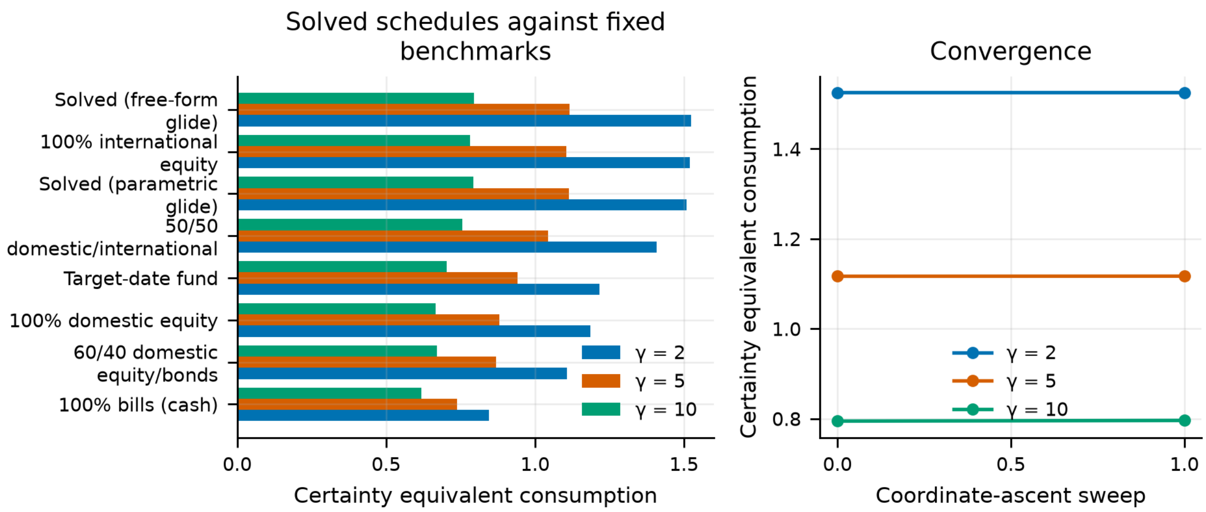

Figure 34.

Certainty equivalent of the solved schedules against the benchmarks, at each risk aversion. The solved advantage over a fixed all-equity portfolio is small; the advantage over the glide path is not.

19.2 The dip belongs to the withdrawal rule

The solved schedule is not quite flat: under the baseline 4% rule it dips at the retirement date and recovers afterwards. That is worth explaining rather than smoothing away, because the obvious reading -- that the model has rediscovered the glide path after all -- is the wrong one.

A 4% rule sets the whole of retirement spending as a fixed fraction of wealth on *one date*. Wealth at 63 is therefore not another point on the wealth path; it is the single number that fixes consumption for the next thirty years. De-risking briefly around a date like that is rational for the same reason nobody holds their house deposit in equities the month before completion. If that is the explanation, the dip should vanish under a rule that anchors on no single date -- so the schedule is re-solved under two that do not.

Table 63. The free-form schedule re-solved under each withdrawal rule, γ = 5.

Withdrawal rule	Equity at retirement	Equity elsewhere	Dip (pp)	Domestic share, working	Domestic, retired	Solved CEC
4% rule (anchors on wealth at 63)	80%	98.9%	+18.9	4.2%	14.7%	1.1170
Constant % of portfolio	100%	99.7%	-0.3	9.5%	16.7%	1.3142
Life expectancy / RMD	100%	99.7%	-0.3	9.5%	20.0%	1.2774

Notes. Each row is a complete re-solve: equity share per year and domestic share per band, optimised against that rule. The certainty equivalents are not comparable across rows, because the rules spend different amounts; the shapes are.

The dip is a property of the withdrawal rule, not of the investment problem. It appears only under the rule that anchors on wealth at a single date, at 19 percentage points. Under the other rules that condition on the portfolio as it stands -- a percentage of the balance, and a life-expectancy divisor -- the schedule is flat at 100% equity from twenty-five to death and the dip does not appear at all.

That is a practical result, and it is not the one glide-path marketing describes: **if your withdrawal policy anchors on your balance at one date, de-risk briefly around that date; if it does not, do not de-risk at all.** A conventional target-date fund does neither — it de-risks slowly across decades and then stays de-risked.

The home/abroad split moves with the rule too, and in the direction the ruin mechanism of Section 31 explains. The working years take 4%–9% domestic and the retired years 15%–20%. Both the level and the phase gap are small next to the equity decision, which is why the search gives the split coarser bands.

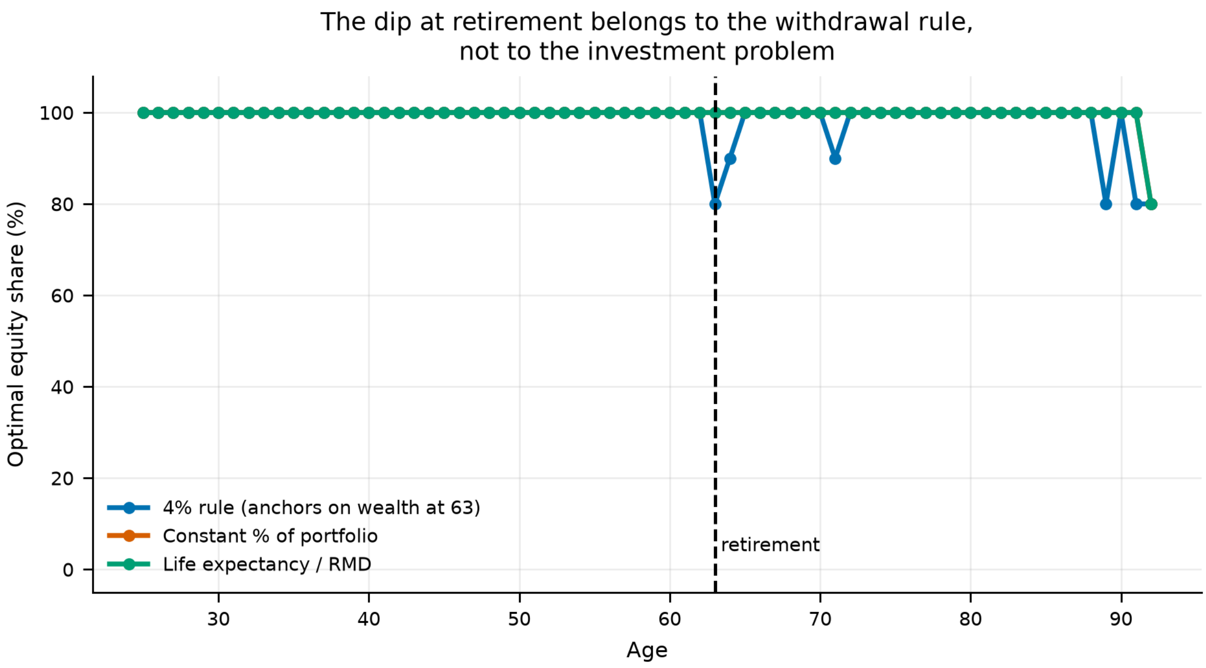

Figure 35.

The solved equity schedule under each withdrawal rule. The dip at the retirement date appears only under the rule that fixes spending from wealth on that one date; the rules that read the portfolio as it stands solve flat.

The conclusion of this section is stronger than the one the benchmark comparison supports. It is not merely that the target-date glide path is worse than an all-equity portfolio on this panel; it is that when the schedule is allowed to be anything, the optimiser does not choose a glide path at all — and the one age-related feature it does choose belongs to the spending rule rather than to the market.

20. The Whole Allocation, Solved

Section 19 solves for the equity share at every age and for the domestic split on five-year bands, and finds the optimum sits at or near the all-equity corner. But it holds the fixed-income sleeve at a fixed 70/30 bond/bill mix. That restriction was made for search cost, not for principle, and it matters for the interpretation: an optimiser that is told what to hold *inside* a sleeve it barely uses will look more decisive about that sleeve than it actually is.

This section removes the restriction. The decision variable is the full weight simplex at every year of the lifecycle — 68 points in the 3-simplex, 204 free parameters — solved directly against certainty-equivalent consumption. Nothing is held fixed: domestic equity, international equity, bonds and bills compete freely at every age, subject only to non-negativity and to summing to one.

20.1 The search

Under common random numbers the objective is a deterministic function of the schedule, so a search over one age at a time is exact for that age and every sweep is monotone. Two stages are used, because a lattice fine enough to be precise is too large to sweep at every age and a purely local search is too easy to trap:

- a **coarse lattice** sweep, evaluating every composition of the simplex at a step of 25% — the 35 lattice points — at each age in turn; then
- a **fine local** sweep over the twelve single-step pairwise exchanges around the incumbent at a step of 5%, repeated until nothing improves.

A pairwise exchange moves weight from one asset to another and so stays on the simplex by construction, which is what makes it a valid local move there; exchanges that would drive a weight negative are dropped rather than clipped, so every candidate evaluated is feasible. Both stages accept a move only if it clears a relative improvement threshold, for the reason given in Section 4.4.

Table 64. Convergence of the full-simplex search at risk aversion 5

Stage	Sweep	CEC	Gain (%)	Cumulative evaluations
start	-1	0.90340	—	1
coarse	0	1.11231	23.1255	2,381
coarse	1	1.11604	0.3353	4,761
fine	0	1.11888	0.2544	5,111
fine	1	1.12001	0.1013	5,530
fine	2	1.12015	0.0123	5,958
fine	3	1.12030	0.0130	6,383

Notes. The coarse stage does almost all of the work; the fine stage moves the objective by a fraction of a percent. That pattern is itself informative — the surface has a broad flat top.

20.2 The solved schedule

Table 65. The solved four-asset schedule at risk aversion 5, every fifth year

Age	Phase	Dom. equity	Intl. equity	Bonds	Bills	Equity
25	working	0.0%	100.0%	0.0%	0.0%	100.0%
30	working	0.0%	100.0%	0.0%	0.0%	100.0%
35	working	5.0%	95.0%	0.0%	0.0%	100.0%
40	working	0.0%	100.0%	0.0%	0.0%	100.0%
45	working	10.0%	90.0%	0.0%	0.0%	100.0%
50	working	5.0%	95.0%	0.0%	0.0%	100.0%
55	working	10.0%	90.0%	0.0%	0.0%	100.0%
60	working	15.0%	85.0%	0.0%	0.0%	100.0%
65	retired	25.0%	65.0%	0.0%	10.0%	90.0%
70	retired	20.0%	80.0%	0.0%	0.0%	100.0%
75	retired	20.0%	80.0%	0.0%	0.0%	100.0%
80	retired	20.0%	80.0%	0.0%	0.0%	100.0%
85	retired	20.0%	80.0%	0.0%	0.0%	100.0%
90	retired	45.0%	55.0%	0.0%	0.0%	100.0%

Notes. Weights are shares of the whole portfolio and sum to one in each row. The full schedule is in the project's result tables.

Table 66. Average solved weights by lifecycle phase and risk aversion

Risk aversion	Phase	Years	Dom. equity	Intl. equity	Bonds	Bills	Equity
2	working	38	0.0%	100.0%	0.0%	0.0%	100.0%
2	retired	30	12.7%	85.5%	0.0%	1.8%	98.2%
5	working	38	6.1%	93.9%	0.0%	0.0%	100.0%
5	retired	30	13.5%	83.8%	0.7%	2.0%	97.3%
10	working	38	12.9%	85.8%	1.2%	0.1%	98.7%
10	retired	30	13.0%	79.3%	2.8%	4.8%	92.3%

At the baseline preference the solved portfolio averages **9.3% domestic equity and 89.5% international equity**, with 0.3% in bonds and 0.9% in bills. The equity share averages 100.0% through the working life and moves by +0.0 percentage points between its first and last thirds — which is to say it is flat.

The fixed-income sleeve is barely used at any age. That is the direct answer to the question this section exists to ask: the 70/30 bond/bill split Section 19 imposed was fixing the composition of something the optimiser does not want to hold in the first place, so the restriction was not binding. Freeing it buys 0.30% over the best fixed benchmark.

The domestic/international split carries the same caveat as Section 6.2 and should not be read as advice. The international leg in this model is a 15-country leave-one-out average, better diversified than any tradeable international index and available to no individual investor without also holding their own market. A solved schedule that wants 89% international is saying the model prefers *more* diversification than the 50/50 headline strategy provides, not that this particular number is attainable.

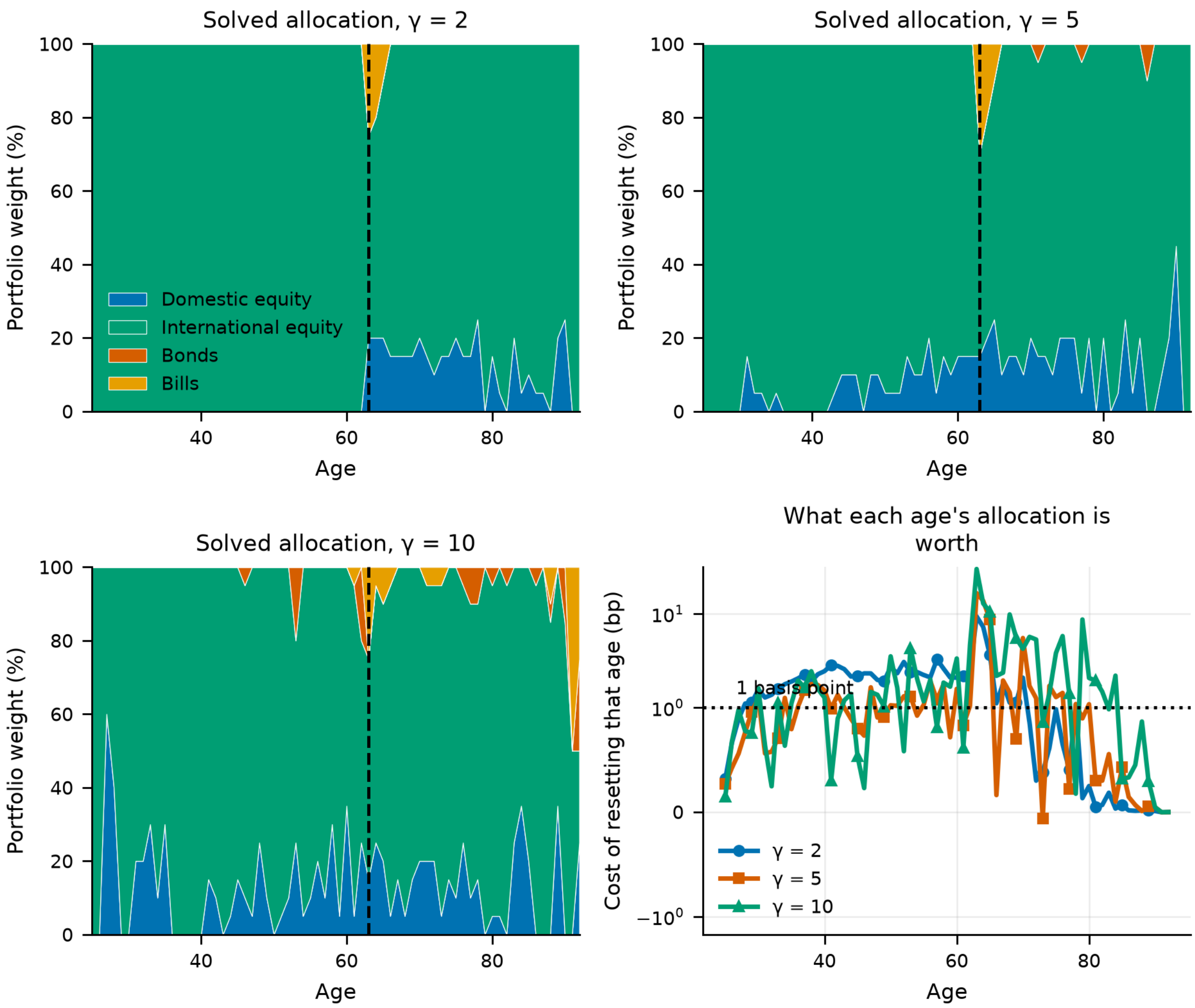

Figure 36.

The solved four-asset schedule at each risk aversion, and what each age of it is worth. The stacked areas are portfolio weights; the right-hand panel resets each age to the schedule's own average and measures the certainty-equivalent cost in basis points, on a log scale with one basis point marked.

20.3 Against the benchmarks

Table 67. Solved and benchmark schedules at risk aversion 5

Schedule	CEC	Gap to best (%)
full simplex optimal	1.1203	0.00
docs07 free form	1.1170	-0.30
docs07 parametric	1.1147	-0.50
100% international equity	1.1060	-1.28
50/50 domestic/international equity	1.0449	-6.73
Target-date fund (glide path)	0.9410	-16.01
100% domestic equity	0.8807	-21.39
60/40 domestic equity/bonds	0.8702	-22.33
100% bills (cash)	0.7380	-34.13

Notes. "Full simplex optimal" is this section's solution. The comparison also includes the schedules solved in Section 19 under the 70/30 restriction, where those are available.

The solved schedule leads the best fixed benchmark (docs07 free form) by **0.30%**. Two things about that number deserve saying plainly.

It is **small**. Freeing 204 parameters and searching them properly buys a fraction of what switching from a target-date glide path to a fixed all-equity portfolio buys in Section 5. The allocation decision has a broad flat top, and almost all of its value is captured by the first choice — whether to hold diversified equity at all — rather than by any refinement of it.

It is also **still not a glide path**. The unrestricted solution differs from the restricted ones of Section 19 in the composition of a sleeve that carries almost no weight, not in the age profile of the equity share. Two independent searches over different parameter spaces reach the same shape, which is a stronger statement than either makes alone.

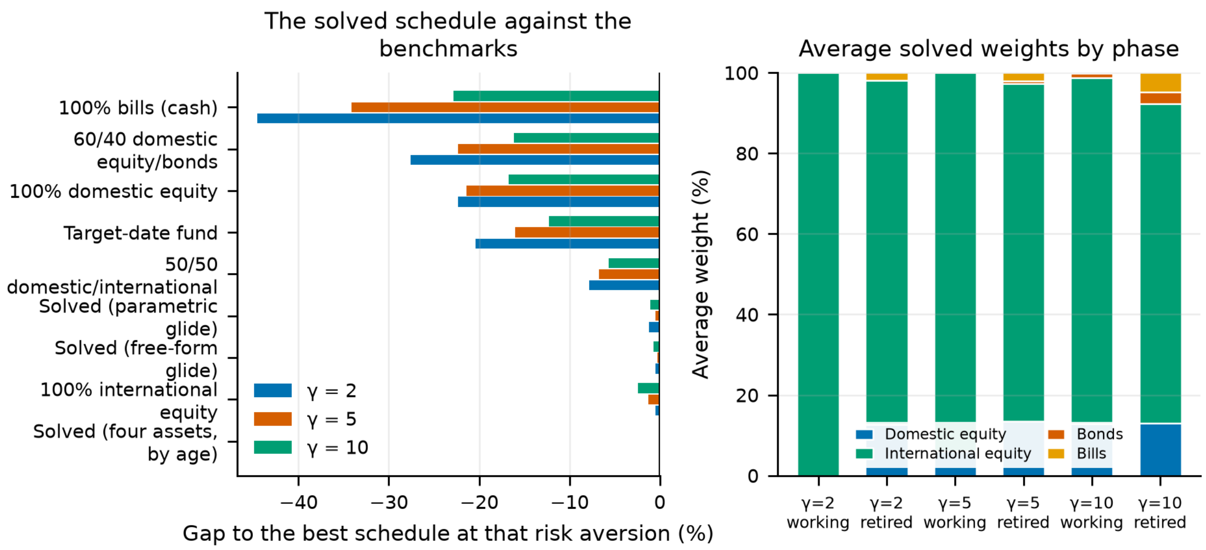

Figure 37.

Left: the solved schedule against the benchmarks at each risk aversion. Right: the average solved weights by lifecycle phase, showing how little of the portfolio the fixed-income sleeve ever carries.

20.4 Where the allocation decision actually matters

A solved schedule always looks structured, and this one looks more structured than the equity-share solve because it has three dimensions to wander in. The deviation profile tests whether the structure is real: each age's allocation is reset to the schedule's own average and the certainty-equivalent cost measured in basis points. Of the 68 ages, **33** move the objective by more than a single basis point.

That is a very different picture from Section 19, where most ages were worth nothing at all, and it is worth reading carefully rather than assuming either conclusion.

Table 68. The eight ages whose allocation is worth the most

Age	Phase	Cost of resetting (bp)	Dom. equity	Intl. equity	Bonds	Bills
63	retired	16.50	15.0%	55.0%	0.0%	30.0%
64	retired	13.66	20.0%	60.0%	0.0%	20.0%
65	retired	8.64	25.0%	65.0%	0.0%	10.0%
70	retired	5.49	20.0%	80.0%	0.0%	0.0%
67	retired	1.94	15.0%	85.0%	0.0%	0.0%
56	working	1.92	20.0%	80.0%	0.0%	0.0%
38	working	1.78	0.0%	100.0%	0.0%	0.0%
58	working	1.77	15.0%	85.0%	0.0%	0.0%

Notes. Cost of resetting that year's allocation to the schedule's own lifecycle average, holding every other year fixed.

The cost is concentrated around the retirement date. The single most valuable age is 63 — the retirement year itself — at 16.5 basis points, and the ten years from 61 to 70 carry 49% of the total cost while being 15% of the lifecycle. That is the sequence-of-returns window Section 28 identifies from a completely different direction, arrived at here through the allocation rather than through the retirement date: what you hold matters most in the years when the portfolio is largest and the withdrawals are about to start.

Working years each cost about 1.1 basis points and retired years 2.0, but the retired average is carried almost entirely by the first few. **The magnitudes remain small in absolute terms** — the largest single age is worth 16.5 basis points and the whole schedule beats the best fixed benchmark by 0.30% — so the honest summary is that the *timing* of the allocation decision is concentrated even though the decision itself is worth little. An investor who got the allocation right in the decade around retirement and used the lifecycle average everywhere else would give up a few basis points.

20.5 Is this a local optimum?

Coordinate ascent cannot escape a local optimum in principle. Re-solving from three different corners of the simplex — an equal split, all equity, and all fixed income — tests for one in practice.

Table 69. Restarting the search from three corners of the simplex

Starting allocation	Solved CEC	Gap to best (%)	Mean dom. equity	Mean intl. equity	Mean bonds	Mean bills
0.25 / 0.25 / 0.25 / 0.25	1.12030	0.0000	9.3%	89.5%	0.3%	0.9%
0.50 / 0.50 / 0.00 / 0.00	1.12012	-0.0156	10.7%	88.5%	0.0%	0.9%
0.00 / 0.00 / 0.50 / 0.50	1.12007	-0.0205	10.1%	88.9%	0.2%	0.8%

Notes. Each restart runs the full two-stage search from the stated starting allocation.

The restarts agree to within **0.021%** of each other. That is not a proof of global optimality — no coordinate search offers one — but it is the check that would have caught the obvious failure, and it did not fire.

21. Borrowing to Invest

Every allocation in this paper so far is long-only and fully invested: the weights are non-negative and sum to one. That is a constraint, not a result, and it rules out a policy with a serious literature behind it. If the case for equities rests on a horizon long enough for diversification across countries and decades to work, then an investor whose financial balance is still small is under-exposed to the very risk they are being told to take, and the natural remedy is to borrow.

The question is never whether leverage raises expected wealth; it obviously does when the expected asset return exceeds the borrowing rate. The question is what it is worth to a *risk-averse* investor at the price they can actually borrow at. This section sweeps that price.

One scoping note, because the answer below is largely negative and the negative does not generalise as far as it first appears. What is levered here is the *whole portfolio*: the ratio scales every weight at once. Borrowing aimed at a single asset is a different policy with a different answer, and Section 25.5 gives it — at the same borrowing cost, a mortgage on one sleeve is worth several times what scaling everything is worth. Read what follows as a verdict on margin, not on debt.

21.1 Mechanics, stated rather than buried

An allocation x over the four assets sums to one. A leverage ratio L means holding L units of that sleeve per unit of equity capital, funding the difference at the real bill rate plus a spread c:

$$r^p_h = L \cdot (x \cdot r_h) - (L - 1) \cdot (r^{bill}_h + c) \quad (4)$$

Two choices in that line are load-bearing.

The borrowing rate floats. It is the realised real bill return plus a constant spread, so an investor who borrows is exposed to the same rate their cash would have earned. A fixed real borrowing rate would be a different — and more favourable — assumption.

Limited liability. The portfolio return is clipped at −100%: the lender takes what is left and the investor's equity goes to zero, but they never owe more than they have. That is a margin call, and it is *generous to leverage* — a real levered investor faces forced liquidation at a threshold, not at zero. The tables therefore report how often the clip binds rather than leaving it implicit. The portfolio is rebalanced annually to maintain both the allocation and the leverage ratio, exactly as every unlevered strategy here is rebalanced to maintain its weights.

21.2 Optimal leverage by the price of credit

For each borrowing spread we search jointly over the leverage ratio and the allocation, taking the allocation from the same coarse simplex lattice Section 20 uses. The optimum is therefore a leverage ratio *and* the portfolio that goes with it, rather than a leverage ratio bolted onto a portfolio chosen elsewhere.

Table 70. The optimal leverage ratio and allocation at each price of credit

Borrowing spread	Optimal leverage	CEC	vs unlevered (%)	Equity	Dom. equity	Intl. equity	Bonds	Bills	Wipeout share
0.00%	1.75x	1.1629	5.10	75%	0%	75%	25%	0%	0.0000%
0.50%	1.25x	1.1426	3.27	100%	25%	75%	0%	0%	0.0000%
1.00%	1.25x	1.1285	2.00	100%	25%	75%	0%	0%	0.0000%
1.50%	1.1x	1.1162	0.88	100%	25%	75%	0%	0%	0.0000%
2.00%	1.1x	1.1105	0.37	100%	25%	75%	0%	0%	0.0000%
3.00%	1x	1.1064	0.00	100%	25%	75%	0%	0%	0.0000%
5.00%	1x	1.1064	0.00	100%	25%	75%	0%	0%	0.0000%

Notes. The spread is an annual real cost over the realised real bill rate. "Wipeout share" is the fraction of path-years in which the limited-liability clip binds — the assumption most favourable to leverage, reported rather than assumed away.

At zero cost, leverage is worth having: the optimum is 1.75× and it is worth +5.10% of certainty-equivalent consumption against the unlevered portfolio.

Note *what* the optimiser levers. At a zero spread it holds 75% equity and levers it 1.75×, for an effective equity exposure of 131% — it borrows against a *diversified* portfolio rather than concentrating into an undiversified one. That is the same preference for diversification the unlevered sections keep finding, expressed through a different instrument.

The interesting number is where it stops. **The break-even spread is 3.00%**: above roughly that annual cost over the real bill rate, no leverage ratio on the grid beats staying unlevered, and the optimum is already 1× by a spread of 3.00%.

That threshold is close to what a household can actually obtain — a retail margin account runs somewhere around one to two percent over the benchmark — so the model does not dismiss leverage out of hand. But the formal break-even overstates how far the case extends. **The advantage falls below a tenth of a percent by a spread of 3.00%**, well before it reaches zero. Over most of the plausible range of borrowing costs, leverage on this panel is not so much *harmful* as *pointless*: it takes on the tail risk documented in Section 21.3 in exchange for a gain that rounds to nothing.

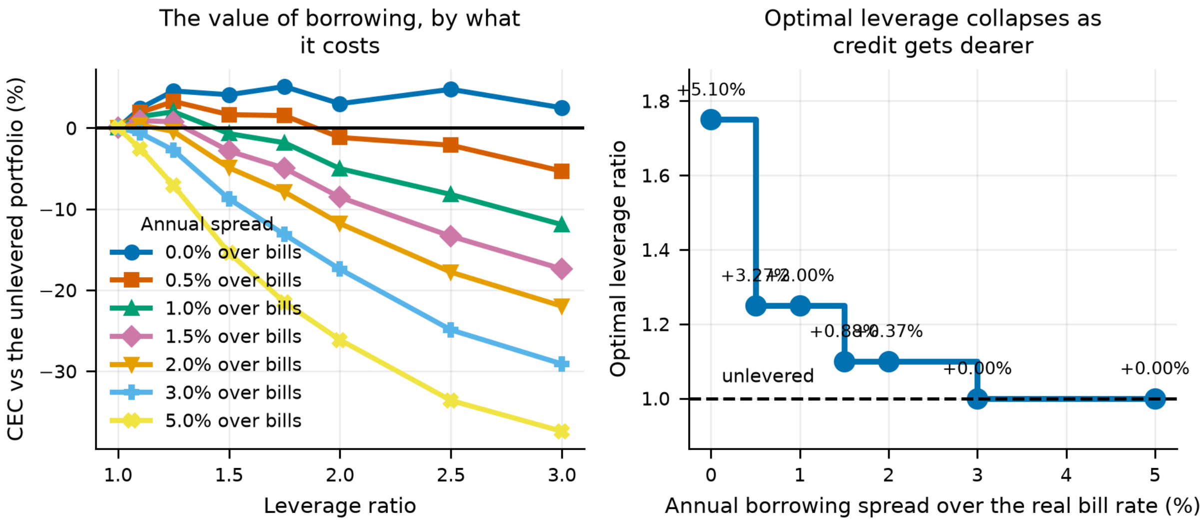

Figure 38.

Left: the value of leverage across the ratio, one line per borrowing spread. Right: the optimal ratio as the price of credit rises, annotated with what it is worth.

21.3 What leverage does to the shape of the outcome

A certainty equivalent alone cannot show what borrowing does to the distribution, and the distribution is the whole argument.

Table 71. Distributional detail at a 1.0% borrowing spread

Leverage	CEC	vs unlevered	P(ruin)	5th pct cons.	Median cons.	95th pct cons.	P(zero bequest)	Wipeout share
1x	1.1064	0.00	8.4%	0.790	1.564	3.701	9.1%	0.0000%
1.5x	1.0987	-0.70	12.6%	0.806	1.951	7.186	13.2%	0.0000%
2x	0.9218	-16.68	24.0%	0.691	2.002	14.004	24.6%	0.0466%
3x	0.7356	-33.52	51.1%	0.539	1.419	37.586	52.0%	1.0090%

Notes. All rows hold the same allocation — the one optimal at that spread — and vary only the leverage ratio, so the differences are attributable to borrowing rather than to reallocation.

Going from unlevered to 3× moves the fifth percentile of retirement consumption by **-31.8%**, the median by -9.3% and the ninety-fifth percentile by +915.5%. The ruin probability moves from 8.4% to 51.1%.

This is the mechanism the certainty equivalent is pricing. Leverage widens both tails, and a risk-averse investor weighs the left one more heavily. The borrowing spread then makes the trade progressively worse, because it is paid in every state of the world including the ones where the leverage did not help.

Note the last column. The clip binds in up to 1.01% of path-years — years in which a real levered investor would have been liquidated rather than merely marked down. It binds only on the high ratios and only when the allocation is held fixed rather than re-optimised, which is why the sweep of Section 21.2 shows none: there the optimiser retreats into bonds and bills precisely to avoid it. Where it does bind, the certainty equivalents overstate what leverage is worth by an amount this model cannot price.

21.4 Should leverage decline with age?

Ayres and Nalebuff (2010) argue that a young investor's financial balance is small relative to the lifetime saving still to come, so a constant *share* of a small balance is a small share of lifetime exposure — and that the remedy is to lever early and delever later. That is a testable claim, and the machinery here tests it directly by solving a leverage ratio at every age, holding the allocation fixed.

Table 72. A leverage ratio solved for every age

Borrowing spread	Leverage at 25	Mean while working	At retirement	Mean in retirement	Solved CEC
0.0%	3x	1.83	1.25x	1.12	1.2263
1.0%	3x	1.73	1x	1.10	1.1766
2.0%	3x	1.57	1x	1.02	1.1428

Notes. The allocation is held at the one optimal for that spread, so the schedule is purely about when to borrow.

The per-age solution is jittery, because the surface is flat enough that the search finds tiny improvements moving a single year between adjacent grid values. Aggregating to decades reports the trend the schedule genuinely carries:

Table 73. Solved leverage by decade of age, at a zero borrowing spread

Decade of age	Years	Mean leverage	Min	Max
20s	5	3.00	3.00	3.00
30s	10	2.10	1.50	3.00
40s	10	1.50	1.50	1.50
50s	10	1.48	1.25	1.50
60s	10	1.12	1.00	1.25
70s	10	1.10	1.00	1.25
80s	10	1.15	1.00	1.25
90s	3	1.17	1.00	1.50

Notes. Aggregated from the per-age solution. The min and max columns show how much the raw schedule moves within each decade — which is where the jitter lives.

The solved schedule **declines with age**, which is the Ayres–Nalebuff prescription arrived at from the other direction: borrow while the financial balance is small relative to the lifetime saving still to come, and delever as it grows. Note that this is the one place in the paper where a genuinely age-varying policy emerges from an unconstrained search — and it is a schedule for *leverage*, not for the equity share. The decade means fall monotonically through the whole working life, so this is not an artefact of the two endpoints.

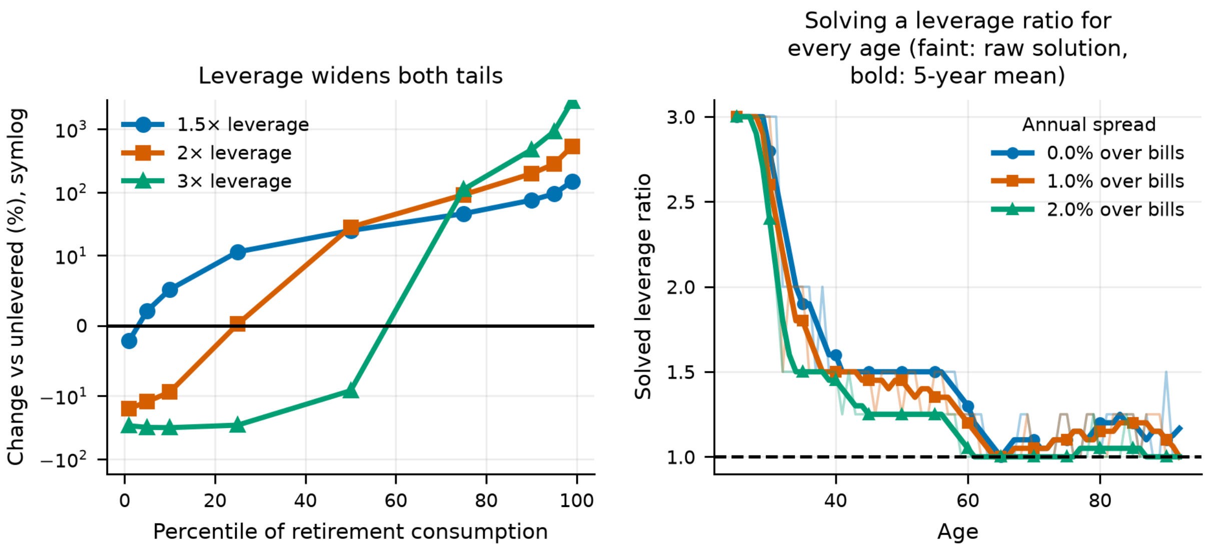

Figure 39.

Left: what leverage does to each percentile of retirement consumption. Right: the leverage ratio solved for every age, at three prices of credit.

21.4.1 Is the shape real, or 68 parameters of overfit?

The schedule above is solved and scored on the same paths, with one free parameter per age. That is exactly the setting in which a solved policy flatters itself, so its certainty equivalent alone cannot establish that *declining* leverage beats a constant ratio — the gain could be nothing but optimisation gain. The discriminating test is how much survives with a **single** knob: hold a ratio while working and unlever at retirement.

Table 74. One knob against sixty-eight: constant, two-level and per-age leverage

Borrowing spread	Best constant	Constant, vs unlevered	Best working ratio	Two-level, vs unlevered	Per age, vs unlevered
0.0%	1.25x	+4.56%	1.5x	+7.91%	+10.84%
1.0%	1.25x	+2.00%	1.5x	+4.19%	+6.34%
2.0%	1x	+0.00%	1.25x	+1.57%	+3.28%

Notes. Every row is scored on the same paths and the same allocation, so the comparison is about the leverage policy alone. The two-level policy holds one ratio while working and unlevers at retirement.

The two-level policy beats the best constant ratio at every price of credit tested, keeping roughly 62% of what the free 68-parameter solve claims. **The declining shape is structural; the per-age detail is mostly optimisation gain.** It also survives where the constant ratio does not: at the higher spreads holding one ratio for life is worth nothing, while unlevering at retirement is still worth having.

21.5 What this changes

- A **constant** leverage ratio is not free money. It is worth +5.10% when credit is free, breaks even at a spread of 3.00%, and is worth under a tenth of a percent from 3.00% upward — which covers most of the range a household actually borrows in.
- That verdict does **not** carry over to a **declining** ratio, which is what Ayres and Nalebuff actually prescribe. Section 21.4.1 prices the two separately on the same paths, and the declining policy is worth more at every spread tested — so the familiar null result from levered-portfolio backtests is in part an artefact of holding the ratio fixed for life.
- The result is driven by the **left tail**. At a moderate 1.5× the median rises +24.7% while the fifth percentile falls +2.1%; push the ratio to 3× and the median falls too (-9.3%). The certainty equivalent prices that trade at the investor's risk aversion, and a less risk-averse investor would take it at a higher spread.
- What gets levered is a **diversified** portfolio. The optimiser does not respond to cheap credit by concentrating, which is the same answer the unlevered sections give in a different form.
- The limited-liability assumption is **generous to leverage**, so the conclusion is conservative in the direction that matters: a model with forced liquidation would like borrowing less than this one does.

22. What the Solved Schedule Costs to Trade

Section 13 prices the expense ratio and finds the differential that would undo the headline. It says nothing about the other cost of running a portfolio, which is trading it, and that omission lands hardest on the part of this paper most exposed to it. A fixed 50/50 portfolio trades because its assets drifted apart. A schedule solved age by age over the whole weight simplex trades for that reason *and* because it decided to hold something different this year — and the optimiser that chose the difference was charged nothing for it.

22.1 Measuring the trade

Turnover is reported one-way: half the sum of absolute weight changes, so selling one asset to buy another counts once. It splits three ways. **Total** is the trade actually required each year on simulated paths. **Drift** is what a portfolio whose target never moved would have had to trade anyway — the floor no schedule can get under. **Schedule** is the deterministic move the target makes between two ages: what following the plan would cost in a world where nothing ever drifted.

Table 75. One-way turnover, averaged over paths and years.

Strategy	Traded a year	Drift alone	Schedule alone	Excess over drift	Over a lifetime
solved simplex	7.96%	1.14%	7.72%	+6.82%	5.4×
Target-date fund	4.14%	3.85%	0.88%	+0.29%	2.8×
60/40 domestic equity/bonds	3.61%	3.61%	0.00%	+0.00%	2.5×
50/50 domestic/international	3.39%	3.39%	0.00%	+0.00%	2.3×
100% domestic equity	0.00%	0.00%	0.00%	+0.00%	0.0×
100% international equity	0.00%	0.00%	0.00%	+0.00%	0.0×
100% bills (cash)	0.00%	0.00%	0.00%	+0.00%	0.0×

Notes. A single-asset portfolio never drifts away from itself, so it trades nothing. A constant-weight portfolio's total and drift columns are equal by construction, which is the check that the decomposition is doing what it claims.

The solved schedule turns over 8.0% of the portfolio a year. The benchmark it has to beat is a single-asset portfolio, which never drifts away from itself and so never rebalances at all — it trades nothing, ever. Almost all of the schedule's trading is it choosing to move rather than drift forcing it: the excess over the drift floor is 6.8% a year, which is the cost the optimiser never saw.

The three columns do not add up, and the reason is worth stating. A schedule that cuts equity in a year equity outperformed is trading *with* the drift rather than against it, so its total can sit below its drift counterfactual. A glide path is partly self-rebalancing, which is a point in its favour that a naive turnover count would miss.

22.2 Charging for it

The cost is proportional to the value turned over and taken at the rebalance, before that year's return compounds on what is left. Every strategy pays it on the same paths, so the question is never whether costs hurt — they hurt everybody — but which portfolio they hurt more. The comparison is against *100% international equity*, which is the best fixed portfolio judged before any cost is introduced; letting the incumbent be re-chosen at each level would slide it to meet the challenger and make the break-even vacuous.

Table 76. γ = 5, 50,000 lifetimes per level.

One-way cost (basis points)	Solved schedule over the best fixed portfolio (%)	Best strategy
0	+1.07	solved simplex
5	+1.04	solved simplex
10	+1.01	solved simplex
25	+0.92	solved simplex
50	+0.76	solved simplex
100	+0.45	solved simplex
200	-0.17	100% international equity
300	-0.80	100% international equity

The solved schedule's edge is thin and it is spent on trading. Its advantage over the best fixed portfolio is 1.07% before costs, and reaches zero at **173 basis points** one-way. That is above an index fund's spread but inside what a retail investor trading small parcels pays, and it is a small enough number that the advantage should be read as an upper bound rather than as a result.

Section 23 reaches the same place by a different road. There the solved schedules are asked to work on data they were not fitted to; here they are asked to pay for the trades they make. Both find that a constant mix, held unchanged and fitted to nothing, is a harder benchmark than the in-sample gains suggest.

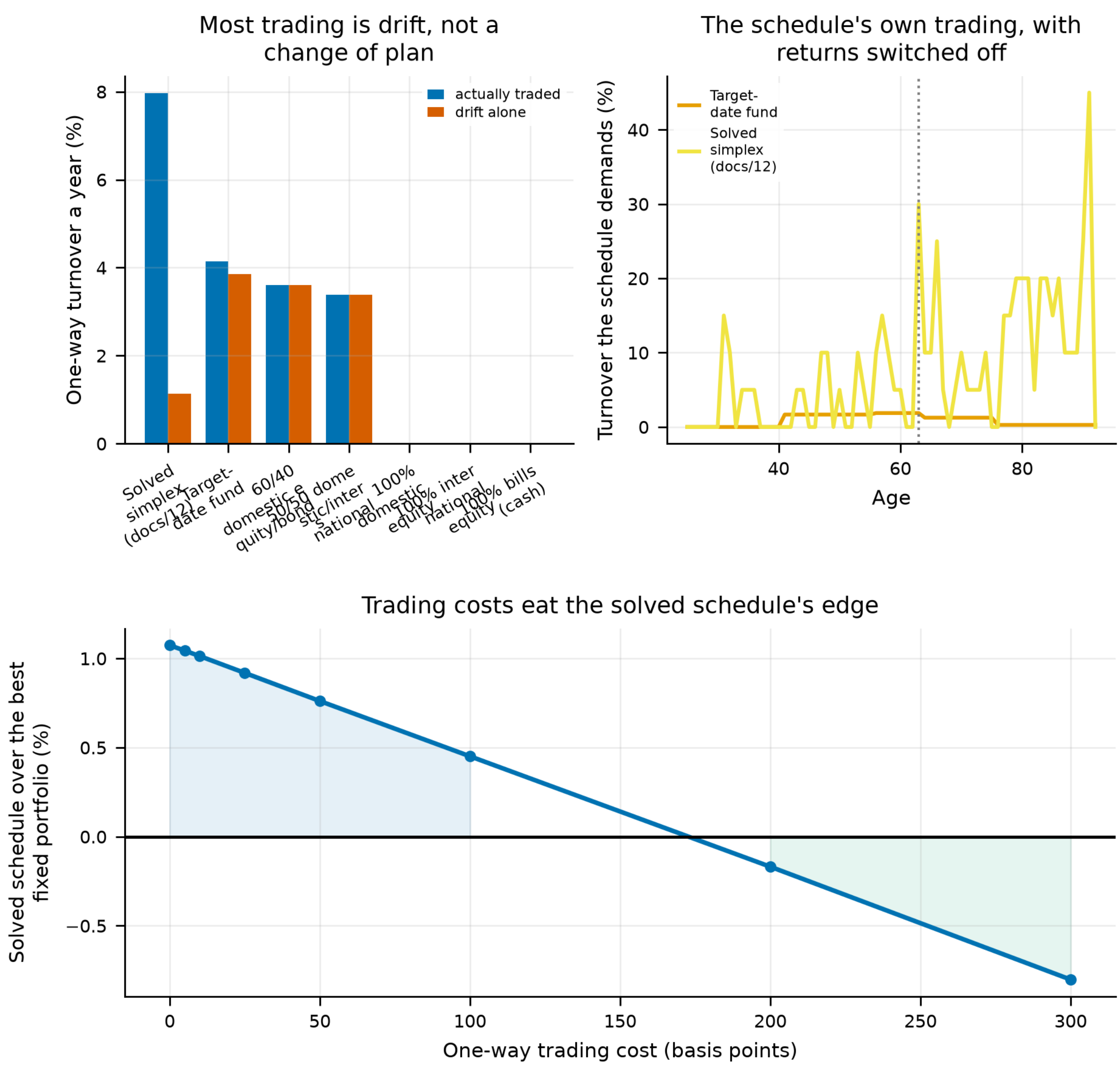

Figure 40.

Left: what each strategy trades a year, against the drift it could not have avoided. Middle: the trading each schedule demands of itself with returns switched off. Right: the solved schedule's lead over the best fixed portfolio as trading gets more expensive.

22.3 What this changes

- The gains in section 20 should be read net of a cost they were solved without. The break-even is 173 basis points one-way.
- The headline of this paper is unaffected. The strategies it compares are constant mixes, and their turnover is drift alone — the same trade every rebalanced portfolio makes.
- **What is not modelled**: costs are proportional and symmetric, with no bid-ask asymmetry, no market impact and no minimum ticket. Contributions during accumulation are invested pro-rata rather than steered toward the underweight asset, which overstates turnover for a real saver — so the break-even here is, if anything, reached too easily. Tax on realised gains is a cost of trading and is absent, along with every other tax in this paper.

23. Do the Solved Schedules Survive Data They Did Not See?

Sections 19, 20 and 21 each hand a coordinate-ascent search sixty-eight free parameters and report what it found. Every one of those gains is measured on the same history the search was given. That is the standard way to overstate a result: with sixty-eight parameters and the 16 independent lifetimes section 7 counts, a search can fit a good deal of noise, and an in-sample gain is an upper bound on what an investor standing at the start could have had.

The design splits the calendar record at 1955, solves on one half and scores on the other, against three references: the gain *where it was solved* — the number those sections report today; the gain *where it was scored*, against the best fixed strategy on the untouched half, which is what an investor of the period could actually have held; and a *ceiling*, the same family solved on the test half itself. Both directions are run, because the two halves of this panel are not interchangeable: one contains two world wars and the other contains the post-war expansion.

23.1 What transferred

Table 77. Solved schedules against the best fixed strategy, γ = 5, 20,000 lifetimes per window.

Family	Solved on	Scored on	Gain where solved (%)	Gain where scored (%)	Ceiling (%)
Glide path (equity share by age)	1890-1955	1956-2020	+1.09	+0.36	+1.72
Glide path (equity share by age)	1956-2020	1890-1955	+1.72	-0.65	+1.09
Full simplex (four assets by age)	1890-1955	1956-2020	+1.34	+0.30	+2.42
Full simplex (four assets by age)	1956-2020	1890-1955	+2.42	-1.85	+1.34

Notes. A negative middle column is a schedule that would have been worse than holding a constant mix.

The solved schedules transfer in 2 of 4 runs, and which ones is the finding. A schedule learned on 1890-1955 and applied to 1956-2020 keeps +0.33%; the same exercise run backwards — learned on 1956-2020, applied to 1890-1955 — delivers -1.25%. The in-sample gain averages 1.64% either way.

That asymmetry has a reading, and it is not a flattering one for the optimiser. The earlier half contains two world wars, several market closures and the Depression; the later half contains the post-war expansion. A schedule fitted to the turbulent window carries something forward into the calm one. A schedule fitted to the calm window learns a world that never came back, and applying it to the turbulent half costs 1.25% against simply holding a constant mix. An investor solving today would be in the second position, not the first.

The benchmark the solved schedules are measured against is *100% international equity* in every run. The strategy that does transfer is a constant mix, held unchanged for a lifetime — which is the strategy this paper is actually about.

23.2 Are the two halves the same world?

Table 78. Every fixed strategy on each half of the record.

Window	Strategy	CEC	Rank
1890-1955	100% International Equity	1.0165	1
1890-1955	50/50 Domestic/International Equity	0.9708	2
1890-1955	Target-Date Fund (glide path)	0.8543	3
1890-1955	100% Domestic Equity	0.8198	4
1890-1955	60/40 Domestic Equity/Domestic Bonds	0.8011	5
1890-1955	100% Bills (cash)	0.7160	6
1956-2020	100% International Equity	1.1329	1
1956-2020	50/50 Domestic/International Equity	1.0692	2
1956-2020	Target-Date Fund (glide path)	1.0207	3
1956-2020	60/40 Domestic Equity/Domestic Bonds	0.9450	4
1956-2020	100% Domestic Equity	0.9170	5
1956-2020	100% Bills (cash)	0.7767	6

Notes. If the fixed ranking moves across the split, some of what fails to transfer is the world changing rather than the search overfitting.

The fixed strategies do **not** keep their order across the halves — 2 positions move, though the winner is the same. Some of what fails to transfer above is therefore the world changing between the halves rather than the search overfitting, and this test cannot separate the two.

One split is one experiment, and that is the limit of this test. The record is cut once, at 1955, because there is only one place to cut it that leaves enough calendar time on each side to solve 68-year schedules. That leaves no distribution to judge the result against: we cannot say how often a split of this record would show an asymmetry this large by chance, because there is no second split to compare it with. Nor can a single cut separate the two explanations the table admits — a schedule that learned noise, and a world that changed underneath it — because both predict the same pattern. Rolling-origin splits, or a placebo distribution over randomly chosen cut years, would separate them; neither fits inside a 131-year record that already has to hold two disjoint working lives. Read this section as a caution with a direction, not as a test with a size.

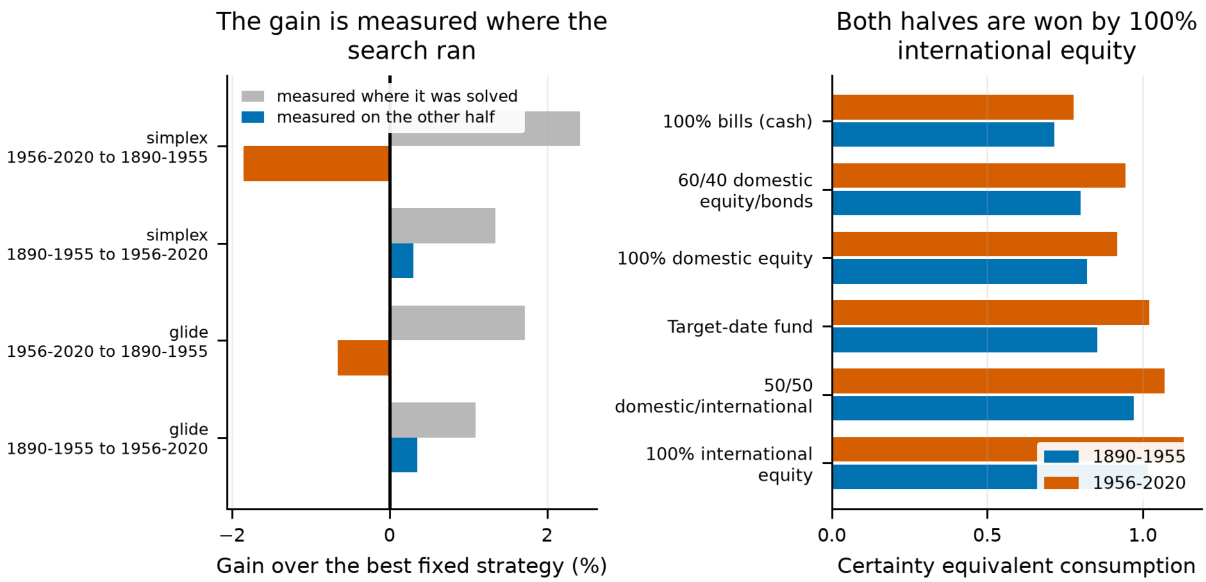

Figure 41.

Left: the gain each solved schedule reports where it was solved, against what it delivers on the half it never saw. Right: the fixed strategies on both halves of the record.

23.3 What this changes

- The gains reported in sections 19, 20 and 21 should be read as **upper bounds**. The transferable part averages -0.46% against 1.64% in sample.
- **The benchmark that keeps winning is *100% international equity*, in every one of the 4 runs.** That is the most useful thing in this section and it deserves stating plainly rather than as an aside: a constant mix, held unchanged for a lifetime and fitted to nothing, is what a solved schedule has to beat on data it did not see, and mostly does not. Section 22 reaches the same conclusion by a different route, by charging the solved schedule for the trades it makes.
- The headline of this paper does not depend on any of it. The comparison that carries the paper is between fixed strategies, none of which is fitted to anything.
- What this test cannot do is prove that a failure to transfer is overfitting rather than a changed world. The two halves are not draws from one process, and the benchmark table is the evidence.

24. Housing as a Fifth Asset

The historical sources behind this study measure four asset classes. Three of them — equity, bonds and bills — form the investable set everywhere above. The fourth is housing, measured just as carefully across 16 countries and 1,805 country-years, and excluded from every result so far. This section asks what including it would do.

Two obstacles stand in the way, and both are addressed rather than assumed away.

24.1 The index is smoothed

House prices come from transactions and valuations rather than from a continuous auction, so this year's index still carries part of last year's level. The published series therefore understates the volatility an owner actually bears, and comparing it with a traded series would flatter it — the classic error in this literature. We invert the smoothing country by country using each country's own estimated first-order coefficient, since index construction differs between them and a pooled coefficient would over-correct the cleanly measured series and under-correct the rest. A country whose returns are not positively autocorrelated is left alone: there is nothing to undo.

Table 79. Appraisal smoothing in the housing series, by country

Country	Years	Lag-1 autocorr.	Mean (%)	SD published (%)	SD de-smoothed (%)	Equity SD (%)
FRA	130	0.63	6.28	10.29	21.53	23.09
JPN	75	0.59	6.57	8.13	15.63	25.79
FIN	99	0.51	9.42	15.24	25.73	30.00
CHE	119	0.47	5.67	6.53	10.79	18.92
NLD	130	0.46	7.45	8.98	14.87	21.18
DNK	130	0.42	8.05	7.75	12.10	17.20
SWE	130	0.35	8.11	8.89	12.75	20.34
GBR	118	0.34	5.39	8.97	12.74	18.68
BEL	119	0.34	7.93	15.15	21.73	22.92
PRT	73	0.33	6.65	8.52	11.51	26.04
NOR	130	0.17	7.83	8.63	10.27	20.13
ESP	120	0.03	5.26	11.75	12.19	20.75
ITA	86	-0.01	4.70	9.34	9.34	26.65
AUS	120	-0.02	6.22	11.73	11.73	16.22
DEU	96	-0.16	6.56	8.45	8.45	33.53
USA	130	-0.17	6.22	7.90	7.90	18.73

Notes. De-smoothing preserves the mean in expectation and restores the variance the filter removed. Countries with non-positive autocorrelation are unchanged.

This leaves housing with an equity-like average return and materially less volatility than equity even after the correction — which is exactly why the second obstacle cannot be waved through.

24.2 A building costs money to hold

A share certificate does not. Rates, insurance, maintenance and management fall on a property owner every year whether the asset rose or fell, and the right figure is specific to the investor and the jurisdiction. Rather than choose one, we sweep it: the whole allocation is re-solved over the five-asset simplex at each annual holding cost, charged on value rather than on gains. The source builds its housing total return from capital gains plus a rental yield it describes as net of depreciation and maintenance, so the swept figure is best read as *additional* to whatever that construction already deducts — the taxes, management, voids and amortised transaction costs it does not. For scale, Chambers, Spaenjers and Steiner (2021), working from property-level records of four institutional portfolios over 1901–1983, find operating costs cut net yields to about two-thirds of gross; against this panel's 4.9% median rental yield that is roughly 160 basis points. If the published series is in fact grosser than that, the break-even below overstates how much extra cost housing can bear; the direction of that error is known even though its size is not, which is why we report the whole curve rather than a single recommended weight.

Housing is recorded for fewer country-years than equity is, so the study runs on the intersection; filling the gaps would mean inventing returns. The four-asset control is re-solved on that same restricted panel, against the same lifetimes and the same income draws, so the restriction cancels out of every comparison. Its own optimum is 10% domestic equity and 90% international.

Housing enters as a **domestic** asset: each simulated investor holds their own country's housing index, drawn on the same calendar years, countries and blocks as their equity and bonds, so the cross-asset correlation the bootstrap exists to preserve is preserved. There is no international housing sleeve — people buy property where they live, and the leave-one-out construction that gives equity a foreign leg has no counterpart a household could execute in bricks.

Table 80. The optimal portfolio at each annual housing holding cost

Holding cost (%)	Housing (%)	Dom. equity (%)	Intl. equity (%)	Bonds (%)	Bills (%)	Gain over four assets (%)
0.0	50.0	0.0	50.0	0.0	0.0	13.20
1.0	40.0	0.0	60.0	0.0	0.0	7.30
2.0	27.5	0.0	72.5	0.0	0.0	3.27
3.0	15.0	5.0	80.0	0.0	0.0	0.95
4.0	2.5	7.5	90.0	0.0	0.0	0.04
5.0	0.0	10.0	90.0	0.0	0.0	0.00
6.0	0.0	10.0	90.0	0.0	0.0	0.00

Notes. One allocation held for life, re-solved at each cost under common random numbers, so a difference between rows is the cost and nothing else.

Housing is worth holding below an annual cost of 4.6% and not above it. Offered free it takes 50% of the portfolio and adds 13.2% to certainty-equivalent consumption over the best four-asset portfolio on the same paths. That gain erodes as the cost rises and vanishes at 4.6% a year.

The weight comes out of **international equity**, which gives up 40% of the portfolio between the dearest case and the free one — more than any other asset. Bonds, bills are untouched at every cost, so housing is not substituting for the portfolio at large — it is competing with the single holding that already does this job.

24.3 Two checks on that answer

The result rests on two choices that could each be driving it, so both are tested rather than defended.

Is it an artefact of the de-smoothing? Re-running the whole sweep on the raw, still-smoothed series puts 55% in housing at zero cost against 50% once the smoothing is undone, with a break-even cost of 4.80% against 4.60%. The naive treatment therefore **overstates** the case for housing, which is the direction the mechanism predicts: smoothing hides volatility, and hidden volatility reads as risk-adjusted return. The correction is doing real work, and it works against housing rather than for it.

Is it an artefact of holding one allocation for life? Re-solving the full age-by-asset schedule over the five-asset simplex — seeded at the constant-mix optimum, so the search can only improve on it — moves the mean housing weight by at most 4% of the portfolio and buys at most 2.28% more certainty-equivalent consumption. That is large enough that the constant-mix figures above should be read as indicative of the level rather than as the optimum itself.

The *shape* of that schedule is worth naming whatever its size. Housing is the one asset in this paper whose optimal weight **rises** with age: at a 3% holding cost the solved schedule holds 4% of it while working and 30% in retirement. That is the opposite of the declining glide path Sections 19 and 20 searched for and did not find in equities, and it has a reading: a retiree drawing on the portfolio wants the asset with the best return per unit of volatility, and once the appraisal smoothing is undone housing is still that asset. It is also the sharpest illustration of this section's caveat — the schedule is only implementable by someone who can rebalance into and out of property annually, which is nobody.

24.4 What this is not

The asset priced here is a liquid, continuously rebalanced, nationally diversified claim on a country's housing stock, because that is what a national house price index measures. **It is not a house.** A single owner-occupied property is concentrated in one street rather than spread across a country, cannot be rebalanced, is bought with leverage, carries transaction costs measured in percent rather than basis points, and pays part of its return as accommodation rather than as cash. None of those differences is in the numbers above. The section prices an asset class; it is not advice about a mortgage.

The cost is also modelled as a constant annual percentage of value. Real holding costs are lumpy, partly fixed, and correlated with the cycle. A flat charge is the tractable approximation, not the truth.

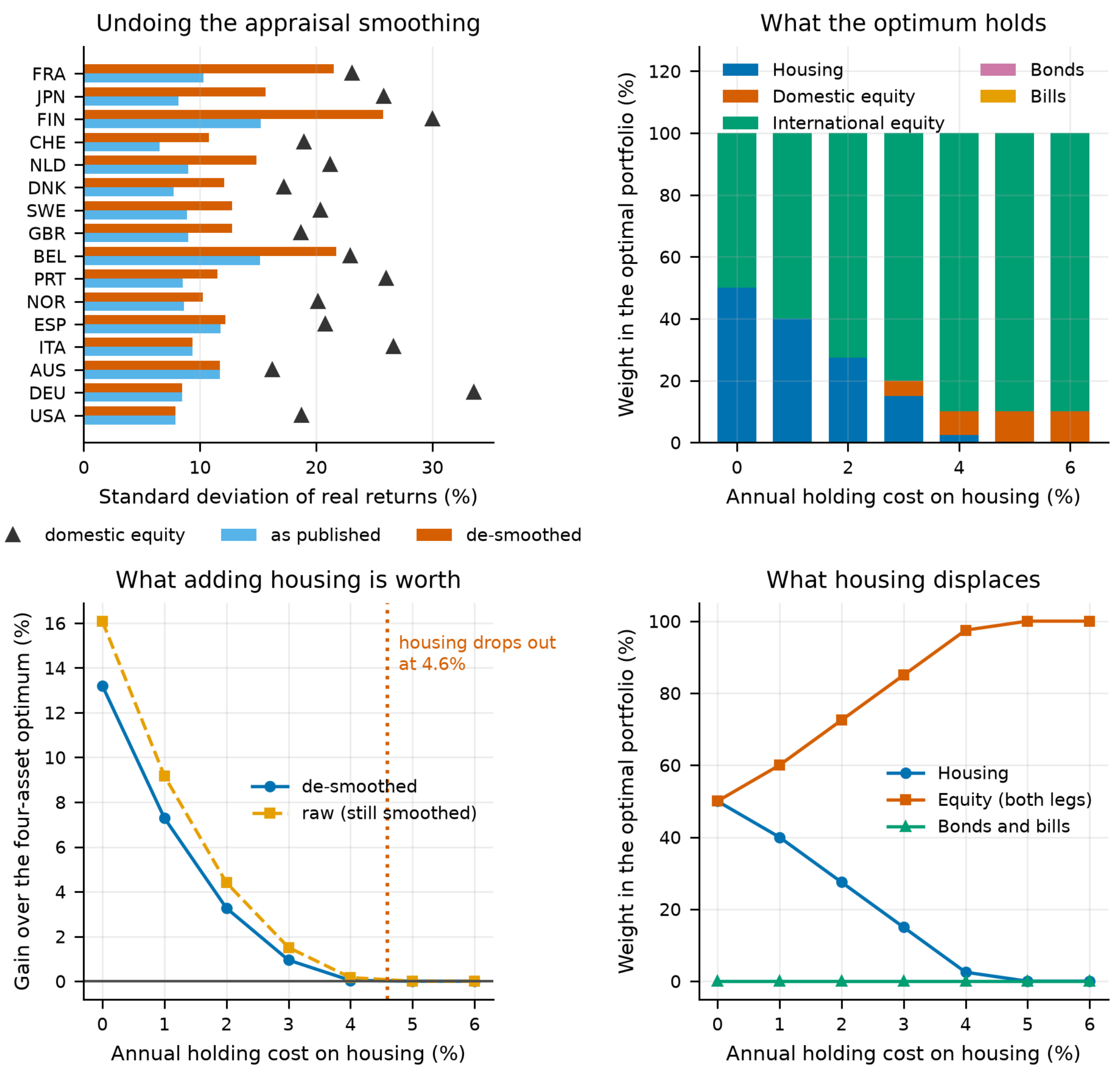

Figure 42.

Top left: the volatility the appraisal smoothing hides, by country, against the same country's equity. Top right: the optimal portfolio at each holding cost. Bottom left: what adding housing is worth, with and without the de-smoothing correction. Bottom right: which sleeve housing displaces as its cost falls.

25. The Mortgage

Section 24 prices housing as an asset owned outright. No household owns it that way. A house is the one asset an ordinary person can borrow 80% against, at a rate close to their own government's, secured on the thing itself — and leaving that out understates what housing does to a lifetime as surely as leaving the holding cost out overstates it. This section puts the mortgage in and asks the two questions that matter: how much, and when.

25.1 How the loan is modelled

The decision variable is the **loan-to-value ratio**, because that is the number a lender quotes and a borrower chooses. A property funded with equity E and a loan at ratio λ returns, on that equity, (r_H − λ·i) / (1 − λ), where r_H is the real return on the property and i the real mortgage rate. That is the leverage multiple 1/(1 − λ) applied to housing alone, so the arithmetic is the same function Section 21 uses for portfolio borrowing and the two remain consistent.

The rate is the borrower's *own country's* real short rate plus a spread, drawn on the same block as every other series, so a lifetime that lives through high real rates pays them. The spread is swept rather than assumed. Equity is **wiped out, not driven negative**: the levered return is floored at total loss, which is the non-recourse assumption and the one most favourable to borrowing. How often that floor binds is reported rather than buried.

25.2 How much

Table 81. Certainty-equivalent consumption by loan-to-value ratio, held for life

LVR	Gross exposure per unit of equity	CEC	Gain over no mortgage (%)	Path-years wiped out (%)
0%	1.00	1.189	0.00	0.00
10%	1.11	1.194	0.37	0.00
20%	1.25	1.199	0.81	0.00
30%	1.43	1.205	1.34	0.00
40%	1.67	1.213	1.96	0.00
50%	2.00	1.222	2.73	0.19
60%	2.50	1.232	3.62	0.35
70%	3.33	1.244	4.58	0.84
80%	5.00	1.252	5.29	3.02

Notes. At the spread reported in the accompanying analysis document, with the allocation re-solved alongside the mortgage.

The search goes to the 80% ceiling and would go further if allowed. The reported figure is the constraint rather than an optimum, and the honest reading is that this model does not price whatever stops real households borrowing more — mortgage insurance, servicing tests, and the concentration risk of a single property, none of which are here.

25.3 When

The solved schedule **declines with age**: 75% while working against 49% in retirement. That is what households actually do — borrow heavily against the first house and pay the loan down over a career — and here it falls out of the optimisation rather than being imposed on it. The mechanism is human capital: a working investor has decades of future earnings a mortgage cannot reach, and that income is what makes the leverage bearable. It is also the mirror image of the equity glide path this paper argues against, and the two are consistent — what should decline with age is the *borrowing*, not the equity.

That is a statement about the *level and the slope* and nothing finer. Subjecting the schedule to the same deviation profile Section 20 applies to solved allocations — resetting one age at a time to the schedule's own average and pricing what is lost — leaves 20 of 68 ages carrying a decision worth more than five basis points, the largest worth 18 and the median age worth 2. The rest sits on a flat part of the surface where the search moves the ratio for free. The plotted line is jagged; the evidence underneath it is not.

25.4 At what price of credit

Table 82. The joint optimum at each price of mortgage credit

Spread	Mean LVR	LVR working	LVR retired	Housing equity	Gross housing (× wealth)	Gain over no mortgage (%)	Wiped out (%)
0%	72%	80%	63%	25%	1.09	25.6	2.02
1%	71%	80%	60%	20%	0.86	14.3	2.21
2%	69%	80%	56%	18%	0.73	9.8	2.11
2%	64%	75%	49%	18%	0.62	5.9	1.62
2%	55%	67%	39%	18%	0.53	2.9	1.34
3%	38%	47%	25%	22%	0.48	1.0	0.66
4%	7%	4%	10%	27%	0.33	0.1	0.12

Notes. Spread over the borrower's own country's real short rate. Every row shares the same paths, income draws and search, so a difference between rows is the price of the loan and nothing else.

The optimal ratio falls from 72% at a 0% spread to 7% at 4% without reaching zero inside the grid, so no break-even price is reported — extrapolating would invent the number the sweep failed to find. What the table does establish is that the answer is highly sensitive to the margin, which is the practical point: a household's mortgage rate, not the return on housing, is what decides this.

25.5 Why this differs from Section 21

Section 21 concluded that borrowing is barely worth doing. This section concludes that borrowing against a house is worth a great deal. Both use the same arithmetic and the same real bill rate plus a swept spread, so the two results have to be reconciled rather than left side by side.

Table 83. The same borrowing cost, two places to spend it

Spread over the real short rate	Lever the portfolio (%)	Mortgage the housing sleeve (%)
0.0%	5.10	25.56
1.0%	2.00	14.28
1.5%	0.88	9.82
2.0%	0.37	5.88
3.0%	0.00	0.96

Notes. Each column is that study's own gain over its own unlevered optimum, so both measure the value of the leverage rather than the value of the asset.

It is not the price of credit. At identical spreads the mortgage is worth several times what portfolio leverage is worth, so holding the borrowing rate fixed does not close the gap and cannot be the explanation.

Table 84. Housing against the assets it competes with

Asset	Mean real	s.d.	Return per unit of risk	Correlation with housing
Housing	4.79%	14.53%	0.33	1.00
Domestic equity	6.63%	22.98%	0.29	0.19
International equity	8.18%	22.61%	0.36	0.09
Bonds	1.77%	12.22%	0.14	0.20
Bills	0.48%	8.23%	0.06	0.32

Notes. Housing is net of the 2% holding cost this section charges; the others are gross. For reference, the two equity legs correlate at 0.40 with each other.

It is diversification. Housing's standalone return per unit of risk is 0.33 against international equity's 0.36 — *worse*, so the gap cannot be a story about housing being the superior asset. What housing has is a correlation of 0.09 with the international equity sleeve, where the two equity legs correlate at 0.40 with each other. Levering the portfolio scales risk the investor already holds. Levering one asset changes what the portfolio is made of, and here it buys more of the one holding that moves independently of the rest — while tying up less capital in it.

That distinction matters beyond this paper. The case for lifecycle leverage (Ayres and Nalebuff, 2010) is usually made for levered *equity*, and on this panel that is the version that does not pay once credit is priced realistically. The declining-with-age shape their argument predicts survives; the instrument it is usually attached to does not. That the cheapest leverage available to a household happens to be secured against the diversifying asset is a convenience, not the mechanism.

25.6 What this is not

The schedule is rebalanced annually, like everything else in this paper. For a mortgage that means costlessly redrawing the loan every year to hit a target ratio. Real mortgages amortise on a fixed schedule, cost several percent of the property to refinance, and are called on missed payments rather than on a drifting loan-to-value. **This is the value of the leverage, not a financing plan.**

Three further gaps deserve naming. There is no mortgage insurance, so the 80% ceiling is a wall rather than a price. There is no tax: in several countries mortgage interest is deductible and owner-occupied capital gains are untaxed, both of which would favour borrowing more than this shows. And the asset is a national housing index, not a house — the concentration risk of a single leveraged property is precisely what makes a real mortgage dangerous, and none of it is in these numbers.

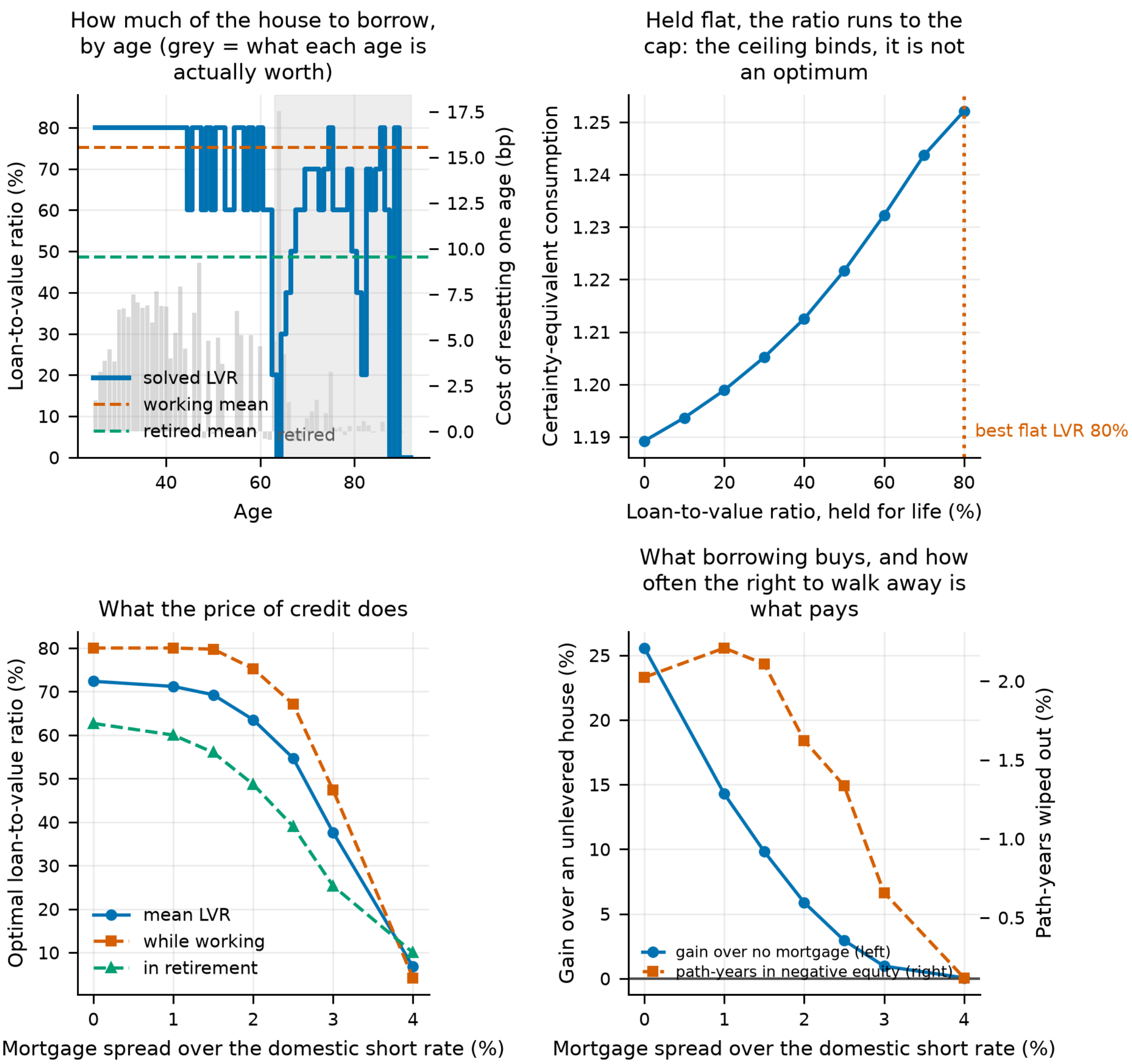

Figure 43.

Top left: the solved loan-to-value ratio by age, with the grey bars showing what each age's choice is actually worth — where they are invisible the line above them carries no information. Top right: certainty-equivalent consumption against a ratio held for life. Bottom left: how the optimal ratio responds to the price of credit, split by working life and retirement. Bottom right: what borrowing buys against an unlevered house, and how often the borrower's right to walk away is what pays for it.

26. Conditioning the Savings Rate

Section 11 began the part of this paper that leaves the portfolio alone: it asked what the market charged on the day a lifetime opened, and found that the answer moves the outcome without changing the allocation. This section and the three after it carry that further, in the order a life meets them — how much you save, what that saving should respond to, when you stop, and how you draw down. The first question is whether the savings rate should vary over a career at all, and whether it should respond to the portfolio.

Two quite different things could make the rate vary, and a comparison that does not separate them will credit the wrong one. **Shape** is variation with age alone: labour income here is hump-shaped, and a fixed rate makes consumption track income exactly, so a 25-year-old consumes least precisely when they are poorest. **Conditioning** is variation with state: whether wealth is ahead of or behind an age-appropriate target. The first uses no market information at all; the second is what "you should have six times salary by fifty" is reaching for.

26.1 A caveat that has to come first

The model cannot identify the savings *level*. Swept continuously, the certainty equivalent peaks at a constant rate of 5% and falls above it. That number should not be believed and it is worth being explicit about why.

The savings level is a trade between consuming now and consuming later. In this model that trade is settled almost entirely by the discount factor — $\beta = 0.96$, which over a 38-year working life discounts retirement consumption by a factor of 0.21 — and by risk aversion acting on the left tail of consumption, which with a floored retirement and risky labour income sits in working life rather than in retirement. Neither of those is something a panel of historical returns has any view on.

So the level is not identified here and this paper does not claim it. What the return panel *can* speak to is the shape of the profile and the value of conditioning it, both holding the average rate fixed. Everything below pins the career-average savings rate at 10% and asks only when, and on what, to save it.

26.2 Shape: when should you save?

We solve for a free savings rate at each of the 38 working years by coordinate ascent over a per-age multiplier, renormalising after every move so that the career average never drifts. The answer is not the same at every risk aversion, and that is the interesting part.

Table 85. Solved savings profile by risk aversion, career average pinned

Risk aversion	Shape	First quarter	Middle half	Last quarter	Peak age	Gain vs flat (%)
$\gamma = 2$	hump-shaped	3.5%	13.2%	9.3%	38	0.48
$\gamma = 5$	hump-shaped	5.8%	13.8%	6.1%	35	0.62
$\gamma = 10$	front-loaded	15.8%	11.7%	1.3%	25	2.48

Notes. The profile is summarised by the average rate in the first quarter, middle half and last quarter of the working life. All three profiles have the same career average by construction, so the comparison is purely about shape.

At moderate risk aversion the solved profile is **hump-shaped**: save least when young, most in peak-earning years, taper into retirement. That is consumption smoothing doing what theory says it should. A 25-year-old sits at the bottom of a hump-shaped income profile, and taking a further tenth of that income away is expensive in utility terms precisely because there is so little of it. This is the "save more later, when you earn more" pattern, and the model produces it without being told to.

At high risk aversion it **inverts** to front-loaded. Two motives compete and risk aversion picks the winner. Smoothing wants saving where income is highest, which is mid-career. Precaution wants a buffer built early, so that it has the whole remaining career to compound against a bad income or market draw. At $\gamma = 10$ precaution wins outright. The model cannot settle which kind of investor a given reader is; what it can say is that both beat a flat rate and that getting the shape right matters more the more risk-averse you are.

Unlike the glide path of Section 19, this structure is real. The deviation profile shows 32 of 38 working years moving the objective by more than a basis point when reset to the career average.

26.3 Conditioning: your position beats the market's direction

Layered on top of the solved shape, and scored against a constant rate interpolated at each rule's own realised career average, we test two conditioning rules: an *on-track* rule that raises the rate when wealth is below an age-appropriate wealth-to-income target, and a *return-responsive* rule that raises it after a weak market year.

Table 86. Conditioning rules at a matched career-average savings rate

Rule	Realised mean rate	CEC	Matched constant CEC	Value (%)
On-track k=0.05	10.62%	0.817	0.783	4.32
On-track k=0.02	10.02%	0.813	0.787	3.29
On-track k=0.01	9.83%	0.808	0.788	2.51
On-track k=0.005	9.81%	0.803	0.789	1.88
Solved shape	10.00%	0.792	0.788	0.62
On-track k=0	10.00%	0.792	0.788	0.62
Return-responsive k=0	10.00%	0.792	0.788	0.62
Return-responsive k=0.25	8.46%	0.799	0.796	0.45
Return-responsive k=-0.25	11.93%	0.773	0.773	-0.00
Return-responsive k=0.5	7.87%	0.794	0.799	-0.59

Notes. The matched comparison is essential: a rule that saves more when behind will, in a world where most paths fall behind, quietly save more overall. Interpolating the constant-rate frontier at each rule's own realised mean strips that out.

Saving more when behind an age-appropriate wealth target is worth several times what the age profile alone is worth. Saving more after a bad market year is worth almost nothing. The sign check behaves as it must: reversing the on-track rule — saving *less* when behind — is strongly negative, which is the reassurance that the machinery measures what it claims.

The interpretation is that a bad market year is a poor proxy for being behind. Wealth relative to an age-appropriate target is the sufficient statistic; the market's recent direction is a noisy shadow of it. Section 27 takes this apart in detail and finds that even this reading needs qualifying.

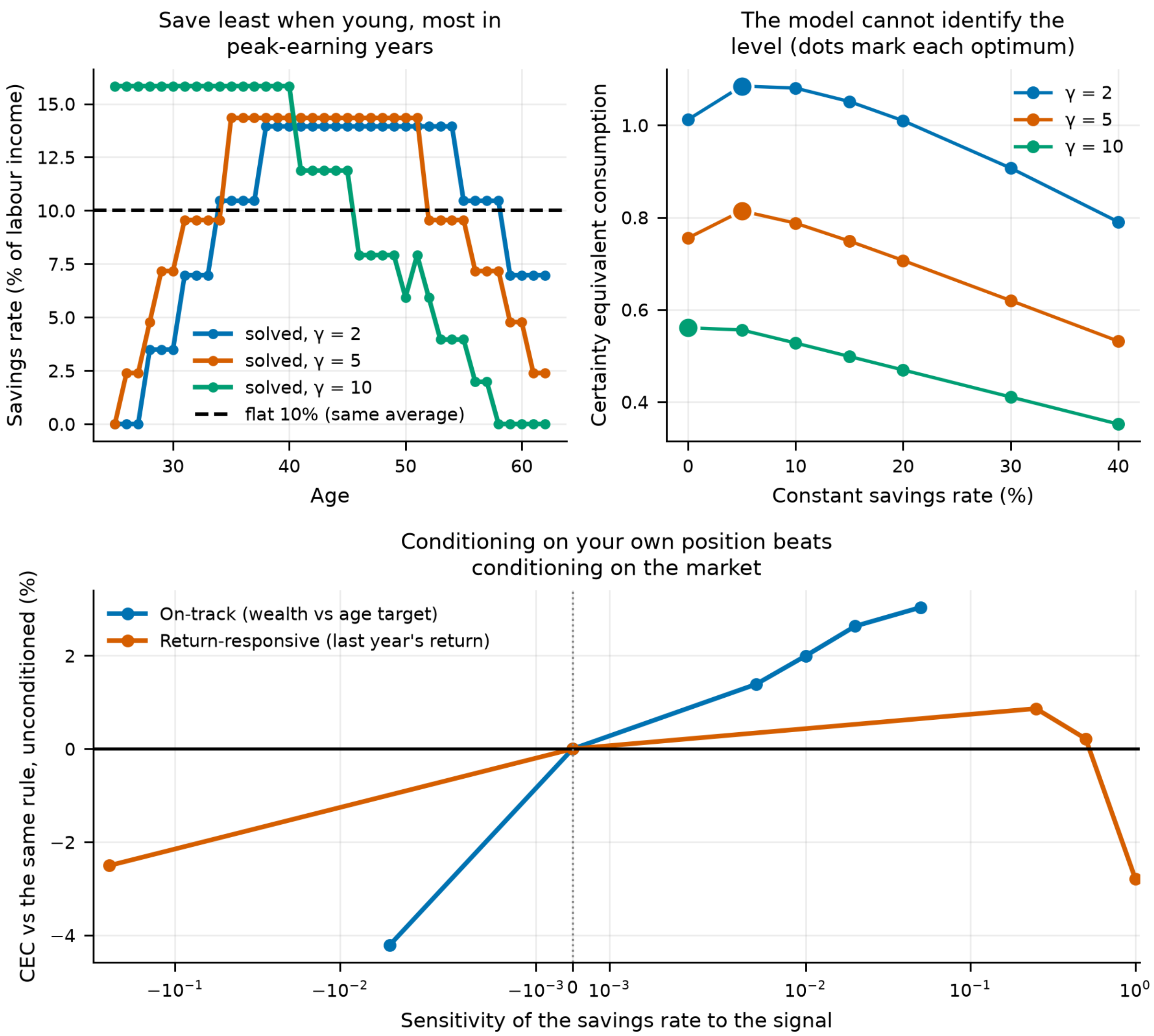

Figure 44.

The solved savings profile by risk aversion, the unidentified constant-rate frontier, and the value of conditioning on wealth against conditioning on returns.

26.4 Do the savings and retirement gains add?

Section 28 below conditions the retirement date on the portfolio; this section conditions the savings rate on it. A natural question is whether an investor should do both.

Table 87. Savings conditioning and retirement conditioning, separately and together

Configuration	CEC	vs neither (%)	Mean rate	Mean retirement age	P(ruin)
Neither	0.788	0.00	10.00%	63.000	11.7%
Savings conditioning only	0.817	3.68	10.62%	63.000	11.7%
Retirement conditioning only	0.805	2.20	10.00%	64.203	11.1%
Both	0.814	3.34	11.71%	63.346	11.8%

Notes. Whole-lifetime utility. Both rules read the same underlying signal, so their gains overlap rather than add.

They do not add. Savings conditioning alone is worth 3.68%, retirement conditioning alone 2.20%, and both together 3.34% — less than the better of the two alone. Both rules respond to the same ahead-or-behind signal, so running them together over-corrects. This is a practical result: an investor choosing between the two levers should choose, not stack.

27. The Accumulation Signal, Decomposed

Section 26 established that conditioning the savings rate on the funded ratio is worth 4.3% of certainty-equivalent consumption at its best setting, measured against a constant rate matched on the same realised career average. Section 27.5 re-scores it on a common basis against eight competing signals and puts it at 4.4%; the two differ because they hold different things fixed, and both are reported rather than reconciled into one. That result rests on five choices that were made once and never varied: the gap was measured in income multiples, the target was the model's own median path, one coefficient governed both directions, the rate was free to move anywhere between 0% and 40%, and the rule ran for the whole career. Each is a modelling decision, and a number that survives only one setting of them is not worth acting on.

This section varies all five, races five further state variables against the funded ratio, tests whether the two best combine, and asks where in the distribution and at which ages the value lands. 195 rule evaluations at 20,000 paths each, all under common random numbers.

Two scoring conventions apply throughout and both matter. Each number is a percentage gain over a *constant* savings rate interpolated at the rule's own realised career average. And conditioning is layered on top of a solved age profile that is itself worth +0.62% against that matched constant, so every figure quoted below is *net* of the shape's own contribution.

27.1 Which units is "behind" measured in?

Being "two times salary short" means something entirely different at 30 and at 60, because the target itself grows from roughly nothing to around ten times income over a career. A response linear in that gap is therefore inert when young and violent when old, whether or not anyone intended it. We test two scale-free alternatives that measure the shortfall as a fraction of the target instead.

Table 88. Three ways of measuring the shortfall, each at its own best coefficient

Gap measured as	Best k	Extra saving when 25% behind (pp)	Value (%)	Net of shape (%)	Mean rate	P(ruin)
Funded ratio (1 − W/Y ÷ target)	0.400	10.00	5.05	4.43	9.56%	11.7%
Log funded ratio (log target ÷ W/Y)	0.200	5.75	5.02	4.40	9.63%	11.7%
Level gap (target − W/Y, income multiples)	0.100	8.16	4.40	3.78	11.22%	11.7%

Notes. The three forms measure the gap in different units, so their coefficients are not comparable. The third column translates each into a common scale: percentage points of income added for a path a quarter short of target at mid-career.

Ranked at their own optima the two scale-free forms are indistinguishable and both beat the level gap. But that comparison is not quite fair, because the three grids do not reach equally far. Interpolating every curve at the strongest response *all* of them can produce separates "this form is better" from "this form's grid let it go further".

Table 89. Functional forms compared at matched response strength

Form	At the same move (pp)	Value there (%)	Best anywhere on its grid (%)	Furthest its grid reaches (pp)
proportional	10.00	5.05	5.05	10.00
log	10.00	4.86	5.02	11.51
level	10.00	4.32	4.40	12.23

Notes. Each form's curve interpolated at the largest contribution move that all three grids can produce. This is the comparison the text relies on; the previous table's ranking partly reflects how far each grid reaches.

At matched strength 0.73 percentage points still separate the forms, so the ranking is about the shape of the response and not only about its strength. A rule linear in income multiples is inert exactly when the investor has time to respond and violent exactly when they do not, and it pays for that.

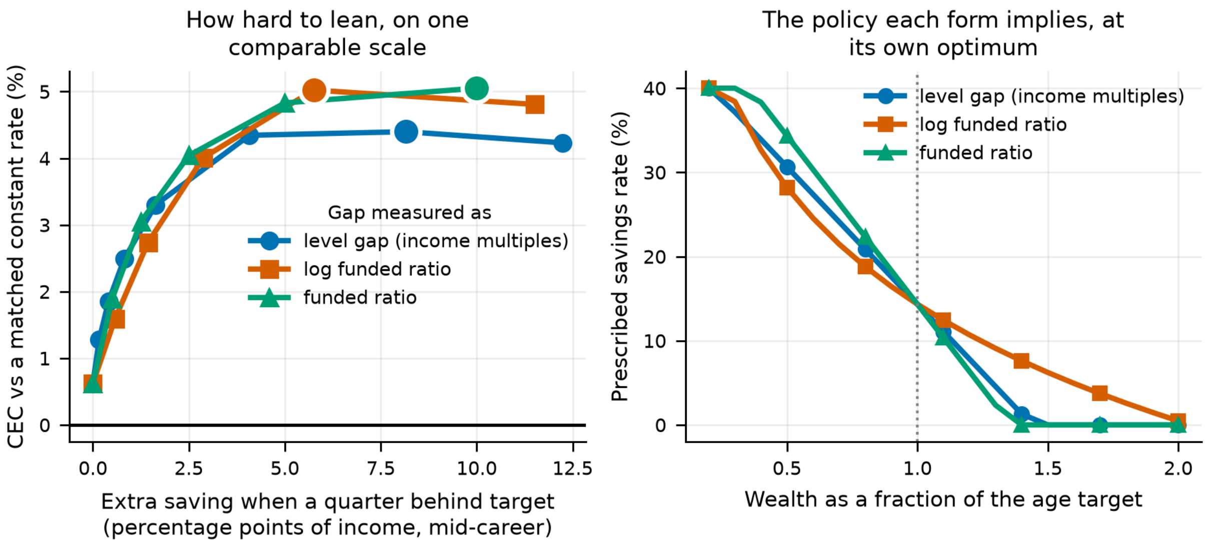

Figure 45.

Left: the value of conditioning against the contribution move it implies, which puts three differently-scaled coefficients on one axis. Right: the policy each form implies at its own optimum, over the range of funded ratios paths actually visit.

27.2 Catching up and easing off are separate policies

A symmetric rule does two things at once: it saves more when behind and less when ahead. There is no reason those should carry the same coefficient, and separating them is the question a real saver faces — almost all published advice is about catching up, and almost none is about easing off.

Table 90. Which half of the conditioning rule earns its keep

Rule	k behind	k ahead	Value (%)	Net of shape (%)	Realised mean rate
Neither (age profile only)	0	0	0.62	0.00	10.0%
Catch up only (never ease off)	0.40	0	3.09	2.47	12.8%
Ease off only (never catch up)	0	0.40	3.25	2.63	7.4%
Both, free coefficients	0.40	0.10	5.26	4.64	10.5%

Notes. Each half is optimised separately over the same grid. Note the realised mean rates: catch-up alone raises the career average well above target and ease-off alone cuts it well below, so only the matched-rate comparison makes them comparable.

Net of the age profile, catching up alone is worth +2.47% and easing off alone +2.63% — a ratio of 1.06, which on any reasonable reading is a tie. That is itself the finding, because the two halves are not symmetric in anything except their value: catch-up alone raises the career savings rate to 12.8% and ease-off alone cuts it to 7.4%. They arrive at the same place from opposite directions. On raw certainty equivalent the one that saves more would simply look better; only the matched comparison reveals the tie.

Run together they reach +4.64%, against 5.10% for the two summed. The halves are strongly sub-additive: much of what each earns alone is the same correction, arrived at from opposite sides. The free search picks different coefficients, so there is no case here for a deliberately asymmetric rule.

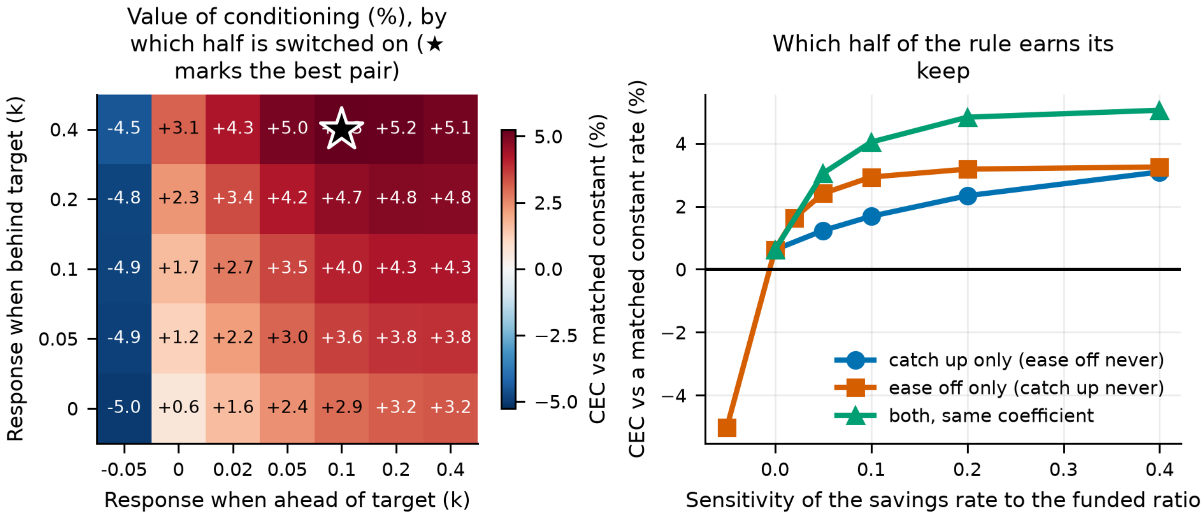

Figure 46.

Left: the value of conditioning over the two coefficients separately, with the best pair starred. Right: each half of the rule switched on alone against both together.

27.3 The version a person could actually follow

A continuous response asks for a freshly computed contribution rate every year. Section 30 finds that coarse guardrail rules give up surprisingly little on the spending side, and the accumulation-side analogue is the same idea run forwards: check once a year and move the contribution by a fixed step only if the funded ratio is more than a dead band away from target.

Table 91. Guardrail savings rules

Dead band (± of target)	Rate step	Value (%)	Net of shape (%)	Mean rate	P(ruin)
10%	5%	3.20	2.58	9.70%	11.7%
20%	5%	3.14	2.52	9.67%	11.7%
35%	5%	2.85	2.23	9.60%	11.7%
10%	3%	2.36	1.73	9.85%	11.7%
20%	3%	2.29	1.67	9.82%	11.7%
35%	3%	2.09	1.47	9.76%	11.7%
10%	2%	1.85	1.23	9.91%	11.7%
20%	2%	1.79	1.17	9.89%	11.7%

Notes. A two-number rule: check the funded ratio once a year, and move the contribution by the step only if the ratio is more than the dead band away from target.

The best guardrail — move 5% of income once you are more than 10% off target — is worth +2.58%, or 58% of what the tuned continuous rule earns. It wants the narrowest dead band and the largest step the grid offers, which is the search saying it would rather be continuous; the table prices that compromise rather than pretending there is none. Even so, keeping most of the value for a rule with two numbers and no arithmetic is the trade most savers should take.

27.4 Does the target have to be right?

The target is the weakest link in the construction: it is the model's own median wealth path, which no investor could know in advance. We test two alternatives that an investor could: an "N times salary by age X" ladder of the kind large fund managers publish, and a flat multiple with no age content at all.

The ladder is worth stating in full rather than gesturing at, since it is the alternative a real saver is most likely to have been handed. Fidelity's widely circulated guideline sets four anchors — one times salary by 30, three by 40, six by 50, eight by 60 and ten by 67. We interpolate linearly between them, and add intermediate anchors at 35, 45 and 55 to keep the interpolation from running straight across a decade; those three are ours, not Fidelity's, and sit on the line the published anchors imply.

Table 92. The "N times salary" ladder, in full

Age	30	35	40	45	50	55	60	67
Wealth, × salary	1×	2×	3×	4×	6×	7×	8×	10×
Published?	yes	no	yes	no	yes	no	yes	yes

Notes. Linear interpolation between anchors; flat at 10× beyond 67 and zero before 30. The published anchors are Fidelity's retirement savings guideline (1× by 30, 3× by 40, 6× by 50, 8× by 60, 10× by 67), which assumes roughly 15% saved a year, retirement at 67, and a target of maintaining pre-retirement lifestyle. The three unpublished anchors are our own interpolation aids.

Table 93. Three wealth-to-income targets, unscaled

Target	Median multiple asked for	Value (%)	Net of shape (%)	Realised mean rate
model median path	11.4	5.05	4.43	9.6%
published ladder	7.7	1.76	1.14	9.6%
flat 8x income	8.0	-7.47	-8.09	15.6%

Notes. The ladder is the published rule of thumb tabulated above; the flat multiple holds the same multiple at every age and serves as a deliberate straw man.

The published ladder captures 26% of the model-implied target's value — useful, but not a substitute. The flat multiple is worth -8.09%: **worse than not conditioning at all**. Telling a 28-year-old they should already hold eight times salary leaves them behind target for most of a career and drives the career average savings rate far above the pinned target. That is not conditioning, it is just saving more, and the matched comparison charges for it.

The ranking says two separate things. A target that rises with age is necessary — the flat one is not merely weaker, it is harmful. But rising is not sufficient: the ladder rises and still gives up a substantial share of the value, so how it rises matters too.

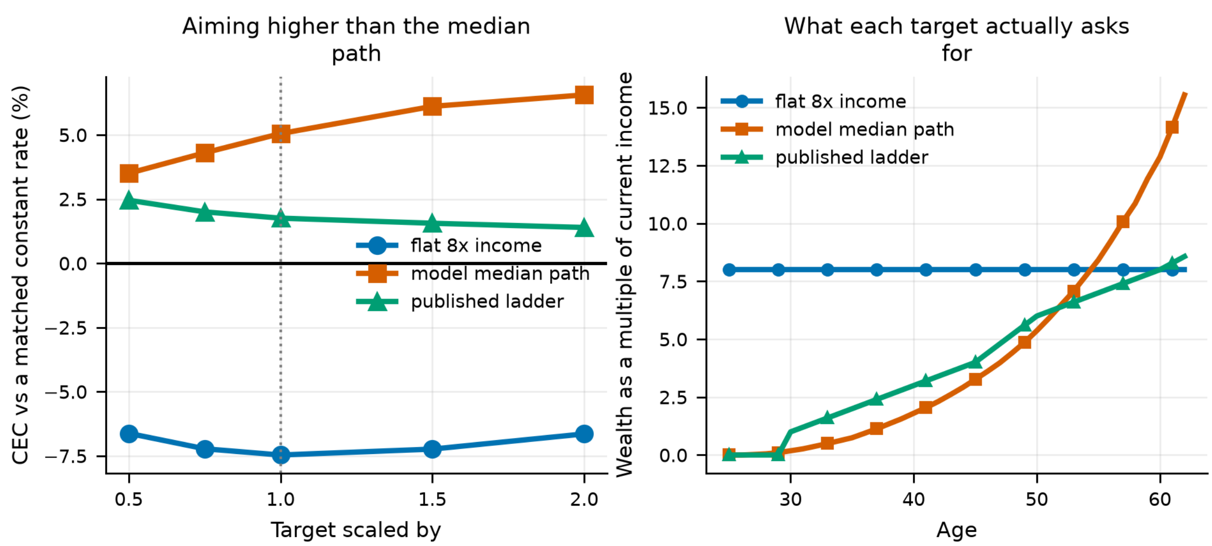

Figure 47.

Left: the value of conditioning as each target is scaled up and down. Right: what each target actually asks the investor to hold at each age.

27.5 Which signal, out of everything available?

Eight candidate state variables, all signed so that positive means "save more", all bounded so that one sensitivity grid is meaningful across them, all swept over the same grid and scored the same way. They fall into three kinds: *stock* signals that say where the investor is, *flow* signals that say what the market just did, and the pay cheque.

Table 94. The signal horse race

Signal	Kind	Best k	Value (%)	Net of shape (%)	Mean rate	P(ruin)	5th pct cons.
Income vs its expected path	income	0.40	7.07	6.45	11.3%	11.9%	0.522
Funded ratio (wealth vs age target)	stock (balance vs target)	0.40	5.05	4.43	9.6%	11.7%	0.772
Raw balance (wealth ÷ income)	stock (balance)	0.40	2.08	1.46	4.2%	11.7%	0.626
Investment gain (balance vs contributions)	stock (market's share of the balance)	0.40	0.97	0.35	1.3%	11.7%	0.489
Trailing 5-year return	flow (market)	0.10	0.65	0.03	9.4%	11.7%	0.722
Trailing 10-year return	flow (market)	0.10	0.65	0.03	9.4%	11.7%	0.720
Last year's return	flow (market)	0.02	0.63	0.01	9.9%	11.7%	0.730
No conditioning (age profile only)	none	0.00	0.62	0.00	10.0%	11.7%	0.732

Notes. Each signal at its own best sensitivity. The "no conditioning" row appears once for every point on the grid and spans zero percentage points across them, which is the determinism check that the common random numbers are holding.

The winner is not a portfolio signal at all. Saving more in years when the pay cheque runs above its expected path — and less when it runs below — is worth +6.45%, against +4.43% for the funded ratio and +1.46% for the raw balance. This is the accumulation-side version of consumption smoothing: an income shock is observed *before* it is spent, so responding to it costs the investor very little utility, whereas responding to a portfolio shortfall means cutting consumption that was already planned. It is also the easiest signal in the list to act on, because it needs no target, no balance and no arithmetic.

Every flow signal is worth essentially nothing. Trailing one-, five- and ten-year returns come in at +0.01%, +0.03% and +0.03% respectively. At that size the ordering within the group is not meaningful and should not be read as one. A return signal knows what the market did but not whether it left *this* investor short — it is the same market for a 30-year-old with nothing saved and a 60-year-old with ten times salary, and they should not respond alike.

Layering the two leaders together reaches +6.92%, against +6.45% for the better one alone and 10.88% if they were additive. They overlap — a run of weak income and a run of weak markets both show up as a balance behind target — but they do not overlap completely, because only one of them is visible before the money is spent.

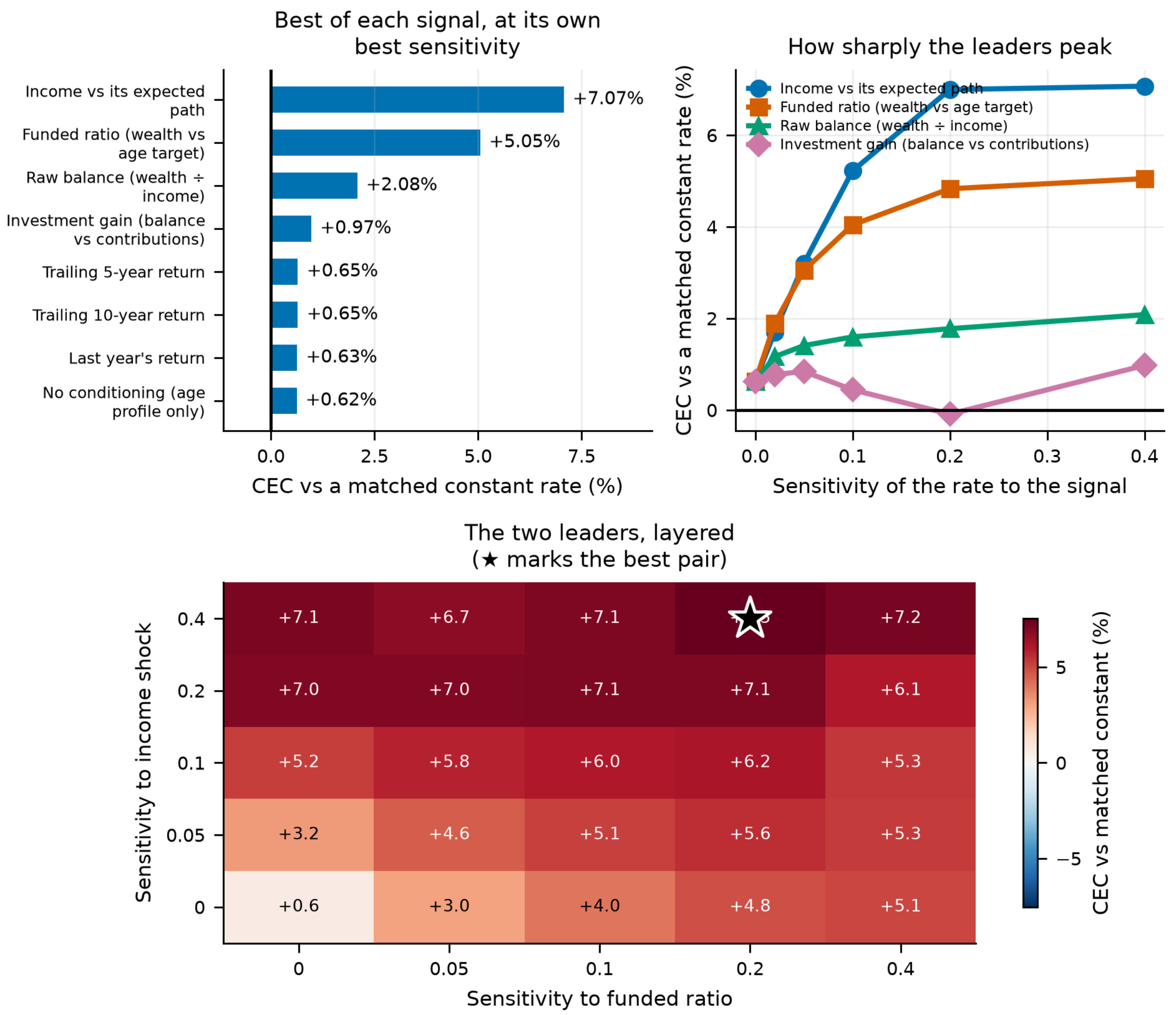

Figure 48.

Top left: the best of each signal at its own optimal sensitivity. Top right: how sharply the leaders peak. Bottom: the two leaders layered, showing partial but incomplete overlap.

27.6 How far does the contribution have to move?

The unconstrained optimum may send the savings rate anywhere in the permitted range. No household budget works like that. Re-pricing the rule with the rate confined to progressively narrower bands around its average is the cheapest test of whether the finding survives contact with a real household.

Table 95. The value of conditioning under a constrained contribution

± allowed move	Floor	Cap	Net value (%)	Share of un constrained (%)	Mean rate	P(ruin)
0%	10%	10%	-0.62	-14	10.0%	11.7%
1%	9%	11%	-0.05	-1	9.7%	11.7%
2%	8%	12%	0.54	12	9.5%	11.7%
3%	7%	13%	1.14	26	9.5%	11.7%
5%	5%	15%	2.24	51	9.4%	11.7%
8%	2%	18%	3.36	76	9.3%	11.7%
10%	0%	20%	4.06	92	9.1%	11.7%
15%	0%	25%	4.31	97	9.4%	11.7%
30%	0%	40%	4.43	100	9.6%	11.7%

Notes. The rate is confined to the career average plus or minus the stated width. At zero width the rule collapses onto the constant rate it is scored against, which is why the first row is exactly zero.

There is no cheap corner here. Up to about ten points the value is close to proportional to the flexibility given — ±3 points of income buys 26% of the unconstrained value, ±5 points 51%, ±10 points 92% — and only then does it saturate. A household that can flex its contribution by a couple of points does *not* get most of the benefit; it gets roughly its share. That is the least convenient version of this result and it is the one the table supports.

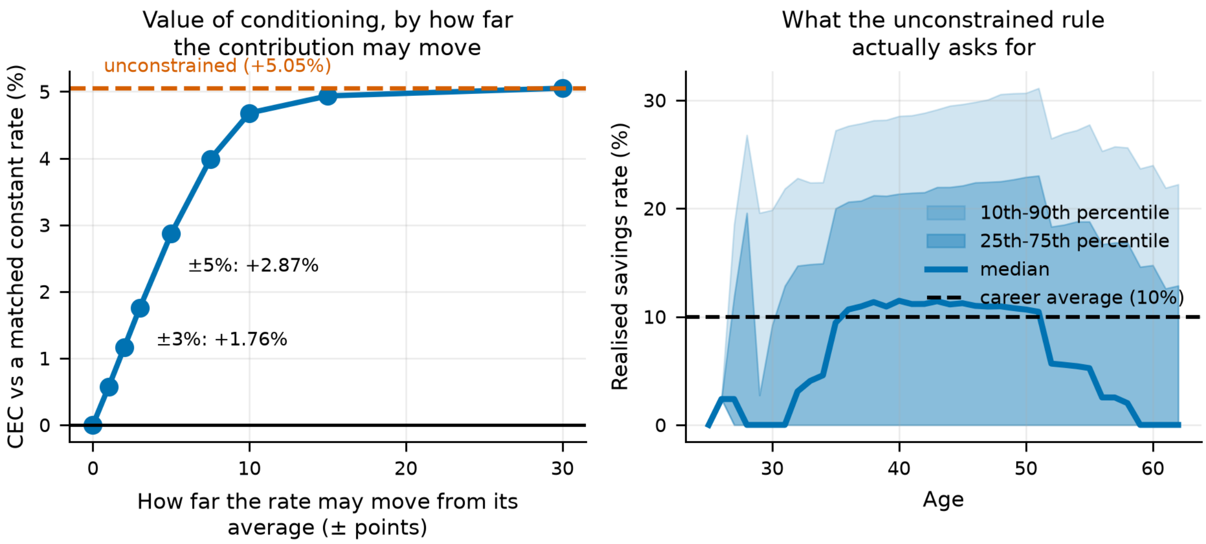

Figure 49.

Left: the value of conditioning as a function of how far the contribution may move. Right: the distribution of prescribed savings rates by age under the unconstrained rule.

27.7 Where the gain lands, and who wants it

A certainty equivalent is one number. It cannot say whether a rule lifted the middle of the distribution or insured the bottom of it, and those are very different products.

Table 96. Retirement consumption by quantile, with and without conditioning

Quantile	No conditioning	Conditioned	Change (%)
p1	0.579	0.604	4.43
p5	0.732	0.772	5.40
p10	0.835	0.892	6.86
p25	1.077	1.153	7.07
p50	1.459	1.582	8.43
p75	2.018	2.224	10.20
p90	2.831	3.085	8.98
p95	3.514	3.821	8.75
p99	5.417	5.703	5.28

Notes. Average real retirement consumption per path, sorted. The rule is funded by contributions taken from consumption on the paths that were going to be comfortable, which is why the top percentile is the one place the change turns negative.

This is not left-tail insurance, and it is not a free lunch either. The gain is positive across almost the whole distribution and is largest in the middle (+8.57% against +5.56% at the bottom decile and +7.67% at the top). The bottom of the distribution is where the rule has least to work with — a path that is behind *because* labour income collapsed cannot save its way out — and the top is where the rule takes its payment. We report this because the intuition that motivated the test was that conditioning would be tail insurance, and it is not.

Across preferences the value rises steeply with risk aversion: +2.13% at γ = 2, +4.43% at γ = 5 and +5.58% at γ = 10. Every preference picks the same coefficient, so what changes is not the rule but how much the investor will pay for it. Conditioning is a risk product, and its price is set by how much the buyer dislikes risk.

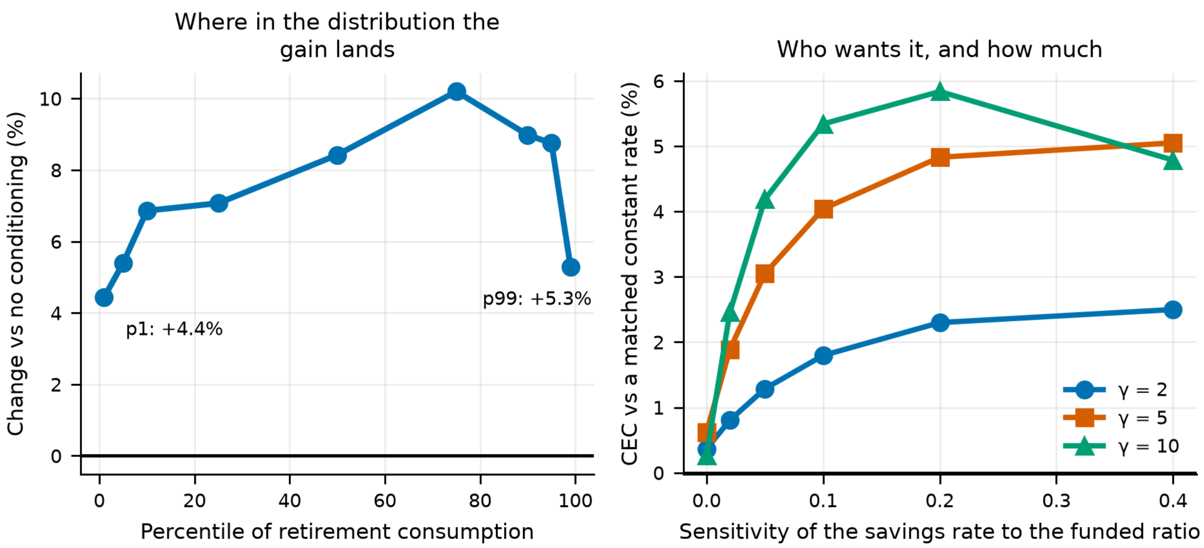

Figure 50.

Left: where in the distribution of retirement consumption the gain lands. Right: the value of conditioning by risk aversion, each at its own coefficient.

27.8 When in a career is the balance worth reading?

Switching the conditioning on only for part of the career prices each stretch of working life separately. Outside the window the rule falls back to the age profile exactly, so these are clean subtractions. The sweep includes overlapping spans because halves and thirds answer different questions; the text below uses only the three non-overlapping windows that tile the career, because comparing a 25-year window against a 13-year one says more about length than about timing.

Table 97. The value of conditioning by career stretch

Conditioning active for ages	Net value (%)	Mean rate	P(ruin)
25-62	4.43	9.6%	11.7%
40-62	3.35	10.2%	11.7%
25-49	2.69	9.1%	11.7%
50-62	2.13	10.9%	11.7%
40-49	1.58	9.7%	11.7%
25-39	1.19	9.6%	11.7%

Notes. Each row switches the rule on only for the stated ages. The three non-overlapping windows tile the career and are the ones the discussion relies on.

The value rises monotonically with age across the three tiles. The balance is close to uninformative when there is barely any of it and decades left to recover; it becomes informative once a shortfall is large in absolute terms and there is little time left to fix it. The three tiles sum to approximately what the whole career is worth, so unlike the two directions of Section 27.2 these stretches really are close to separable.

How hard the rule leans at each age is a different question from whether leaning there is worth anything, and the two do not track each other one for one: per year of conditioning the value varies more across the career than the size of the adjustment does. The early career is not just quieter, it is worth less than its quietness would suggest.

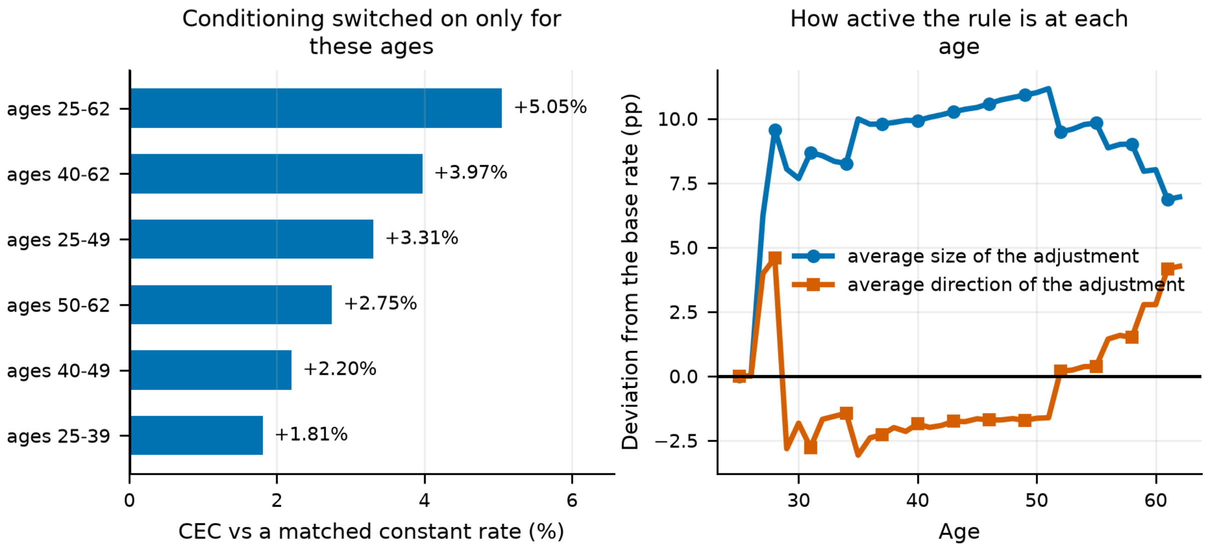

Figure 51.

Left: the value of conditioning when switched on only for the stated ages. Right: how far the rule moves the contribution at each age, and in which direction.

27.9 What the value depends on

Table 98. The value of conditioning by what the money is invested in

Portfolio	Net value (%)	Mean rate	P(ruin)
bills only	5.74	15.8%	55.9%
target date fund	4.66	10.4%	21.9%
sixty forty	4.60	11.8%	24.0%
balanced all equity	4.43	9.6%	11.7%

Notes. Each portfolio is scored against its own constant-rate frontier, so these are not contaminated by differences in the level of the outcome.

Conditioning is worth most to the investor whose plan is going worst. Across the four portfolios the value tracks the ruin probability almost exactly: bills-only, which ruins 56% of the time, gets +5.74%, while all-equity, which ruins 12% of the time, gets +4.43%. Note that this runs *opposite* to portfolio volatility: the safest portfolio benefits most. What the signal is worth is not driven by how much the balance bounces around, but by how often it ends up somewhere the investor cannot live on.

Table 99. The value of conditioning by how risky the pay cheque is

Income shock s.d. (× baseline)	Net value (%)	Mean rate	P(ruin)
0.5×	2.05	9.9%	11.7%
1×	4.43	9.6%	11.7%
1.5×	4.90	9.3%	11.7%

Notes. Labour-income shock standard deviations scaled up and down, with the target and the constant-rate frontier both recomputed for each arm.

Labour-income risk cuts the other way from the portfolio result, and in the direction the dispersion story predicts: more income risk means more dispersion in where a path ends up relative to target, and more for the signal to correct. Taken together, the two interactions say that the funded ratio is worth reading in proportion to how far off target a path can drift and how badly that drift ends — not in proportion to how much the portfolio moves year to year.

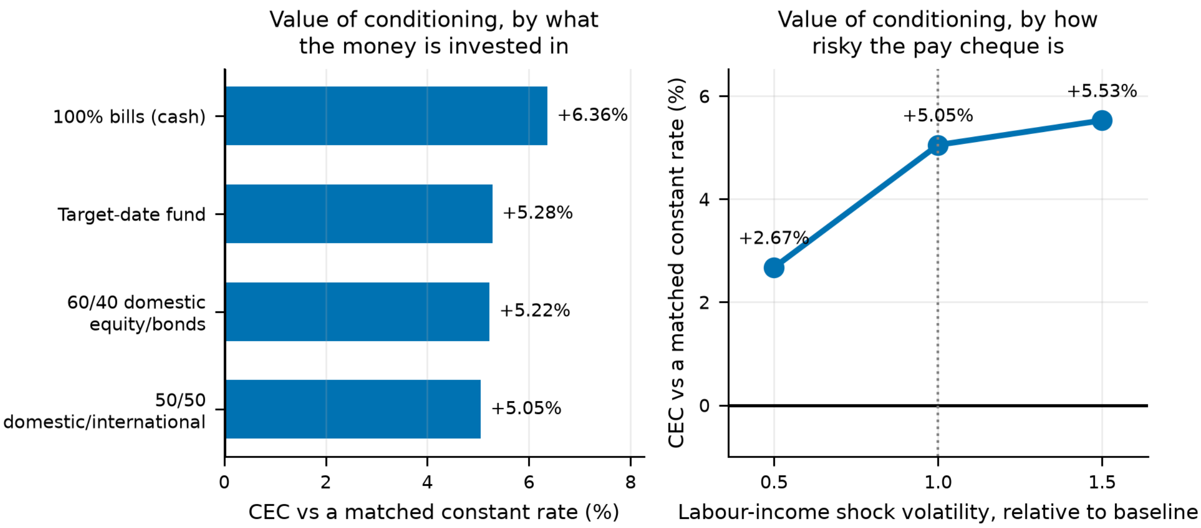

Figure 52.

The value of conditioning by portfolio and by labour-income volatility. The two point in opposite directions with respect to "risk", which is why the mechanism is about outcomes rather than variance.

28. Endogenous Retirement Timing

Every result so far retires the investor on a birthday. Real people do not. They retire when the balance looks big enough — which means, mechanically, that they retire disproportionately after good markets. That is worth testing rather than assuming, because a balance that looks big after a bull run is also a balance bought at high valuations, and the same market move that triggers the decision may be lowering the returns that have to fund it.

This section makes the retirement date a path-dependent decision and asks two questions: whether a wealth trigger beats a fixed date, and how much of the outcome the decade around the retirement date explains.

28.1 A wealth trigger against a date

Utility here is evaluated over the whole lifetime rather than the retirement window, because a rule that retires people early buys them leisure that a retirement-only window would charge for without crediting. We also introduce a working-life income floor of 25% of average earnings. That is not decoration either: retirement in this model carries a real consumption floor through the progressive social-security schedule and working life, at zero, carries none. Wherever compared policies share the same working-life consumption the asymmetry cancels, but here it does not — and left uncorrected it rewards retiring early purely for reaching the safety net sooner.

Table 100. Fixed retirement dates against wealth triggers

Rule	CEC γ=5	Median age	Mean age	S.d. age	P(ruin)	Median wealth at retirement	Median cons.
Fixed age 60	0.804	60.000	60.000	0.000	14.0%	17.3	1.325
Fixed age 63 (baseline)	0.790	63.000	63.000	0.000	11.7%	21.3	1.485
Fixed age 66	0.772	66.000	66.000	0.000	9.2%	25.9	1.674
Fixed age 70	0.745	70.000	70.000	0.000	5.8%	33.6	1.975
Flexible +/-3 years, 25x income	0.799	65.000	63.755	2.505	11.5%	23.3	1.552
Wealth trigger 20x income	0.815	62.000	62.743	5.368	12.4%	22.0	1.493
Wealth trigger 25x income	0.807	65.000	64.223	5.213	11.1%	24.3	1.582
Wealth trigger 30x income	0.799	67.000	65.336	4.929	10.3%	26.2	1.659

Notes. A wealth trigger retires the investor once financial wealth reaches the stated multiple of current income, within an age window. Whole-lifetime utility, working-income floor at 25% of average earnings.

Read naively this table says wealth triggers beat fixed dates. That reading is wrong, and the reason is instructive enough to spend a paragraph on. The model contains no disutility of labour: working an extra year is costless and produces income, so the certainty equivalent falls monotonically with the fixed retirement age for purely mechanical reasons. A wealth trigger that retires people later on average will therefore look better than age 63 without any conditioning value at all.

The correct comparison holds the mean retirement age fixed. We interpolate the fixed-date frontier at each rule's own realised mean retirement age and score against that.

Table 101. The value of conditioning, against a date matched on the same mean age

Rule	Mean retirement age	CEC	Matched fixed-date CEC	Value of conditioning (%)
Flexible +/-3 years, 25x income	63.755	0.799	0.786	1.65
Wealth trigger 20x income	62.743	0.815	0.791	3.03
Wealth trigger 25x income	64.223	0.807	0.783	3.11
Wealth trigger 30x income	65.336	0.799	0.776	2.98

Notes. This is the number that isolates conditioning from timing. The unmatched comparison in the previous table conflates the two.

Matched, the value of conditioning is 3.11% at the best trigger — real, but roughly 62% smaller than the same comparison without the working-income floor, where it reaches 8.14%. Both numbers are reported because the difference between them is a model artefact we found and removed, not a result.

28.2 The retirement-date lottery

How much of a retirement outcome is decided by when you happened to be born? We regress the outcome on the annualised real portfolio return over the decade centred on each path's own retirement date. That window explains 5.9% of the variation in retirement consumption, against 16.8% for the whole 68-year lifetime. A single decade — 15% of the horizon — carries 35% of the explanatory power of the whole thing.

That is the sequence-of-returns problem stated as a number, and it is larger than the difference between any two allocation strategies in this paper. It is also the strongest argument for the flexible retirement date of the previous subsection: the one lever that acts directly on the lottery is the ability to choose which decade you retire into.

Table 102. Retirement outcomes by decile of the retirement-decade return

Decile	Mean Window Return	Median Retirement Consumption	Prob Ruin	Median Retire Age	Median Wealth At Retirement
1.000	-0.028	1.229	0.345	70.000	17.309
2.000	0.017	1.446	0.191	67.000	21.275
3.000	0.039	1.477	0.137	66.000	21.840
4.000	0.056	1.546	0.109	65.000	23.221
5.000	0.070	1.629	0.085	65.000	25.040
6.000	0.084	1.682	0.083	64.000	26.286
7.000	0.098	1.680	0.064	64.000	26.239
8.000	0.114	1.708	0.049	64.000	26.810
9.000	0.132	1.709	0.030	64.000	26.865
10.000	0.172	1.814	0.019	63.000	29.336

Notes. Paths sorted by the annualised real return over the ten years centred on their own retirement date, then split into deciles.

Do people retire into bull markets, and does it hurt?

Under a wealth trigger the answer to the first question is yes: the correlation between retirement age and the preceding market run-up is -0.310, so a strong market pulls the retirement date forward. Early retirees experience a mean pre-retirement run-up of 12.1% against 7.1% for late ones.

The second question is the interesting one, and the answer is reassuring for the trigger. The correlation between the run-up and the *subsequent* return is -0.068 — essentially zero. On this panel, retiring after a good decade does not systematically buy you a bad one. The folk warning that bull markets lure people into retiring at the worst possible time is not supported here: the mean subsequent return for early retirees (6.0%) is if anything slightly higher than for late ones (6.0%). We report this because it cuts against the intuition that motivated the test.

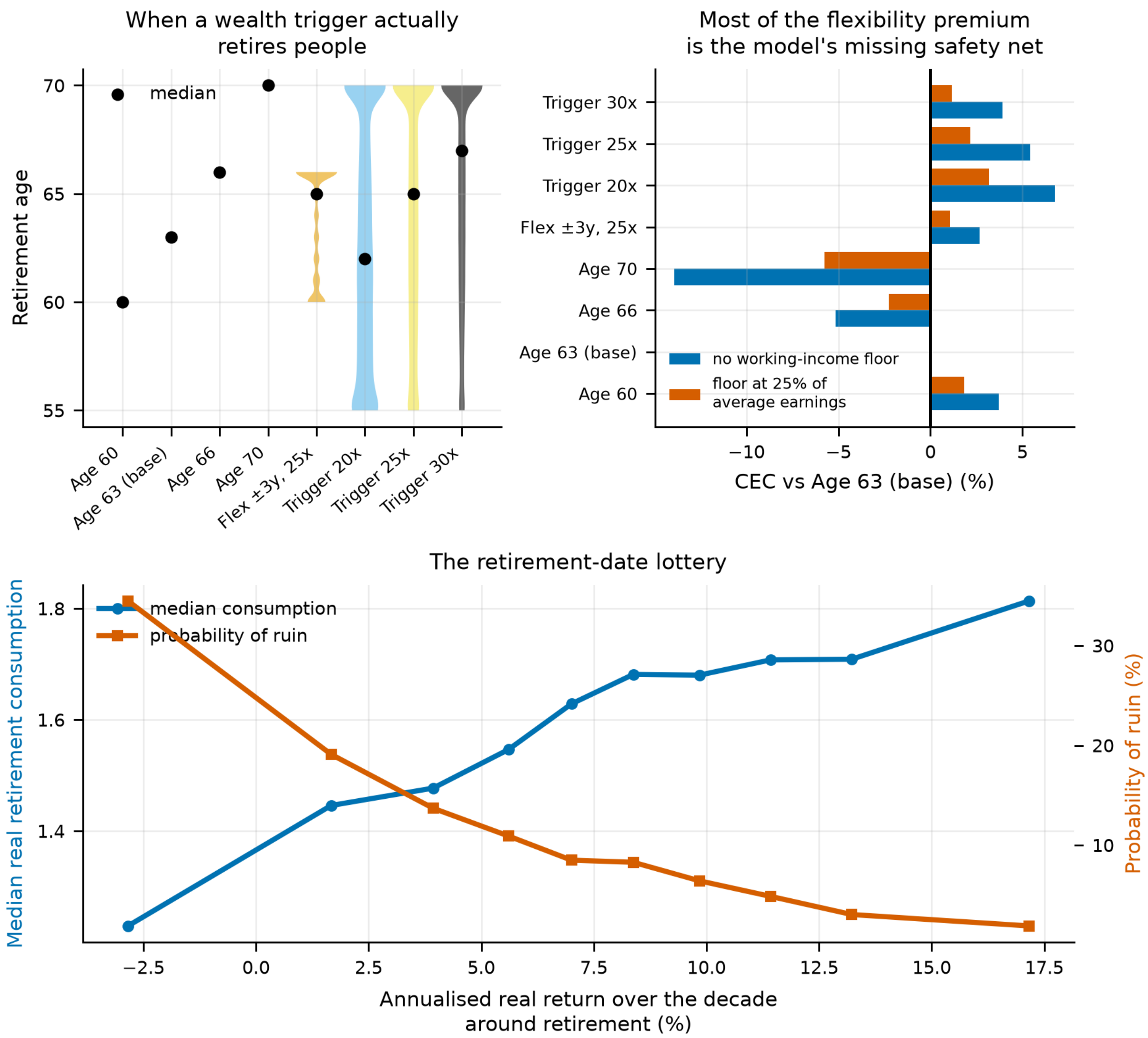

Figure 53.

Endogenous retirement timing. The distribution of retirement ages under a wealth trigger, the value of conditioning against a matched fixed date, and the retirement-date lottery.

29. Sequence-of-Returns Risk, Isolated

Section 28 establishes that the single decade around a person's retirement date explains more of their outcome than the entire allocation question, and Section 12 shows that reading a state variable at the retirement date rather than the birth date turns a null into an eight-point effect. Both are symptoms of the same thing. Neither measures it.

The experiment that does is simple enough to state in a sentence. Take a simulated lifetime, keep its 68 annual returns exactly as drawn, and shuffle the order. Same multiset, same mean, same everything a return distribution can describe. Anything that changes is sequence.

For a lump sum nothing would change at all: the product of gross returns is commutative, so a buy-and-hold investor with no cash flows ends at the same wealth whatever the order. That is verified here to machine precision, and it is the point — sequence risk exists *only* through cash flows. A contribution made before a crash buys more; a withdrawal made after one sells more. Every series of a year moves together under the permutation, so the cross-asset covariance the bootstrap exists to preserve survives it.

29.1 The decomposition

Each original path is a bag of returns. Reordering it makes the outcome a random variable, so the total variance splits exactly into the average variance *within* a bag — which is sequence, since the bag is held fixed — and the variance of the bag means, which is the risk of having drawn that bag at all. Estimating the first term needs several orderings of the same path, which is what the 8 replications are for.

Table 103. Variance of mean retirement consumption for *50/50 domestic/international*, split into ordering and returns, γ = 5.

What was shuffled	Share of variance that is ordering	SD from ordering	SD from the returns	CEC	P(ruin)
nothing	0.0%	0.000	1.012	1.0485	11.7%
the working years	17.9%	0.456	0.978	1.0464	11.7%
the retired years	1.9%	0.142	1.009	1.0426	11.8%
the whole lifetime	49.3%	0.837	0.848	1.0287	12.6%

Notes. The first row is the control. Shuffling nothing must give a sequence share of zero, and the pipeline fails loudly if it does not — without that check the decomposition could be measuring anything.

Ordering is a minority of the risk, and not a small one. Reshuffling a fixed bag of returns accounts for 49.3% of the variance in retirement consumption; the rest is the risk of having drawn that bag at all.

And it is an accumulation phenomenon, which is the opposite of where the literature points. Shuffling only the working years accounts for 17.9% of the variance; shuffling only the retired years accounts for 1.9%. The next subsection explains why, and the explanation is not about returns.

The whole-lifetime figure exceeds the two phases added together, and that is not an inconsistency. Shuffling the whole lifetime lets a year lived at eighty land at twenty-six, so it captures the interaction between the phases as well as the ordering inside each — which the phase-restricted shuffles hold fixed by construction.

29.2 The withdrawal rule decides where it lands

The result above is a property of the withdrawal rule as much as of the returns, and reporting it under one rule alone would be reporting the rule. A withdrawal fixed in real terms is computed from wealth *at* retirement and never revisited, so it refuses to let retirement-phase returns reach the retiree's consumption at all — until the money runs out, at which point they reach it all at once. A percentage-of-portfolio rule passes every return straight through to what the retiree eats. The two should therefore put sequence risk in completely different places, and the decomposition is repeated under three rules to find out.

Table 104. Where sequence risk sits, rule by rule.

Withdrawal rule	Ordering risk in the working years	in the retired years	ratio	Cost in CEC (%)	Cost in ruin (points)
Fixed real withdrawal	17.9%	1.9%	0.11	-1.89	+0.89
Dynamic, with guardrails	17.8%	10.0%	0.56	-2.09	+1.28
Fixed percentage of the portfolio	16.2%	14.1%	0.87	-1.41	+0.00

Notes. The last two columns are what a random ordering costs relative to the drawn one under each rule — in consumption, and in the probability of running out.

The rule relocates the risk rather than removing it. Under a fixed real withdrawal the retired years carry 0.11 times the ordering risk of the working years; under a percentage-of-portfolio rule they carry 0.87 times — 8× more. Nothing about the returns has changed between those rows. What changed is whether the retiree is allowed to feel them.

That is not a free lunch, and the last column says so. The fixed real rule buys its smooth consumption by absorbing the ordering into *ruin* instead: a random order costs it +0.89 points of failure probability against +0.00 for the percentage rule, which simply cuts spending instead of running out. The choice between them is a choice about which form the sequence risk takes, not about how much of it there is.

29.3 Does it change which portfolio wins?

A decomposition says how much of the dispersion is ordering. It does not say whether ordering changes which portfolio an investor should hold, and those are different questions with potentially different answers.

Table 105. *50/50 domestic/international* against *Target-date fund* under each shuffle.

What was shuffled	Lead of the pair (%)	Best of the whole menu	P(ruin), challenger	P(ruin), incumbent
nothing	+11.00	100% international equity	11.7%	21.9%
the working years	+10.79	100% international equity	11.7%	21.9%
the retired years	+9.24	100% international equity	11.8%	20.4%
the whole lifetime	+7.25	100% international equity	12.6%	19.8%

Notes. Two comparisons sit side by side. The lead column is the headline pair only — *50/50 domestic/international* over *Target-date fund*. The next column is the best of all 6 strategies, which need not be either of them.

The ranking is unmoved by the ordering. *100% international equity* wins under every shuffle. Sequence risk is large, it is real, and it does not discriminate between these portfolios — which is why it belongs in this part of the paper rather than among the allocation sections.

The lead does narrow, and by enough to be worth stating: +11.00% on the drawn order against +7.25% on a random one, so shuffling absorbs 34% of the gap between the two portfolios. Ordering does not reverse the comparison, but it is not neutral to it either: a saver unlucky in the order banks less of the advantage the allocation offers.

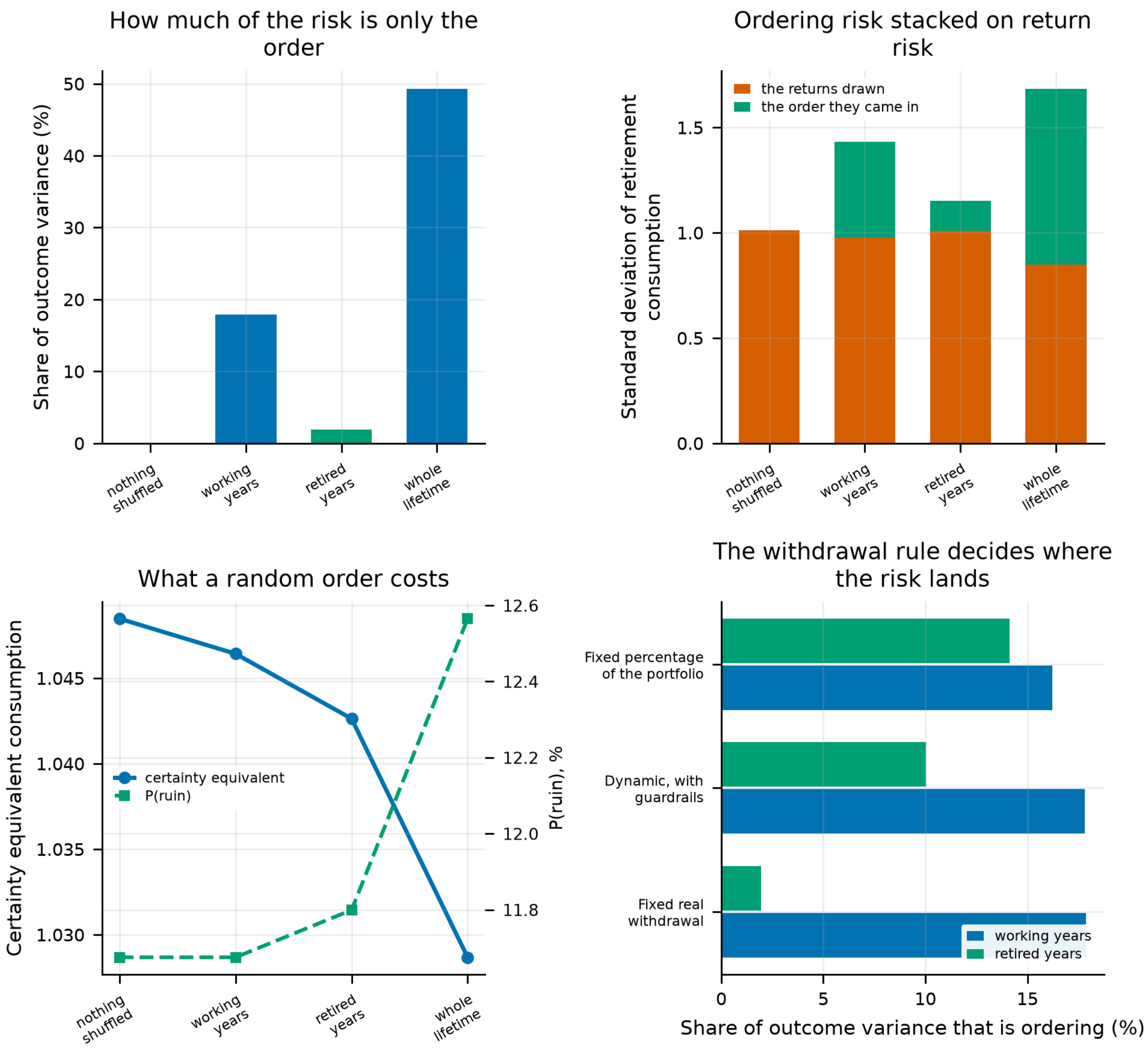

Figure 54.

Top left: the share of outcome variance that is nothing but the order. Top right: the same in consumption units, ordering risk stacked on return risk. Bottom left: what a random order costs in certainty-equivalent consumption and in failure probability. Bottom right: where the ordering risk lands under each withdrawal rule.

29.4 What this changes

- **Sequence risk is measurable, and it is large.** Ordering accounts for 49.3% of the variance in retirement consumption on this panel — a number the paper has been gesturing at since Section 28 without putting a figure to it.
- **Where it lands is a property of the withdrawal rule, not of the market.** A fixed real withdrawal pushes it almost entirely into the accumulation phase and pays for that in ruin; a percentage rule lets it into consumption instead. That reframes Section 30: the rules there are not competing on how much risk they take but on what shape it takes.
- **It does not touch the allocation answer.** The headline pair keeps its order under every shuffle, which is worth knowing precisely because the effect is so large elsewhere.
- **What is not modelled**: any response to the sequence. The investor here does not spend less after a bad year, delay retirement, or hold a cash buffer, and each of those would reduce the figures above. This measures the exposure, not what a thoughtful retiree would do about it — so it is an upper bound on the

damage and, read the other way, a lower bound on the value of the flexible rules in Section 30.

30. Retirement Spending Rules

The baseline draws down the portfolio with a constant real withdrawal — the four-percent rule and its variants. That is one policy among many, and it is a poor one: it ignores the portfolio entirely after the first year. This section compares 8 spending policies drawn from 8 families, each optimised over its own rate so that no family is handicapped by a badly chosen parameter.

- **Constant real.** Fixed initial percentage, inflation-adjusted thereafter. The status quo.
- **Percentage of portfolio.** A fixed fraction of the current balance each year — never runs out, but consumption is as volatile as the portfolio.
- **Guardrails.** Guyton–Klinger style: hold the real withdrawal constant until the implied rate drifts outside a band, then cut or raise it by a fixed step.
- **Floor and ceiling.** A percentage-of-portfolio rule bounded above and below relative to last year's withdrawal.
- **Endowment smoothing.** A weighted average of last year's spending and a target percentage of the current balance.
- **Required minimum distribution.** Balance divided by a regulatory life-expectancy divisor.
- **Amortisation.** The annuity payment that would exhaust the balance over the remaining horizon at an assumed return.
- **Actuarial.** As amortisation but using Gompertz life expectancy recomputed each year rather than a fixed horizon.

Table 106. Best spending rule of each family, at γ = 5

Best of family	CEC	CEC (no bequest)	P(ruin)	Median cons.	5th pct cons.	Cons. volatility	Median worst cut	Median bequest
Amortisation (annuity payment) (4% assumed return)	1.257	1.274	0.0%	2.196	0.865	0.253	40%	0.0
Actuarial (Gompertz life expectancy)	1.256	1.246	0.0%	2.132	0.846	0.265	45%	4.4
Constant % of portfolio	1.205	1.196	0.0%	1.903	0.788	0.223	48%	14.2
Endowment smoothing (light smoothing)	1.205	1.195	0.0%	1.902	0.786	0.219	45%	13.8
Guyton-Klinger guardrails (tight band)	1.200	1.192	4.1%	1.945	0.792	0.200	34%	11.3
Life expectancy / RMD	1.192	1.203	0.0%	2.242	0.849	0.427	30%	0.0
Vanguard dynamic (ceiling/floor) (asymmetric)	1.133	1.126	8.2%	1.746	0.755	0.137	18%	25.7
Constant real (4%-rule style)	1.049	1.042	17.3%	1.567	0.691	0.070	0%	35.5

Notes. Each family is evaluated at its own optimal rate. "Median worst cut" is the largest single-year real spending reduction the median path experiences — the discomfort the rule imposes in exchange for its ruin protection.

Amortisation (annuity payment) (4% assumed return) ranks first at γ = 5, with a certainty equivalent of 1.257 and a ruin probability of 0.0%. The adaptive families as a group dominate the constant-real rule, and the mechanism is not subtle: a rule that never looks at the portfolio cannot avoid exhausting it, and the certainty equivalent is extremely sensitive to the paths where it does.

The cost of that protection is spending volatility, and the table reports it explicitly rather than burying it. The best-performing rules ask the median retiree to absorb a real spending cut of tens of percent at some point in retirement. Whether that trade is acceptable is a question about the retiree, not about the model; the certainty equivalent prices it under one particular utility function and a reader with a different one should read the volatility and worst-cut columns instead.

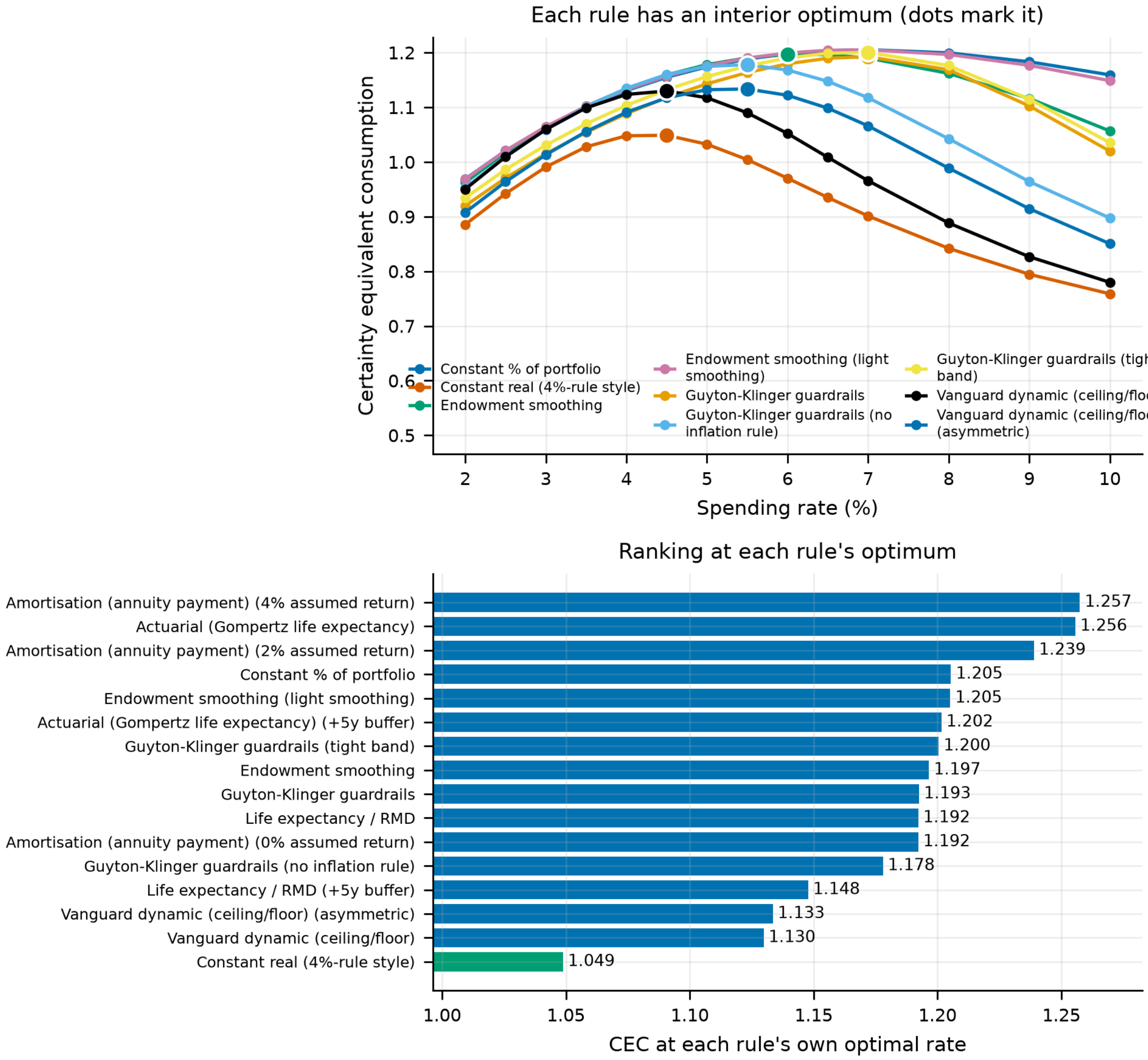

Figure 55.

Certainty equivalent as a function of the withdrawal rate for each spending family. Each family is compared at the peak of its own curve rather than at a common rate.

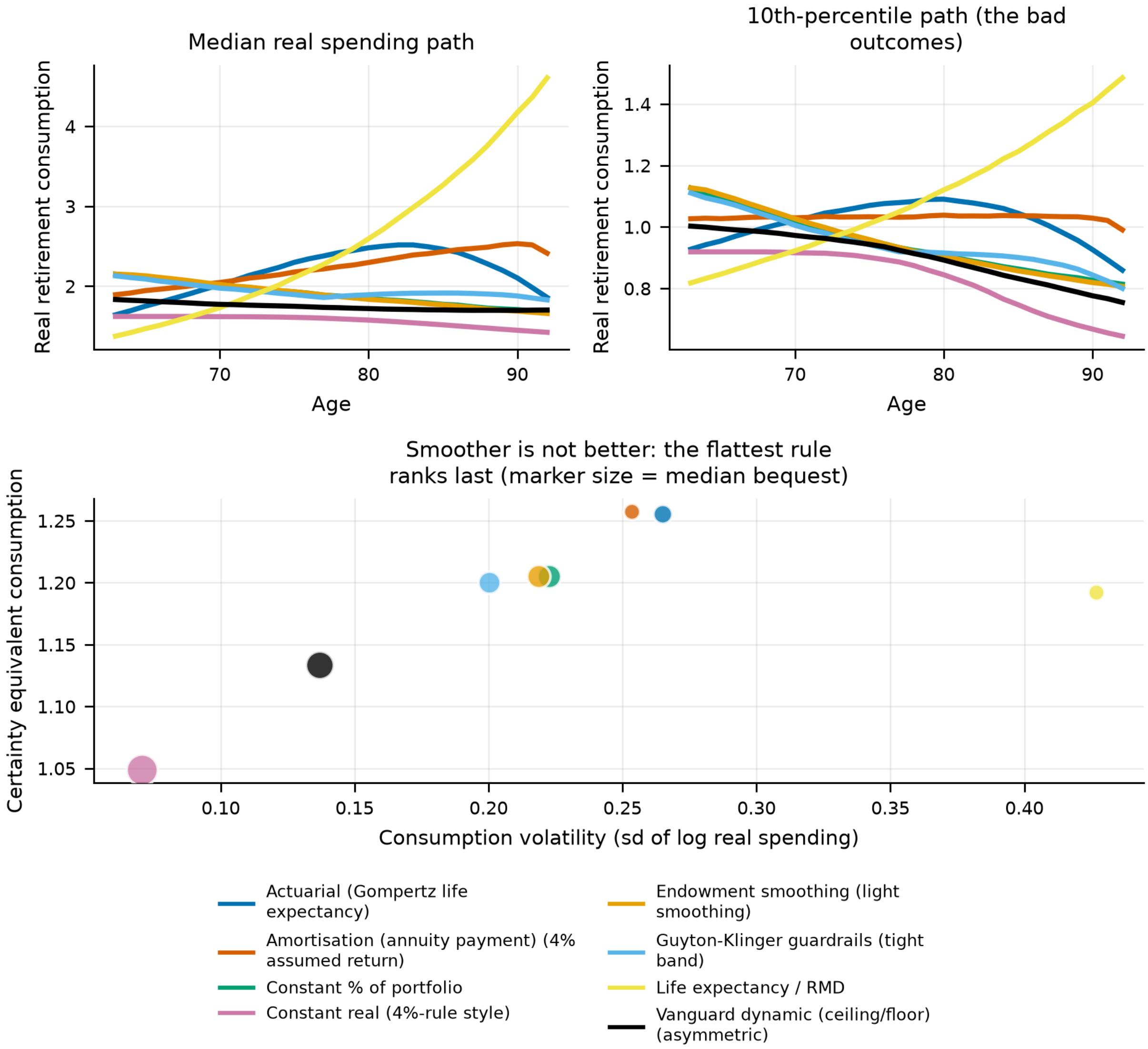

Figure 56.

Realised spending paths under each rule for a common set of market histories. The adaptive rules trade a smooth path for a solvent one.

30.1 The bequest pivot

The ranking is not invariant to the bequest motive, and this is the most interesting finding in the section. With the bequest weight switched off, the best rule is **Amortisation (annuity payment) (4% assumed return)**; with it switched on at $\theta = 2$, the ranking shifts toward rules that leave something behind. Rules that deliberately exhaust the portfolio — amortisation, actuarial — maximise consumption precisely by planning to leave nothing, which is optimal if and only if nothing is what the investor wants to leave.

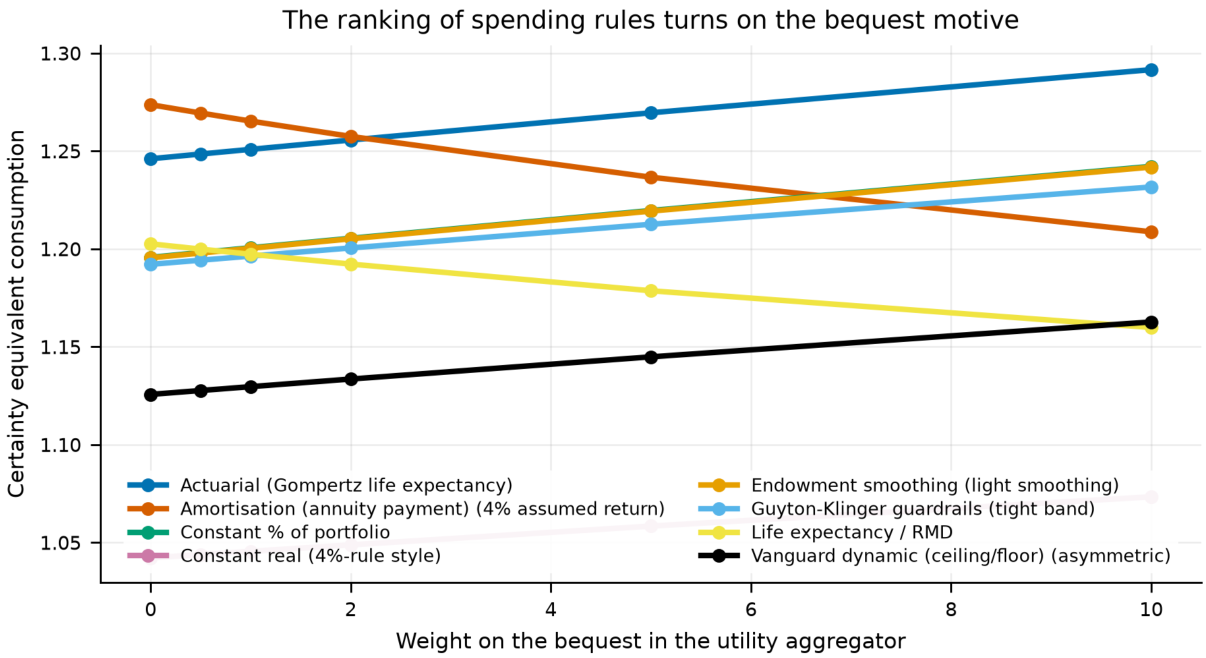

Figure 57.

Certainty equivalent by spending rule across the bequest weight. The lines cross: a rule that is optimal for an investor with no bequest motive is not optimal for one who has a strong one.

Across the 6 bequest weights tested the identity of the best rule changes, which means the question "what is the best withdrawal policy?" is not well posed without stating a bequest motive. That is a modelling result rather than a market result, but it is one that a great deal of withdrawal-rate advice ignores.

31. The Plan and the Portfolio, Solved Together

Section 19 solves the allocation schedule with the withdrawal rule pinned to a 4% rule. Section 30 ranks withdrawal rules with the allocation pinned to a 50/50 portfolio. Each then checks that its answer survives the thing it held fixed, and concludes that it does — the spending rule and the asset allocation are close to separable, and can be chosen independently.

That claim deserves a harder test than either section gave it, because both tested it on a *ranking of six candidate portfolios*. A ranking of six candidates cannot resolve a grid step, and the location of an optimum is not rank information. This section tests it on the optimum, and then removes the assumption altogether by solving both decisions at once.

Each rule is run at its own best rate: a percentage-of-portfolio rule at 4% is not the same policy as a fixed real rule at 4%, and comparing them there compares spending levels rather than rule shapes. Every rated rule turns out to have an *interior* optimum on a grid running to 15%, which is the check that the objective disciplines spending rather than rewarding whatever spends fastest — if the certainty equivalent simply rose with the withdrawal rate, nothing in this section would mean anything.

31.1 The ranking separates; the optimum does not

Table 107. The domestic share held from the retirement date, solved under each withdrawal rule, γ = 5.

Withdrawal rule	Optimal domestic share	CEC at the optimum	Margin over runner-up (%)
constant real	20%	1.0678	0.11
vanguard dynamic	20%	1.1628	0.01
guyton klinger	10%	1.2251	0.16
endowment	10%	1.2332	0.20
constant percent	10%	1.2482	0.11
life expectancy	10%	1.2177	0.11

Notes. Certainty equivalents are not comparable across rows — the rules spend different amounts — but the argmaxes are.

The optimum moves with the withdrawal rule. It runs from 10% to 20% domestic across the 6 rules, a spread of 10 percentage points on a decision Section 30 treats as independent of the rule. Nothing in that section is wrong: it was reporting rankings, and the rankings hold. It is the resolution that was the problem.

Table 108. Why the optimum moves: the two objectives the allocation can serve, and which rules let it choose between them.

Withdrawal rule	Ruin observed	Maximises CEC at	Minimises ruin at	Agree	Lowest ruin reachable
constant real	yes	20%	20%	yes	8.23%
vanguard dynamic	yes	20%	20%	yes	10.20%
guyton klinger	yes	10%	30%	no	3.42%
endowment	yes	10%	40%	no	0.18%
constant percent	no	10%	—	no	0.00%
life expectancy	no	10%	—	no	0.00%

Notes. Whether ruin is reachable is read off the simulated paths rather than off the rule's description. Where it is not, the ruin column is flat at zero and its argmin would be an artefact of the grid order, so it is left undefined.

The mechanism is the one Section 29 identified. Under a rule that fixes real spending from wealth on the retirement date, retirement-phase returns cannot reach consumption until the money runs out — so the allocation's only remaining job is keeping the portfolio alive, and it is chosen to minimise ruin. Ruin is reachable under 4 of the 6 rules; 2 of those put the certainty-equivalent maximum and the ruin minimum at the same share. The 2 rules that cannot run out have no ruin to minimise, and every one of them sits at 10% — the return-maximising corner. Same market, same investor, different objective.

31.2 Solving both at once

The search alternates: pick the best plan for the current schedule, re-solve the free-form schedule for that plan, repeat. It stops when a round returns the plan it started with — the fixed point, where neither decision would change given the other.

Table 109. What each degree of freedom is worth, against a 4%-rule investor holding 50/50, γ = 5.

What was free to move	Plan chosen	CEC	Gain (%)
neither	constant_real at 4.0%, retire at 63	1.0407	+0.00
allocation only	constant_real at 4.0%, retire at 63	1.1129	+6.94
plan only	constant_percent at 7.0%, retire at 63	1.1929	+14.62
both	constant_percent at 7.0%, retire at 63	1.3100	+25.88

Notes. The interaction is the joint gain less the two one-sided gains. It is zero exactly when the two decisions can be chosen independently.

Solving them together is worth more than solving them apart. The interaction is +4.32% of certainty-equivalent consumption — the joint search beats the sum of the two one-sided searches, because the allocation that suits a rule which cannot run out is not the allocation that suits one which can. An adviser who picks the portfolio first and the withdrawal policy afterwards leaves that on the table.

The alternation reached its fixed point in 2 rounds, which is itself informative: the plan chosen for a flat starting schedule survived being re-chosen for the solved one.

31.3 The one decision this model cannot price

Retirement age is the third leg of a retirement plan and it is left out of the search above deliberately. Scored on its own, the certainty equivalent rises monotonically with the retirement date and the optimiser takes the oldest age on the grid.

Table 110. The retirement date, scored against the solved schedule.

Retirement age	CEC at the joint optimum's rule
58	1.1549
60	1.2153
63	1.3100
66	1.4109
68	1.4855
70	1.5666

That is not a finding about retirement; it is the model's costless labour showing through. Nothing here charges the investor for the years they spend working, so another one is free and the search takes every one it is offered — which is why Section 28 requires a matched comparison and why the date is held fixed above. Reporting the corner is the point: a joint optimisation is the cleanest way to find out which decisions a model can rank and which it merely appears to. Section 32 supplies the missing side of that ledger and the corner does not survive it.

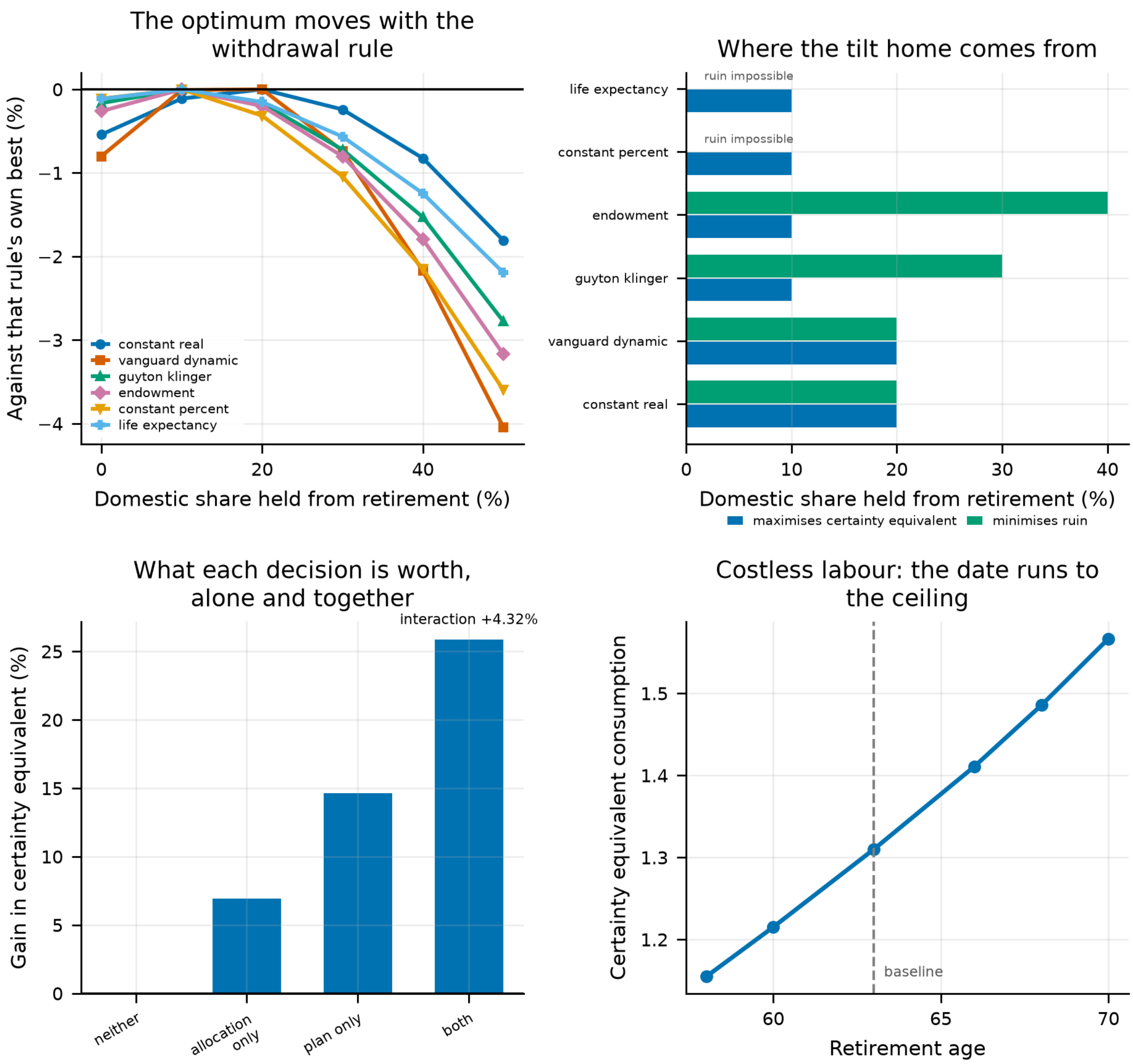

Figure 58.

Top left: the retirement-phase domestic curve under each withdrawal rule, each drawn against its own best so the shapes can be compared — the argmax moves. The window stops at 50% because every optimum lies below 30% and the curves decline monotonically beyond it; each is still normalised against its best over the whole grid. Top right: the share that maximises the certainty equivalent beside the share that minimises ruin, which is the mechanism. Bottom left: what each degree of freedom is worth alone and together. Bottom right: the retirement date, running to the ceiling because labour is free.

31.4 What this changes

- **Separability is a claim about rankings, not about optima.** Section 30's conclusion survives as stated and fails as usually read. Both halves matter: an adviser choosing between six model portfolios can pick the rule first, and one solving for an allocation cannot.
- **The withdrawal rule decides what the portfolio is for.** A rule that can run out makes the allocation a ruin problem; a rule that cannot makes it a return problem. That is the same mechanism Section 29 finds in the variance decomposition, arriving here as a difference in the argmax.
- **The two decisions are worth more together than apart**, by +4.32% of certainty-equivalent consumption.

- **What is not modelled**: the plan is chosen once, at the start, and never revised. A real retiree re-reads their balance every year and may change rule as well as rate, so the gains here are a lower bound on what a genuinely adaptive plan would earn. And there is no disutility of labour, which is why 31.3 exists.

32. What a Year of Retirement Is Worth

Section 31 solves the withdrawal rule and the allocation together, and finds that the third leg of a retirement plan — the date — cannot be solved at all. Left free it runs to the oldest age on the grid and stays there, because nothing in this model charges anyone for the years they spend working. Another year is more contributions, a shorter drawdown and a larger benefit, at no cost whatever. This section supplies the missing side of that ledger.

Rather than invent a disutility of labour in utils — a quantity nothing here can calibrate — the cost is written as a *consumption equivalent*. Let L be the number such that a retired year at consumption c is worth exactly as much as a working year at $L \cdot c$. Working years are then evaluated at c/L and retired years left alone, so $L = 1$ charges nothing and reproduces every other result in this paper exactly.

The aggregation here runs from age 25, not from the retirement date. A retirement-window objective cannot price a retirement date: it charges the investor for the years they worked and credits them nothing for the ones they did not. These certainty equivalents are therefore not comparable with those elsewhere in the paper.

32.1 The pension you claim early

There is a larger thing to fix first. Every other section holds the retirement date fixed, so the benefit starts on the same birthday for every strategy and its start date cancels. It does not cancel here: left alone, this model pays whoever stops work at fifty a full unreduced pension for forty-three years. That is not a preference for leisure, it is a gift, and it is worth more than any plausible value of leisure.

Real systems reduce a benefit claimed early and raise one deferred, by roughly enough to leave its expected present value alone. The multiplier that does that exactly is the ratio of annuity factors, A(reference)/A(age), derived from this model's own Gompertz survival curve and discount factor rather than taken from a statute.

Table 111. The actuarially fair claiming adjustment, measured against age 63.

Retire at	Benefit multiplier	Adjustment (%)	Per year from the reference (%)
50	0.432	-56.8	-6.25
52	0.485	-51.5	-6.36
55	0.582	-41.8	-6.54
58	0.706	-29.4	-6.73
60	0.807	-19.3	-6.88
62	0.930	-7.0	-7.04
63	1.000	+0.0	—
65	1.164	+16.4	-7.30
67	1.366	+36.6	-7.50
70	1.771	+77.1	-7.84

Notes. Fair by construction in this model rather than approximately fair in someone else's: it is solved from the same survival law Section 17 uses.

That works out at 6.3% to 7.8% a year away from 63. The US schedule reduces a benefit by about 6.7% for each of the first three years claimed early and raises it about 8% for each year deferred past full retirement age. Nothing here was fitted to that; the agreement is a check on the survival law, not a coincidence to lean on.

The claiming rule decides the answer before leisure gets a say. Unadjusted, the best date is 52 even when working costs nothing at all. Adjusted, it is 60. Everything below is the adjusted arm.

32.2 When to stop

Table 112. The best retirement date at each value of leisure, γ = 5.

A retired year is worth	Retire at	Lifetime CEC	Runner-up	Margin (%)
+0%	60	0.7668	58	0.89
+5%	60	0.7345	58	0.39
+10%	58	0.7045	60	0.02
+15%	58	0.6787	60	0.37
+20%	58	0.6543	60	0.66
+25%	58	0.6313	60	0.91
+30%	58	0.6095	60	1.12
+40%	58	0.5695	55	0.28
+50%	55	0.5365	58	0.47
+75%	55	0.4685	52	1.16
+100%	52	0.4153	55	0.34

Notes. An optimum sitting on either end of the age grid would be the grid's answer rather than the model's; the grid runs from 50 to 70 for that reason.

Pricing the cost of working gives the date an interior optimum. It runs from 60 when a retired year is worth no more than a working one to 52 at the top of the grid. Section 31's corner is gone — not because the model changed its mind, but because both sides of the decision are now priced.

Weighting by survival pulls it earlier still, at 2 of the 11 values of leisure tested and by up to 3 years. Deferring is a bet that you will be there to collect, and the Gompertz law of Section 17 prices that bet where a certain ninety-third birthday cannot.

32.3 The break-even, which is the number to carry

An optimal date depends on a calibration nobody can hand you. A break-even does not. For each date earlier than the one an investor would choose if working cost nothing, this is the value of leisure at which it becomes worthwhile.

Table 113. What each year of earlier retirement costs to justify, against age 60.

Retire at	Years earlier	Leisure must be worth	which is a consumption drop of
58	2	9.7%	9%
55	5	31.3%	24%
52	8	60.4%	38%
50	10	81.2%	45%

Notes. The reference is the date that wins when working costs nothing, so what is priced is the earlier retirement rather than the model's own lean toward it.

That turns the question into one a reader can settle for themselves. Stopping 2 years early is a claim that a year of your own time is worth at least 10% of a year's consumption — about the size of the consumption drop actually observed at retirement. Stopping 10 years early is a much larger claim: 81%. Whether either is true, this paper does not say.

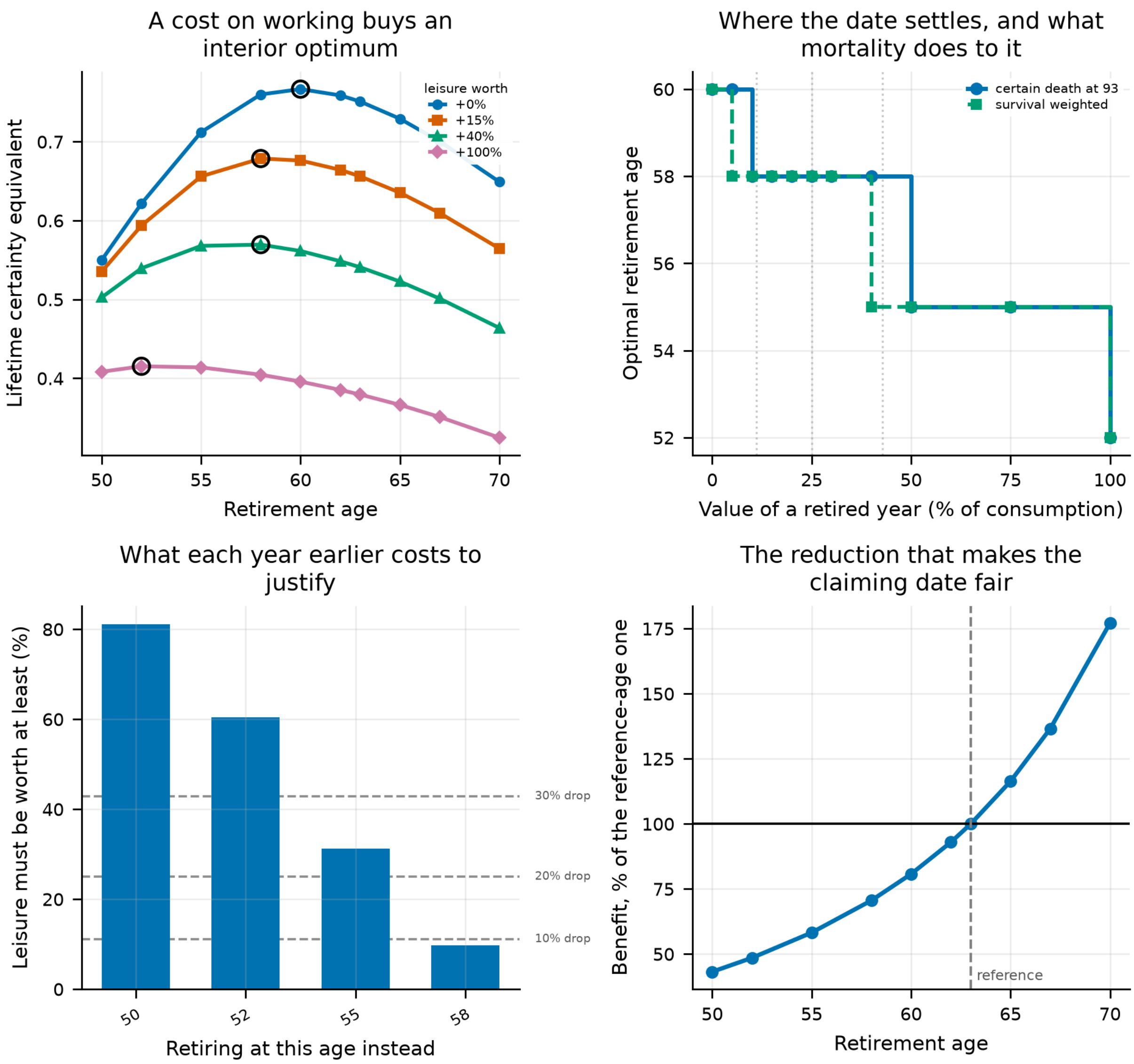

Figure 59.

Top left: lifetime certainty equivalent against the retirement date, one line per value of leisure, with each line's maximum circled — a cost on working buys an interior optimum. Top right: how that optimum moves, with and without survival weighting. Bottom left: the break-even value of leisure for each earlier date, against the consumption drops observed at retirement. Bottom right: the claiming adjustment that makes the date of the pension worth nothing either way.

32.4 The same question under Australia's pension

Everything above pays an American pension: earnings-related, payable from the day work stops, and adjusted for the age it starts at. Australia's is shaped differently in two ways that pull against each other. The Age Pension is payable at 67 however early somebody stopped, so there is no claiming choice to adjust and no bridge — an Australian retiring at 55 funds twelve years alone before a cent arrives. And the Superannuation Guarantee puts 12% of earnings into the portfolio on top of voluntary saving, which is what pays for crossing that bridge. Running the pension alone and then the pension with the guarantee separates the two.

Table 114. The retirement date under each pension, γ = 5.

Pension system	Best date, leisure free	at the top	Earlier dates any leisure justifies	Nearest earlier date costs	per year
US social security	60	52	4 of 4	9.7%	4.9%
Age Pension only	67	60	6 of 8	16.0%	8.0%
Age Pension + super guarantee	62	52	5 of 5	4.2%	2.1%

Notes. Break-evens are measured against each system's own zero-leisure date, so they price the earlier retirement rather than the system's own lean.

Australia's pension puts the date 7 years later. On the same voluntary saving the best date moves from 60 under the American schedule to 67. Nothing about the investor changed. But the two systems differ in *two* ways at once, so that number belongs to the pair and not to either of them — which is what Section 32.5 is for.

Compulsory saving takes some of it back. Adding the Superannuation Guarantee moves the date from 67 to 62, recovering 5 of the 7 between the two systems. That one is a single change — the guarantee is on or off and nothing else moves — so unlike the row above it can be read as a cause.

Net of both, an Australian's best date is 2 years later than an American's at the same voluntary saving (62 against 60). And it moves the *slope* as well as the start. A year earlier than the best date costs an Australian 2.1% of lifetime certainty-equivalent consumption against an American's 4.9%, a factor of 0.44. The Australian system does not shift one trade-off later so much as pose a softer one: the date is later *and* each year taken off it is cheaper. Both follow from the same gate — with the pension fixed at 67, a year of retirement bought before then is funded entirely from the portfolio, so it costs more than a year bought against a benefit that moves with it.

This arm needs a caveat the others do not. A retiree who exhausts the portfolio before the pension age receives, here, a means-tested payment at 60% of the full rate, standing in for the working-age safety net. At zero it would be literally nothing, and a CRRA aggregator punishes a single year of that without limit — one path in ten thousand at the consumption floor is enough to decide a certainty equivalent on its own. That floor is a judgement, and the earliest dates are sensitive to it in a way the later ones are not. Nor is preservation age modelled: this portfolio can be drawn at 50, where Australian super cannot.

32.5 Which of the two differences does the work

The comparison above changes two things at once. Australia's pension starts on a fixed birthday rather than when work stops, *and* it is worked out by a means test rather than from a career of earnings. Attributing the gap to either without crossing them is guesswork, so this is the 2×2: each feature alone, then both, against the same baseline.

Table 115. Each feature of Australia's pension against the American baseline, γ = 5.

What changes	Best date	Years later	CEC	vs baseline
when the benefit starts	58	-2	0.7754	+0.0086
how the benefit is worked out	67	+7	0.6292	-0.1376
both together	67	+7	0.6292	-0.1376
interaction	—	+2	—	-0.0086

Notes. The baseline is the American schedule: earnings-related, payable from retirement, actuarially reduced for claiming early.

It is how the benefit is worked out, not when it starts. Changing only the start date moves the best date by -2 years. Changing only the formula — a means-tested pension replacing an earnings-related one, still payable the day work stops — moves it +7. The eligibility gate moves the date the *opposite* way to the system it belongs to, and the reason is worth stating: a pension payable at a fixed age is not reduced for stopping early, whereas an actuarially adjusted one is. The gate removes a bridge and a penalty at once, and on this panel the penalty is worth more than the bridge.

The two do not add. The interaction is +2 years against a joint +7, or 29% of it — which is what one feature disarming the other looks like. Once the means test has taken the pension away, the birthday it would have arrived on stops mattering.

And the formula does so much because this household is not poor enough to be paid. The assets test begins withdrawing the pension at 3.0× average earnings and reaches zero at 6.8×. The median retiree here holds 13.8× — 2.0 times the cut-off — so 83% of them sit where the Age Pension has been tapered away entirely. Against a full rate worth 29% of career-average income, what is actually paid averages 3.9%; the American schedule, means-tested against nothing, pays 43%. So this is only partly a comparison of pension *designs*. It is substantially a comparison between a household that receives a pension and the same household receiving almost none.

32.6 The withdrawal rule is half the comparison

Everything above spends by this project's own rule: a fixed real amount set as a share of the portfolio at retirement. That rule cannot answer the question a pension is *for*. It sets the standard of living from the portfolio, so a larger balance spends more rather than lasting longer, and the pension is spent on top of the withdrawal rather than funding part of it. The alternative fixes the target instead — a set share of pre-retirement income for life, the pension netted off, the portfolio paying the rest.

Table 116. Each pension system under each withdrawal rule, γ = 5, retiring at 63.

Rule	Pension system	CEC	Ruin (%)	Mean c	5th pct c
fixed_real_rule	US social security	1.0422	11.7	1.729	0.708
replace_50	US social security	0.6008	2.5	0.852	0.438
replace_75	US social security	0.6706	11.7	1.136	0.454
replace_100	US social security	0.7521	24.8	1.373	0.500
fixed_real_rule	Age Pension only	0.4191	11.7	1.129	0.341
replace_50	Age Pension only	0.3389	10.3	0.755	0.255
replace_75	Age Pension only	0.4877	26.1	1.014	0.379
replace_100	Age Pension only	0.5737	41.0	1.190	0.449
fixed_real_rule	Age Pension + super guarantee	0.7185	11.7	2.520	0.452
replace_50	Age Pension + super guarantee	0.3327	1.9	0.797	0.250
replace_75	Age Pension + super guarantee	0.4950	6.8	1.162	0.374
replace_100	Age Pension + super guarantee	0.6352	13.4	1.485	0.449

Notes. The replacement rules target a share of average income over the final working years, held in real terms.

Under the portfolio rule all three systems ruin at the same rate, to the last decimal. That is an artefact, not a finding, and it is worth naming because this project has leaned on that rule throughout: withdrawals set as a share of wealth at retirement scale with the portfolio, so doubling the balance doubles the spending and exhausts it in the same year. Ruin under that rule is a property of the return sequence alone, and says nothing about how much was saved.

Hold the target still and the Superannuation Guarantee finally shows up where it should. At 100% replacement the Australian household runs out 11.4 percentage points less often than the American one. Compulsory saving buys safety — but only under a rule that permits safety to be bought.

And the certainty equivalent still favours the American schedule at every target. Both facts hold at once, and the combination is the result: the Australian household runs out *less often* and is *poorer when it does*. What remains once the portfolio is gone is a flat means-tested pension rather than an earnings-related one, and a CRRA aggregator at this risk aversion prices the depth of the floor above the frequency of reaching it. A reader who cares about the probability of running out and a reader who cares about the worst case will rank these two systems differently, and neither is misreading the table.

32.7 What this changes

- **The retirement date is not unpriceable — it was unpriced.** Charging for the years spent working turns Section 31's corner into an interior optimum between 52 and 60, depending on what a year is worth.
- **The claiming rule matters more than leisure does.** An unreduced pension starting whenever work stops is worth more than any plausible value of leisure and would have decided the answer on its own. That is worth stating for its own sake: a lifecycle model that lets the retirement date move must price the pension's start date, or it is not measuring preferences at all.
- **The output is a break-even, not a recommendation.** The value of your own time is not something this panel of returns can measure. What it can do is say what you must believe to justify a given date, and leave the believing to you.
- **What is not modelled**: partial retirement, which is what most people actually do; any change in the value of leisure with age or health, when both plainly change; an earliest claiming age, so the adjusted benefit here can start at fifty where no real system would pay it; and the possibility that work is worth something positive — purpose, company, structure — which would push the date later and which this parameterisation cannot represent, since L is bounded below at one.

33. The Tax Each System Actually Charges

Every result so far is tax-free, and until Section 32 that was harmless. A charge every strategy pays alike cancels out of a comparison between strategies, so the ranking of portfolios is unchanged by leaving it out. Comparing two *countries* is where the argument fails, because what differs between the American and Australian retirement systems is precisely which income is taxed, and when.

The asymmetry is the reason it could not be reasoned about. Australia charges early and lightly — on fund earnings through accumulation, and far less than the 15% headline implies — and then not at all: superannuation drawn after 60 from a taxed fund is not merely tax-free but *not assessable income*, so it cannot affect the tax on anything else. The Age Pension is assessable, but the seniors and pensioners offset lifts the effective threshold above the full single rate, so a pensioner living on the pension pays nothing. The United States charges late, progressively, and interactively: social security is taxable on a sliding scale whose thresholds have not been indexed since 1993, and a withdrawal from a traditional account is ordinary income *and* counts toward the provisional income deciding how much of the benefit is taxed.

33.1 The torpedo, measured

At the withdrawal this household actually makes, the American retiree faces a marginal rate of 22.2% while sitting in a 12% bracket — a factor of 1.85, which is the arithmetic ceiling: every dollar drawn drags up to eighty-five cents of benefit into the tax base behind it. The Australian retiree faces no such thing, because the withdrawal that would do the dragging is not assessable income.

The shape matters as much as the level. The marginal rate rises, then *falls* as the 85% inclusion completes, then rises again into the next bracket — so it is not monotone in income, and a retiree who looked up their bracket would be wrong about the cost of their next withdrawal across most of the range this household occupies.

33.2 What it does to the comparison

Table 117. Each system tax-free and under its own schedule, γ = 5.

Pension system	Tax regime	CEC	Ruin (%)	Tax paid (% of gross)
US social security	No tax	1.0409	11.9	0.0
US social security	Roth	1.0409	11.9	0.0
US social security	Traditional	1.0620	11.9	9.4
Age Pension only	No tax	0.4269	11.9	0.0
Age Pension only	Australia, super after 60	0.4269	11.9	0.0
Age Pension + super guarantee	No tax	0.7223	11.9	0.0
Age Pension + super guarantee	Australia, super after 60	0.7181	11.9	0.0

Notes. The two American rows are the honest pair: a Roth is what every earlier section implicitly assumed, and a traditional account is what superannuation deserves to be compared with, since both take contributions from pre-tax earnings.

Tax is worth +2.0% of certainty-equivalent consumption to an American saving in a traditional account, and costs 0.6% to an Australian. To an American saving in a Roth it costs nothing at all: with no other assessable income the benefit alone stays below thresholds frozen since 1993, and the torpedo needs other income to fire. The gap between the two countries moves from -30.6% to -32.4%, and the ranking holds.

The gap widens, by 1.8 percentage points — and which side moved matters, because a gap can widen from either end. 78% of the movement is the American arm *gaining* rather than the Australian arm losing: a pre-tax contribution grows by the tax not paid on it, and what is owed later is owed on a lower income, so the deferral is worth more than the retirement tax costs. Australia's charge is small at both ends — nothing in retirement, and very little during accumulation once unrealised gains, the capital-gains discount and franking credits are counted, which the next subsection sets out.

33.3 What the Australian fund actually pays

The statutory rate on fund earnings is 15%, and charging that against the return would overstate the charge by more than an order of magnitude. Three things stand between the headline and what is paid, and on this portfolio they very nearly cancel it. **Unrealised gains are not income**: a fund that holds owes nothing on appreciation, and because earnings in the retirement phase are exempt outright, a gain carried across that boundary is never taxed at all. **A realised gain held beyond a year is discounted by a third**, so the rate on it is ten per cent. And **a franked dividend carries a credit worth more than the liability** — Section 15 derives it on this same panel — so a domestic dividend is a refund that subsidises the tax on the international sleeve rather than adding to it.

Together they leave a drag of 0.073% a year against the 1.20% the statutory rate on the return would imply, a factor of 16. At a domestic share of 50% the income side comes out positive: the credit more than covers the fund's tax on the rest of the portfolio.

Table 118. The Australian charge by how much the fund is made to sell.

Sold each year	Drag a year	CEC	Ruin (%)	Against holding to the pension phase (%)
0.00%	+0.1282%	0.7297	11.9	+0.0
3.35%	-0.0727%	0.7181	11.9	-1.6
10.00%	-0.4718%	0.6956	11.9	-4.7
25.00%	-1.3718%	0.6483	11.9	-11.1
50.00%	-2.8718%	0.5788	11.9	-20.7

Notes. Holding until the pension phase is the reference. Section 22 measures 3.35% a year for this strategy, all of it forced by drift rather than chosen.

Holding until the pension phase is not a fanciful case — it is what an index fund left alone does. At the 3.35% a year Section 22 measures for this strategy, the charge costs 1.6% of the certainty equivalent.

What is still missing, in rough order of how much it would move this: the Australian marginal tax on earnings held outside super, since voluntary saving is treated as a Roth in both countries to isolate the retirement wrapper; capital gains tax and its one-third discount inside a fund, which would push the 15% on fund earnings down; US state income taxes, which would push the American charge up; and any filing status but single.

34. Discussion

34.1 What the replication does and does not establish

The central claim reproduces cleanly and survives an unusually wide robustness exercise. That is a real result and it should update anyone whose prior rested on i.i.d. US-only simulation. But it is worth being precise about what has been established.

What has been established is a statement about a particular return-generating process: if the future resembles a block-resampled draw from the developed-market twentieth century, then a diversified all-equity portfolio dominates a declining glide path on essentially every metric a lifecycle investor would use. What has *not* been established is that the future will resemble that process. Section 11 removes part of that gap by conditioning on the valuation a lifetime starts at, which is the piece of the objection an investor can actually observe. What remains outside the model is the rest of it: the secular decline in real yields, and whether the equity premium the twentieth century delivered is a structural feature or a realised surprise.

The robustness exercise is worth naming, because surviving one is a claim that has been made for weaker evidence. Four things were varied and none overturned the ranking: the preference and lifecycle parameters (Section 6), the membership of the panel one country at a time and the era it is drawn from (Section 8), the weighting of the international sleeve across five constructions (Section 9), and the fund costs of implementing it (Section 13). That exercise also produced the paper's most uncomfortable number. The delete-one jackknife of Section 8.4 puts a standard error on the headline gap wide enough that the interval barely excludes zero, and it applies to every other comparison here: sixteen countries is sixteen countries however many bootstrap paths are drawn through them. The direction of the results in this paper is better established than their magnitudes, and the magnitudes are quoted to more decimal places than the panel can support because that is what the simulation resolves, not what the evidence does.

That distinction matters most for the sceptic's strongest objection, which we take seriously: the sample is a sample of survivors at the level of the *system* even where it is not at the level of the country. Every country in the panel ended the century with a functioning capital market. A world in which that was not true is outside the support of the sampler, and no amount of block resampling can put it back in.

34.2 The hierarchy of levers

Reading the extensions together produces a ranking of what actually moves a retirement outcome, and it is not the ranking the industry's attention implies.

- **When you retire dominates.** The decade around the retirement date explains 5.9% of the variation in retirement consumption — more than the gap between any two allocation strategies tested here. It is also the one dimension no allocation rule can diversify.
- **How much you save dominates what you hold.** The savings-rate dimension of the tornado analysis moves the outcome by more than most allocation choices, and conditioning the savings rate is worth more than every allocation refinement in Sections 10 to 21 combined.
- **How you spend it matters more than expected.** The gap between the best and worst spending rules in Section 30 is comparable to the gap between the best and worst allocation strategies in Section 5.
- **What you hold matters, but mostly through diversification.** The all-equity advantage is real, and it is driven by the international leg rather than by equity exposure as such.
- **Currency hedging is not a lever but a leak**: Section 10 finds it loses certainty-equivalent consumption at every ratio tested before a single basis point of cost is charged, so there is no price at which it becomes worth doing. The fine structure of the glide path is worth less than a basis point at most ages, and freeing every portfolio weight at every age adds less than the difference between two adjacent spending rules.
- **Levering the portfolio is barely a lever** at household borrowing costs. The advantage rounds to nothing across most of the plausible range of spreads, and what it does buy is bought out of the left tail. **Levering one asset is a different question**, and Section 25.5 shows it has a different answer: at the same borrowing cost, a mortgage on the housing sleeve is worth several times what scaling the whole portfolio is worth, because it buys more of the holding that moves independently of the rest rather than more of what the investor already owns.
- **Borrowing is where age structure actually lives.** The two policies an unconstrained search makes genuinely age-varying in this paper are both borrowing schedules — the leverage ratio of Section 21 and the loan-to-value of Section 25 — and both decline. The equity share does not. The glide path the industry applies to equities has the right shape attached to the wrong instrument: what should fall as an investor ages is the debt, not the risk.
- **The valuation you start at is not an allocation lever at all.** It moves what a portfolio delivers without changing which portfolio to hold, so it belongs in a reader's expectations and their withdrawal planning rather than in their asset mix.
- **Widening the opportunity set is a bigger lever than refining it.** Adding a fifth asset class buys more, at a low enough holding cost, than every refinement within the four existing sleeves combined — and loses it all at a holding cost a real owner would recognise. Which asset classes are available is a more consequential question than how finely the weights among them are tuned.

A reader looking for the highest-value change to their own plan should read that list from the top, not from the bottom. The profession's attention is allocated in roughly the reverse order.

34.3 Why the pay cheque beats the portfolio

The single most surprising result in the paper is that the best state variable for a savings rule is labour income relative to its expected path, not the portfolio balance. The mechanism is worth spelling out because it generalises.

Both signals identify the same underlying problem — this path is falling behind — but they differ in *when* the information arrives relative to the consumption decision. An income shock is observed at the moment income is received and before it has been committed to consumption; saving a larger fraction of a windfall costs almost nothing in utility because the counterfactual consumption was never planned. A portfolio shortfall is observed after the consumption plan is set, so responding to it means cutting consumption that the household expected to have. Under a concave utility function those two are not the same transaction.

The practical implication is unusually clean. "Save a fixed fraction of every raise and every bonus" is a rule that requires no target, no balance lookup and no arithmetic, and on this model it outperforms the wealth-target advice that dominates financial planning.

34.4 Implications for default design

If the results here were taken at face value by a plan sponsor, the implied redesign would not be "raise the equity share of the target-date fund", though that is what the headline suggests. It would be a reordering of what the default does at all.

- **Default the contribution schedule, not just the portfolio.** Auto-escalation tied to pay increases captures the strongest signal in Section 27 and requires no member decision.
- **Default the drawdown policy.** Section 30 shows the choice of withdrawal rule is worth as much as the choice of portfolio, and it is almost never defaulted.
- **Stop quoting four percent.** On this panel the sustainable rate at a five-percent ruin tolerance is 3.2% even on the best strategy tested. A default that plans around four percent is planning around a one-in-seven failure.
- **Treat the retirement date as a plan variable.** The largest single source of outcome variance is the one members are given the least help with.

We state these as implications of the model, not as advice. The model has no disutility of labour, no taxes, no fees, no owner-occupied housing and no behavioural constraints, and each of those would temper the conclusions in Section 24's direction.

35. Limitations and Threats to Validity

This section is deliberately long. A replication that reports only the ways in which it succeeded is not much use.

35.1 The cross-section is sixteen countries

This is the largest weakness in the paper, and it is a weakness we chose deliberately. The developed-market universe runs to 38 markets. We cover 16, because the other 22 have no recorded return series in any openly licensed source. Section 3.6 audits what we do use; Section 3.6.1 reports what the excluded countries have and why it is not enough.

Every statement in this paper about international diversification is therefore a statement about an average over 15 other developed markets, weighted by history length and drawn jointly with the domestic leg. That is a real limit on how far the mechanism generalises, and in particular the panel is entirely advanced-economy and entirely survivor: markets that closed and never reopened are not in the source database, so the tail this paper measures is the tail of markets that came back.

Section 8 quantifies the part of this that is quantifiable. No single country carries the headline, and the ranking is stable across every window of history tested — but the delete-one jackknife puts a standard error on the headline gap that is wide enough to matter, and a reader should carry that interval to every other number in the paper.

Nothing here is estimated more precisely than sixteen countries allow, however many decimal places the simulation resolves. What Section 8 cannot address is the survivorship in the sample frame itself: deleting a country that is present says nothing about a country that was never there.

The last five years of the series are separately flagged as unverified in Section 3.6, on a variance test that every country in the sample fails in the same direction.

35.2 What the bootstrap cannot represent

- **No valuation conditioning.** Blocks are drawn without regard to the starting dividend yield or price-earnings ratio, so a simulated lifetime beginning at a market peak is statistically identical to one beginning at a trough. If long-horizon returns are predictable from valuations, the dispersion here is too wide and the mean too high.
- **No regime structure.** Block resampling preserves local persistence but not the possibility that the whole return-generating process changes. Secular declines in real yields, changes in the corporate tax base and the shift toward intangible capital are all invisible.
- **System-level survivorship.** Every country in the panel still had a functioning market in 2020. The set of histories in which that is false has zero probability under this sampler.
- **Blocks respect gaps, and gaps are informative.** A country with a wartime hole contributes fewer long blocks, so the sampler is mildly biased away from exactly the disrupted histories that motivate the exercise. We quantify this rather than correct it.
- **The realised backtest selects on windows, not just on countries.** Section 7 avoids the sampler entirely by running the lifetimes history actually served, but the filter that makes that possible — an unbroken run the length of a working life and a retirement — removes every window that straddles a catastrophe rather than following one. The markets whose equity was destroyed therefore contribute only the lifetimes that begin at the floor of the recovery. That biases the comparison toward domestic equity, so the lead reported there is conservative in direction; it also means the spread across countries in that section is not the spread a randomly chosen saver faced.

35.3 What the lifecycle model omits

- **One tax, and only one.** Section 14 prices foreign dividend withholding, because it is the tax that falls unequally on the two legs this paper compares and is therefore the one that can change the answer — and it does, inside the statutory range. Every other tax is absent: returns are otherwise pre-tax, withdrawals untaxed, and there is no distinction between a taxable and a tax-deferred account. The asymmetric treatment of capital gains and income would change the ranking of spending rules in Section 30 more than it would change the allocation ranking; dividend imputation, which refunds corporate tax to domestic shareholders in Australia and New Zealand, would push the same way as withholding and is not modelled either.
- **Every return is gross.** No fee is charged anywhere in the baseline. Section 13 prices the omission rather than repairing it, by finding the fee differential that would undo the headline, and Section 22 does the same for the cost of trading. Both leave the baseline itself gross, so every level quoted outside those two sections is a number no investor receives.
- **No housing in the baseline.** For most households the primary residence is the largest asset and the mortgage the largest liability, and both interact with inflation in ways the financial portfolio does not. Sections 24 and 25 add housing and leverage against it to the opportunity set, but every other result in this paper is computed without them. Section 3.6.2 audits the 1,805 country-years of observed housing returns behind that, and explains why an appraisal-smoothed index cannot be used as written.

- **No economy-wide wage growth.** The income profile is an age effect only. Section 3.6.3 measures what that leaves out — a median 1.41% a year across 18 countries, which compounds to 1.68× over a career — and finds the bias runs against our conclusion, because less human capital weakens rather than strengthens the case for early equity.
- **No annuities.** A real annuity is the natural competitor to a withdrawal rule and is absent entirely.
- **One country's pension behind sixteen countries' returns.** Every result outside Section 18 pays the US primary-insurance-amount schedule, and that is not a neutral choice: an earnings-related benefit paid regardless of wealth is a risk-free real annuity, which puts a floor under retirement consumption and flatters equity. Section 18 is the one place this is tested, against Australia's means-tested Age Pension and compulsory Superannuation Guarantee, and it moves the level further than any parameter swept anywhere else in this paper. The ranking survives there only because the guarantee lifts the model's saver clear of the means test; a saver inside the taper band gets a different answer, and nothing else in this paper speaks to them.
- **No disutility of labour.** This is why the retirement-timing results of Section 28 require a matched comparison, and it means the model can never say when someone *should* retire, only what conditioning the date is worth.
- **A fixed horizon everywhere but one section.** Death arrives at 93 with certainty in every result except Section 17, which re-weights the headline under four Gompertz survival laws and finds the ranking unmoved. That is a narrower finding than it sounds: it re-weights a policy that does not itself respond to longevity, so a small answer is partly built in. Nothing here prices an annuity, or a spending rule that adapts to a mortality table, and both are where the value of longevity risk actually sits.
- **No behavioural constraints.** Every rule here assumes the investor executes it. Section 27.6 prices the cost of a constrained contribution but nothing prices the probability that a saver abandons an all-equity portfolio in the middle of a 60% drawdown.

35.4 Specification sensitivities we know about

Three modelling choices are load-bearing and a reader should know where they bite.

- **The bequest shift.** Without De Nardi's κ, a single path with exactly zero terminal wealth sends the certainty equivalent to negative infinity at $\gamma \geq 1$. The value of κ is not identified by anything in the data and it changes the ranking of spending rules.
- **The evaluation window.** Allocation comparisons use the retirement window because working-life consumption is identical across strategies by construction; policy comparisons that change working-life consumption use the whole lifetime. Mixing the two would produce incoherent comparisons and we flag every switch.
- **The working-income floor.** Retirement carries a consumption floor through the progressive benefit schedule and working life, at zero, carries none. That asymmetry cancels in the allocation comparisons and does not cancel in the timing ones, where leaving it uncorrected inflates the value of retiring early by roughly a factor of two.

35.5 What the new searches assume

- **The simplex search is a coordinate search.** Section 20 reports restarts from three corners and they agree, but coordinate ascent offers no guarantee of a global optimum in a 204-dimensional space and none is claimed.

- **The opportunity set is still narrow.** Section 24 adds housing as a fifth asset, but there is no credit, no commodities and no inflation-linked bond, and an inflation-linked bond in particular would change the fixed-income result of Section 20 more than any refinement within the sleeves that are present.
- **The valuation conditioning applies to the first block only.** A lifetime is a chain of calendar windows and only the first is a starting condition; the rest are the future, which no investor chooses. Section 11 therefore measures the effect of the opening decade's valuation on a sixty-eight-year outcome, diluted by everything that follows. A design that made the whole chain valuation-dependent would report a larger effect and would be assuming a great deal more.
- **The de-smoothing of housing is itself a model.** Section 24 inverts a first-order filter, which is the standard correction and not necessarily the right one: if house price indices carry higher-order smoothing, the true volatility is higher than the correction restores and housing is worth less than reported. The direction of that error is known even though its size is not.
- **The mortgage is rebalanced annually.** Section 25 redraws the loan every year at no cost to hit a target loan-to-value. Real mortgages amortise on a fixed schedule, cost several percent of the property to refinance, and are called on missed payments rather than on a drifting ratio. The section reports the value of the leverage, not a financing plan, and it prices no mortgage insurance and no tax deductibility in either direction.
- **Housing is priced as an index, not as a house.** The asset in Section 24 is a liquid, continuously rebalanced, nationally diversified claim on the housing stock. A single leveraged owner-occupied property shares almost none of those properties, and nothing in this paper speaks to it.
- **Leverage is modelled with limited liability** and a floating borrowing rate, rebalanced annually. Real margin lending liquidates at a threshold rather than at zero and reprices continuously, both of which would make borrowing less attractive than Section 21 finds it.
- **No borrowing constraint binds by quantity.** The model lets an investor borrow three times their financial capital at a spread; no lender would extend that against a retirement account.

35.6 Statistical caveats

Certainty equivalents are reported without standard errors. Under common random numbers the *differences* between policies are far more precisely estimated than their levels, which is what the comparisons rely on, but a reader wanting a confidence interval on any single level will not find one here. The optimisers are coordinate-ascent procedures on a grid: exact per coordinate under common random numbers, but with no guarantee of a global optimum in the joint space. Multiple restarts are reported where it matters.

- **The cohort interval rests on 16 clusters.** The confidence interval in Section 7 is a cluster bootstrap over countries, and a percentile interval built on 16 clusters has poorer coverage than its width suggests. The cohorts are also unequally distributed across those clusters, so the pooled mean is implicitly weighted by how much unbroken history a country happens to have. Both readings are reported there; the country-equal-weighted one is the conservative one.
- **One split is one experiment.** Section 23 cuts the record once and solves on either side. The asymmetry it finds is real in the sense that it is what the data say, but a single cut cannot separate a schedule that was overfitted from a world that changed, and with one experiment there is no distribution to judge the asymmetry against. It should be read as a caution rather than as a test with a size.

36. Conclusion

We set out to reproduce a specific empirical claim and ended up with a hierarchy. The claim reproduces: on a 16-country panel of the developed-market twentieth century, sampled in blocks so that its persistence and its tails survive, a diversified all-equity portfolio delivers 11.2% more certainty-equivalent retirement consumption than a target-date glide path while failing less often. Solving for the allocation schedule directly rather than testing candidates returns the same answer, and the ranking holds across every parameter dimension we sweep. Freeing all four portfolio weights at every age — 204 parameters on the simplex, with nothing held fixed — reaches the same shape again and adds 0.30%; relaxing the long-only constraint pays well only when credit is nearly free, and breaks even by a 3.00% spread over the real bill rate.

But the extensions matter more than the replication. The single decade around a person's retirement date explains 5.9% of the variation in their retirement outcome — more than the entire allocation question. Conditioning the savings rate on financial position is worth more than every allocation refinement we test. And when we decompose that signal properly, the best state variable available to a saver turns out not to be their portfolio at all but their own pay cheque relative to what they expected to earn.

Two of the extensions relax assumptions the model itself was making. Conditioning on the valuation a lifetime starts at — the objection a reader is most entitled to raise, since they know what their own market costs — leaves the allocation answer where it was and moves the level: valuation tells an investor what to expect, not what to hold. Adding housing to the opportunity set shows an asset that is genuinely competitive once its index is de-smoothed, and that stops being competitive at an annual holding cost low enough that a real owner's rates, insurance and maintenance decide the question.

Several of the results here came out against the intuition that motivated them, and we have tried to report those cases with their diagnostics rather than smoothing them into a tidier story: the glide-path structure that dissolved under a deviation profile; the retirement-timing premium that halved under a matched baseline; the savings conditioning that turned out not to be tail insurance; the bull-market warning that the data did not support. A replication is most useful when it reports what it found rather than what it expected, and the machinery for telling those apart — matched baselines, common random numbers, deviation profiles, grid-edge checks — is as much a contribution here as any individual number.

One assumption deserves to outrank the rest of the caveats, because it moves the numbers further than any parameter we sweep. Every result outside Section 18 pays the same public pension in all sixteen countries — the American one, progressive in career earnings and paid whatever else the retiree owns. That is a risk-free real annuity, and a great deal of what looks like portfolio performance in this paper is standing on it. Replacing it with the Australian pair — a means-tested Age Pension and a compulsory Superannuation Guarantee on top of voluntary saving — raises average retirement consumption and lowers the certainty equivalent at the same time, because the compulsory contribution buys a larger portfolio and the means test removes the floor that portfolio would have sat on. Both halves of that are true, and which one a reader should care about depends on whether they are planning for the average case or the bad one.

The allocation ranking does not survive that substitution, and this is the only place in the paper where it fails. Under the Australian pair the lead of 5.73% becomes -6.42% and the best strategy becomes *Target-date fund* — the de-risking glide path this paper spends most of its length arguing against. The mechanism is the taper, not the returns: inside the assets-tested band every extra dollar of assets costs more pension a year than any asset in this panel reliably earns, so a portfolio that stays smaller keeps an entitlement that a portfolio that grows loses. A glide path does exactly that, and under a means test it is being rewarded for it. Nothing about the return panel has changed; the objective function has.

The lesson is not about countries. It is that this paper, like the study it replicates, models an investor whose retirement income is a portfolio plus an unconditional annuity, and its conclusions are conclusions about that investor. A saver whose public entitlement is withdrawn at the margin as their balance grows faces a different problem, and the answer to it is different. Which of the two a reader is depends on their balance and their country's schedule, not on anything in the return data.

Two further audits of the optimisation sections point the same way as each other and should be read together. Solving a schedule on one half of the record and scoring it on the other leaves a constant mix as the benchmark to beat in every run; charging the same solved schedule for the trades it makes takes most of what remains. Neither touches the paper's headline, which compares fixed portfolios fitted to nothing — but both say that the gains from solving for an allocation are smaller, and more fragile, than the in-sample numbers make them look.

One caution belongs in the summary rather than the appendix. Deleting each country in turn and rebuilding the panel around its absence leaves the ranking intact every time, and the ranking is stable across every window of history we can stand in — but the same runs, read as a jackknife, put a standard error on the headline gap wide enough that the interval barely excludes zero. Sixteen countries is sixteen countries. Everything above should be read as well established in direction and thinly established in magnitude, and a reader who wants one thing from this paper should take the ordering rather than the size of any gap in it.

The practical summary is short. On this evidence, an investor's attention is best spent, in order, on when they retire, on how much and when they save, on how they spend the portfolio down, and only then on what it holds — and when they get to what it holds, the answer is more diversified equity than the default gives them, for longer than the default holds it.

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Appendix A. Model Parameters

Every parameter below is read from a single YAML configuration file. No value is hard-coded in the simulation modules, which is what makes the sweeps of Sections 6, 21 and 24 mechanical rather than manual.

Table 119. The complete baseline parameter set

Parameter	Symbol	Value	Where it is used
Start age	—	25	All
Retirement age	—	63	All; varied in §6.4 and made endogenous in §28
Death age	—	93	All; varied in §6.4
Savings rate	s	10%	All; solved in §26 and conditioned in §27
Income profile (linear)	b_1	0.045	Labour income
Income profile (quadratic)	b_2	-0.0009	Labour income
Permanent shock s.d.	σ_p	0.10	Labour income; scaled in §27.9
Transitory shock s.d.	σ_t	0.25	Labour income; scaled in §27.9
Social-security formula	—	progressive	Retirement income floor
PIA bend point 1	—	0.21	× economy-wide average earnings
PIA bend point 2	—	1.28	× economy-wide average earnings
PIA replacement rates	—	90% / 32% / 15%	Successive tranches of career-average earnings
Withdrawal rule	—	fixed_real_rule	Baseline; eight families compared in §30
Withdrawal rate	—	4%	Baseline; swept in §5.4 and §6.4
Discount factor	β	0.96	Utility
Risk aversions	γ	2, 5, 10	Baseline γ = 5
Elasticities of substitution	ψ	0.5, 1.5	Epstein–Zin only
Bequest weight	θ	2	Utility; pivoted in §30.1
Bequest shift	κ	1	De Nardi (2004) specification
Consumption floor	—	0.0001	Numerical guard only
Evaluation window	—	retirement	Allocation comparisons; whole lifetime in §28–§27
Paths per strategy	N	100,000	All
Horizon	H	68	All
Mean block length	—	10 years	Bootstrap; swept in §4.1
Block length range	—	1–40 years	Bootstrap
Country draw	—	per_lifetime	Bootstrap; varied in Appendix C
Country weighting	—	history	Bootstrap; varied in Appendix C
Master seed	—	20240115	Reproducibility

Notes. Section references indicate where each parameter is varied. Parameters specific to a single extension (hedging cost, retirement trigger multiple, savings-rate grids, accumulation response grids) are documented in the configuration file and in the relevant section.

Appendix B. The Country Panel

The full panel, with real domestic equity statistics for each. Every country listed carries complete Jordà–Schularick–Taylor equity, bond and bill total returns over the years shown.

Table 120. Real domestic equity returns, every country in the panel

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	Kurtosis	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	0.4	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.0	0.08
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	1.8	0.04
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	5.2	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	3.7	0.15
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	4.5	-0.01
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	2.9	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	1.7	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	2.9	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	1.6	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	2.4	0.25
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	2.1	0.32
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.5	0.13
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.4	0.10
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	5.7	-0.05
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.1	0.00

Notes. Annual real total returns deflated by each country's own consumer price index. Kurtosis is excess kurtosis.

B.1 Structural gaps

The 16 contiguous runs of missing data in the panel's countries, preserved as gaps rather than interpolated. The bootstrap never draws a block that crosses one.

Table 121. Structural gaps in the empirical panel

Country	From	To	Years	Missing series
Australia	1890	1899	10	bond
Australia	1945	1947	3	bill
Belgium	1914	1919	6	bond, bill
Switzerland	1890	1899	10	dom_eq, bond
Switzerland	1915	1915	1	bond
Germany	1923	1923	1	bill
Germany	1944	1949	6	bond, bill
Denmark	1915	1915	1	bond
Spain	1890	1899	10	dom_eq, bond
Spain	1936	1940	5	bond, bill
Spain	2018	2018	1	bill
Finland	1890	1895	6	dom_eq
France	1915	1921	7	bill
Italy	1915	1921	7	bill
Japan	1946	1947	2	dom_eq
Netherlands	1890	1899	10	dom_eq

Notes. Concentrated around the two world wars, which is precisely the period a survivorship-prone sample would drop.

Appendix C. Supplementary Tables

C.1 Bootstrap sampling variations

The two structural choices in the sampler — drawing the country once per lifetime rather than per block, and weighting countries by history length rather than uniformly — are varied here. Neither reverses any ranking.

Table 122. Redrawing the country at every block

Strategy	Strategy	Cec Γ=5	P(Ruin)	Median Bequest
50/50 domestic/international equity	50/50 domestic/international equity	1.046	11.8%	43.028
100% domestic equity	100% domestic equity	0.878	25.9%	14.222
100% international equity	100% international equity	1.105	9.1%	63.743
60/40 domestic equity/bonds	60/40 domestic equity/bonds	0.866	24.9%	9.002
Target-date fund (glide path)	Target-date fund (glide path)	0.939	22.8%	8.726
100% bills (cash)	100% bills (cash)	0.737	57.5%	0.000

Table 123. Weighting countries uniformly rather than by history length

Strategy	Strategy	Cec Γ=5	P(Ruin)	Median Bequest
50/50 domestic/international equity	50/50 domestic/international equity	1.047	11.8%	44.715
100% domestic equity	100% domestic equity	0.881	25.4%	16.672
100% international equity	100% international equity	1.109	9.0%	64.113
60/40 domestic equity/bonds	60/40 domestic equity/bonds	0.870	24.2%	10.156
Target-date fund (glide path)	Target-date fund (glide path)	0.942	22.3%	9.336
100% bills (cash)	100% bills (cash)	0.739	56.4%	0.000

C.2 The income and benefit schedules

Table 124. The deterministic labour-income profile

Age	Real income	Multiple of age-25 income
25	1.000	1.000
30	1.224	1.224
35	1.433	1.433
40	1.604	1.604
45	1.716	1.716
50	1.755	1.755
55	1.716	1.716
60	1.604	1.604

Notes. Before permanent and transitory shocks. The hump peaks in the early fifties, which is what makes the solved savings profile of §26.2 hump-shaped at moderate risk aversion.

Table 125. The progressive social-security benefit schedule

Career-average earnings	Annual benefit	Replacement rate
0.307	0.276	90.0%
0.628	0.388	61.8%
0.949	0.491	51.7%
1.270	0.593	46.7%
1.591	0.696	43.8%
1.911	0.799	41.8%
2.232	0.856	38.3%
2.553	0.904	35.4%
2.874	0.952	33.1%
3.195	1.000	31.3%
3.516	1.048	29.8%
3.837	1.096	28.6%

Notes. US primary-insurance-amount bend points. The falling replacement rate is what supplies a genuine real consumption floor; a flat rate would scale with the outcome and insure nothing.

C.3 Long-horizon bootstrap outcomes

Table 126. Distribution of 68-year annualised and cumulative real returns

Series	Mean	1st pct	5th pct	Median	95th pct	99th pct	1st pct cumulative	Median cumulative
Domestic equity	4.43%	-5.02%	-1.53%	4.82%	8.89%	10.70%	-0.97	23.5
International equity	6.61%	1.66%	3.26%	6.65%	9.80%	11.21%	2.07	78.8
Bonds	1.33%	-6.68%	-3.29%	1.65%	4.66%	6.07%	-0.99	2.0
Bills	0.24%	-7.22%	-4.01%	0.69%	2.84%	3.67%	-0.99	0.6
Inflation	4.62%	0.80%	1.59%	3.96%	10.05%	13.51%	0.72	13.0

Notes. Annualised figures are geometric. The first-percentile cumulative multiple for domestic equity is below zero in real terms — a lifetime of holding a single national market that ended with less purchasing power than it started with. That outcome is the reason the international leg matters.

Appendix D. Computational Design

Every number, table and figure in this paper is produced by one program from the raw source files, and the document itself resolves its quoted figures from that program's output when it is typeset. Nothing is transcribed by hand. A rerun that changed a result would change the paper rather than leave it silently contradicting its own evidence.

This appendix describes how the computation is organised and how its correctness is established, because both bear on how much weight the results can carry.

D.1 Structure

The work is split into stages that each own one idea, so that an extension can reuse the baseline machinery instead of reimplementing it. Reuse is the point: every extension in this paper is compared against a baseline, and a comparison between two different implementations of the same model measures the implementations as much as the economics.

Table 127. How the computation is organised

Stage	What it owns	Paper section
Panel construction	Deflation to real returns, the leave-one-out international leg, currency-hedged legs, coverage and gap diagnostics, per-cell provenance	§3
Sampler	The stationary block bootstrap with gap-respecting admissibility; moment and persistence diagnostics	§4.1
Lifecycle simulator	The wealth recursion, income process, social-security schedule and strategy definitions	§4.2
Preferences	CRRA and Epstein–Zin certainty equivalents, the bequest term, shortfall and ruin metrics	§4.3
Sweep engine	Parameter sweeps under common random numbers; tornado and crossover analysis	§6
Spending rules	Eight families of withdrawal policy behind a single interface	§30
Glide-path solver	A batched evaluator and coordinate ascent over the age-by-asset schedule	§19
Allocation solver	The weight simplex solved at every age: lattice search then pairwise-exchange ascent, over four assets or five	§20, §24
Leverage	A levered evaluator, the cost-of-credit sweep and the age-varying leverage schedule	§21
Hedging	Covered-interest-parity hedged legs, break-even cost and optimal hedge ratio	§10
Path-dependent engine	Endogenous retirement dates and state-conditioned saving	§28–§27
Savings rules	Rule families, the fixed-mean shape solver and matched-rate scoring	§26–§27
Valuation	The look-ahead-free trailing dividend yield, the structural no-leak check and the valuation buckets	§11
Housing	De-smoothing the appraisal index, and the five-asset simplex re-solved at each holding cost	§24
Mortgage	Leverage applied to the housing sleeve alone, and the loan-to-value schedule solved by age	§25

D.2 Performance

Portfolio returns for a whole cohort are formed as a batched matrix product and the wealth recursion runs vectorised across paths. That is what makes a hundred thousand lifetimes per strategy — and the tens of thousands of policy evaluations the optimisers require — tractable on a single core, which in turn is what makes the sensitivity analysis of Section 6 affordable enough to run exhaustively rather than selectively.

D.3 Verification

Correctness rests on 1244 automated tests. The ones that carry the most weight are equivalence tests between simulators, because every extension here is built on an engine that must agree with the one that produced the baseline:

- The path-dependent engine reproduces the fixed-date engine *bit for bit* given a fixed retirement age and a constant savings rate.
- The batched glide-path evaluator reproduces the reference simulator to a stated floating-point tolerance.
- Zero-sensitivity conditioning rules reproduce their own base age profile exactly, which is what makes the incremental values in §27 meaningful.
- The block sampler never draws across a data gap, verified against the run-length structure directly.
- A savings rule is never handed a return it has not yet lived through — the no-lookahead property is asserted, not assumed.
- Every available cell of the panel is an observation, checked against a per-cell record of where each number came from.

A further discipline runs through the prose. Where this paper states a direction — that one quantity exceeds another, that a ranking survives, that an effect has a sign — the statement is classified from the computed table when the document is typeset rather than written down in advance. The mechanism exists so that a result which moves takes its description with it, rather than leaving prose that quietly contradicts the table beneath it.

D.4 List of figures

D.5 List of tables